

# Scalp, Temple, and Face

The scalp, temple, and face are important areas of the head, and therefore, need to be studied thoroughly. The injuries are frequently inflicted in these areas. The nature of injuries varies from superficial wounds to deep cuts. The scalp and temple are covered by thick hair; hence lesions at these sites go unnoticed for quite sometime by the people. Infection from these regions can travel inside cranial cavity through venous channels leading to fatal consequences. The face is the commonest site for plastic surgery done to enhance the beauty or to repair congenital defects and defects produced by the injuries. The simple inspection of face provides a substantial clue to a number of underlying bodily diseases. Also, the face being the exposed part of the body is prone to sun allergy and skin cancer.

## SCALP

The term *scalp* is applied to the soft tissues covering the vault of skull.

### Extent

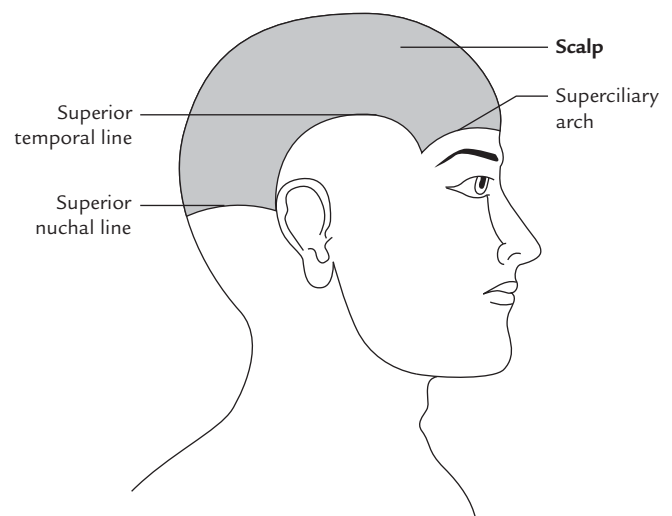
It extends anteriorly up to the eyebrows (superciliary arches), posteriorly up to the superior nuchal lines, and laterally on each side up to superior temporal line (Fig. 3.1).

**N.B.** According to some authorities, scalp extends laterally on each side, up to the zygomatic arch.

### LAYERS

The scalp consists of five layers. From superficial to deep these are as follows (Fig. 3.2):

1. Skin.
2. Connective tissue (superficial fascia).
3. Aponeurosis (occipitofrontalis muscle and its aponeurosis).



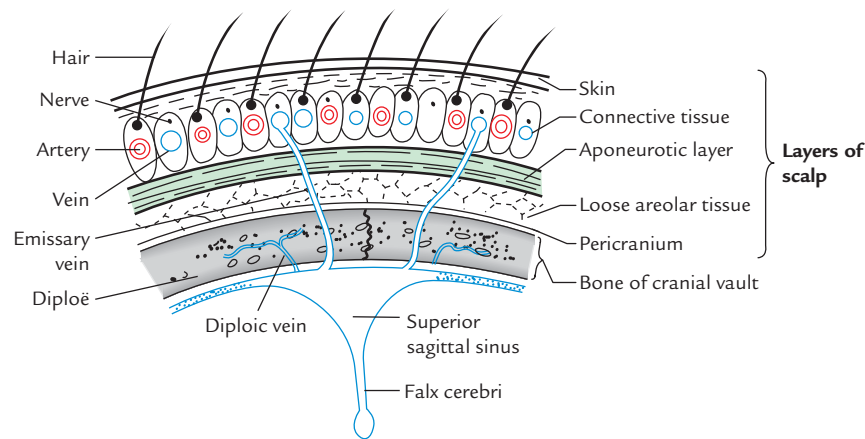
**Fig. 3.1** Extent of the scalp.

4. Loose areolar tissue.
5. Pericranium.

The sequence of the layers of scalp can be easily remembered by a perfect **mnemonic SCALP** derived by using initial letter of each layer.

**Skin:** The skin of scalp is thick and hairy except over the forehead. It is firmly adherent to epicranial aponeurosis by dense connective tissue of superficial fascia, as in palms and soles. Being hairy it contains maximum number of hair follicles and associated sebaceous glands. As a result, scalp is the commonest site of **sebaceous cysts**. It also contains numerous sweat glands.

**N.B.** It has been estimated that there are about 1,20,000 hair on the scalp of an adult individual. About 20–100 hair are lost daily and replaced concomitantly. The baldness (loss of hair) mainly affects males and few elderly females with high level of androgenic hormones in their blood.



**Fig. 3.2** Layers of the scalp.

**Connective tissue (superficial fascia):** The superficial fascia of the scalp is made up of dense fibrous connective tissue that firmly binds the skin to the underlying occipitofrontalis and its aponeurosis. Fibrous septa divide this layer into numerous small pockets containing lobules of fat. The blood vessels and nerves of the scalp lie in this layer. They enter it from below at the periphery.

The walls of the vessels are adherent to the fibrous network; hence when blood vessels are torn or cut during an injury, they are unable to retract and cause *profuse bleeding*. The bleeding, however, can be stopped by pressing against the underlying bone.

**Aponeurosis:** This layer is formed by occipitofrontalis muscle and its aponeurosis. The occipitofrontalis muscle consists of four small bellies: two frontal bellies and two occipital bellies. Since greater part of this layer is formed by aponeurosis, it is called *aponeurotic layer*. The aponeurosis of occipitofrontalis muscle is also called *epicranial aponeurosis* or *galea aponeurotica* (Latin: *galea* = helmet).

The wounds of the scalp do not gape unless epicranial aponeurosis is cut transversely because the aponeurosis is under tension in the anteroposterior direction by the tone of occipitofrontalis muscle.

**Loose areolar tissue:** As the name indicates, this layer is made of loose areolar tissue. It serves as a natural plane of cleavage during craniotomy. This layer is traversed by emissary veins connecting veins in the second layer of scalp with intracranial dural venous sinuses.

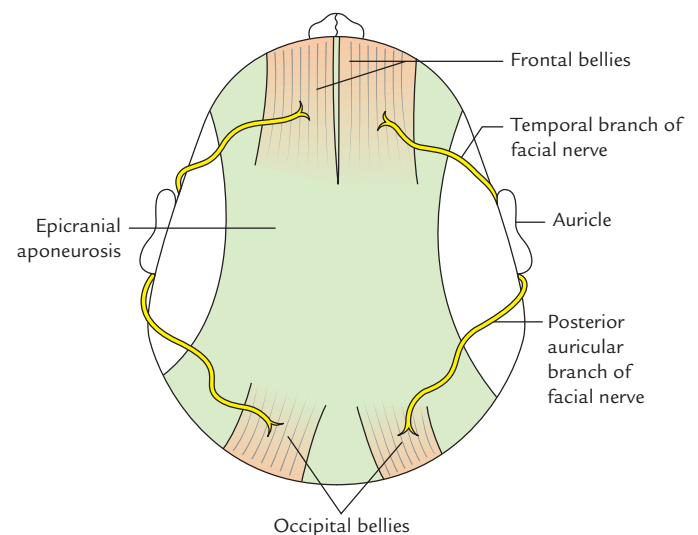
**Pericranium:** The fifth layer of scalp is formed by the periosteum of bones of vault of skull called *pericranium*. It is loosely attached to the bones and can be easily stripped, but at sutures it is firmly attached to sutural membrane, which in turn attaches it to the endocranium (the periosteum covering inner aspect of the skull bones).

### Further Details

#### **Occipitofrontalis muscle (Fig. 3.3)**

##### *Origin and insertion:*

1. The frontal bellies arise from skin and subcutaneous tissue over the eyebrows and root of the nose, run backwards to be inserted into epicranial aponeurosis in front of coronal suture. They have no bony attachment of their own. The deeper fibres of frontal belly merge with procerus, corrugator supercilii, and orbicularis oculi muscles. The front bellies are longer, wider, and partly united with each other along their medial borders.
2. The occipital bellies arise from lateral two-third of the superior nuchal lines and extend forwards to be inserted into the epicranial aponeurosis. The two occipital bellies are small and separated from each other by a considerable gap. This gap is filled by epicranial aponeurosis.



**Fig. 3.3** Occipitofrontalis muscle.

**Nerve supply:** The occipital belly on each side is supplied by *posterior auricular branch of facial nerve* whereas the frontal belly is supplied by *temporal branch of the facial nerve*.

#### Actions

1. Alternate contractions of frontal and occipital bellies move the scalp forwards and backwards over the vault of the skull.
2. Acting from above, the contraction of frontal bellies raise the eyebrows as in surprise.
3. Acting from below, the contractions of frontal bellies produce transverse wrinkles on the forehead as in fright.

### Clinical correlation

- **Surgical layers of the scalp:** First three layers of the scalp, i.e., skin, connective tissue layer, and aponeurotic layer are firmly adhered to each other and cannot be separated from each other. These layers are termed *surgical layers of the scalp* and form the *scalp proper*.

The layer of loose areolar tissue beneath the aponeurotic layer accounts for the free mobility of the scalp proper on the underlying bone. Further, it provides an easy plane of cleavage in injury and a plane in which blood from severed blood vessels can spread for a long distance. When the hairs are caught in machinery, the scalp proper is avulsed. The Red Indians of bygone days used to remove the scalp (proper) of their victims as punishment. It is in this plane that surgeons mobilize scalp flaps.

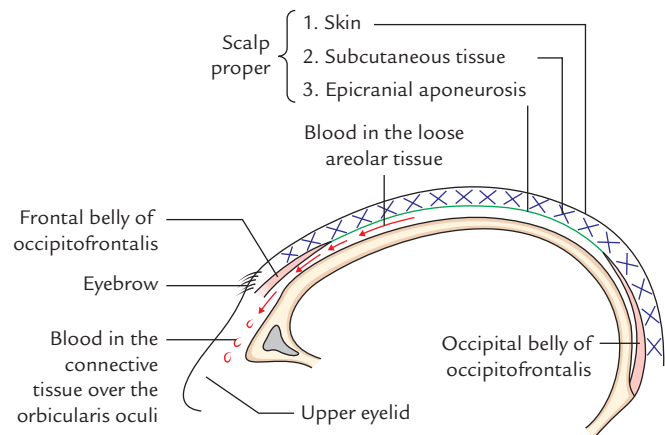
- **Black eye:** The blood and fluid collecting in the layer of loose areolar tissue following a blow on head tracks freely under the scalp producing generalized swelling over the dome of the skull, but cannot pass into either occipital or temple regions because of the bony attachments of the occipitofrontalis. The blood and fluid can, however, track forward into the eyelids because occipitofrontalis has no bony attachment anteriorly (Fig. 3.4). This leads to formation of hematoma few hours after a head injury or cranial operation causing black discoloration of skin around the eyes, a condition called **black eye**.

It is important to note that the commonest cause of black eye is local violence, such as fist fight causing subcutaneous extravasation of blood into the eyelids.

- **Dangerous area of the scalp:** The layer of loose areolar tissue is called *dangerous layer of scalp* because blood and pus freely tend to collect in this layer. If pus collects in this layer, the infection may travel readily along emissary veins into the intracranial dural venous sinuses leading to their thrombosis, which may be fatal.
- **Safety-valve hematoma, cephalhematoma and caput succedaneum**
  - Fracture of cranial vault in children may be associated with the tearing of dura mater and pericranium. In such cases the blood from intracranial hemorrhage communicates with the subaponeurotic space of the scalp through the fracture lines. The signs of cerebral compression do not develop until the subaponeurotic space is fully filled with blood. For this reason the collection of blood in the fourth layer is called a *safety-valve hematoma*.

If cerebrospinal fluid collects in the subaponeurotic space, the condition is called *cephalhydrocele*.

- **Cephalhematoma** is a subperiosteal collection of blood. Since the periosteum of skull loosely covers the bones of skull except at the sutural lines where it is firmly attached to the sutural membranes, the hematoma is bound by suture lines and assumes the shape of related bones. It is firm and its edges are well-defined. A cephalhematoma is commonly found in the parietal region.
- **Caput succedaneum** is a subcutaneous edema over the presenting part of the head at delivery. It takes place during the passage of head through the birth canal due to interference of the venous return. It is the most common form of birth trauma of the scalp and usually occurs over the occiput and crosses the suture lines. The affected parts of the scalp feel soft and margins are partly defined. Generally the edema subsides, in a few days.



**Fig. 3.4** Sagittal section of the skull cap and overlying soft tissues. Note the migration of blood from 4th layer of the scalp (layer of loose areolar tissue) to the scanty subcutaneous tissue over orbicularis oculi.

### NERVE SUPPLY

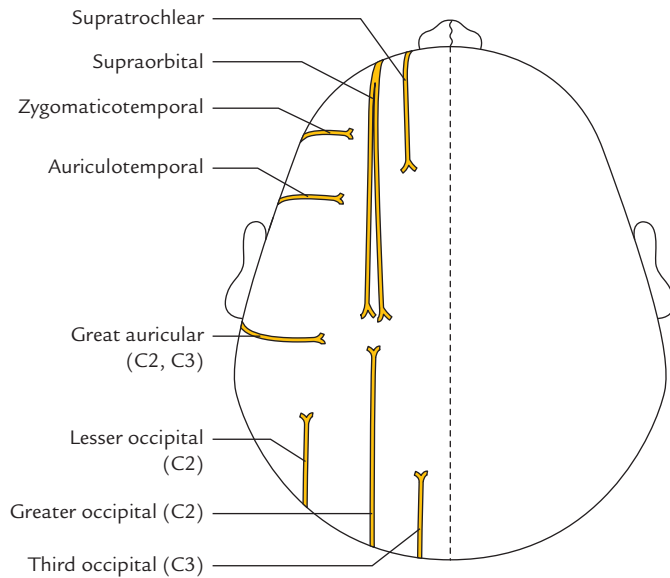
#### A. Sensory supply of the scalp (Fig. 3.5)

The scalp on each side of the midline is supplied by eight sensory nerves: four in front of the auricle and four behind the auricle. The nerves in front of the auricle are derived from trigeminal nerve, whereas those behind the auricle are derived from the 2nd and 3rd cervical nerves (Table 3.1).

#### B. Motor supply of the scalp

The scalp on each side of the midline is supplied by two motor nerves: one in front of the ear and one behind the ear; both these nerves are derived from the facial nerve (Fig. 3.3).

- Nerve in front of the ear is *temporal branch of the facial nerve*. It supplies frontal belly of occipitofrontalis muscle.



**Fig. 3.5** Sensory supply of the scalp (shown in left half only).

- Nerve behind the ear is the *posterior auricular branch of the facial nerve*. It supplies occipital belly of occipitofrontalis muscle.

### ARTERIAL SUPPLY

The scalp has very rich blood supply. On each side of midline, it is supplied by *five arteries*: three in front of the auricle and two behind the auricle (Table 3.2 and Fig. 3.6).

#### Site of Entry of Arteries in the Scalp

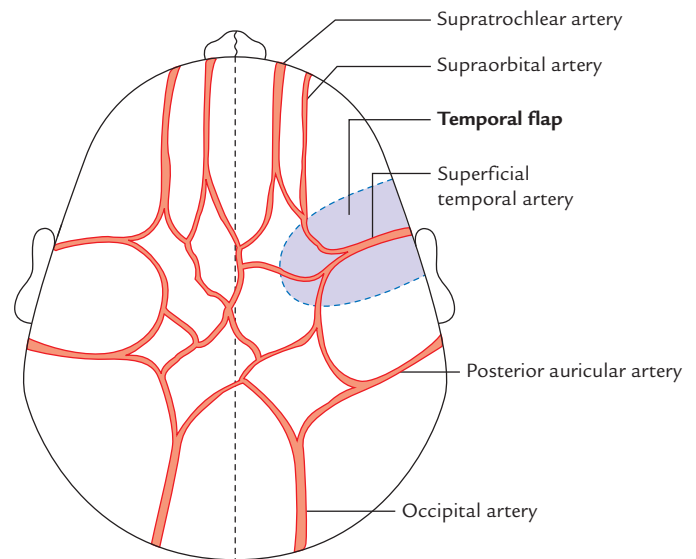
1. **Supratrochlear artery**, enters the scalp around supraorbital margin one-finger breadth lateral to the midline.

**Table 3.1** Sensory nerves of the scalp

| In front of the auricle (from before backwards)                                             | Behind the auricle (from before backwards)                                |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Supratrochlear, a branch of frontal nerve from ophthalmic division of trigeminal nerve      | Great auricular, derived from ventral rami of 2nd and 3rd cervical nerves |
| Supraorbital, a branch of frontal nerve from ophthalmic division of trigeminal nerve        | Lesser occipital, derived from ventral ramus of 2nd cervical nerve        |
| Zygomaticotemporal, a branch of zygomatic nerve from maxillary division of trigeminal nerve | Greater occipital, derived from dorsal ramus of 2nd cervical nerve        |
| Auriculotemporal, a branch of mandibular division of trigeminal nerve                       | Third occipital, derived from dorsal ramus of 3rd cervical nerve          |

**Table 3.2** Arteries supplying the scalp

| In front of the auricle (from before backwards)                            | Behind the auricle (from before backwards)                      |
|----------------------------------------------------------------------------|-----------------------------------------------------------------|
| Supratrochlear, a branch of ophthalmic artery from internal carotid artery | Posterior auricular artery, a branch of external carotid artery |
| Supraorbital, a branch of ophthalmic artery from internal carotid artery   | Occipital artery, a branch of external carotid artery           |
| Superficial temporal, a branch of external carotid artery                  |                                                                 |



**Fig. 3.6** Arterial supply of the scalp. The area enclosed by blue dotted line shows the temporal flap.

2. **Supraorbital artery**, enters the scalp through supraorbital notch 2–3 cm lateral to the supratrochlear artery.
3. **Superficial temporal artery** (the smaller terminal branch of the external carotid), enters the scalp in front of the root of zygoma and divides into anterior and posterior branches.
4. **Posterior auricular artery**, enters the scalp behind the root of the ear.
5. **Occipital artery**, enters the scalp midway between the ear and the external occipital protuberance.

**N.B.** The first two arteries are the branches of ophthalmic artery from internal carotid artery while later three arteries are the branches of external carotid artery.

The arteries of one side freely anastomose with one another. There is also cross anastomosis between the



arteries of two sides. *Thus the scalp is also the site of potential collateral circulation between the external and internal carotid arteries.*

### Clinical correlation

- The **scalp wounds bleed profusely but heal quickly** due to high vascularity. The avulsed portions of scalp, therefore, should not be cut away rather they should be placed in position and stitched.
- Since all the arteries supplying the scalp enter it from below at the periphery, (i.e., from face and neck) the flaps of scalp during craniotomy are made by incisions given in the centre and reflected towards the periphery; thereby preserving their blood supply. To ensure the blood supply of the scalp, the skin incisions should be placed in such a way that the base of the flap incorporates the stem of one of the arteries supplying the scalp.
- Since all the superficial arteries supplying the scalp ascend from face and neck. Therefore *life-threatening scalp-hemorrhage* can be stopped as a first-aid measure, by encircling the head just above the ears and eyebrows with a string or strong-strap of cotton and tied tightly.

## VENOUS DRAINAGE

The scalp on each side of the midline is drained by five veins. The veins of the scalp accompany the arteries and have similar names. These are as follows:

1. **Supratrochlear and supraorbital veins:** They join each other at the medial angle of the eye to form the **angular vein**, which continues downwards as the facial vein behind the facial artery.
3. **Superficial temporal vein:** It descends in front of tragus to enter the parotid gland where it joins the maxillary vein to form the **retromandibular vein**, which terminates by dividing into anterior and posterior divisions. The anterior division unites with the facial vein to form *common facial vein*, which drains into the internal jugular vein.
4. **Posterior auricular vein:** It descends behind the auricle and unites with the posterior division of the retromandibular vein to form the *external jugular vein*, which drains into the subclavian vein.
5. **Occipital vein:** It terminates in the *suboccipital venous plexus*.

The veins of the scalp communicate with intracranial dural venous sinuses through emissary veins.

### Clinical correlation

In infants, the veins of the scalp are easily seen deep to the skin, hence they are the favored sites for **intravenous infusion**.

## Emissary Veins

The veins connecting the veins outside the cranium with the intracranial dural venous sinuses by passing through foramina in the cranium are called *emissary veins*.

### Emissary Veins in the Region of the Scalp

On each side of the midline in the region of the scalp two sets of emissary veins are encountered, viz.

1. **Parietal emissary vein**, which passes through parietal foramen and communicates with the superior sagittal sinus.
2. **Mastoid emissary vein**, which passes through mastoid foramen and communicates with the sigmoid sinus.

## Diploic Veins

The diploic veins are described in detail in Chapter 21.

### Clinical correlation

Diploic veins are thin walled. They are devoid of valves and drain the diploë of some cranial bones. They begin to develop with diploë at about 2 years of age.

Radiologically they may be seen as relatively transparent bands, 3–4 mm wide.

## LYMPHATIC DRAINAGE

The lymphatics from anterior part of the scalp except the lower half of the forehead drain into *preauricular (or superficial parotid) lymph* nodes situated on the surface of the parotid gland.

The lymphatics from posterior part of the scalp drain into *posterior auricular (mastoid) and occipital lymph nodes*.

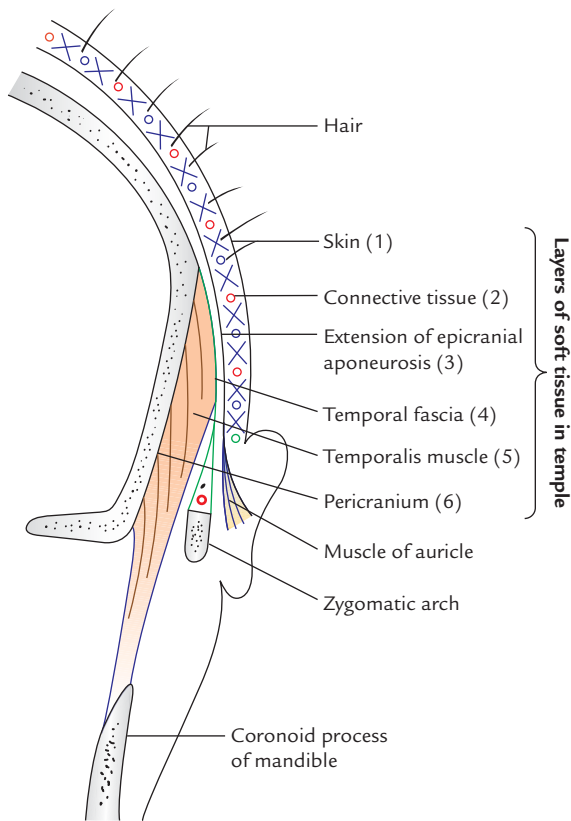
## TEMPLE

The area on the side of the skull between the superior temporal line and zygomatic arch is popularly known as *temple*. The name temple is supposedly derived from the fact that with age (i.e., time) greying of hair occur first in this area (tempus = time).

## LAYERS OF SOFT TISSUE IN THE TEMPLE

There are *six layers* of soft tissue in the region of temple. From superficial to deep, these are (Fig. 3.7):

1. Skin.
2. Connective tissue.
3. Extension of epicranial aponeurosis.
4. Temporal fascia.
5. Temporalis muscle.
6. Pericranium.



**Fig. 3.7** Layers of the soft tissue in the temple.

**Skin and connective tissue:** These two layers are same as in the region of the scalp.

**Extension of epicranial aponeurosis:** It gives origin to the auricularis anterior and auricularis superior muscles of the auricle.

**Temporal fascia:** It forms a roof over temporal fossa. It is attached above to the temporal line and below to the zygomatic arch. Inferiorly the temporal fascia splits into two layers, which are attached to the inner and outer lips of the zygomatic arch. A small gap between the two layers contains (a) fat, (b) a branch from superficial temporal artery, and (c) zygomaticotemporal nerve.

The superficial surface of temporal fascia is related to: (a) superficial temporal vessels, (b) auriculotemporal nerve, and (c) temporal branches of the facial nerve.

The deep surface of temporal fascia gives origin to some fibres of temporalis muscle.

**Temporalis muscle:** This large fan-shaped muscle arises from the floor of the temporal fossa and inner aspect of the temporal fascia. Its fibres converge inferiorly and pass deep to the zygomatic arch to be inserted into the coronoid process of mandible. (It is described in detail in Chapter 10.)

**Pericranium:** This layer is same as in the region of the scalp.

### Clinical correlation

The temporal fascia is silvery-white thick fibrous sheet, which is more or less aponeurotic in character. It is the thickest fascia in the body. Morphologically, it represents a shell of a bone. In some species (e.g., tortoise), it is replaced by bone to make the temporal fossa a bony tunnel.

The temporal fascia is as a graft used by ENT surgeons for **tympanoplasty** (repair of the tympanic membrane).

## FACE

The face is the front aspect of the head possessing eyes, nose, and mouth. It extends superiorly up to the hair line, inferiorly up to the chin and base of the mandible and on each side up to the auricle. The forehead is common to both the face and the scalp. For proper understanding of anatomy of the face, one should first study the features on the anterior aspect of the skull (norma frontalis) described in Chapter 2 and living anatomy described in Chapter 1.

### SKIN OF THE FACE

The skin of the face is thick, elastic, and very vascular. It contains large number of sweat and sebaceous glands. The sweat glands help to regulate the body temperature whereas sebaceous glands keep the face oily by their secretion. The skin of face is lax except on the nose where it is firmly attached to the underlying cartilages and provides insertion to the muscles of facial expression.

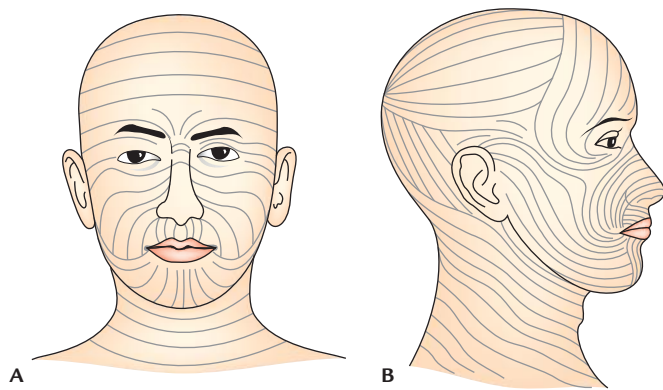
### Clinical correlation

Since the blood supply to the skin of the face is profuse, therefore it is rare in plastic surgery for skin flaps to necrose in this region. The laxity of greater part of skin facilitates rapid spread of edema in the region of the face. Face is also the common site for acne due to the presence of large number of sebaceous glands in this region.

### Cleavage Lines of Skin in the Face (Fig. 3.8)

The direction of cleavage lines in the face varies regionally. However, these lines frequently (but not always) coincide with natural wrinkle lines of the face. The natural wrinkle lines result from repeated folding of the skin perpendicular to the long axis of the underlying contracting muscles of facial expression. They become prominent in the elderly due to loss of youthful skin elasticity.

When the lesions of skin, viz. scars, pigmented patches, skin cancers are excised. It is important to give incisions along the long axis of natural wrinkle lines and the lesion should be enclosed in an ellipse. If the resulting scar is to be aesthetically acceptable.



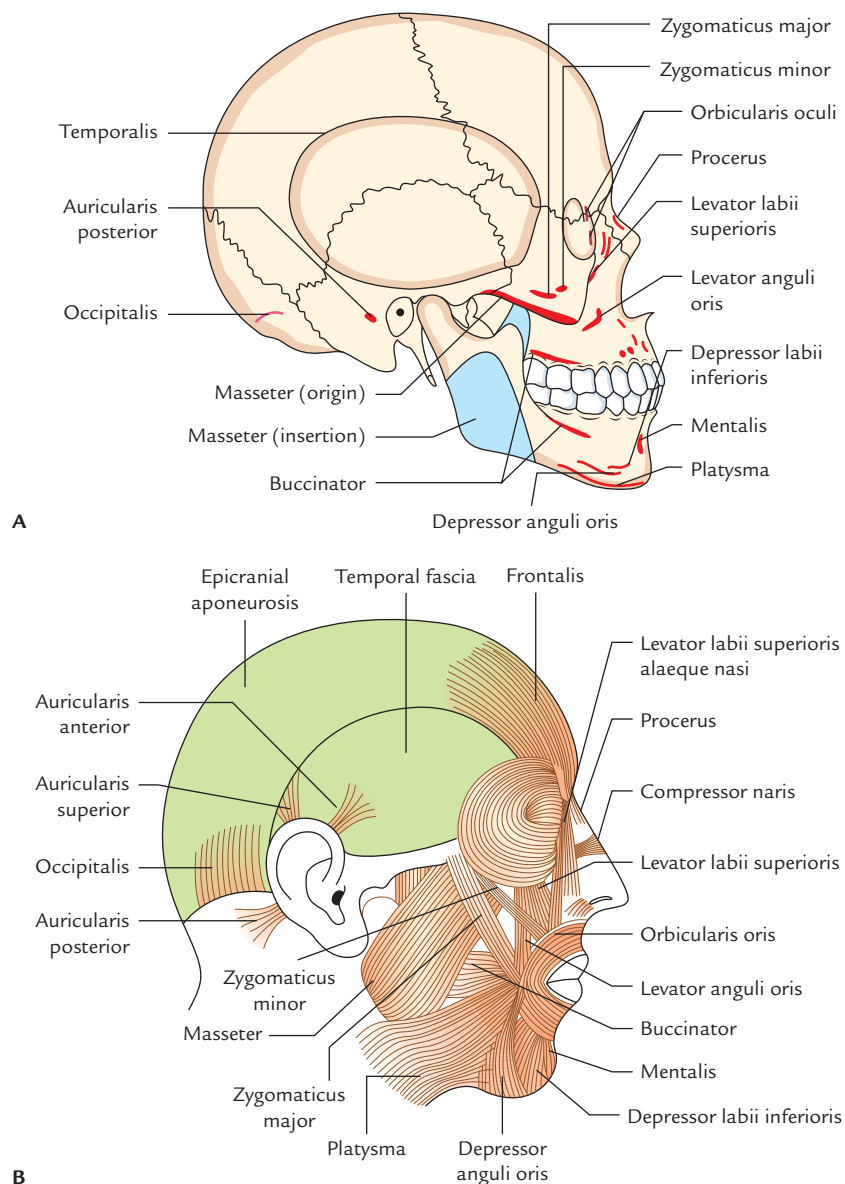
**Fig. 3.8** Lines of cleavage in the head and neck region: A, front view; B, lateral view.

### Superficial Fascia

It contains muscles of facial expression, vessels and nerves, and variable amount of fat. The fat is absent in the eyelids but is well-developed in cheeks forming *buccal pad of fat*, which provides rounded contour to cheeks. The buccal pads of fat are very prominent in infants in whom they help in suckling the milk and are called *suctorial pad of fat*.

### Deep Fascia

The deep fascia is absent in the region of face except over the parotid gland and masseter muscle, which are covered by *parotidomasseteric fascia*. The absence of deep fascia in the face is essential for the facial expression.



**Fig. 3.9** Muscles of the facial expression: A, sites of bony attachment; B, position in the face. Masseter (muscle of mastication) is also shown.

## MUSCLES OF FACIAL EXPRESSION

The muscles of facial expression are embedded in the superficial fascia. Most of them arise from bones of the skull and are inserted into the skin. They bring about different types of facial expressions, hence the name *muscles of facial expression*, the actions of many of them are implied by their names (Fig. 3.9).

### Characteristic Features of the Muscles of Facial Expression

The characteristics of the muscles of facial expressions are as follows:

1. They lie in superficial fascia and inserted into the skin.
2. Morphologically, they represent the specialized members of the subcutaneous muscle ( *panniculus carnosus*), of lower animals.
3. Embryologically, they develop from mesoderm of 2nd pharyngeal arch, hence supplied by facial nerve, the nerve of 2nd arch.
4. Functionally, they perform all important functions of non-verbal communication in addition to closing and opening the orifices in the region of the face.

### Location and Function

The facial muscles are arranged in groups around the orifices of mouth, eye, and nose as sphincters and dilators of these orifices.

In addition to regulating the opening and closing of these orifices, as mentioned earlier, they produce different types of facial expressions. To perform fine movements of facial expressions the facial muscles have *small motor units*.

Though all these muscles are important for facial expression, the students are advised not to burden their memory with their attachments. However, they should appreciate the importance of *orbicularis oculi*, *orbicularis oris*, and *buccinator* for the serious consequences following their paralysis.

**N.B.** In addition to muscles around the eye, nose, and mouth, the muscles of facial expression also include the muscles of scalp, auricle and the subcutaneous muscle of the neck—the platysma.

### Muscles Around the Orifice of the Eye

These include:

1. Orbicularis oculi.
2. Corrugator supercilii.
3. Frontalis.
4. Levator palpebrae superioris.

**Orbicularis oculi** It consists of three parts—orbital, palpebral, and lacrimal.

1. **Orbital part**—arises from medial palpebral ligament, frontal process of maxilla, and adjoining part of the frontal bone. The fibres form complete elliptical loops on and around the orbital margin, without interruption before returning to their point of origin.

*It closes the eye tightly to protect the eye from intense light and dust particles. It is also used by people for winking.*

2. **Palpebral part**—arises from medial palpebral ligament and sweeps laterally over the upper and lower eyelids to be inserted into the lateral palpebral ligament. A small group of fibres called *ciliary bundle* lie along the margins of both eyelids.

*It closes the eyelids gently as in sleep or in blinking.*

3. **Lacrimal part**—lies behind the lacrimal sac. It arises from posterior lacrimal crest and lacrimal fascia forming a sheath of lacrimal sac that passes laterally in front of tarsal plates of both eyelids to be inserted into the lateral palpebral raphe.

*It dilates the lacrimal sac by exerting traction on the lacrimal fascia, thus helping in the drainage of lacrimal fluid.*

### Clinical correlation

- **Crow's feet:** The contraction of entire orbicularis oculi draws the skin of forehead, temple, and cheek towards the lateral angle of the eye, producing radiating skin folds from the lateral angle of the eye, which may be a permanent feature in some old people forming the so called *crow's feet*.
- **Ectropion:** The paralysis of orbicularis oculi results in drooping of the lower eyelid (ectropion) causing spilling of tear on the cheek (epiphora).

**Corrugator supercilii:** It arises from the medial end of the superciliary arch, passes laterally and upwards to be inserted into the skin of the eyebrow above the middle of the supraorbital margin.

It drags the eyebrow medially and downwards producing vertical wrinkles on the forehead as in frowning, an expression of annoyance.

**Frontalis:** It is already described in the section on scalp. The frontalis elevates the eyebrows and produces transverse wrinkles on the forehead as an expression of surprise, horror or fright.

**Levator palpebrae superioris:** It is not a muscle of the face but one of the orbital muscles, hence described in detail under orbit (Chapter 19). Levator palpebrae superioris is an antagonist to the sphincteric action of palpebral part of orbicularis oculi. It elevates the upper eyelid.

### Muscles Around the Nasal Cavity

The muscles associated with nasal cavity are as follows:

1. Procerus.

### Transverse Facial Artery

It is a small artery that arises from superficial temporal artery, within the parotid gland. After emerging from parotid gland it runs forwards on the masseter between the zygomatic arch and the parotid duct accompanied by buccal branch of the facial nerve.

### Arteries Accompanying the Cutaneous Nerves

They are small and usually go unnoticed but some of them can be easily seen, *viz.* infraorbital artery, buccal artery, and mental artery. Note, they all are derived from maxillary artery.

### VENOUS DRAINAGE (Fig. 3.18)

The venous blood from the face is drained by two veins (Fig. 3.18), *viz.*

1. Facial vein.
2. Retromandibular vein.

### Facial Vein

It is the largest vein of the face. It is formed at the medial angle of the eye by the union of supratrochlear and supraorbital veins. After formation, it runs straight downwards and backwards behind the facial artery to reach the anteroinferior angle of the masseter. Here it pierces the deep fascia, crosses superficial to submandibular gland and joins the anterior division of *retromandibular vein* below the angle of the mandible to form the *common facial vein*, which drains into the internal jugular vein.

The **tributaries** of facial vein correspond to the branches of facial artery.

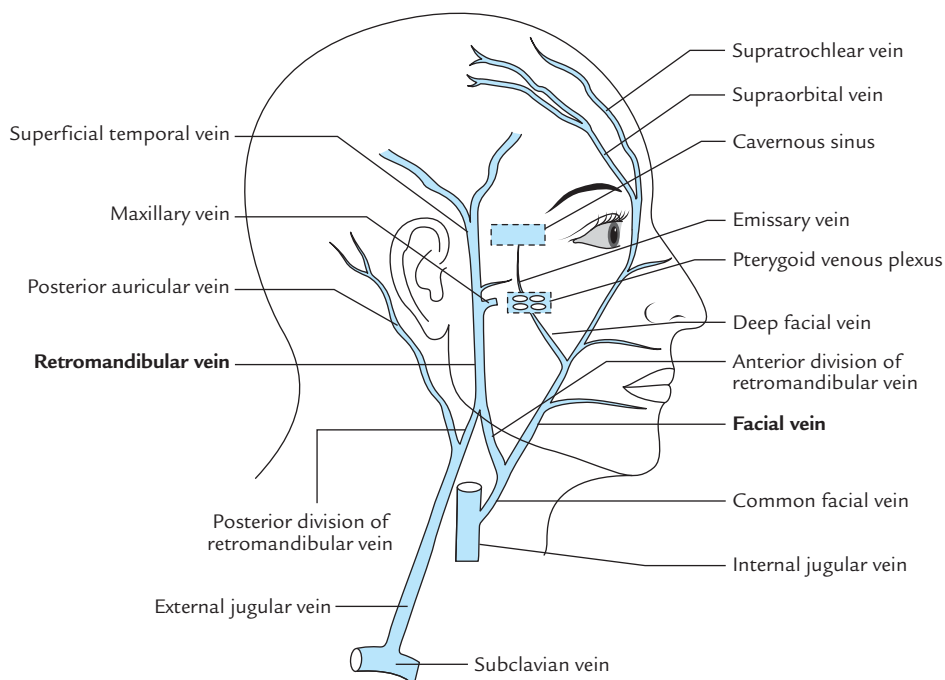
### Deep Connections

The facial vein communicates with the cavernous sinus through the following two routes:

1. **At the point of commencement**, the facial vein communicates with the superior ophthalmic vein, which passes backwards within the orbit and drains into cavernous sinus.
2. **In the cheek**, the facial vein is joined to the pterygoid venous plexus by the *deep facial vein*. The deep facial vein passes backwards over the buccinator deep to the ramus of the mandible and communicates with the *pterygoid venous plexus* around the lateral pterygoid muscle, which in turn communicates with the cavernous sinus through an emissary vein.

### Clinical correlation

**Dangerous area of the face (Fig. 3.19):** The facial vein and its communications are devoid of valves in their lumens. Since facial vein rests directly on the muscles of facial expression, the movements of these muscles may facilitate the spread of septic emboli from infected area of the lower part of the nose, upper lip, and adjoining part of the cheek in retrograde direction through deep facial vein, pterygoid venous plexus, and emissary vein into the cavernous sinus leading to meningitis and cavernous sinus thrombosis. For this reason, this portion of the face is called *dangerous area of the face*.



**Fig. 3.18** Venous drainage of the face.



## Nerve Supply

The conjunctiva is supplied by the following nerves:

1. Palpebral conjunctiva of upper eyelid and ocular conjunctiva—by ophthalmic nerve.
2. Palpebral conjunctiva of lower eyelid—by maxillary nerve.

## Arterial Supply

The conjunctiva is supplied by palpebral and anterior ciliary arteries derived from ophthalmic artery.

## Venous Drainage

The venous blood from palpebral conjunctiva is drained into facial vein while from ocular conjunctiva into ophthalmic veins.

## Lymph Drainage

The lymph from conjunctiva is drained into preauricular lymph nodes.

## LACRIMAL APPARATUS (Fig. 3.25)

### STRUCTURE

The structures concerned with secretion and drainage of lacrimal (tear) fluid together form the *lacrimal apparatus* (Fig. 3.25).

The lacrimal apparatus consists of the following structures:

1. Lacrimal gland.
2. Ducts of lacrimal gland.
3. Conjunctival sac.
4. Lacrimal puncta.

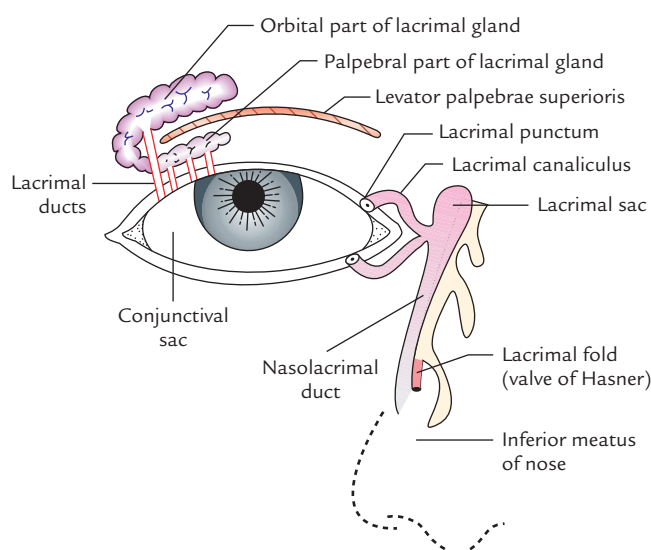


Fig. 3.25 Lacrimal apparatus.

5. Lacrimal canaliculi.
6. Lacrimal sac.
7. Nasolacrimal duct.

**Lacrimal gland:** The lacrimal gland is a (J-shaped) *serous gland*. It consists of upper large **orbital part** and lower small **palpebral part**. The two parts are continuous with each other around the lateral margin of the levator palpebrae superioris.

The *orbital part* is almond shaped and situated in the lacrimal fossa in the anterolateral part of the roof of the bony orbit.

The *palpebral part* is one-third of the size of the orbital part and is situated in the lateral part of the upper eyelid below the levator palpebrae superioris and extends up to the superior fornix of conjunctiva.

**Ducts of the lacrimal gland:** The ducts of lacrimal gland are approximately 12 in number, about 4 or 5 from orbital part and 6–8 from palpebrae part. They open into the lateral parts of the superior fornix of the conjunctival sac.

Lacrimal gland secretes the *lacrimal (tear) fluid* and its ducts convey it to conjunctival sac.

About 1 ml of lacrimal fluid is secreted per day. Half of it evaporates and remaining half is drained into lacrimal sac.

*Functions of lacrimal fluid*

1. Flushes the conjunctiva and keeps the cornea moist.
2. Provides nourishment to cornea.
3. Serves to express emotions—shedding of tears during grief and sorrow.
4. Prevents infection (for it contains bactericidal lysozyme).

### Clinical correlation

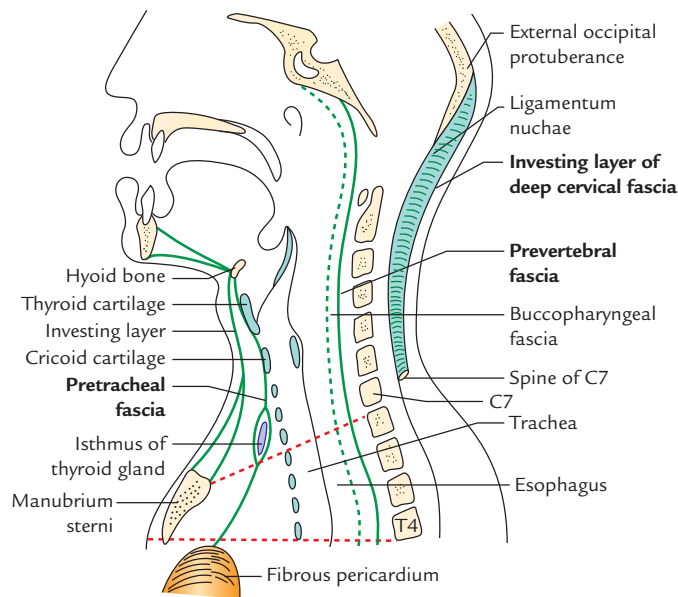
When lacrimal gland fails to secrete lacrimal fluid as in case of Bell's palsy, the artificial tears are instilled in the eye at regular intervals to prevent the cornea from drying and desiccation.

Ducts from orbital part traverse the palpebral part to open into conjunctival sac. Therefore, removal of palpebral part of gland is functionally equivalent to removal of the entire lacrimal gland.

**Accessory lacrimal glands (glands of Krause):** These are very small serous glands located beneath the palpebral conjunctiva near the fornices. They are about 35–40 in the upper eyelid and 6–8 in the lower eyelid.

### Clinical correlation

The removal of entire lacrimal gland does not lead to dryness of conjunctiva because it is kept moist by the secretions of accessory lacrimal glands.



**Fig. 4.8** Vertical extent (disposition) of the three layers of the deep cervical fascia. Buccopharyngeal fascia is also shown. Note the continuity of fascial spaces of the neck into the mediastinum.

loose areolar tissue produced by the movements of these structures.

These laminar condensations take different names in different regions and are continuous with each other either directly or indirectly through loose areolar tissue.

The loose areolar tissue forms fascial sheets or laminae in the same way as unwoven cotton forms sheets of cloth after spinning.

These laminae ensheath muscles and other moving structures. This permits movement between the sheaths and movements of structures within them.

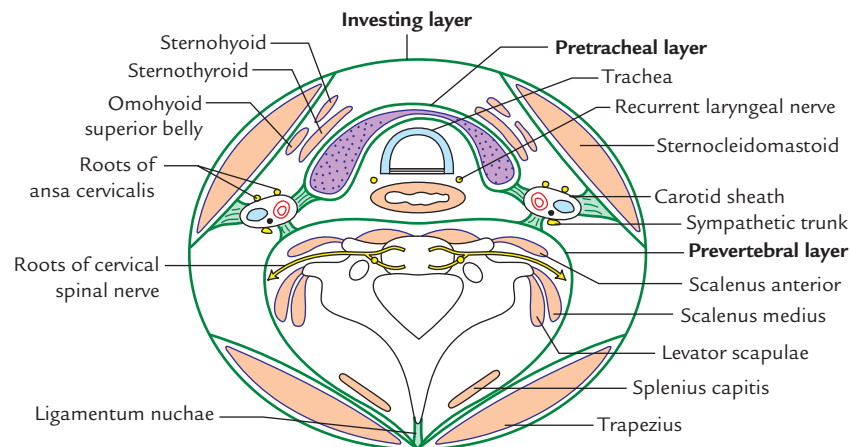
**Investing fascia** This layer encircles the neck like a collar, deep to platysma and superficial fascia. In doing so it encloses the sternocleidomastoid and trapezius—the two large superficial muscles on either side of the neck.

#### Attachments

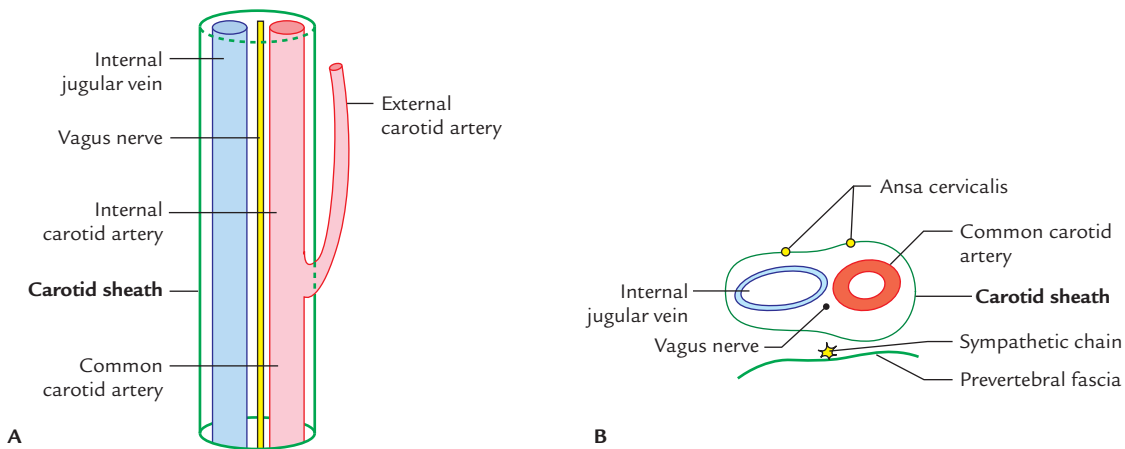
1. **Superiorly** it is attached to external occipital protuberance, superior nuchal line, mastoid process, and lower border of mandible from behind forwards.
2. **Inferiorly** it is attached to the spine of scapula, acromion process, upper aspect of clavicle, and jugular notch of manubrium sterni from behind forwards.
3. **Anteriorly** across the midline, it becomes continuous with its counter part of the other side. In the anterior midline it is attached to symphysis menti, hyoid bone jugular notch from above downwards.
4. **Posteriorly** it is attached to ligamentum nuchae and spine of 7th cervical vertebra.

#### Tracing of the investing layer

**Vertical tracing:** When traced upwards above the hyoid bone, it splits to enclose the submandibular salivary gland (Fig. 4.10). The superficial layer covers the superficial surface of the gland and it is attached to the lower border of the body of the mandible. The deep layer covers the medial surface of the gland and is attached to the mylohyoid line of the mandible. Posterior to the submandibular gland, at the lower pole of the parotid gland, it splits to enclose this gland also (Fig. 4.10).



**Fig. 4.9** Diagrammatic transverse section through neck at the level of the 6th cervical vertebra to show the horizontal disposition of the three layers of deep cervical fascia.



**Fig. 4.14** Carotid sheath: A, surface view; B, sectional view.

### Clinical correlation

- The infections of retropharyngeal space can pass down behind the esophagus and may extend into the posterior mediastinum, of course, through superior mediastinum.
- The retropharyngeal abscess due to involvement of retropharyngeal lymph nodes produces paramedian unilateral bulge in the posterior pharyngeal wall due to fusion of buccopharyngeal fascia to the prevertebral fascia in the median plane.
- Tubercular caries of cervical vertebra produce abscess in the prevertebral space which often extends into the retropharyngeal space.
- Pus collected in the prevertebral space due to caries of cervical vertebrae may travel forward and cause midline bulge in the posterior pharyngeal wall. Pus may travel laterally first into the posterior triangle deep to fascial carpet formed by prevertebral fascia and then along the axillary sheath into the axilla or even into the arm.

### Other Derivatives of the Deep Cervical Fascia

These include:

1. Carotid sheath.
2. Buccopharyngeal fascia.

**Carotid sheath** It is a tubular condensation of deep cervical fascia around the (a) common carotid and internal carotid arteries, and (b) internal jugular vein and vagus nerve (Fig. 4.14). The vagus nerve lies posteriorly between the vein and arteries. The carotid sheath is wedged between the three layers of the deep cervical fascia, i.e., investing layer, pretracheal fascia, and prevertebral fascia, and attached to all

these layers by loose areolar tissue. The sheath is thick around common and internal carotid arteries but thin over the internal jugular vein in order to allow its expansion during increased venous return.

Carotid sheath extends from the base of the skull above to the arch of aorta below. At the base of the skull, it is attached to the margins of the carotid canal and jugular fossa. At the arch of aorta, it blends with its tunica adventitia.

### Relations

1. The ansa cervicalis is embedded in the anterior wall of the carotid sheath.
2. The cervical sympathetic chain is closely related to posterior wall of the sheath, plastered to the prevertebral fascia.

### Clinical correlation

The malignant and tuberculous deep cervical lymph nodes frequently become adherent to the internal jugular vein. Therefore, during block dissection of the neck, the sheath is exposed and a portion of the vessel is resected to facilitate removal of these lymph nodes.

**Buccopharyngeal fascia** It is a delicate and distensible layer of fascia that covers the constrictor muscles of pharynx and buccinator muscle. The buccopharyngeal fascia extends from the base of the skull to the esophagus.

**N.B.** Together with similar layer of fascia on the internal surfaces of the constrictor muscles, the *pharyngobasilar fascia* closes the gaps in the muscular wall of the pharynx.

### Boundaries (Fig. 5.6)

- Anterior:** Posterior border of sternocleidomastoid muscle.
- Posterior:** Anterior border of trapezius muscle.
- Inferior (base):** Superior aspect of middle third of the clavicle.
- Superior (apex):** Meeting point of sternocleidomastoid and trapezius muscles at the superior nuchal line of the occipital bone.

### Roof

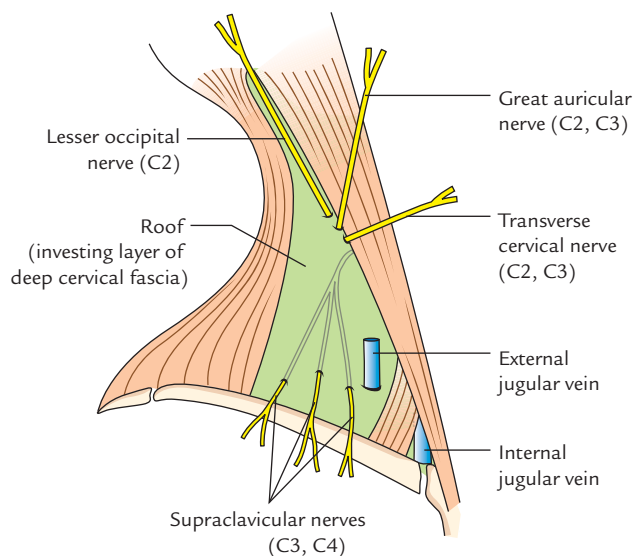
It is formed by the investing layer of the deep cervical fascia, stretching between the sternocleidomastoid and trapezius muscles.

The **superficial fascia overlying the roof** contains platysma, external jugular and posterior jugular veins, and cutaneous nerves and vessels.

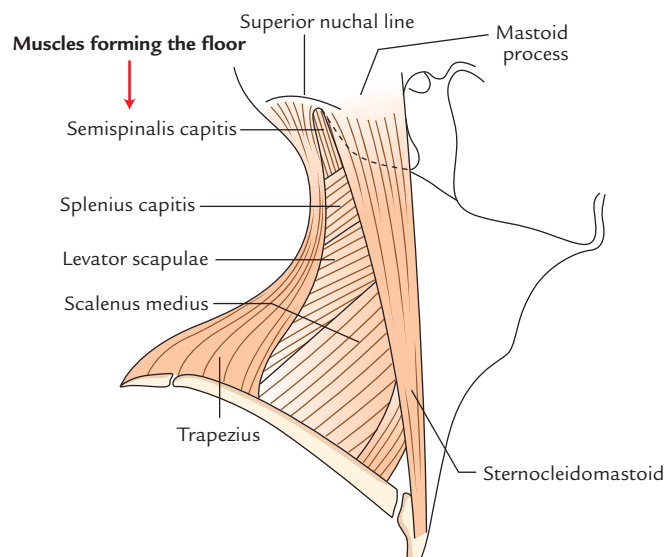
**Structures piercing the roof of the posterior triangle** (Fig. 5.5) are:

1. **Four cutaneous branches of cervical plexus, viz.**
  - (a) Lesser occipital nerve (C2)
  - (b) Great auricular nerve (C2, C3)
  - (c) Transverse cervical nerve (C2, C3)
  - (d) Supraclavicular nerves (C3, C4).

They pierce the roof near the middle of the posterior border of the sternocleidomastoid muscle (for details see page 83).
2. **External jugular vein:** It begins just below the angle of mandible, runs downwards and backwards crossing the sternocleidomastoid obliquely and under the cover of platysma (for details see page 70).



**Fig. 5.5** Structures piercing the roof of the posterior triangle.



**Fig. 5.6** Boundaries and floor of the posterior triangle.

### Floor (Fig. 5.6)

The floor of posterior triangle is muscular and is formed from above downwards by the following muscles:

1. Semispinalis capitis.
2. Splenius capitis.
3. Levator scapulae.
4. Scalenus medius.
5. First digitation of serratus anterior (sometimes).

### Fascial carpet of the posterior triangle

The muscular floor of posterior triangle is covered by prevertebral layer of deep cervical fascia, which forms the **fascial carpet of the floor of the posterior triangle**. It forms axillary sheath around subclavian artery and brachial plexus travelling from the root of the neck to the upper limb.

The lower part of the posterior triangle is crossed by inferior belly of omohyoid superficial to the fascial carpet.

### Clinical correlation

Pus collected in the posterior triangle deep to its fascial carpet from tubercular cervical vertebrae may track downwards and laterally along the axillary sheath to first appear in the axilla or even in the arm subsequently.

### Subdivisions (Fig. 5.7)

The posterior triangle is subdivided into two parts by the inferior belly of the omohyoid, which crosses the lower part of the triangle obliquely upwards and forwards (a) a larger upper part called **occipital triangle**, and (b) a small lower part called **subclavian (supraclavicular) triangle**. These parts are so named because they contain occipital and subclavian arteries, respectively.

### Contents (Figs 5.7 and 5.8)

1. In the occipital triangle (i.e., above the omohyoid)
  - (a) Spinal accessory nerve
  - (b) 3rd and 4th cervical nerves providing branches to levator scapulae and trapezius muscles
  - (c) Dorsal scapular nerve (C5)
  - (d) Four cutaneous branches of cervical plexus (initial parts)
  - (e) Superficial transverse cervical artery
  - (f) Occipital artery.
2. In the subclavian/supraclavicular triangle (i.e., below the omohyoid)
  - (a) 3rd part of the subclavian artery
  - (b) Subclavian vein
  - (c) Terminal part of external jugular vein
  - (d) Trunks of brachial plexus
  - (e) Superficial (transverse) cervical, suprascapular, and dorsal scapular arteries
  - (f) Lymph nodes.

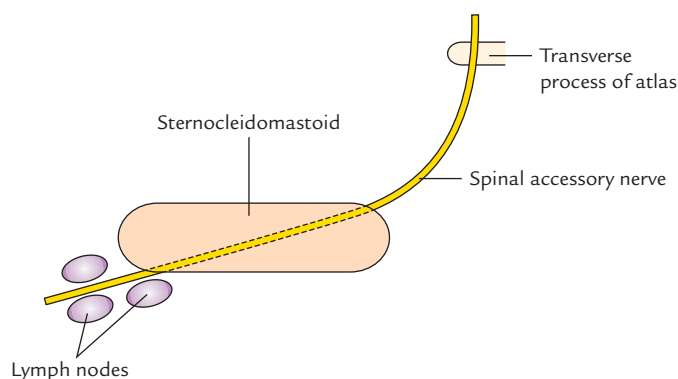
#### N.B.

- The most important contents of posterior triangle are:
  - (a) third part of subclavian artery, (b) brachial plexus (cervical part), (c) spinal accessory nerve, and (d) lymph nodes.
- All the important contents of the posterior triangle lie deep to the fascial carpet of the floor except spinal accessory nerve, which lies just underneath the roof. In operations on the posterior triangle all the structures except spinal accessory nerve are safe, provided fascial carpet of posterior triangle is left intact.

### Relevant Features of Some of the Contents

- **Spinal accessory nerve:** This nerve emerges in the posterior triangle by piercing the posterior border of the sternocleidomastoid (a little above the middle of this border). In this situation, it is related to lymph nodes of the upper deep cervical chain (Fig. 5.9). The nerve then crosses the posterior triangle by running downwards and laterally over and parallel to the fibres of levator scapulae muscle to disappear underneath the anterior border of trapezius (about 5–6 cm above the clavicle) and supplies trapezius muscle. In the posterior triangle it is adherent to the deep aspect of the fascial roof of this triangle.

**N.B.** More proximally, the nerve lies in front of transverse process of atlas vertebra runs downwards and laterally to enter the anterior border of sternocleidomastoid muscle between its upper two quarters and supplies it.

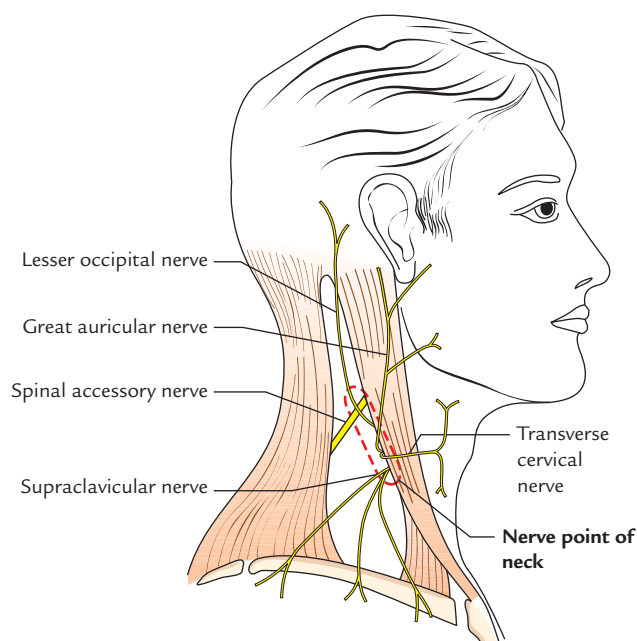


**Fig. 5.9** Course of spinal accessory nerve right through sternocleidomastoid muscle. Note the relation of lymph nodes to this nerve.

### Clinical correlation

The spinal accessory nerve may be damaged in operations involving the removal or biopsy of lymph nodes in the posterior triangle of the neck.

- **Four cutaneous branches of cervical plexus (Fig. 5.10):** Although cervical plexus is located deep to the sternocleidomastoid but its cutaneous branches emerge at the midpoint or just above the midpoint of the posterior border of the sternocleidomastoid by piercing the deep cervical fascia.



**Fig. 5.10** Cutaneous branches of the cervical plexus. Note the location of 'nerve point of the neck' (dotted red circle).



# Anterior Region of the Neck

The triangular area on the front of the neck between the two sternocleidomastoid muscles is called *anterior region of the neck*. It is limited above by lower border of the body of mandible and line extending on either side from the angle of the mandible to the mastoid process, and below by the suprasternal notch. This area includes both **anterior triangles**, and **suprahyoid** and **infrahyoid** areas (Fig. 6.1).

The anterior region of the neck is divided by the anterior median line (extending from symphysis menti to the jugular notch) into two anterior triangles of the neck.

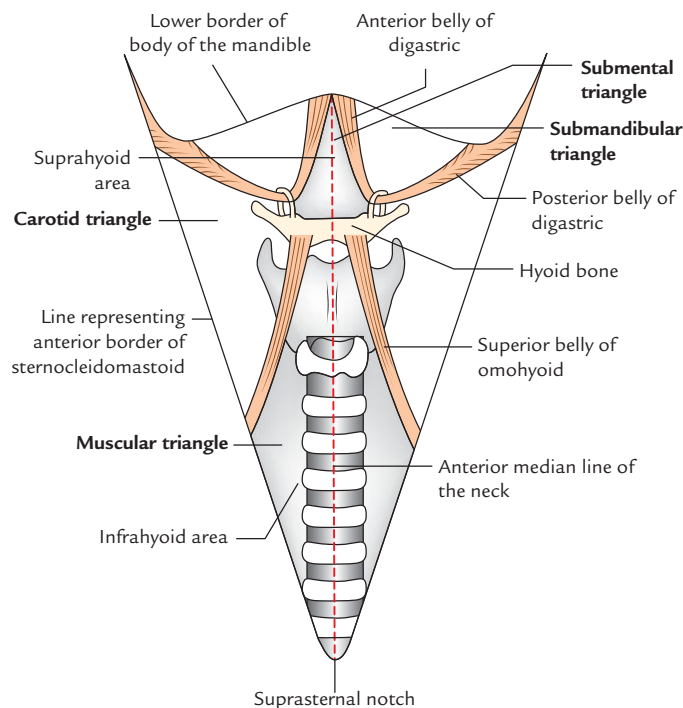
Before going into details of anterior triangles of the neck, it is of great clinical importance to know the structures in the anterior median line of the neck. The identification of these

structures is essential while performing tracheostomy. It is also useful in clinical examination of midline swellings of the neck.

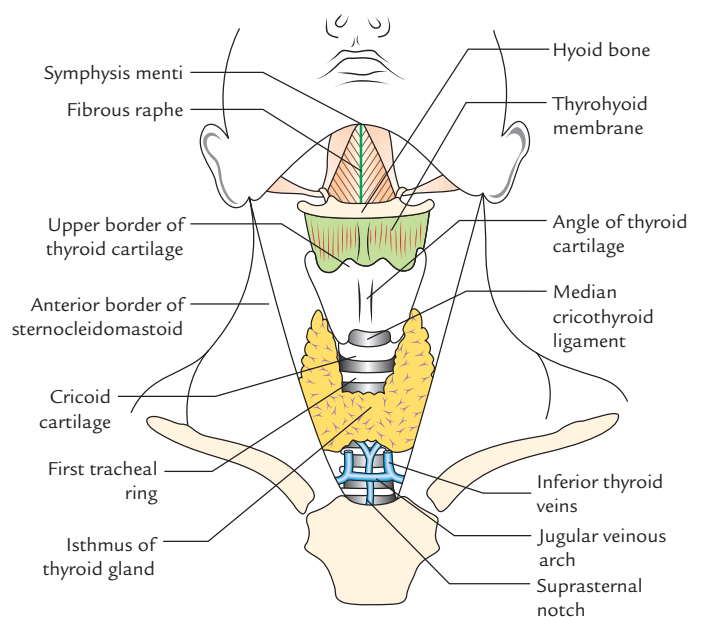
## STRUCTURES IN THE ANTERIOR MEDIAN REGION OF THE NECK

The anterior median region of the neck is 2–3 cm wide strip extending from symphysis menti to the suprasternal notch. Deep to skin and underlying the superficial and deep cervical fasciae, the anterior median region presents the following structures from above downwards (Fig. 6.2):

1. **Symphysis menti:** Its lower border can be felt where the two halves of the body of the mandible unite in the midline.



**Fig. 6.1** Schematic diagram of anterior median region of the neck showing suprahyoid and infrahyoid area.



**Fig. 6.2** Structures in the anterior median line of the neck.

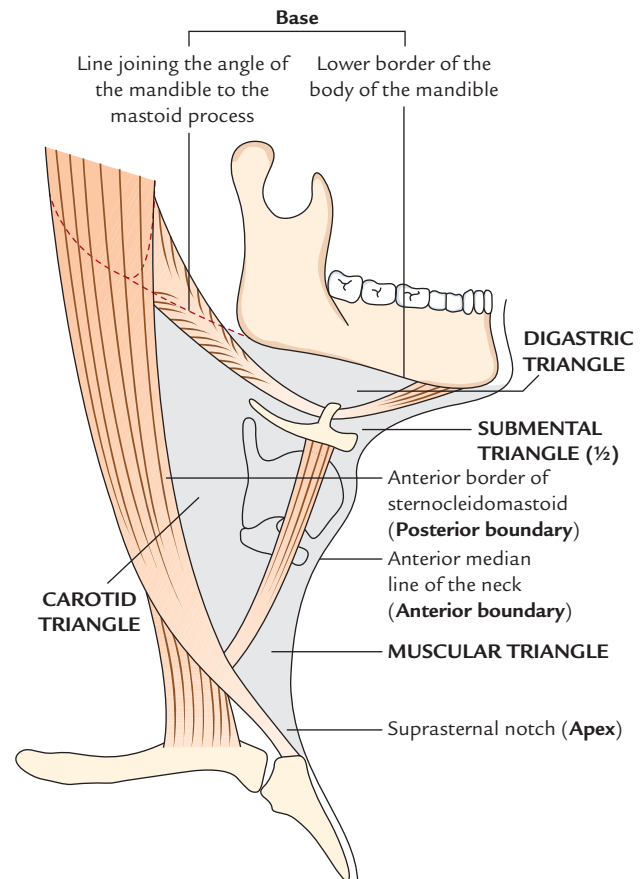
2. **Fibrous raphe:** It extends from symphysis menti to the hyoid bone. Mylohyoid muscles of both sides meet along this raphe to form the floor of the mouth (diaphragma oris).
3. **Hyoid bone:** It is a horseshoe-shaped bone suspended inferior to mandible by stylohyoid ligaments like children's swing. Hyoid bone does not articulate with any other bone and has only muscular and ligamentous attachments. It lies at the level of C3 vertebra. The regions above and below the hyoid bone are called **suprahyoid region** and **infrahyoid region**, respectively.
4. **Median thyrohyoid ligament:** It is the midline thickening of the thyrohyoid membrane that connects the upper border of thyroid cartilage to the lower border and greater cornu of the hyoid bone. The membrane is separated from the hyoid bone by a **subhyoid bursa**.
5. **Upper border of the thyroid cartilage:** It is notched and can be felt. It lies opposite the body of T4 vertebra.
6. **Angle of the thyroid cartilage:** It forms the laryngeal prominence, which is more prominent in adult males and is called **Adam's apple**.
7. **Median cricothyroid ligament:** It is the midline thickening of the cricothyroid membrane extending between upper border of cricoid cartilage and lower border of the thyroid cartilage.
8. **Cricoid cartilage:** It lies at the lower border of C6 vertebra and is the most important surface landmark on the front of the neck because a number of important structures are located at this level, such as (a) junction of larynx and trachea, (b) junction of pharynx and esophagus, and (c) anterior tubercle of transverse process of C6 vertebra (carotid tubercle) against which common carotid artery can be compressed.
9. **First tracheal ring:** It can be felt by gentle pressure just above the isthmus of the thyroid gland.
10. **Isthmus of the thyroid gland:** It lies in front of the 2nd, 3rd, and 4th tracheal rings.
11. **Inferior thyroid veins:** After emerging from isthmus they run downwards to lie in front of the 5th, 6th, and 7th tracheal rings.
12. **Thyroidea ima artery:** When present, it ascends in front of the trachea to the isthmus of the thyroid; it usually arises from the brachiocephalic artery.
13. **Jugular venous arch:** This traverses the venous arch, which connects the anterior jugular veins just above the suprasternal notch.
14. **Suprasternal notch:** It can be felt between the anterior ends of the clavicle. It lies opposite the lower border of the body of T2 vertebra. Occasionally, left brachiocephalic vein and brachiocephalic artery may lie in front of trachea in the suprasternal notch.

### Clinical correlation

- The **most common anterior midline swellings of the neck** are: (a) enlarged submental lymph nodes, (b) thyroglossal cyst, and (c) goitre.
- The **tracheostomy** is most commonly done in the retrothyroid region after cutting the isthmus of thyroid gland (for detail see page 164).
- The **cut-throat wounds** are most commonly observed just above or below the hyoid bone. The main vessels of the neck, viz. common carotid, internal and external carotid arteries, and internal jugular vein, usually escape the injury as they are automatically pushed backwards to the deeper plane, when neck is inadvertently extended by the culprit, before being cut.

### ANTERIOR TRIANGLE OF THE NECK

The anterior triangle of the neck is a large triangular space on each side of the neck, in front of sternocleidomastoid with its apex directed downwards and base directed upwards (Fig. 6.3).



**Fig. 6.3** Boundaries and subdivisions of the right anterior triangle.

**Boundaries (Fig. 6.3)**

**Anterior:** Anterior median line of the neck.

**Posterior:** Anterior border of the sternocleidomastoid muscle.

**Base:** Lower border of the body of mandible and a line extending from the angle of mandible to the mastoid process.

**Apex:** Suprasternal notch at the meeting point between anterior border of sternocleidomastoid and anterior median line.

**Roof:** Investing layer of deep cervical fascia.

The superficial fascia over the fascial roof contains platysma in the upper and anterior parts. The cervical branch of facial nerve and ascending and descending branches of transverse cervical cutaneous nerve traverse the plane between the platysma and the fascial roof of the triangle.

**SUBDIVISIONS**

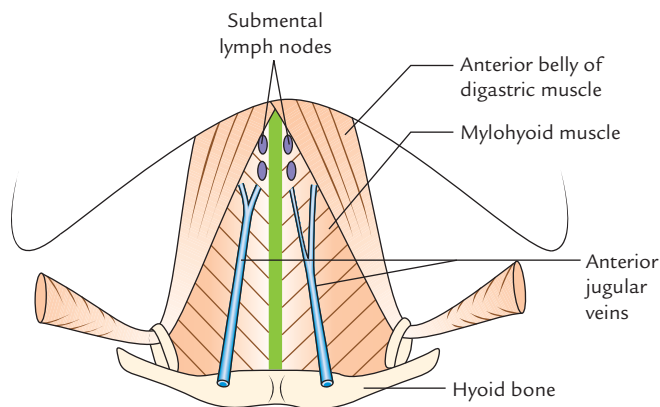
For convenience of description, the anterior triangle has been subdivided, by the digastric muscle and superior belly of omohyoid muscle, into the following  $3\frac{1}{2}$  triangles (Fig. 6.3):

1. Submental triangle (half only).
2. Digastric (submandibular) triangle.
3. Carotid triangle.
4. Muscular triangle.

Before studying the subdivisions of anterior triangle, it is appropriate for students to know the attachments of digastric and omohyoid muscles, which are used for its subdivision.

**Digastric muscle (Fig. 6.5A):** It has two bellies – posterior and anterior. The triangular posterior belly arises from the digastric notch on the medial surface of the base of the mastoid process, and runs downwards and forwards, towards the hyoid bone. The narrow anterior belly arises from digastric fossa on the base of the mandible near symphysis menti and runs downwards and backwards, towards the hyoid bone. Above the hyoid bone the two bellies are connected by an intermediate tendon, which is anchored to the hyoid bone by a fascial sling of investing layer of deep cervical fascia.

**Omohyoid muscle (Fig. 6.9):** It consists of superior and inferior bellies joined at an angle by an intermediate tendon. The inferior belly arises from upper border of scapula, near the suprascapular notch, it then passes behind the sternocleidomastoid and ends there in an intermediate tendon. The superior belly begins at the intermediate tendon and passes almost vertically upwards to be attached to the lower border of the hyoid bone. The intermediate tendon lies at the level of arch of cricoid cartilage and anchored to the clavicle by fascial sling derived from investing layer of deep cervical fascia.



**Fig. 6.4** Boundaries and contents of the submental triangle.

**Submental Triangle (Fig. 6.4)**

It is actually a median triangle on the front of neck below the chin and above the hyoid bone. It thus occupies the upper portions of both the anterior triangles.

**Boundaries**

**On each side:** Anterior belly of digastric muscle.

**Base:** Body of hyoid bone.

**Apex:** Chin or symphysis menti.

**Floor**

It is formed by the two mylohyoid muscles meeting at the median fibrous raphe extending from symphysis menti to the hyoid bone.

**Roof**

Investing layer of deep cervical fascia.

**Contents**

1. Submental lymph nodes (3 or 4 in number). Efferents from these nodes pass to the submandibular lymph nodes.
2. Submental veins and commencement of anterior jugular veins.

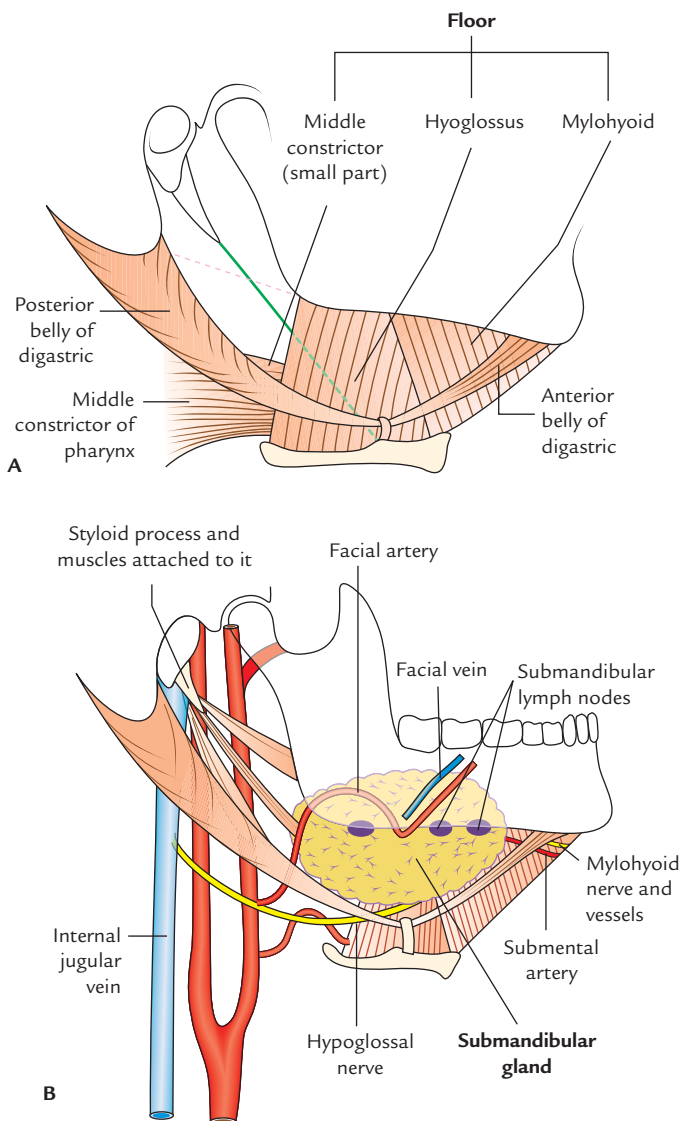
**N.B.** The contents of the triangle actually lie in the superficial fascia over the roof of the submental triangle.

**Clinical correlation**

Infections of floor of mouth, tip of tongue, and incisor teeth cause painful enlargement of submental lymph nodes because lymph from these structures is drained into these nodes.

**Digastric (Submandibular) Triangle (Fig. 6.5 A, B)**

It is so named because it is located between the two bellies of the digastric muscle and below the base of the mandible.



**Fig. 6.5** Digastic (submandibular) triangle: **A**, boundaries and floor; **B**, boundaries and contents.

### Boundaries

- Anteroinferiorly:** Anterior belly of digastric muscle.  
**Posteroinferiorly:** Posterior belly of digastric muscle, supplemented by stylohyoid muscle.  
**Base:** It is formed by the base of the mandible and imaginary line joining the angle of the mandible to the mastoid process.  
**Apex:** It is formed by the intermediate tendon of the digastric muscle, being bound down to hyoid bone by a fascial sling derived from investing layer of deep cervical fascia.

### Floor

It is formed by mylohyoid (anteriorly), hyoglossus, and small part of the middle constrictor (posteriorly).

### Roof

It is formed by investing layer of deep cervical fascia, which splits to enclose the submandibular salivary gland. In the superficial fascia over the roof lies platysma, cervical branch of facial nerve, and ascending branch of transverse cervical nerve.

### Contents

The digastric triangle is subdivided into anterior and posterior parts by the stylomandibular ligament, which extends from the tip of the styloid process to the angle of the mandible. The posterior part of the triangle is continuous above with the parotid region.

*Contents in the anterior part of the triangle:*

1. Submandibular salivary gland.
2. Submandibular lymph nodes.
3. Hypoglossal nerve.
4. Facial vein (lies superficial to the gland).
5. Facial artery (lies deep to the gland).
6. Submental artery.
7. Mylohyoid nerve and vessels.

*Contents in the posterior part of the triangle:*

1. External carotid artery.
2. Carotid sheath and its contents.
3. Structures passing between the external and internal carotid arteries (for details refer to page 241).

All these structures are discussed in detail in the submandibular region (Chapter 9, p. 131).

### Clinical correlation

**Submandibular triangle is the common site of swellings** due to: (a) involvement of submandibular lymph nodes as they drain lymph from widespread areas such as anterior two-third of tongue (except tip), floor of mouth, frontal, ethmoidal, and maxillary air sinuses, etc. and (b) enlargement of the submandibular gland. The efferents from submandibular lymph nodes mostly pass to jugulo-omohyoid and partially to jugulodigastric nodes situated along the internal jugular vein.

### Carotid Triangle (Fig. 6.6)

It is so called because it contains all the three carotid arteries, viz. common carotid, internal carotid, and external carotid.

### Boundaries

- Superiorly:** Posterior belly of digastric supplemented by stylohyoid.  
**Anteroinferiorly:** Superior belly of omohyoid.  
**Posteriorly:** Anterior border of sternocleidomastoid.  
**Roof:** It is formed by investing layer of deep cervical fascia. The superficial fascia over the roof contains platysma, cervical branch

of facial nerve and transverse cervical nerve (a cutaneous branch of the cervical plexus).

*Floor:*

It is formed by the following four muscles:

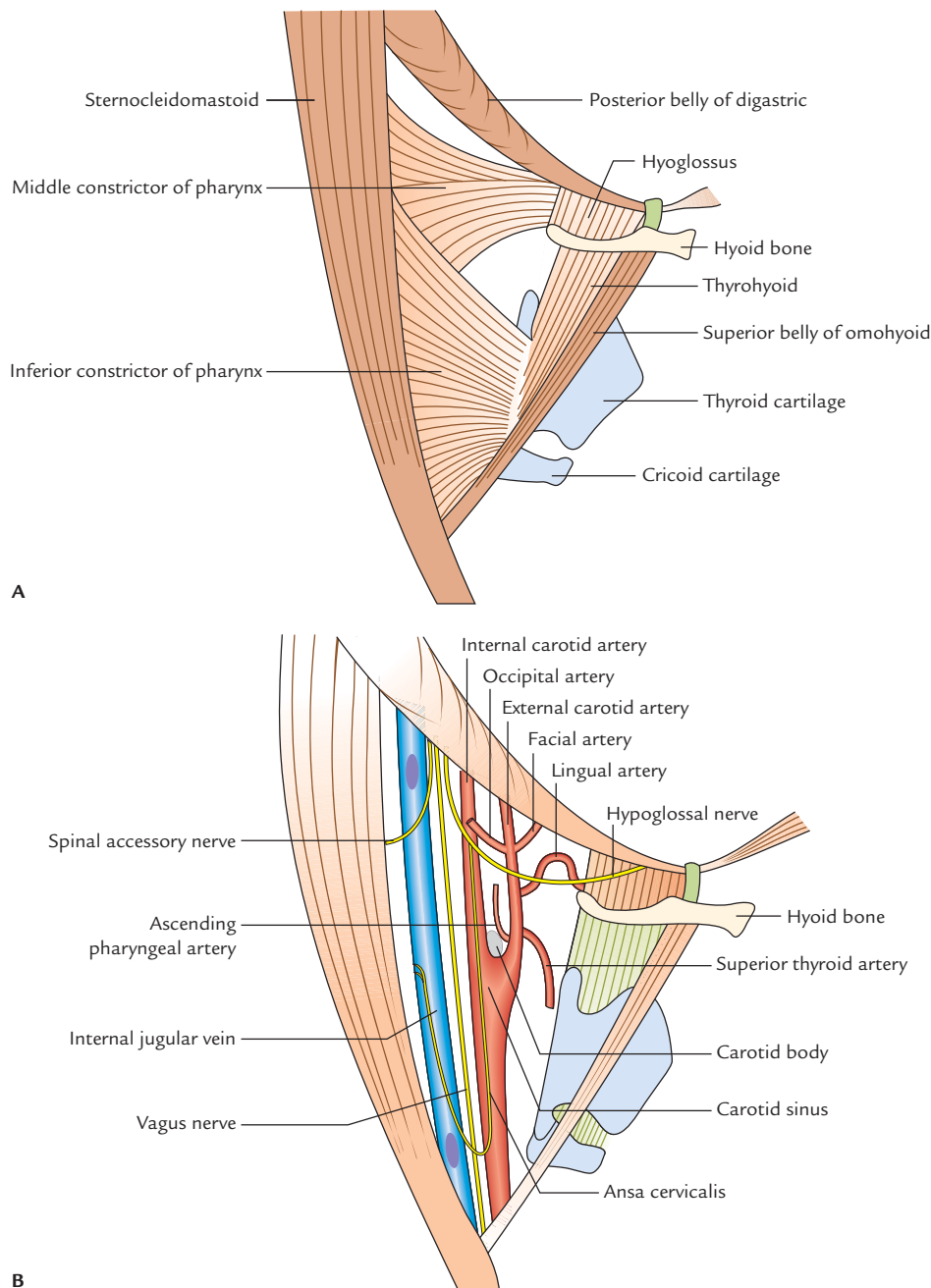
1. Thyrohyoid.
2. Hyoglossus.
3. Middle constrictor.
4. Inferior constrictor.

### Contents

#### 1. Carotid arteries

- (a) Common carotid artery
- (b) Internal carotid artery

- (c) External carotid artery and its first five branches.
2. Carotid sinus and carotid body.
3. Internal jugular vein.
4. Last three cranial nerves
  - (a) Vagus nerve
  - (b) Spinal accessory nerve
  - (c) Hypoglossal nerve.
5. Carotid sheath
6. Ansa cervicalis.
7. Cervical part of the sympathetic chain.
8. Deep cervical lymph nodes.



**Fig. 6.6** Carotid triangle: A, boundaries and floor; B, boundaries and contents.



### Vagus nerve

It descends vertically downwards, first between the internal carotid artery and internal jugular vein and then between common carotid artery and internal jugular vein. It gives the following two branches in the carotid triangle.

1. **Pharyngeal branch:** It runs inferomedially between the external and internal carotid arteries to join the pharyngeal plexus on the superior constrictor of the pharynx.
2. **Superior laryngeal nerve:** It runs on a deep plane, deep to both internal and external carotid arteries, where it divides into internal and external laryngeal nerves. The internal laryngeal nerve (sensory) passes forwards to disappear deep to thyrohyoid muscle; there it pierces the thyrohyoid membrane to supply the laryngeal mucosa.

The external laryngeal nerve (motor) descends to supply the inferior constrictor and cricothyroid muscles after passing deep to the superior belly of the omohyoid.

### Accessory nerve

It runs downwards and backwards across the upper part of the triangle, superficial to the internal jugular vein to enter the sternocleidomastoid muscle, which it supplies.

### Carotid sheath

It is a fascial sheath which encloses internal jugular vein, and internal and common carotid arteries. The vagus nerve lies in between the vein and the artery on a deeper plane (for details refer to Chapter 4). The ansa cervicalis is embedded in its anterior wall whereas the cervical sympathetic chain lies just deep to its posterior wall on the prevertebral fascia.

### Carotid sinus

It is a fusiform dilatation at the terminal end of common carotid artery or at the beginning of internal carotid artery. It has rich innervation from glossopharyngeal, vagus, and sympathetic nerves. The carotid sinus acts as a baroreceptor (pressure receptor) and regulates the blood pressure in the cerebral arteries.

### Clinical correlation

In an individual with *carotid sinus hypersensitivity*, pressure on carotid sinuses can cause enough slowing of heart rate, fall in blood pressure, and cerebral ischemia that will lead to fainting (syncope). Individuals with hypersensitive carotid sinuses often had sudden attacks of syncope on rotation of head especially when wearing a shirt with tight collar or a tie with tight knot, condition called **carotid sinus syndrome**. Symptoms can be relieved by periarterial neurectomy.

### Carotid body

It is a small oval structure situated behind the bifurcation of the common carotid artery. It is reddish-brown and receives

rich nerve supply from glossopharyngeal, vagus, and sympathetic nerves. It acts as a chemoreceptor and responds to the changes in the oxygen and carbon dioxide content of the blood. The excess of CO<sub>2</sub> and reduced O<sub>2</sub> tension in blood stimulates it causing increase in blood pressure and heart rate.

### Ansa cervicalis (Figs 6.7 and 6.8)

The ansa cervicalis (ansa hypoglossi) is a U-shaped nerve loop present in the region of the carotid triangle embedded in the anterior wall of the carotid sheath. It is derived from

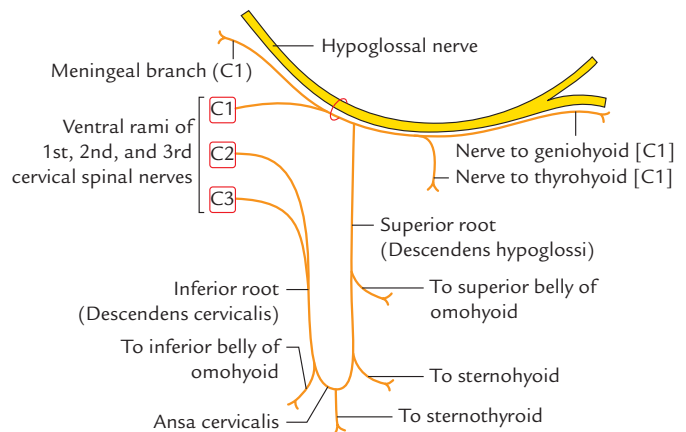


Fig. 6.7 Formation and distribution of the ansa cervicalis.

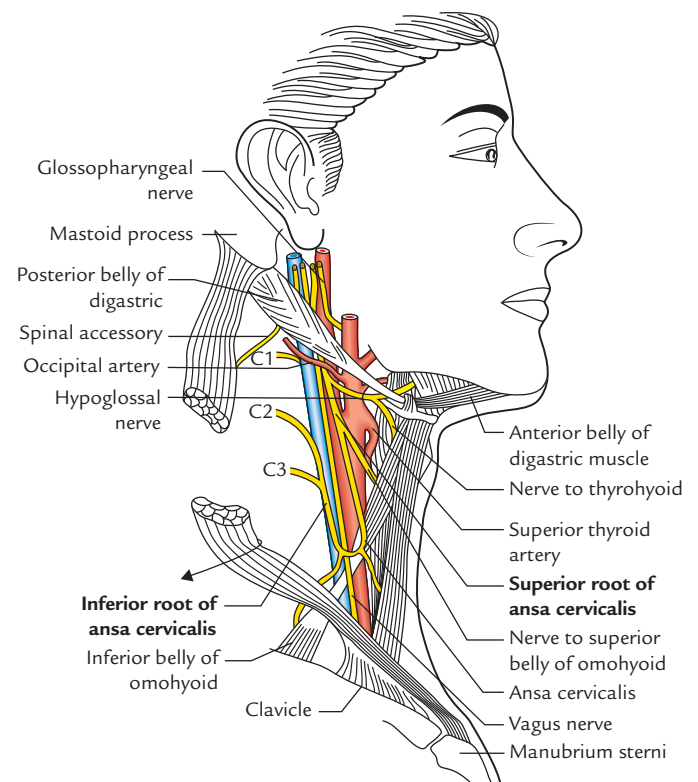


Fig. 6.8 Location and relations of the ansa cervicalis.

## Soft Tissues on the Back of the Neck

The soft tissues on the back of the neck are divided into four layers. From superficial to deep these are:

1. Skin (cutaneous innervation).
2. Superficial fascia.
3. Deep fascia.
4. Muscles.

### SKIN

The skin on the back of the neck is supplied by the medial branches of the dorsal rami of C2, C3, and C4 spinal nerves. (The dorsal ramus of the C1 does not divide into medial and lateral branches and does not give a cutaneous branch.)

The cutaneous nerves derived from the medial branches of dorsal rami of C2, C3, and C4 are (Fig. 7.1):

1. **Greater occipital nerve (C2):** It pierces the deep fascia on the level with the superior nuchal line 2.5 cm lateral to the external occipital protuberance and supply the posterior part of the scalp as far as the vertex. It is the **thickest cutaneous nerve** in the body.
2. **Third occipital nerve (C3):** It pierces the deep fascia medial to the greater occipital nerve, a little below the superior nuchal line. It is a small cutaneous nerve and supplies the skin of the nape of the neck up to the external occipital protuberance.
3. **Cutaneous branches of C4 and C5:** These branches pierce the deep fascia close to the midline in series with the 3rd occipital nerve. They supply the adjacent skin.

### SUPERFICIAL FASCIA

The **superficial fascia** (subcutaneous tissue) of the back is thick and in spite of its fat content, it is very tough. It contains cutaneous nerves and vessels. The cutaneous nerves are greater occipital, third occipital, and cutaneous branch of C4 and C5 (described above).

The cutaneous vessels are **occipital artery** and minute twigs derived from second part of the vertebral artery. The occipital artery appears 2.5 cm from midline and accompanies the greater occipital nerve.

### DEEP FASCIA

The deep fascia of the back is called **nuchal fascia**. It is attached in the median plane to the spines, supraspinous ligaments, and ligamentum nuchae. Extending laterally, it sheaths the muscles of the back.

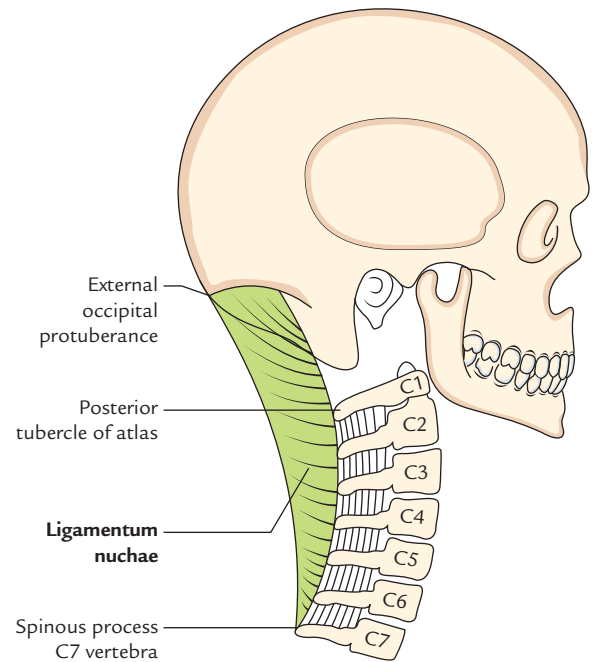


Fig. 7.2 Ligamentum nuchae.

### LIGAMENTUM NUCHAE (Fig. 7.2)

It is a triangular sheet of fibroelastic tissue that forms the median fibrous septum between the muscles of the two sides of the back of the neck.

The ligamentum nuchae presents three borders:

1. **Superior border** attached to the external occipital crest.
2. **Anterior border** attached to the posterior tubercle of atlas and the spines of the cervical vertebrae (C2 to C7).
3. **Free posterior border**, which extends from the external occipital protuberance to the spine of the 7th cervical vertebrae.

**N.B.** In quadrupeds, the ligamentum nuchae is a well-developed powerful elastic structure, which supports the heavy head against the gravity. In human beings, it is not that powerful and contains little elastic tissue.

### MUSCLES

The muscles of the back of the neck on either side of midline are arranged into **superficial and deep groups**:

1. **Superficial group:** The muscles of this group are arranged into two layers:
  - (a) *First (or superficial) layer*—consisting of trapezius muscle.
  - (b) *Second (or deep) layer*—consisting of levator scapulae, rhomboideus minor, and rhomboideus major.

## CHAPTER

# 8

# Parotid Region

The parotid region is the area around the ear, bounded *anteriorly* by anterior border of masseter, *superiorly* by the zygomatic arch, *posteriorly* by mastoid process, and *inferiorly* by line joining the angle of the mandible to the mastoid process.

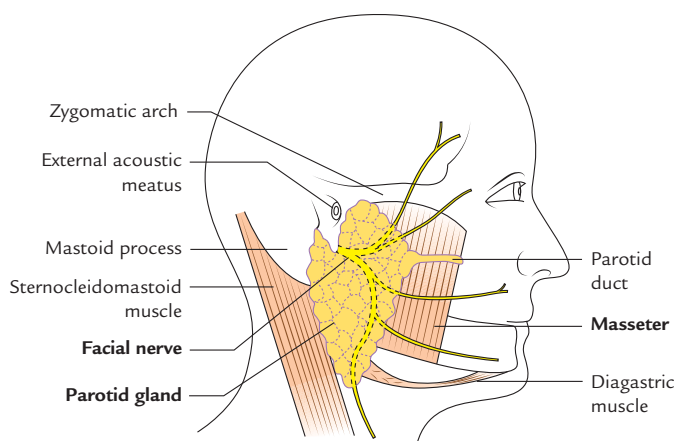
The principal structures in this area are **parotid gland** and **facial nerve**.

The facial nerve comes out of the cranial cavity through the stylomastoid foramen in this region and enters the parotid gland, where it divides into five terminal branches which emerge on the face, underneath the anterior margin of the parotid gland (Fig. 8.1).

The detailed knowledge of the anatomy of parotid region is important in clinical practice particularly while performing surgical procedures on parotid gland.

### PAROTID GLAND

The parotid (para = around, otic = ear) gland is the largest of the three pairs of salivary glands, *viz.* parotid, submandibular, and sublingual. It is composed almost entirely of **serous alveoli**. It is lobulated, yellowish brown, and weighs about 25 g.



**Fig. 8.1** Main features of the parotid region.

### LOCATION

The parotid gland lies in the pyramidal fossa, posterior to the ramus of the mandible called **retromandibular fossa (parotid bed)**.

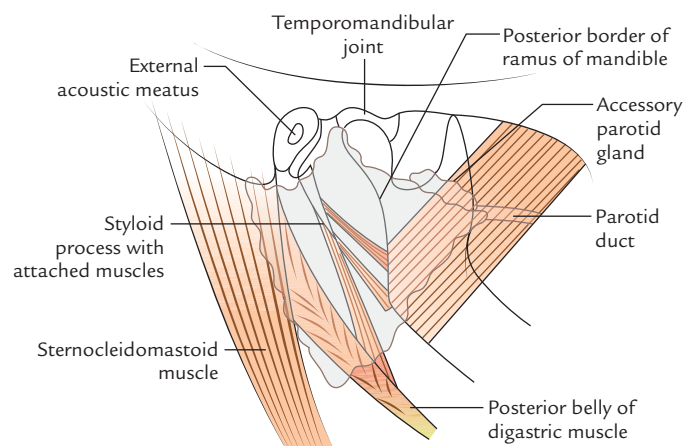
### Boundaries of the Parotid Bed (Fig. 8.2)

It is bounded:

- Anteriorly:* by the posterior border of the ramus of mandible.
- Posteriorly:* by the mastoid process.
- Superiorly:* by the external acoustic meatus and posterior part of temporomandibular joint.
- Medially:* by styloid process.

The parotid bed is lined by muscles, probably to make it soft as under:

1. **Ramus** of the mandible is covered by two muscles: masseter laterally and the medial pterygoid medially.
2. **Mastoid process** is covered by two muscles: sternocleidomastoid laterally and posterior belly of digastric muscle medially.
3. **Styloid process** is enveloped by three slender muscles: styloglossus, stylopharyngeus, and stylohyoid.



**Fig. 8.2** Parotid bed and location of the parotid gland.

The parotid gland is too soft to be palpable in healthy individual but one can identify the bony boundaries of the parotid bed.

### Extent

The parotid gland being soft is not confined only to the parotid bed but extends beyond it. The extent of parotid gland is as follows: it extends from external auditory meatus *above*, to the upper part of the carotid triangle *below*; *medially* it extends to the styloid process (close to the side wall of pharynx) and wraps around the neck of the mandible. *Posteriorly* it overlaps the sternocleidomastoid muscle and *anteriorly* it extends over the masseter for a variable distance. A part of this forward extension is often detached from the rest of the gland and is known as **accessory parotid gland**. The accessory parotid gland lies between the zygomatic arch above and the parotid duct below (Fig. 8.5). Several ducts from accessory gland open into the parotid duct.

### PAROTID CAPSULE (OR PAROTID SHEATH)

The parotid gland is enclosed in a fibrous capsule called **parotid capsule** (Fig. 8.3).

It is formed by the tough investing layer of deep cervical fascia. This fascia splits in the region between the angle of the mandible and mastoid process to enclose the gland. The superficial lamina is thick, strong, unyielding, and adherent to the gland while deep lamina is thin. The superficial lamina blends with the epimysium of masseter to form a thick **parotidomasseteric fascia**, which is attached above to the zygomatic arch. The thin deep lamina is attached to the tympanic plate and styloid process of the temporal bone; it thickens to form **stylomandibular ligament**, which separates the parotid gland from the submandibular gland.

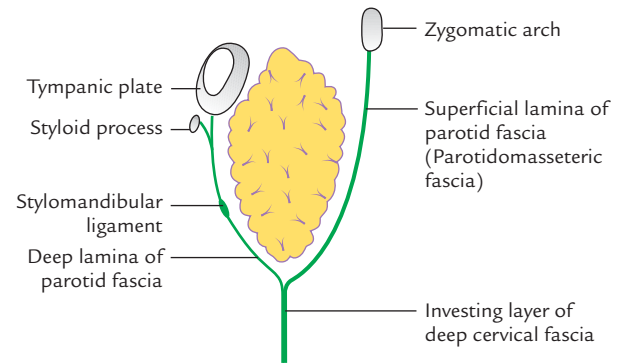


Fig. 8.3 Parotid capsule.

### Clinical correlation

**Infection of the parotid gland.** The parotid gland is commonly infected by the *mumps virus* causing inflammation and swelling of the gland (**mumps**).

The **parotid swellings are very painful** due to unyielding nature of the parotid capsule, any inflammation or tension within the parotid gland will cause severe pain. This is caused by the stretching of the capsule and stimulation of branches of great auricular nerve. The pain is usually exacerbated at meal time when gustatory stimulus to gland results in increased parotid secretion enhancing further turgor within the capsule. However, patient is relieved of pain to some extent after taking meals due to release of secretions. This is seen routinely in patients suffering from mumps (also see Clinical correlation on page 117).

### EXTERNAL FEATURES (Figs 8.4 and 8.5)

The gland resembles a three-sided pyramid with apex directed downwards. It presents the following features:

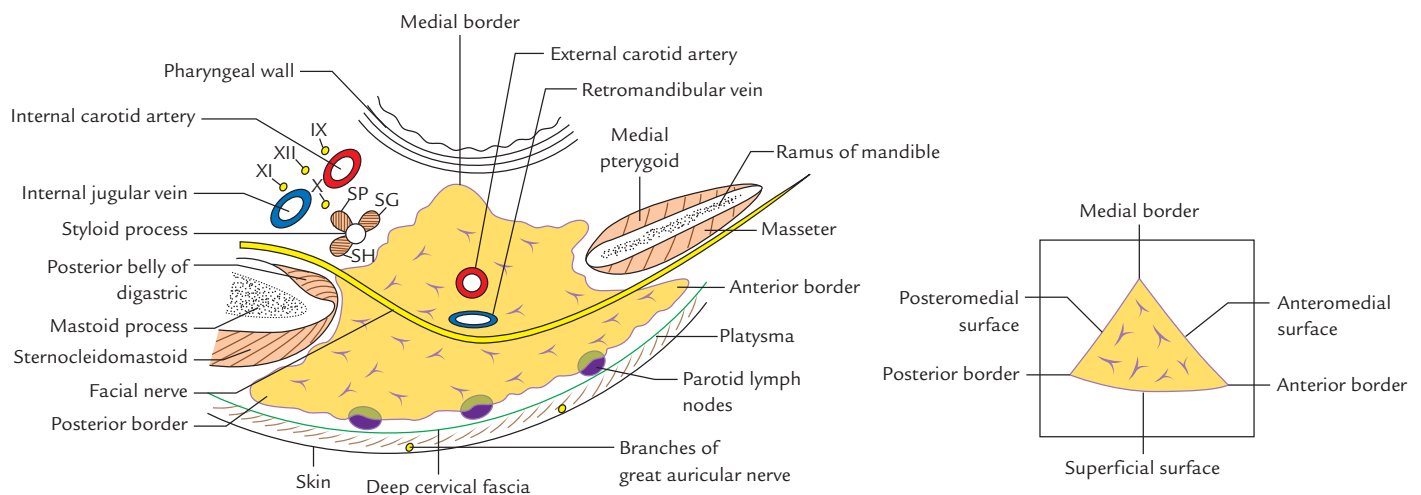
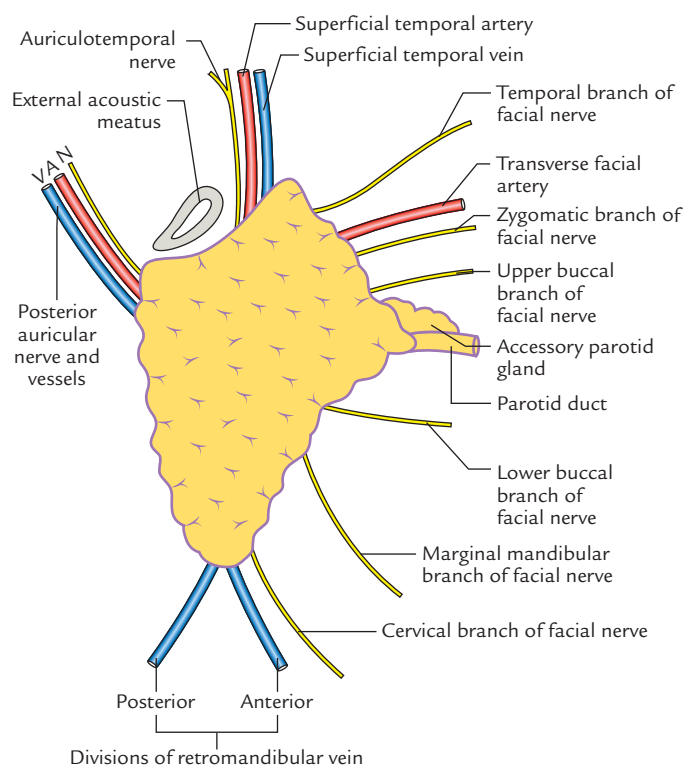


Fig. 8.4 Horizontal section through parotid gland showing its relations and the structures passing through it. The inset figure shows borders and surfaces of the parotid gland (SG = styloglossus muscle, SH = stylohyoid muscle, SP = stylopharyngeus muscle).



**Fig. 8.5** The structures emerging at the periphery of the parotid gland.

- **An apex**
- **Four surfaces**
  - (a) Superior surface or base.
  - (b) Superficial surface.
  - (c) Anteromedial surface.
  - (d) Posteromedial surface.
- **Three borders**
  - (a) Anterior.
  - (b) Posterior.
  - (c) Medial.

## RELATIONS

### Apex

It projects downwards overlapping the posterior belly of digastric muscle and adjoining part of the carotid triangle.

The structures emerging through the apex include (Fig. 8.5):

1. Cervical branch of the facial nerve.
2. Anterior and posterior divisions of retromandibular vein.

### Superior Surface or Base

It is concave and related to the external acoustic meatus and posterior aspect of temporomandibular joint.

The following structures emerge through it (Fig. 8.5):

1. Superficial temporal vessels.
2. Auriculotemporal nerve.

### Superficial Surface

It is the largest of the four surfaces. It is covered from superficial to deep by (Fig. 8.4):

1. Skin.
2. Superficial fascia containing anterior branches of greater auricular nerve, superficial parotid (preauricular) lymph nodes, and platysma.
3. Parotid fascia.
4. Deep parotid lymph nodes embedded in the gland.

### Anteromedial Surface

It is deeply grooved by the posterior border of the ramus of the mandible. It is related to (Fig. 8.4):

1. Masseter.
2. Medial pterygoid.
3. Posterior border of the ramus of the mandible.
4. Lateral aspect of the temporomandibular joint.

The branches of facial nerve emerge on face from underneath the anterior margin of this surface (Fig. 8.5).

### Posteromedial Surface

It is moulded onto the mastoid and styloid processes and their covering muscles. Thus it is related to (Fig. 8.4):

1. Mastoid process, sternocleidomastoid, and posterior belly of digastric.
2. Styloid process and styloid group of muscles.

The styloid process and its muscles separate the gland from internal carotid artery, internal jugular vein, and last four cranial nerves.

The following structures enter the gland through this surface:

1. Facial nerve trunk in its upper part.
2. External carotid artery in its lower part.

### Anterior Border

It separates the superficial surface from the anteromedial surface. The following structures (from above downwards) emerge in a radiating fashion underneath this border (Fig. 8.5):

- |                                                                                                                                                                                                                                         |                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <ol style="list-style-type: none"> <li>1. Temporal branch of the facial nerve.</li> <li>2. Zygomatic branch of the facial nerve.</li> <li>3. Transverse facial vessels.</li> <li>4. Upper buccal branch of the facial nerve.</li> </ol> | Above the<br>parotid duct |
| <ol style="list-style-type: none"> <li>5. Parotid duct.</li> <li>6. Lower buccal branch of the facial nerve.</li> <li>7. Marginal mandibular branch of the facial nerve.</li> </ol>                                                     | Below the<br>parotid duct |



### Posterior Border

It separates the superficial surface from the posteromedial surface.

The following structures emerge underneath this border (Fig. 8.5):

1. Posterior auricular vessels.
2. Posterior auricular branch of the facial nerve.

### Medial Border

It separates the anteromedial surface from the posteromedial surface. It is related to the lateral wall of the pharynx.

### STRUCTURES PRESENT WITHIN THE PAROTID GLAND

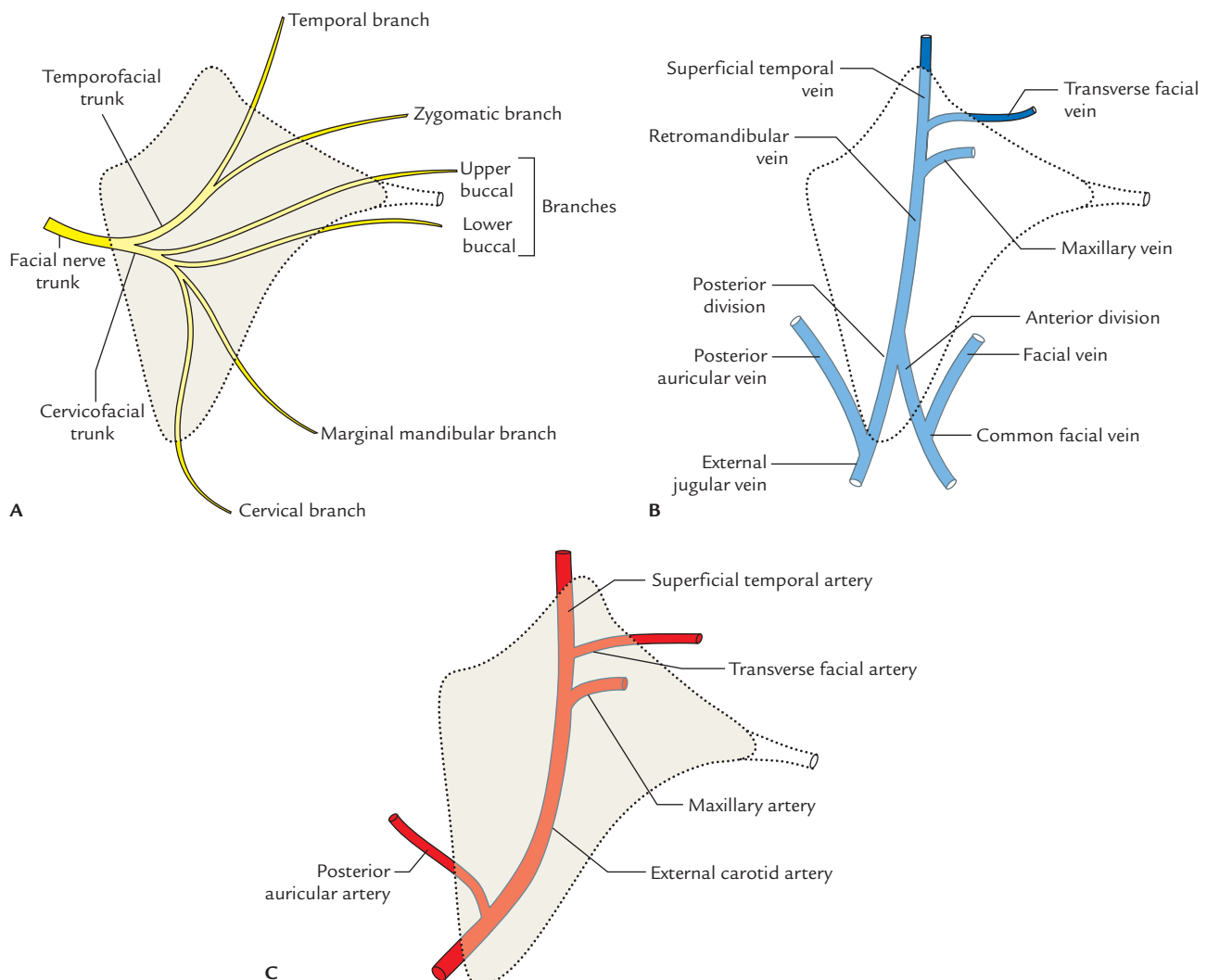
Three main structures either in part or in whole traverse the gland and branch within it (Figs 8.4 and 8.6). From superficial to deep these are:

1. Facial nerve.
2. Retromandibular vein.
3. External carotid artery.

**N.B.** Some members of the deep parotid lymph nodes and filaments of auriculotemporal nerve are also located within the gland.

The **facial nerve** (Fig. 8.6A) is most superficial. It enters the gland through the upper part of the posteromedial surface and divides into its terminal branches within the gland. The branches run horizontally and leave the gland through its anteromedial surface and appear on the face by passing underneath its anterior border.

The five terminal branches of the facial nerve radiate like a goose-foot through the anterior border of the gland and supply the muscles of facial expression. Such branching pattern of the facial nerve is termed **pes anserinus**.



**Fig. 8.6** The structures traversing the parotid gland: A, facial nerve; B, retromandibular vein; C, external carotid artery.

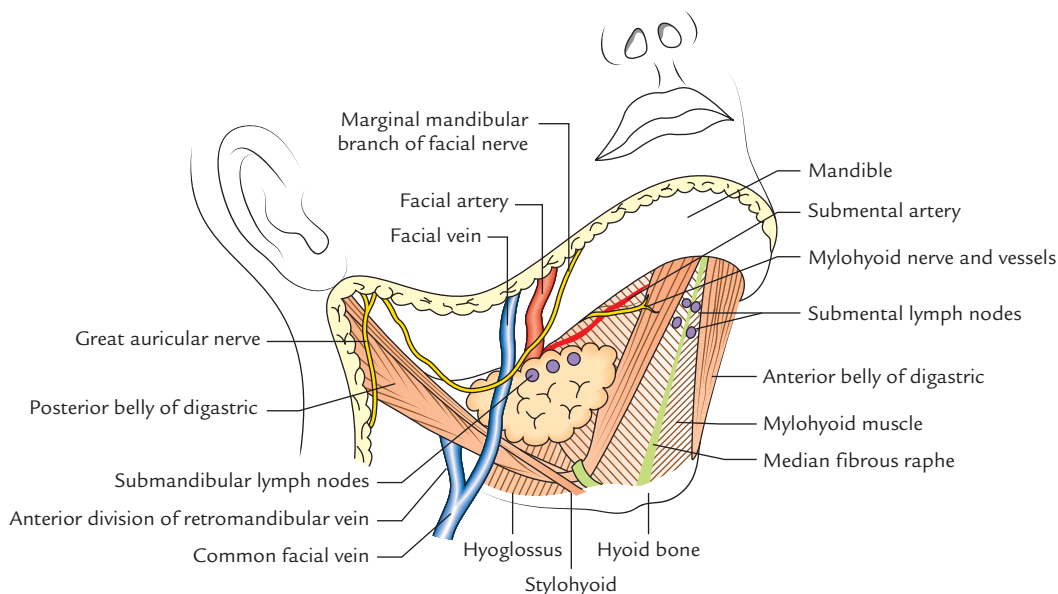
# Submandibular Region

The submandibular region is below and under the cover of the body of the mandible. It extends upwards up to the mylohyoid line and below up to the hyoid bone. Superficially it includes both submandibular and submental triangles (Fig. 9.1).

The deep structures in this region include floor of the mouth and the root of the tongue. Clinically this region is very important because it is the common site for swellings due to enlargement of submandibular lymph nodes and submandibular salivary gland. The inflammatory edema of the floor of the mouth (called **Ludwig's angina**) spreads in this region to cause generalized swelling of the region. The surgical procedures are commonly performed in this region, hence students should study this region very carefully. The term *submandibular* often misleads the students as it gives the impression that the submandibular region deals with the structures below the mandible only.

## SUPERFICIAL STRUCTURES IN THE SUBMANDIBULAR REGION

The submandibular region is crossed superficially by platysma in the superficial fascia. Deep to platysma, the marginal mandibular nerve (lowest branch of facial nerve) crosses the lower border of the mandible near the angle and runs below it across the submandibular region before turning upwards to reach the mental region of the face where it supplies depressors of the lower lip (depressor labii inferioris, depressor anguli oris) and mentalis muscle. The surgical incisions are, therefore, not made along the lower border of the mandible to avoid injury to this nerve. The great auricular nerve crosses the posterior belly of digastric on its way to supply the skin of face and auricle (Fig. 9.1).



**Fig. 9.1** Structures seen in the superficial dissection of the right submandibular region.

## DEEP STRUCTURES IN THE SUBMANDIBULAR REGION

The deep structures in the submandibular region (also called *contents of the submandibular region*) are:

1. Suprahyoid muscles (all paired), viz. digastric, stylohyoid, mylohyoid, and geniohyoid.
2. Extrinsic muscles of the tongue (all paired), viz. hyoglossus, styloglossus, and genioglossus.
3. Submandibular and sublingual salivary glands.
4. Facial and lingual arteries.
5. Lingual, hypoglossal, and glossopharyngeal nerves.
6. Submandibular ganglion.

**N.B.** It is important to know that on either side of midline, the *submandibular salivary gland* is the most prominent structure and the *hyoglossus muscle* is the key muscle of the submandibular region.

## MUSCLES OF THE SUBMANDIBULAR REGION

From the point of view of surgical procedures, the muscles in the submandibular region form four muscular planes; from superficial to deep, these are:

1. **First muscular plane:** formed by the digastric and stylohyoid muscles.
2. **Second muscular plane:** formed by the mylohyoid muscle.
3. **Third muscular plane:** formed by the geniohyoid, hyoglossus, and styloglossus muscles.
4. **Fourth muscular plane:** formed by the genioglossus and a part of superior constrictor of the pharynx.

## MUSCLES OF FIRST MUSCULAR PLANE

### Digastric Muscle (Fig. 9.2)

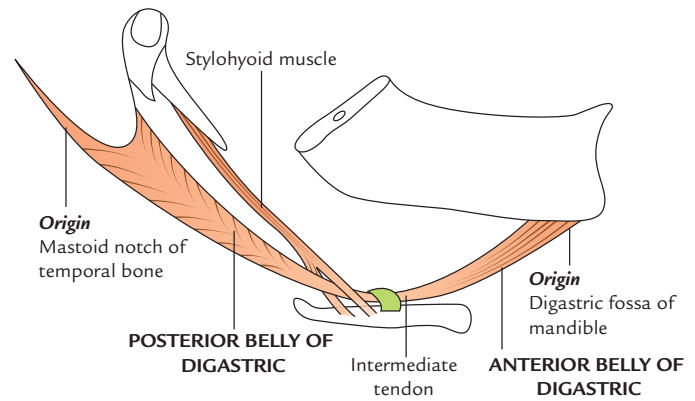
It is a strap-like muscle consisting of posterior and anterior bellies, united by an intermediate tendon. This muscle is so called because it has two bellies.

#### Origin

1. The *bipinnate posterior belly* arises from the mastoid notch of the temporal bone.
2. The *unipinnate anterior belly* arises from the digastric fossa on the lower border of the mandible close to the symphysis menti.

#### Insertion

1. Posterior belly passes downwards and forwards between the carotid triangle below and behind and digastric triangle above and in front and inserted into the intermediate tendon.



**Fig. 9.2** Origin and insertion of the digastric muscle.

2. Anterior belly passes downwards and backwards on the mylohyoid to be inserted into the intermediate tendon.

The intermediate tendon is anchored to the junction of the body and greater cornu of hyoid bone by an inverted U-shaped facial sling of investing layer of deep cervical fascia. The tendon of digastric muscle passes between the two slips of tendon of stylohyoid muscle.

#### Nerve supply

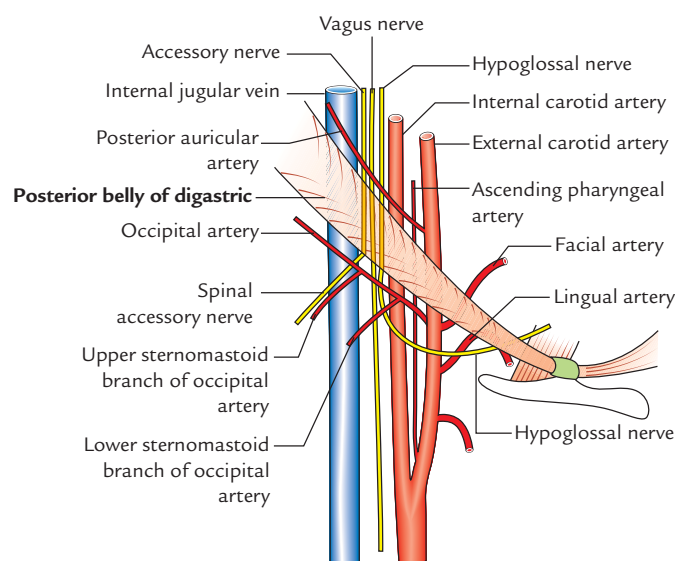
1. The posterior belly develops from mesoderm of the 2nd pharyngeal arch and therefore, supplied by the facial nerve.
2. The anterior belly develops from the 1st pharyngeal arch and, therefore, supplied by the mylohyoid nerve, a branch of inferior alveolar nerve from mandibular nerve.

#### Actions

1. Helps to depress the mandible when the mouth is opened widely against resistance.
2. Pulls the hyoid bone upwards during deglutition.

#### Relations of the posterior belly of the digastric muscle

1. **Superficial:**
  - (a) Skin, superficial fascia, platysma, and investing layer of deep cervical fascia.
  - (b) Mastoid process and sternocleidomastoid muscle.
  - (c) Parotid gland and the angle of the mandible.
2. **Deep (Fig. 9.3):**
  - (a) Neurovascular bundle of neck consisting of internal jugular vein, external and internal carotid arteries 10th, 11th, and 12th cranial nerves.
  - (b) **Upper border:** Stylohyoid muscle and posterior auricular artery run along the upper border of the posterior belly of the digastric.



**Fig. 9.3** Important deep relations of the posterior belly of the digastric muscle.

- (c) **Lower border:** Occipital artery runs along and under the cover of the lower border of digastric.

**N.B.** The relations of posterior belly of digastric are important because three cranial nerves: 10th, 11th, and 12th, and three great blood vessels of neck, viz. internal jugular vein, internal and external carotid arteries pass deep to it.

Differences between the posterior and anterior bellies of the digastric muscles are listed in Table 9.1.

**Table 9.1** Differences between the anterior and posterior belly of the digastric muscle

| Anterior belly                                             | Posterior belly                                         |
|------------------------------------------------------------|---------------------------------------------------------|
| Unipennate                                                 | Bipennate                                               |
| Develops from 1st pharyngeal arch                          | Develops from 2nd pharyngeal arch                       |
| Supplied by mylohyoid nerve (nerve of 1st pharyngeal arch) | Supplied by facial nerve (nerve of 2nd pharyngeal arch) |

### Stylohyoid Muscle (Fig. 9.4)

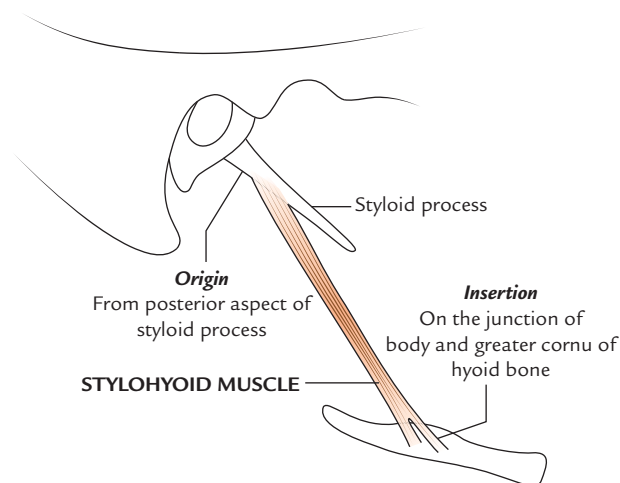
It is a slender muscle that lies along the upper border of the posterior belly of the digastric muscle.

#### Origin

It arises from the posterior surface of the styloid process.

#### Insertion

It is inserted into the hyoid bone at the junction between the body and greater cornu. At the insertion, its tendon splits into two slips that pass one on either side of the intermediate tendon of the digastric muscle (Fig. 9.3).



**Fig. 9.4** Origin and insertion of the stylohyoid muscle.

### Nerve supply

The stylohyoid muscle develops from the 2nd arch and, therefore, it is supplied by the facial nerve.

### Actions

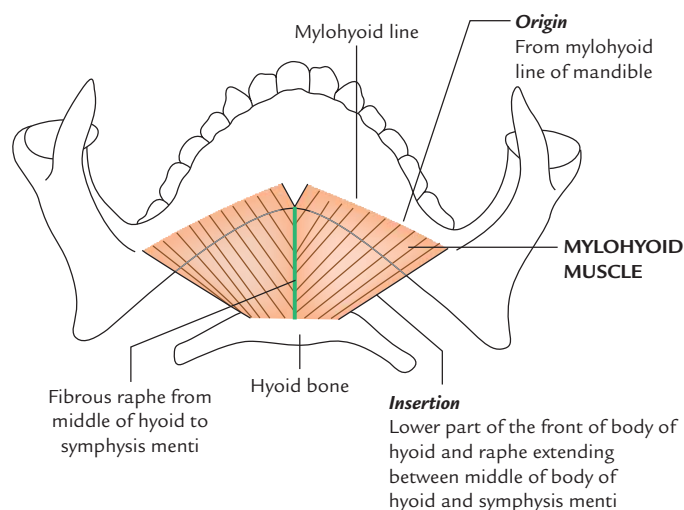
It draws the hyoid bone upwards and backwards, and elongates the floor of the mouth.

**N.B.** The stylohyoid muscle is considered as the delaminated portion of the posterior belly of the digastric muscle.

## MUSCLES OF SECOND MUSCULAR PLANE

### Mylohyoid Muscle (Fig. 9.5)

It is a flat, triangular muscle lying deep to the anterior belly of the digastric muscle. The right and left mylohyoid muscles join in the median fibrous raphe to form the gutter-shaped



**Fig. 9.5** Origin and insertion of the mylohyoid muscle (as seen from below).

floor of the mouth; over which lies the tongue, hence the floor of the mouth is also called **diaphragma oris**.

### Origin

From the mylohyoid line of the mandible.

### Insertion

The fibres run downwards and medially. The posterior fibres are inserted into the body of the hyoid bone. The middle and anterior fibres are inserted into the median fibrous raphe extending from symphysis menti to the hyoid bone.

### Nerve supply

The mylohyoid muscle develops from the first pharyngeal arch, therefore it is supplied by mylohyoid nerve, a branch of inferior alveolar nerve from mandibular nerve.

### Actions

1. The mylohyoid muscle elevates the floor of the mouth and hence the tongue during the first stage of the deglutition.
2. It also helps in the depression of the mandible against resistance.
3. It fixes or elevates the hyoid bone.

## MUSCLES OF THIRD MUSCULAR PLANE

### Geniohyoid Muscle (Fig. 9.6)

It is a narrow muscle that lies alongside the midline deep to the mylohyoid.

### Origin

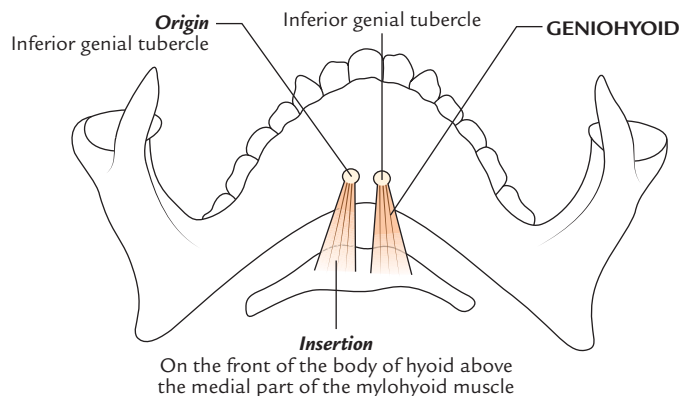
From inferior genial tubercle of the mandible.

### Insertion

The fibres run backwards and downwards to be inserted into the anterior surface (front) of the body of the hyoid bone, above the medial part of the mylohyoid muscle.

### Nerve supply

C1 fibres through hypoglossal nerve.



**Fig. 9.6** Origin and insertion of the geniohyoid muscle.

### Actions

1. The geniohyoid elevates the hyoid bone by pulling it upwards and forwards and thus shortens the floor of the mouth.
2. They may depress the mandible when the hyoid bone is fixed.

### Hyoglossus Muscle (Fig. 9.7)

It is the flat quadrilateral muscle of the tongue.

### Origin

From upper surface of the entire length of the greater cornu and adjacent part of the body of hyoid bone.

### Insertion

Into the side of tongue between styloglossus laterally and inferior longitudinal medially. The fibres of hyoglossus from hyoid bone run upwards and slightly forward and decussate with the fibres of styloglossus.

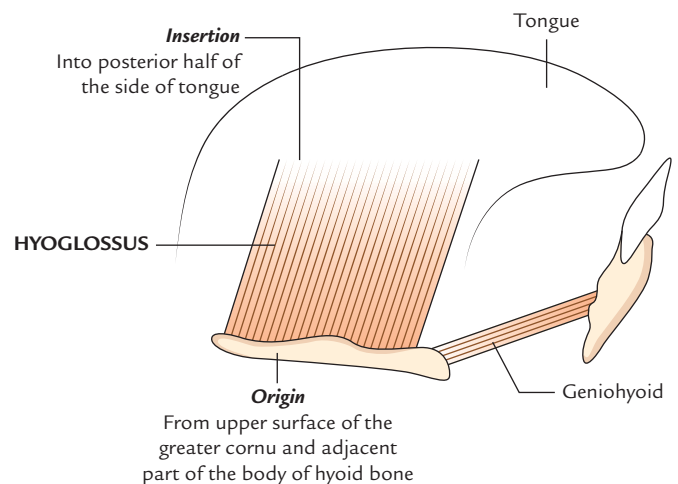
**N.B.** A part of hyoglossus may be attached to the lesser cornu of the hyoid bone and form a separate muscle called *chondroglossus*.

### Relations

The hyoglossus, a quadrilateral sheet of muscle, is the key muscle of the suprahyoid region because it serves as a landmark for neighboring structures in the region. Therefore, its relations are very important to surgeons.

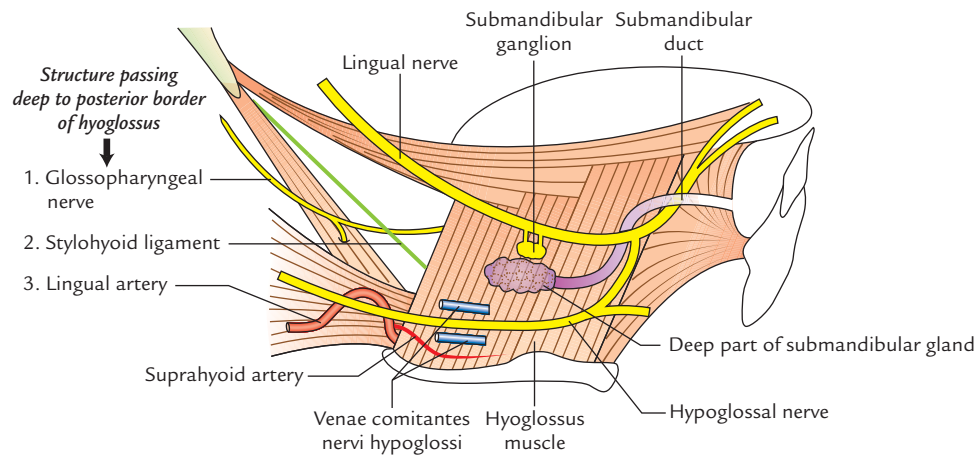
**Superficial relations (Fig. 9.8):** The superficial relations of the hyoglossus are as follows:

1. Hypoglossal nerve, crosses the lower part of muscle from behind forwards.
2. Lingual nerve, crosses the upper part of muscle from behind forwards.



**Fig. 9.7** Origin and insertion of the hyoglossus muscle.





**Fig. 9.8** Superficial relations of the hyoglossus muscle.

3. Deep part of the submandibular gland and submandibular duct. The gland lies in the middle of hyoglossus muscle and the duct lies between the gland and the muscle.
4. Submandibular ganglion lies between the lingual nerve and deep part of the submandibular gland.
5. Styloglossus muscle—interdigitates with hyoglossus.
6. Mylohyoid muscle overlaps the hyoglossus anterosuperiorly.

**Deep relations (Fig. 9.22D):** The deep relations of the hyoglossus are as follows:

1. Inferior longitudinal muscle of tongue.
2. Genioglossus muscle anteriorly.
3. Middle constrictor of pharynx posteriorly.
4. Glossopharyngeal nerve.
5. Stylohyoid ligament.
6. Lingual artery.

**Structures passing deep to posterior border of hyoglossus**

From above downwards, these are (Fig. 9.8) as follows:

1. Glossopharyngeal nerve.
2. Stylohyoid ligament.
3. Lingual artery.

**Nerve supply**

The hyoglossus muscle develops from **occipital myotomes**, therefore, it is supplied by hypoglossal nerve.

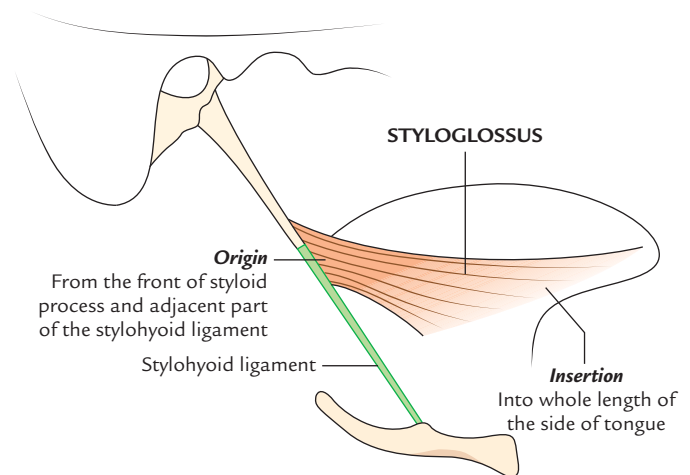
**Actions**

1. Depresses the side of tongue to make the dorsal surface of the tongue convex.
2. Helps in the retraction of the protruded tongue.

**Styloglossus Muscle (Fig. 9.9)**

**Origin**

From front of the tip of styloid process and adjoining part of stylohyoid ligament.



**Fig. 9.9** Origin and insertion of the styloglossus muscle.

**Insertion**

The fibres run forwards to be inserted into the whole length of the side of tongue, interdigitating with the fibres of the hyoglossus muscle.

**Nerve supply**

It is supplied by hypoglossal nerve.

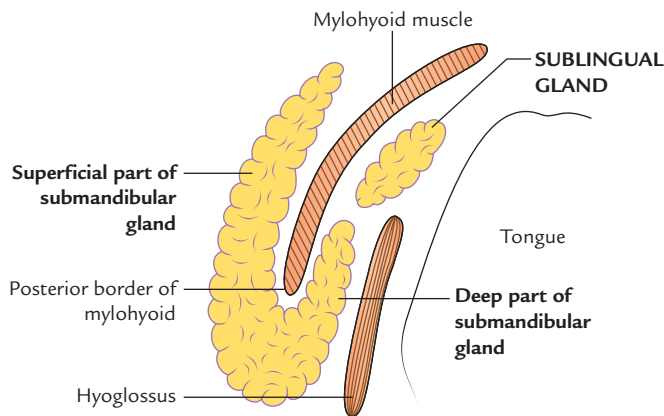
**Actions**

Retracts the tongue upwards and backwards, and thus it is antagonist to the genioglossus. Along with its counterpart of opposite side, it forms a median gutter on the dorsum of tongue for the passage of food.

**MUSCLES OF FOURTH MUSCULAR PLANE**

**Genioglossus Muscle (Fig. 9.10)**

It is a fan-shaped extrinsic muscle of tongue and along with its counterpart of the opposite side forms the most of the bulk of the tongue.



**Fig. 9.13** Horizontal section through submandibular region showing the parts of submandibular gland. The sublingual salivary gland is also seen. Note the relationship of the sublingual gland with the deep part of the submandibular gland.

### Parts

It consists of two parts: (a) a large superficial part and (b) a small deep part. The superficial part lies superficial to the mylohyoid muscle, while deep part lies deep to the mylohyoid muscle. The two parts are continuous with each other around the posterior border of the mylohyoid muscle (Fig. 9.13).

### Superficial Part

This part of the gland is quite large and fills the anterior part of the digastric triangle extending upwards up to the mylohyoid line. The superficial part presents **two ends**—anterior and posterior and **three surfaces**—inferior, lateral, and medial.

The *anterior end* extends up to the anterior belly of the digastric muscle.

The *posterior end* extends up to the stylomandibular ligament, which separates the submandibular gland from the parotid gland. This end presents a groove produced by ascending limb of the cervical loop of the facial artery.

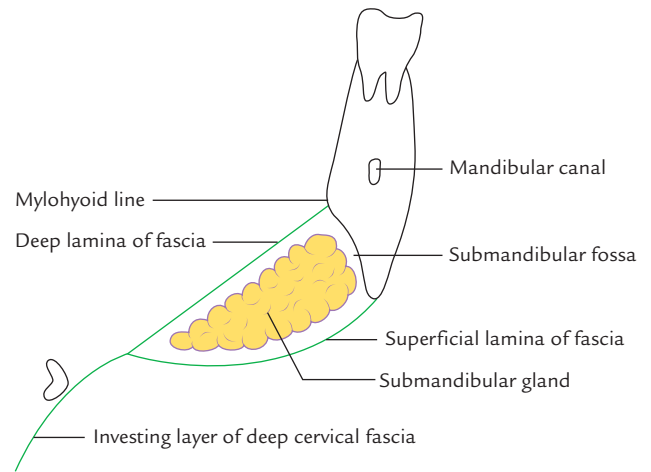
### Fascial covering or capsule (Fig. 9.14)

The superficial part is partially enclosed between the two layers of investing layer of deep cervical fascia. At the greater cornu of hyoid bone the investing layer of deep cervical fascia splits into two laminae to enclose the superficial part. The superficial layer covers the inferior surface of the gland and is attached to the base of the mandible. The deep layer covers the medial surface of the gland and is attached to the mylohyoid line of the mandible.

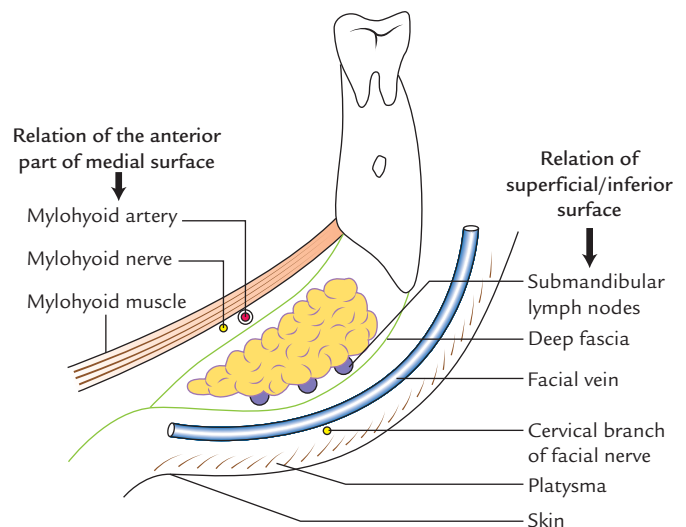
### Relations

The three surfaces of the superficial part have important relations:

- **Superficial surface** (inferior surface) from superficial to deep is covered by the following structures (Fig. 9.15):



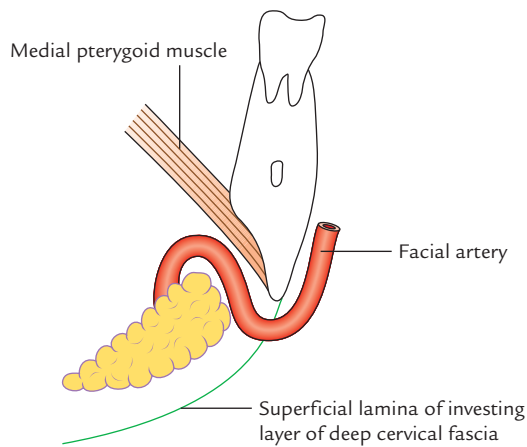
**Fig. 9.14** Capsule of the superficial part of the submandibular gland.



**Fig. 9.15** Relations of the superficial (inferior) surface of the submandibular salivary gland. The relations of anterior part of the medial (deep) surface are also seen.

- Skin.
- Superficial fascia containing platysma and cervical branch of facial nerve.
- Deep fascia.
- Facial vein.
- Submandibular lymph nodes.
- **Lateral surface** is related to (Fig. 9.16):
  - Submandibular fossa on the inner aspect of the body of mandible.
  - Medial pterygoid muscle (near its insertion).
  - Facial artery.

It is important to note that the facial artery loops downwards and forwards between the bone and the gland, and then winds around the lower border of the body of the mandible at the anteroinferior angle of the mandible to reach the face (Fig. 9.16).



**Fig. 9.16** Relations of the lateral surface.

- **Medial surface** is extensive and divided into three parts: anterior, middle, and posterior (Fig. 9.17):
  - (a) *Anterior part* is related to:
    - Mylohyoid muscle.
    - Submental branch of facial artery.
    - Mylohyoid nerve and vessels.
  - (b) *Middle (intermediate) part* is related to:
    - Hyoglossus muscle.
    - Styloglossus muscle.
    - Lingual and hypoglossal nerves.
    - Submandibular ganglion.
  - (c) *Posterior part* is related to:
    - Styloglossus muscle.
    - Stylohyoid ligament.
    - Glossopharyngeal nerve.
    - Wall of pharynx.

### Deep Part

The deep part is small and lies on the hyoglossus muscle deep to the mylohyoid; *posteriorly* it is continuous with superficial part around the posterior border of the mylohyoid, and *anteriorly* it extends up to the sublingual salivary gland (Fig. 9.13).

### Relations (Fig. 9.18)

*Medial:* Hyoglossus.  
*Lateral:* Mylohyoid.  
*Superior:* Lingual nerve and submandibular ganglion.  
*Inferior:* Hypoglossal nerve accompanied by a pair of veins (Venae comitantes nervi hypoglossi).

### Submandibular duct (Wharton's duct; Fig. 9.19)

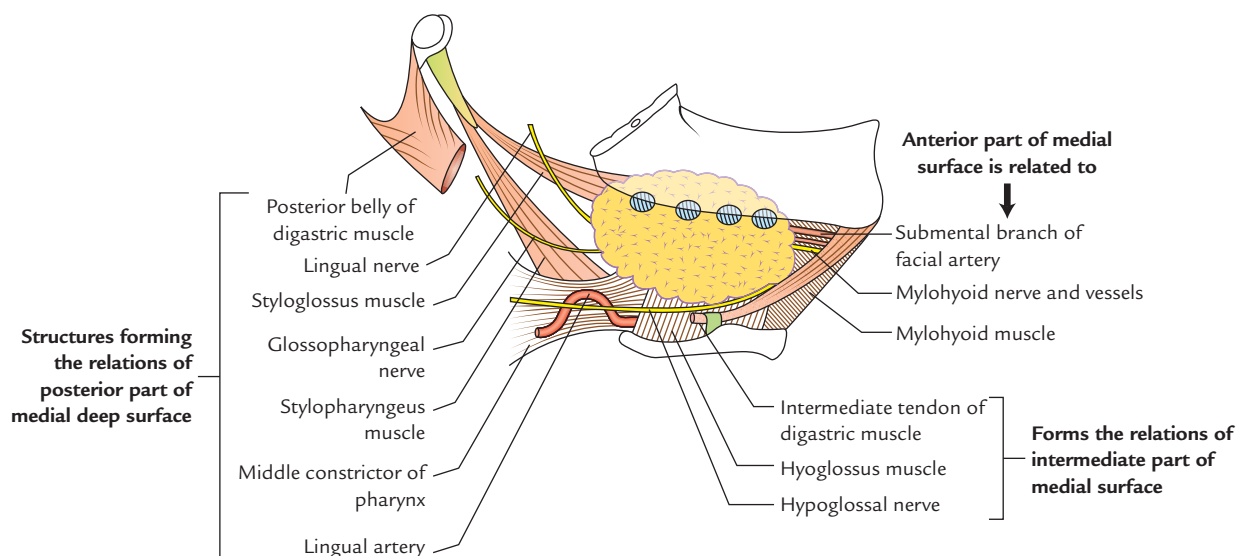
The submandibular duct is about 5 cm long and emerges at the anterior end of the deep part. It runs forwards on the hyoglossus between the lingual and hypoglossal nerves. Near the anterior border of the hyoglossus, it is crossed by lingual nerve. It continues running forward between the sublingual gland and the genioglossus. Here it lies just deep to the mucus membrane of the oral cavity. Finally, it opens into the oral cavity on the summit of a *sublingual papilla* at the side of the frenulum of the tongue.

### Blood supply

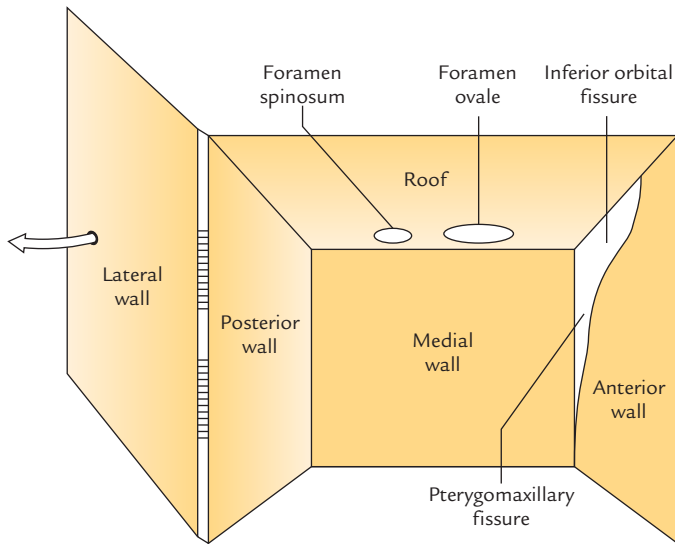
The gland is supplied by sublingual and submental arteries and drained by common facial and lingual veins.

### Lymphatic drainage

The lymphatics from submandibular gland first drain into submandibular lymph nodes and subsequently into jugulo-digastric lymph nodes.



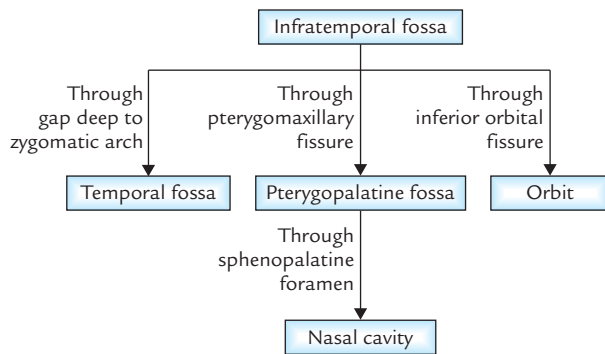
**Fig. 9.17** Relations of the medial (deep) surface of the submandibular gland.



**Fig. 10.3** Schematic diagram to show the boundaries and communications of the infratemporal fossa.

- above with the temporal fossa through a gap, deep to zygomatic arch; and middle cranial fossa through foramen ovale and foramen spinosum,
- below it is continuous with the tissue spaces of the neck lateral to the pharynx,
- with the pterygopalatine fossa through pterygomaxillary fissure, and
- with the orbit through inferior orbital fissure.

All communications are presented in Flowchart 10.1.



**Flowchart 10.1** Communications of the infratemporal fossa.

**N.B.** The infratemporal fossa can be best visualized as a hollow cube without floor (Fig. 10.3).

## CONTENTS

The major structures present in the infratemporal fossa are:

- Muscles:** Lateral pterygoid, medial pterygoid, and tendon of temporalis.

- Blood vessels:** Maxillary artery, maxillary vein, and pterygoid venous plexus.
- Neural structures:** Mandibular nerve, chorda tympani nerve, and otic ganglion.

## MUSCLES

### Lateral Pterygoid (Fig. 10.4)

It is a short, thick conical muscle with its apex pointing backwards. It passes backwards and slightly laterally from the roof and medial wall of the fossa to the neck of the mandible.

#### Origin

The lateral pterygoid consists of two heads, upper and lower:

- The **upper smaller head** arises from the infratemporal surface and crest of the greater wing of the sphenoid bone.
- The **lower larger head** arises from the lateral surface of the lateral pterygoid plate of the sphenoid bone.

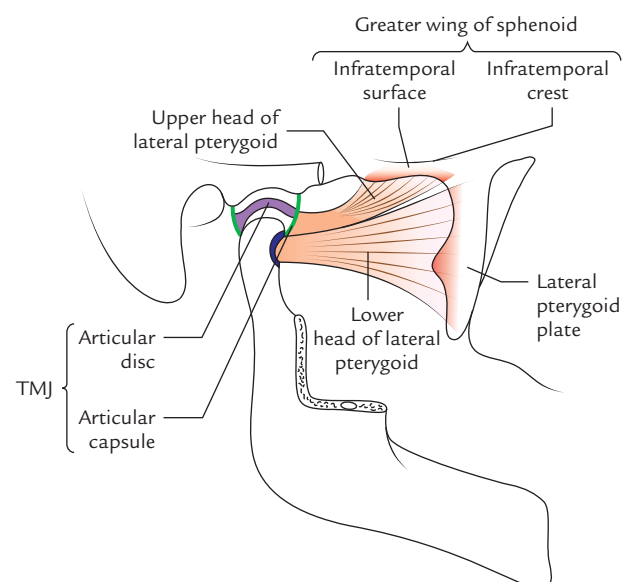
#### Insertion

The fibres of two heads run backwards and laterally, and converge to form a thick tendon, which is inserted into:

- Pterygoid fovea on the front of the neck of the mandible.
- Articular disc and capsule of the temporomandibular joint.

#### Nerve supply

Lateral pterygoid is supplied by a branch of anterior division of the mandibular nerve.



**Fig. 10.4** Origin and insertion of the lateral pterygoid muscle.

### Actions

1. Lateral pterygoids of two sides depress the mandible (opens the mouth) by pulling forward the condylar processes of the mandible and the articular discs of the temporomandibular joints.
2. Lateral and medial pterygoid muscles of two sides acting together protrude the mandible.
3. Lateral and medial pterygoid muscles of the two sides contract alternately to produce side-to-side movements of the lower jaw as in chewing.

### N.B.

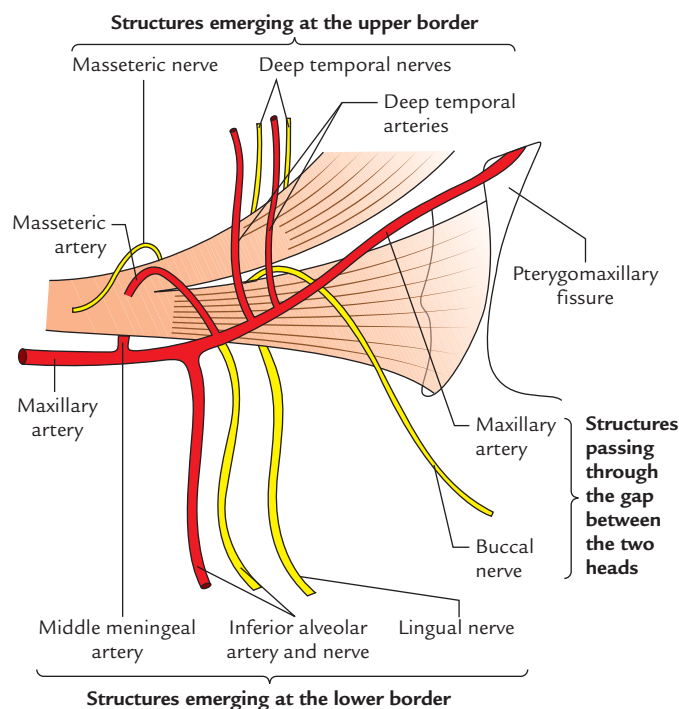
- The lower head of lateral pterygoid passes between the two heads of the medial pterygoid muscle.
- It is the only masticatory muscle, which opens the mouth.
- The articular disc of temporomandibular joint is developmentally a part of tendon of lateral pterygoid muscle.

### Relations (Fig. 10.5)

The lateral pterygoid is regarded as the **key muscle of the infratemporal region** because its relations provide a fair idea about the layout of structures in this region. Its relations are:

#### Superficial:

1. Ramus of the mandible.
2. Masseter.
3. Tendon of temporalis.
4. Superficial head of medial pterygoid.
5. Maxillary artery and its temporal and masseteric branches.



**Fig. 10.5** Relation of the lateral pterygoid muscle.

#### Deep:

1. Mandibular nerve.
2. Middle meningeal artery.
3. Sphenomandibular ligament.
4. Deep head of medial pterygoid muscle.

#### Structures emerging at the upper border:

1. Deep temporal nerves (two in number).
2. Masseteric nerve.

#### Structures emerging at the lower border:

1. Inferior alveolar nerve and artery.
2. Lingual nerve.
3. Middle meningeal artery (it passes up deep to the lower border).

#### Structures passing through the gap between the two heads:

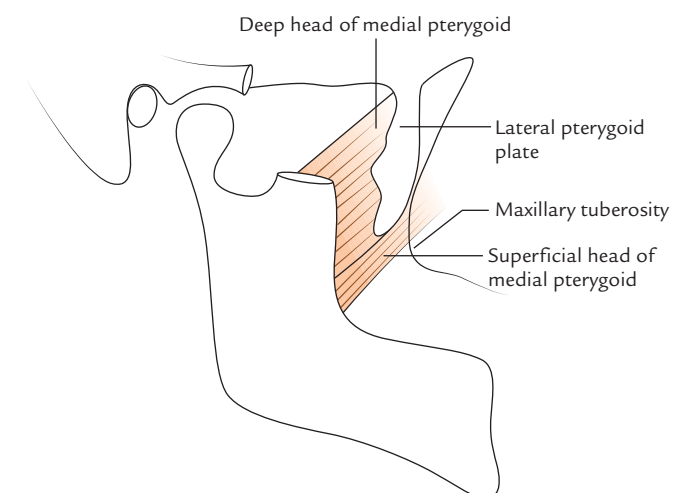
1. **Maxillary artery**, which enters the gap to reach the pterygopalatine fossa through pterygomaxillary fissure.
2. **Buccal nerve**, a branch of mandibular nerve. It comes out through the gap to provide sensory innervation to the skin and mucus membrane of the cheek.

### Medial Pterygoid (Fig. 10.6)

The medial pterygoid is a thick quadrilateral muscle and consists of two heads: superficial and deep.

#### Origin

1. The small **superficial head** (a small slip of muscle) arises from maxillary tuberosity and lateral surface of the pyramidal process of palatine bone.
2. The **large deep head** (forming the bulk of muscle) arises from medial surface of the lateral pterygoid plate and grooved surface of the pyramidal process of palatine bone.



**Fig. 10.6** Origin and insertion of the medial pterygoid muscle.



### Insertion

The fibres run downwards, backwards, and laterally to be inserted by a strong tendinous lamina into a roughened area on the posteroinferior part of the medial surface and angle of ramus of mandible as high as the mandibular foramen and as forwards as the mylohyoid groove (Fig. 9.11).

### Nerve supply

The medial pterygoid is supplied by a *nerve to medial pterygoid*, a branch from the main trunk of the mandibular nerve.

### Relations

*Superficial (Fig. 10.7):*

1. Lingual nerve.
2. Inferior alveolar nerve.
3. Inferior alveolar vessels.

*Deep:*

1. Levator palati and tensor palati muscles.
2. Superior constrictor of pharynx.
3. Styloglossus and stylopharyngeus muscles.

### Actions

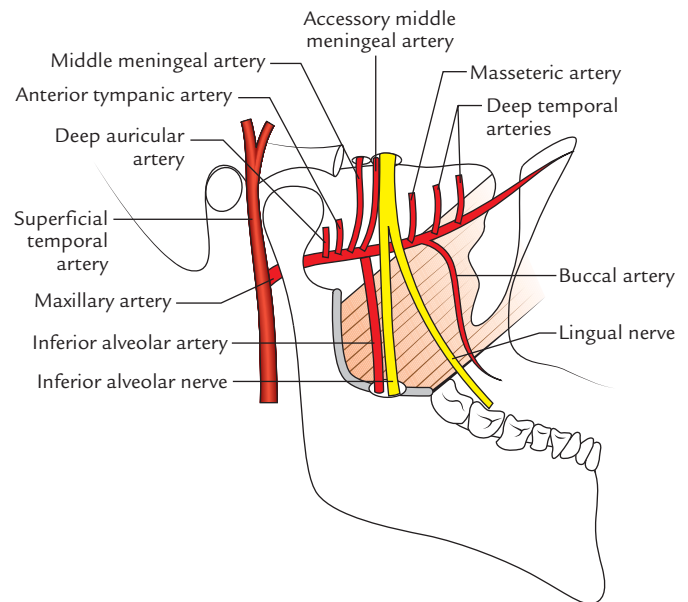
1. Medial pterygoids of two sides elevate the mandible to help in closing of mouth.
2. Acting with lateral pterygoids, the medial pterygoids protrude the mandible.
3. When medial and lateral pterygoids of one side act together, the corresponding side of the mandible is rotated forwards and to the opposite side.
4. Lateral and medial pterygoids of two sides when contract alternately produce side-to-side movements, which are used to grind the food.

## BLOOD VESSELS

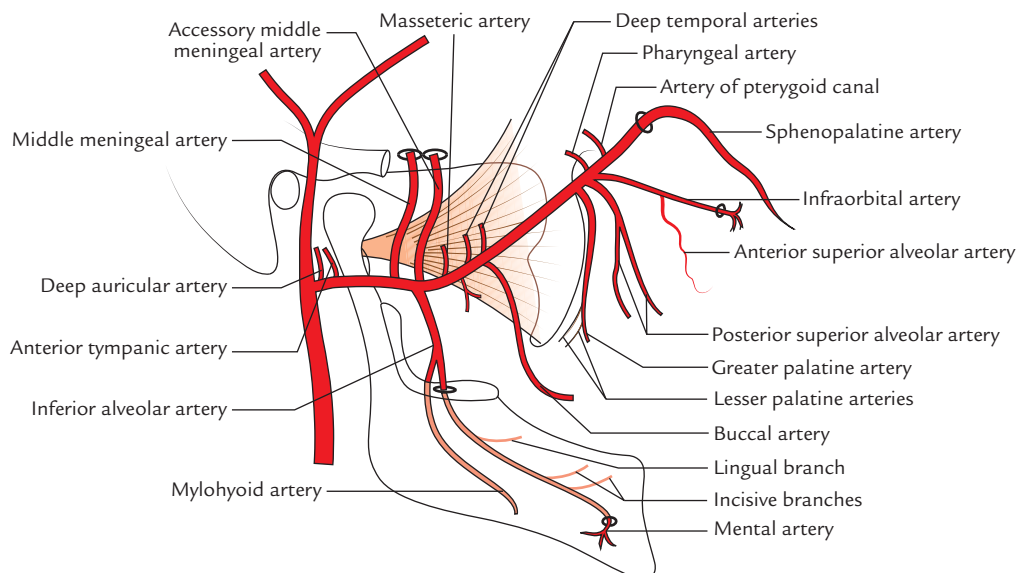
### MAXILLARY ARTERY (Figs 10.7 and 10.8)

The maxillary artery is the larger terminal branch of the external carotid artery.

It arises behind the neck of the mandible, runs horizontally forward up to the lower border of lower head of lateral pterygoid. Now it turns upwards and forwards, crosses the lower head of lateral pterygoid superficially (sometimes deep). After emerging between the two heads of lateral pterygoid it enters the pterygopalatine fossa by passing



**Fig. 10.7** Superficial relations of the medial pterygoid muscle.



**Fig. 10.8** Branches of the maxillary artery.

through pterygomaxillary fissure. Here it ends by giving its terminal branches.

The maxillary artery has a wide territory of distribution. It supplies:

- (a) upper and lower jaws,
- (b) muscles of temporal and infratemporal fossae,
- (c) nose and paranasal sinuses,
- (d) palate and roof of pharynx,
- (e) external and middle ear,
- (f) pharyngotympanic tube, and
- (g) dura mater.

**N.B.** The maxillary artery enters the infratemporal fossa by passing forwards, between the neck of mandible and the sphenomandibular ligament.

### Parts and Relations

The maxillary artery is divided into three parts by the lower head of lateral pterygoid muscle. The parts are:

1. **First part (mandibular part):** From beginning (origin) to lower border of lateral pterygoid. It lies between the neck of the mandible laterally and sphenomandibular ligament medially. The auriculotemporal nerve lies above this part.
2. **Second part (pterygoid part):** From lower border to the upper border of the lower head of lateral pterygoid (i.e., second part lies on or deep to lower head of lateral pterygoid).
3. **Third part (pterygopalatine part):** From upper border of the lower head of lateral pterygoid to pterygopalatine fossa. In pterygopalatine fossa it lies in front of the pterygopalatine ganglion.

### N.B.

- Most of the branches from the first and second parts of maxillary artery accompany the branches of the mandibular nerve.
- Branches from the third part of the maxillary artery accompany the branches of maxillary nerve and pterygopalatine ganglion.
- Branches from the second part of the maxillary artery are muscular only and supply muscles of mastication.
- All the branches (1st and 3rd part) of the maxillary artery pass through bony foramina and fissures except branches from its second part.

### Branches of the Maxillary Artery (Fig. 10.8)

#### I. Branches from the First Part (Five Branches)

1. **Deep auricular artery**—passes upwards and backwards to enter the external acoustic meatus by piercing its floor and supplies:
  - (a) skin of external acoustic meatus, and
  - (b) outer surfaces of tympanic membrane.

2. **Anterior tympanic artery**—enters the tympanic cavity by passing through petrotympanic fissure and it supplies the **inner surface of the tympanic membrane**.
3. **Middle meningeal artery**—is the largest meningeal branch. It supplies **meninges** as well as the **skull bone**.

Clinically it is the most important branch of the maxillary artery.

*The middle meningeal artery* arises from the first part of the maxillary artery. It ascends upwards deep to the lateral pterygoid, behind the mandibular nerve. Passing between the two roots of the auriculotemporal nerve, to enter the cranial cavity through foramen spinosum in company with meningeal branch of mandibular nerve (nervus spinosus).

As it emerges in the cranial cavity, it courses laterally on the floor of the middle cranial fossa and turns upwards and forwards on the greater wing of the sphenoid, where it divides into frontal and parietal branches:

- (a) *Frontal (anterior) branch*, courses up towards the pterion and then curves backwards to ascend towards the vertex, lying over the precentral gyrus of the cerebral hemisphere. In the region of pterion the artery frequently lies in a bony tunnel in the parietal bone for a centimeter or more.
- (b) *Parietal (posterior) branches* arch backwards on the squamous part of the temporal bone, cross the lower border of the parietal bone in front of its mastoid angle; here it divides into branches, which spread out as far as lambda. It lies along the superior temporal gyrus.

*Distribution:* The middle meningeal artery and its branches lie outside the dura and deep to the inner surface of the skull. Both of these are supplied by the artery.

**N.B.** The middle meningeal artery and its branches are accompanied by corresponding veins, which lie between the artery and the bone.

4. **Accessory middle meningeal artery**—runs upwards and enters the cranial cavity through foramen ovale. It supplies **meninges** and structures in the **infratemporal fossa**.
5. **Inferior alveolar/dental artery**—runs downwards between the sphenomandibular ligament and the ramus of the mandible, enters the mandibular foramen, runs through the mandibular canal, supplies molar and premolar teeth and adjoining gum. It then divides into mental and incisive branches.

The incisive branch supplies the **canine and incisor teeth**. The mental artery emerges through the mental foramen to supply the **skin of the chin**. Before entering the mandibular foramen the inferior alveolar artery gives off two branches, namely,

## MAXILLARY VEIN AND PTERYGOID VENOUS PLEXUS

### Maxillary Vein

It is a short venous trunk, which accompanies the first part of the maxillary artery. It is formed by the confluence of veins from the pterygoid venous plexus and passes backwards between the sphenomandibular ligament and the neck of the mandible. Within the parotid gland it unites with the superficial temporal vein to form the retromandibular vein.

### Pterygoid Venous Plexus

It is a network of very small veins that lie around and within the lateral pterygoid muscle.

The pterygoid venous plexus communicates:

- (a) with inferior ophthalmic vein via inferior orbital fissure,
- (b) with cavernous sinus by emissary veins via foramen ovale or foramen of Vesalius, and
- (c) with facial vein through the deep facial vein.

The plexus is drained by maxillary vein which is formed at the lower border of the lateral pterygoid muscle.

**N.B.** The pterygoid venous plexus is sometimes referred to as a **peripheral heart** for during yawning when the mouth is widely open due to contraction of lateral pterygoid muscle, the stagnant venous blood is pumped up into the cavernous sinus and maxillary vein. Possibly this is the reason why people yawn in the morning when they get up from sleep.

## NEURAL STRUCTURES

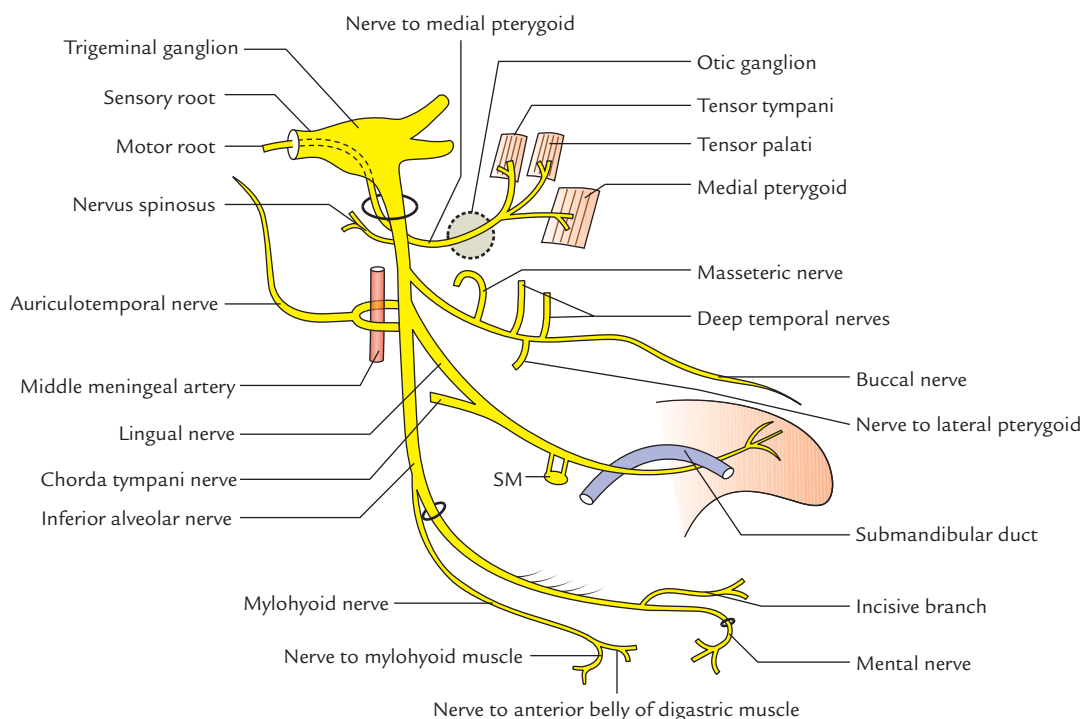
The infratemporal fossa contains the mandibular nerve and its branches, chorda tympani nerve, and otic ganglion.

### MANDIBULAR NERVE (Fig. 10.9)

The mandibular nerve is the largest of the three divisions of the trigeminal nerve. It is a mixed nerve, consisting of both sensory and motor fibres. It is a nerve of the first pharyngeal arch, hence supplies all the structures derived from this arch, e.g., muscles of mastication.

#### Course and Relations

The mandibular nerve is the largest of three divisions of trigeminal nerve. It begins in the middle cranial fossa as two roots: a larger sensory root and a smaller motor root. The **large sensory root** arises from lateral convex part of the trigeminal ganglion and immediately leaves the cranial cavity by passing through foramen ovale to enter the infratemporal fossa. The small motor root arises from the pons, lies deep to the trigeminal ganglion and the sensory root. It also passes through foramen ovale to join the sensory root just below the foramen ovale in the infratemporal fossa, thus forming the main trunk. The mandibular nerve trunk is short and after a short course divides into a small anterior and large posterior division.



**Fig. 10.9** Course and distribution of the mandibular nerve (SM = submandibular ganglion).

**3. Inferior alveolar nerve:** It is the larger terminal branch of the posterior division of mandibular nerve. It is a mixed nerve. It receives all the motor fibres of the trigeminal nerve.

**Course:** It emerges below the lower head of the lateral pterygoid and passes vertically downwards and forward on the medial pterygoid. The nerve lies anterior to inferior alveolar vessels between the sphenomandibular ligament and the ramus of the mandible. Then enters the mandibular foramen in company with inferior alveolar artery, traverses the mandibular canal as far as mental foramen, where it terminates by dividing into mental and incisive branches.

**Branches:**

- Nerve to mylohyoid* arises from the inferior alveolar nerve before it enters the mandibular canal. It pierces the sphenomandibular ligament to reach the mylohyoid groove. It supplies mylohyoid and anterior belly of digastric muscle.
- Dental branches* supply molar and premolar teeth.
- Mental nerve* emerges out through mental foramen to supply skin of the chin and skin and mucus membrane of the lower lip.
- Incisive branch*, supplies canine and incisor teeth.

The branches of the mandibular nerve are summarized in Table 10.2.

**Table 10.2** Branches of the mandibular nerve

| From main trunk                    | From anterior division     | From posterior division |
|------------------------------------|----------------------------|-------------------------|
| Nervus spinosus (meningeal branch) | Masseteric nerve           | Auriculotemporal nerve  |
| Nerve to medial pterygoid          | Deep temporal nerves       | Lingual nerve           |
|                                    | Nerve to lateral pterygoid | Inferior alveolar nerve |
|                                    | Buccal nerve               |                         |

**N.B.** All the branches of posterior division of the mandibular nerve are sensory except nerve to mylohyoid, which is motor.

### Clinical correlation

- Inferior alveolar nerve block (generally called inferior alveolar block or 1A block):** It is the most common nerve block performed in dentistry to carry out dental procedures on the mandibular teeth.

The anesthetic agent is injected slightly superior to the entry of inferior alveolar nerve into the mandibular foramen being overhung by the lingula. While performing inferior alveolar nerve block, if needle is inserted too far posteriorly, it may enter the parotid gland and damage the facial nerve leading to *transient facial palsy*.

- The lingual nerve is at great risk (10%) during surgical removal of the impacted third molar teeth. The nerve is also at risk during surgical removal of the submandibular salivary gland, during which the submandibular duct must be dissected out carefully from the nerve.
- Referred Pain:** It is the pain frequently referred from one branch of mandibular nerve to the other. Thus in patients suffering from tongue cancer, the pain radiates to the ear and to the temporal fossa in the area of distribution of auriculo-temporal nerve. The pain is relieved by dividing the lingual nerve below and behind the last molar tooth. At this site the nerve is in contact with the mandible and covered only by mucus membrane. Similarly, pain from teeth is also referred to the ear and temporal region.

## CHORDA TYMPANI NERVE

The chorda tympani nerve is the slender branch of facial nerve. It is so named because of its intimate relationship to the middle ear (tympanum).

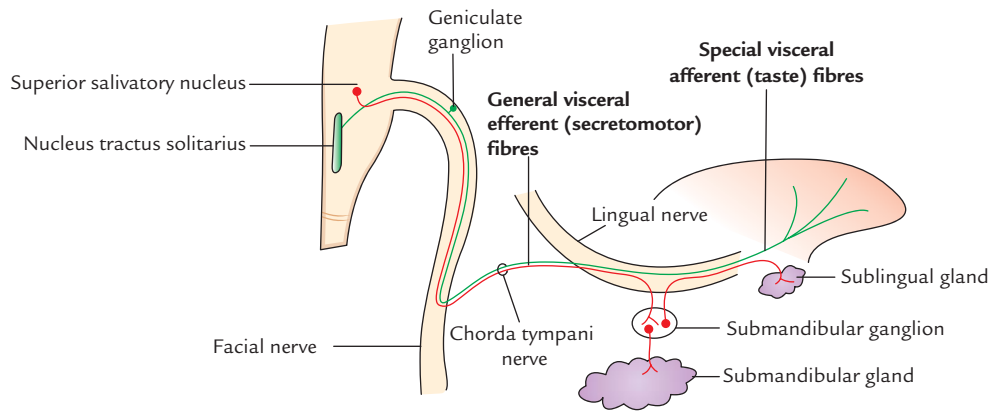
### Function Components (Fig. 10.10)

The chorda tympani nerve contains:

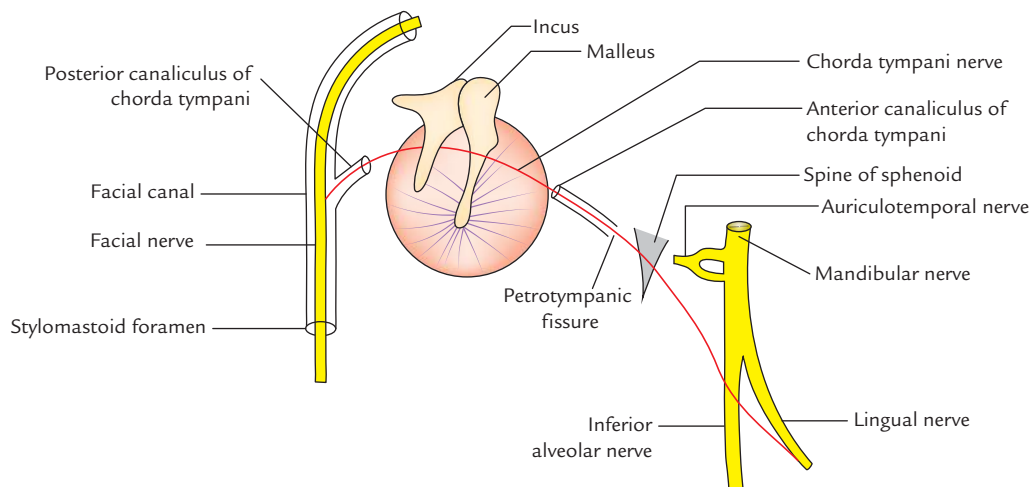
- General visceral efferent fibres:** These are preganglionic parasympathetic (secretomotor) fibres to submandibular and sublingual salivary glands.
- Special visceral afferent fibres:** These fibres carry taste sensations from anterior 2/3rd of tongue (except vallate papillae).

### Origin, Course, and Relations (Fig. 10.11)

It arises from facial nerve in the facial canal about 6 mm above the stylomastoid foramen, within the posterior wall of the tympanic (middle ear) cavity. It enters the middle ear through the posterior canaliculus of chorda tympani in the posterior wall, runs across the lateral wall (tympanic membrane). Here it crosses medial aspect of handle of malleus and lateral aspect of long process of incus. At the anterior margin of tympanic membrane it enters anterior canaliculus in the anterior wall of the middle ear passes through canaliculus and emerges at the base of skull through medial end of petrotympanic fissure. It then goes medially, forwards and downwards, grooves the medial side of the spine of the sphenoid, running anteroinferiorly deep to lateral pterygoid to join the posterior aspect of the lingual nerve about 2 cm below the base of the skull.



**Fig. 10.10** Functional components of the chorda tympani nerve.



**Fig. 10.11** Origin, course, and relations of the chorda tympani nerve.

### Distribution

1. It supplies secretomotor fibres to submandibular and sublingual glands through the submandibular ganglion.
2. It carries taste sensations from anterior two-third of the tongue (except vallate papillae).

### OTIC GANGLION (Fig. 10.12)

It is a small parasympathetic ganglion connected to the mandibular division of trigeminal nerve and provides a relay station to the secretomotor fibres to the parotid gland. Topographically, it is intimately related to the mandibular nerve but functionally it is related to glossopharyngeal nerve.

### Size

Pinhead (2–3 mm in size).

### Location

Infratemporal fossa, just below the foramen ovale.

### Relations

*Lateral:* Mandibular nerve.

*Medial:* Tensor palati muscle.

*Posterior:* Middle meningeal artery.

*Anterior:* Medial pterygoid muscle.

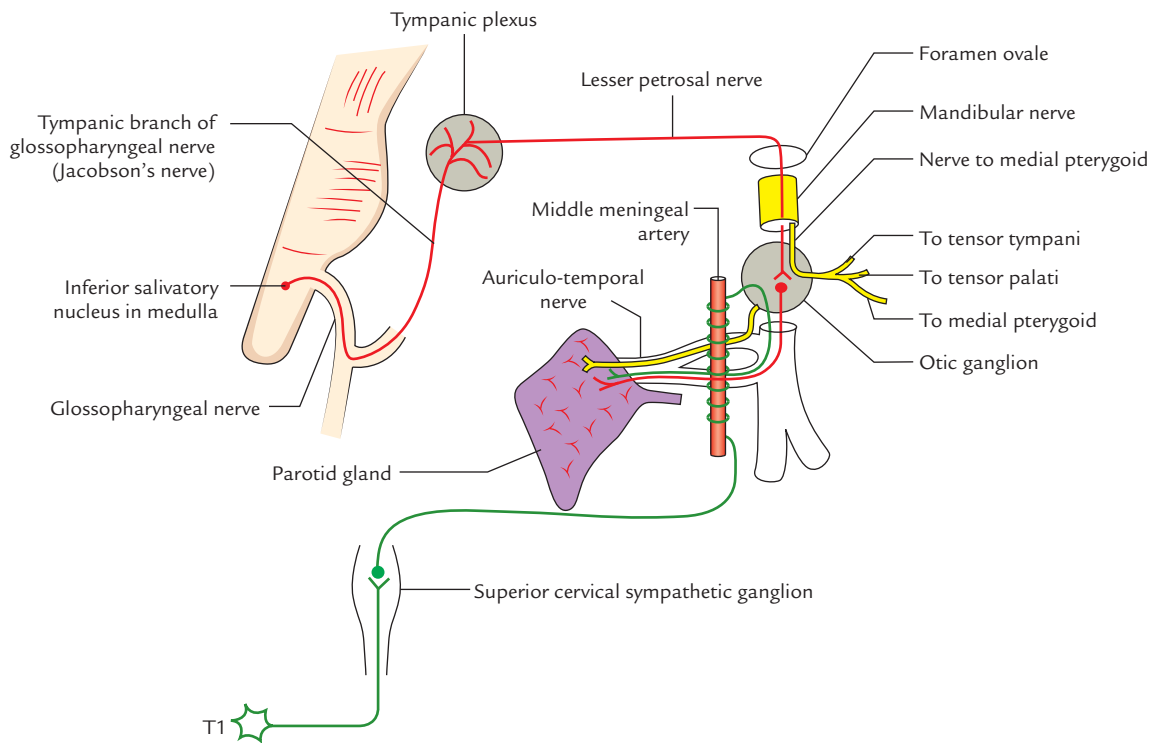
### Roots or Connections

1. **Parasympathetic motor (secretomotor):** From lesser petrosal nerve.

*Preganglionic parasympathetic fibres* arise from inferior salivatory nucleus; pass successively through glossopharyngeal nerve, tympanic branch of glossopharyngeal nerve (Jacobson's nerve), tympanic plexus, and lesser petrosal nerve to relay in the ganglion. *Postganglionic parasympathetic fibres* from ganglion cells pass through auriculo-temporal nerve to supply parotid gland.

2. **Sympathetic:** From sympathetic plexus around the middle meningeal artery.





**Fig. 10.12** Otic ganglion and its connections.

*Preganglionic sympathetic fibres* arise from T1 and T2 spinal segments, enter the cervical sympathetic chain at the level of its inferior ganglion and then ascend to relay in the superior cervical sympathetic ganglion. The postganglionic fibres arise from this ganglion and form plexus around the middle meningeal artery. They then pass through the ganglion without relay to reach the parotid gland via auriculotemporal nerve. They are vasomotor in nature and responsible for thick salivary secretion.

3. **Sensory:** From auriculotemporal nerve.
4. **Somatic motor:** Nerve to medial pterygoid. It passes through ganglion to supply medial pterygoid, tensor palati, and tensor tympani muscles.

### Branches

- |                                                                                                                                                |   |                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. Postganglionic parasympathetic</li> <li>2. Postganglionic sympathetic</li> <li>3. Sensory</li> </ol> | } | All supply parotid gland through auriculotemporal nerve |
|------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------|

### N.B.

- In humans, the *chorda tympani* nerve is connected to the otic ganglion and nerve to pterygoid canal. These

connections provide an alternative pathway of taste sensations from anterior two-third of the tongue.

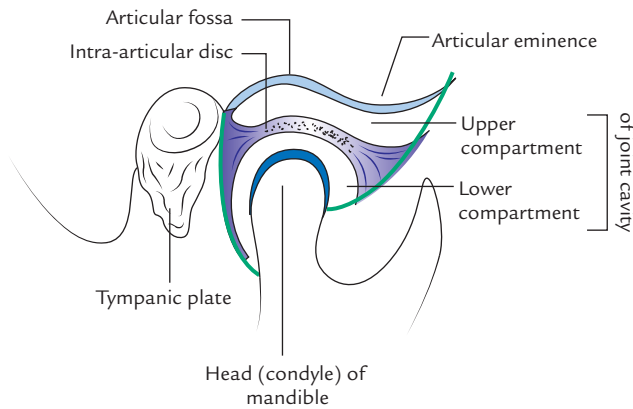
- Clinical evidence suggests that in humans the parotid gland also receives secretomotor fibres through chorda tympani nerve.

## TEMPOROMANDIBULAR JOINT

The temporomandibular joint (TMJ), one on each side of the head, is a joint between temporal bone and mandible that allows the movements of the mandible for speech and mastication.

The mandible is a single bone having two heads, which articulate on each side with temporal bone of cranium. The cranium, with which the mandible articulates, is also, mechanically a single component. The movement cannot take place at one temporomandibular joint without a concomitant movement occurring at the joint on the opposite side. Therefore, temporomandibular joints are the bilateral components of a single **craniomandibular articulation/joint**. The temporomandibular joints are often involved in various disease processes. Therefore medical students, particularly dental students need to understand the anatomy of TMJ, including movements and disorders associated with the joint.

*Type:* It is a synovial joint of condylar variety.



**Fig. 10.13** Articular surfaces of the temporomandibular joint.

### Articular Surface

The **upper articular surface** is formed by the (a) *articular fossa*, and (b) *articular eminence* of the temporal bone. This surface is concavo-convex from behind forwards (Fig. 10.13).

The **lower articular surface** is formed by the head (condyle) of the mandible. This surface is elliptical in shape.

The articular surfaces are **covered by fibrocartilage** and not by hyaline cartilage, hence temporomandibular joint is an *atypical synovial joint* (Fig. 10.13).

### JOINT CAVITY (Fig. 10.14)

The cavity of temporomandibular joint is divided into *upper menisco-temporal* and *lower menisco-mandibular compartments* by an intra-articular disc of fibrocartilage.

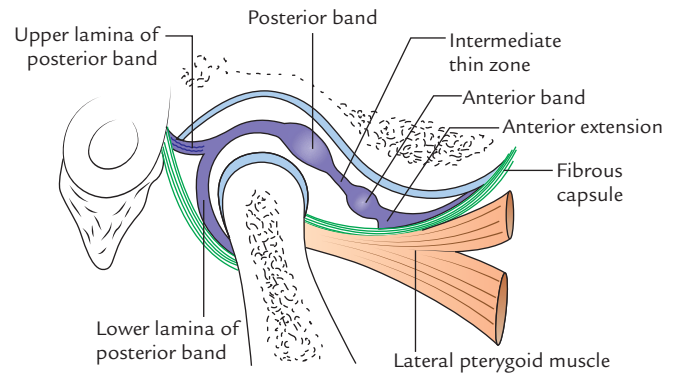
The upper compartment permits gliding movements, whereas lower compartment permits gliding as well as rotational movements.

### Articular Disc

The articular disc is an oval plate of fibrocartilage. Though termed *fibrocartilage*, it consists mainly of collagen fibres with few cartilage cells. It is congruent with both the articular surfaces. Thus its upper surface is concavo-convex (from before backwards) and its inferior surface is concave. The concavo-convex superior surface fits against the articular eminence and the concavity of the articular fossa. The lower concave surface fits with convex head of the mandible.

The periphery of the disc is attached firmly to the fibrous capsule.

The disc has a thick margin, the *peripheral annulus* and a central depression on its inferior surface. In sagittal section, the disc appears to possess a thin *intermediate zone* and thickened anterior and posterior bands (Fig. 10.14). The anterior band extends anteriorly through the capsule to be



**Fig. 10.14** Parts of the intra-articular disc.

continuous with the tendon of lateral pterygoid. The posterior band splits into two laminae: upper and lower. The upper lamina composed of fibroelastic tissue is attached to the squamotympanic fissure. The lower lamina composed of fibrous non-elastic tissue is attached to the back of the condyle. The bilaminar region contains a venous plexus. The central part of the disc is avascular.

The elastic fibres in the posterior part of the disc bring back the disc in the articular fossa when the open mouth is closed.

**N.B.** Parts of articular disc (Fig. 10.14): In sagittal section, the articular disc of TMJ presents five different parts, from before backwards these are: (a) **anterior extension**, (b) **anterior thick band**, (c) **intermediate zone**, (d) **posterior thick band**, and (e) **posterior bilaminar zone**.

### LIGAMENTS

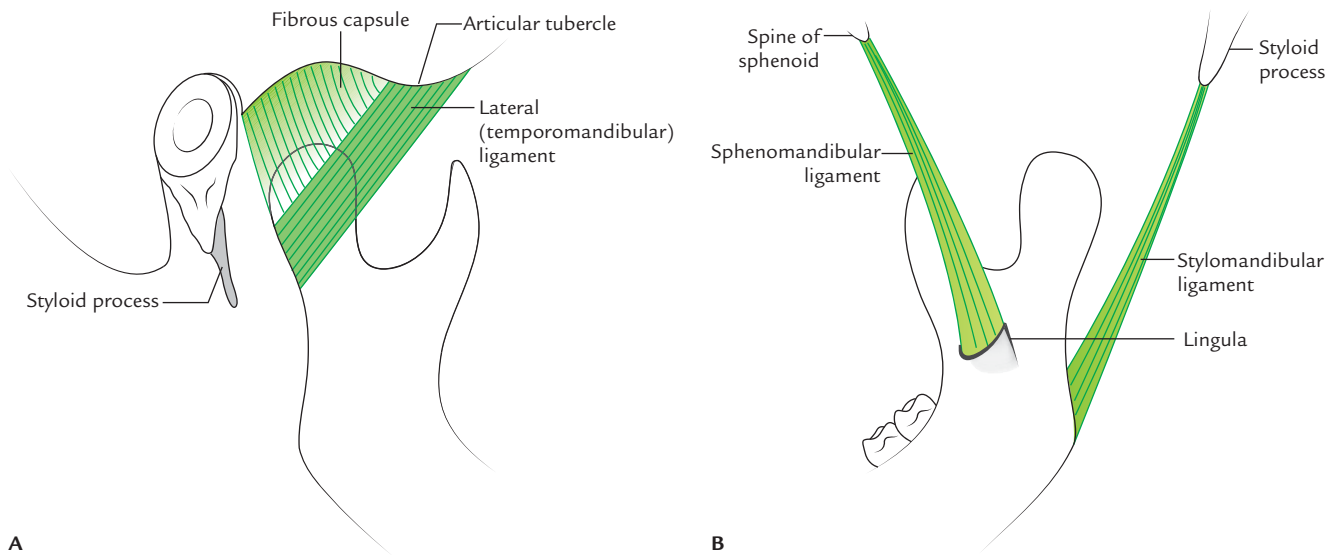
The ligaments are the fibrous capsule, temporomandibular, sphenomandibular, and stylomandibular ligaments. The latter two are accessory ligaments (Fig. 10.15A and B).

1. **Fibrous capsule:** It is a fibrous sac to enclose the joint cavity. It is attached above to the articular tubercle, the circumference of articular fossa, and the squamotympanic fissure; and below to the neck of mandible.

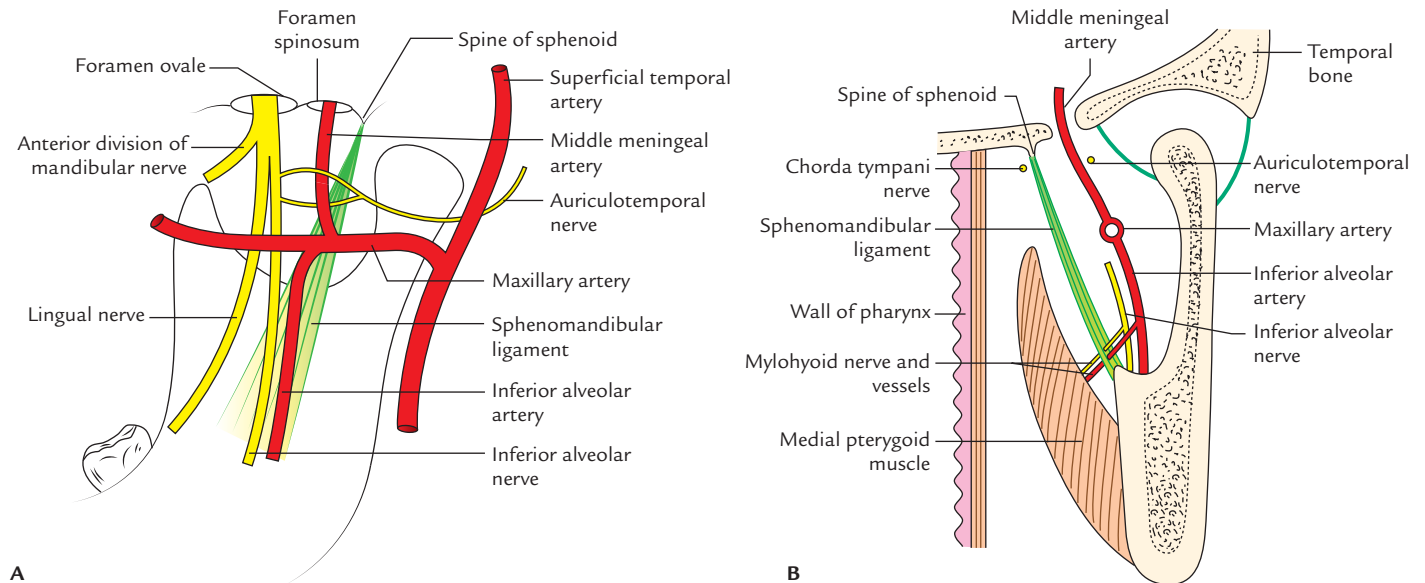
The capsule is loose above the intra-articular disc and tight below it.

The synovial membrane lines the inner aspect of the fibrous capsule and the neck of the mandible.

2. **Lateral (temporomandibular) ligament:** It is a true ligament and formed as a result of thickening on the lateral aspect of the capsular ligament. Its fibres are directed downwards and backwards. It is attached above to the articular tubercle on the root of zygoma and below to the posterolateral aspect of the neck of the mandible. The lateral ligament strengthens the lateral aspect of the capsule.



**Fig. 10.15** Ligaments of the temporomandibular joint: A, fibrous capsule and lateral ligament; B, accessory ligaments.



**Fig. 10.16** Relations of the sphenomandibular ligament: A, superficial relations; B, deep relations.

3. **Sphenomandibular ligament:** It is attached above to the spine of the sphenoid and below to the lingula and lower margin of the mandibular foramen of the mandible. It lies on a deeper plane away from the joint capsule. The sphenomandibular ligament represents the unossified intermediate part of the sheath of the Meckel's cartilage of the first pharyngeal arch. It becomes accentuated and taut when the mandible is protruded.

*Relations of sphenomandibular ligament (Fig. 10.16A and B) are as under:*

*Laterally,* it is related to: (a) lateral pterygoid muscle, (b) auriculo-temporal nerve, (c) maxillary artery, and (d) inferior alveolar nerve and vessels.

*Medially,* it is related to: (a) medial pterygoid, (b) chorda tympani nerve and (c) wall of the pharynx.

**N.B.** Near its lower end the sphenomandibular ligament is pierced by mylohyoid nerve and vessels.

### Clinical correlation

The sphenomandibular ligament is an important landmark for administration of local anesthetic during inferior alveolar nerve block.

4. **Stylomandibular ligament:** It is attached above to the lateral surface of the styloid process and below to the

### Protraction

It is done by lateral and medial pterygoids and masseter.

### Retraction

It is done by posterior fibres of temporalis. It may be assisted by middle and deep fibres of the masseter, the digastric and geniohyoid muscles.

### Side-to-side (Chewing) Movements

These movements are performed by alternate contraction of medial and lateral pterygoids on each side.

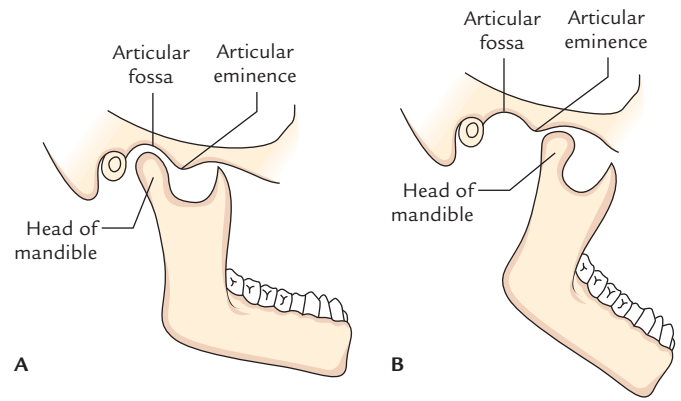
Movements of the temporomandibular joint and muscles producing them are summarized in Table 10.3.

**Table 10.3** Movements of the mandible and muscles producing them

| Movement                              | Muscle                                                                                               |
|---------------------------------------|------------------------------------------------------------------------------------------------------|
| Depression                            | <b>Lateral pterygoid</b><br>Digastric<br>Geniohyoid<br>Mylohyoid<br>Gravity                          |
| Elevation                             | Masseter<br>Medial pterygoid<br>Temporalis                                                           |
| Protrusion<br>(Protraction)           | Medial pterygoid<br>Lateral pterygoid                                                                |
| Retraction                            | Posterior fibres of temporalis                                                                       |
| Side-to-side<br>(chewing)<br>movement | Medial and lateral pterygoids of one side<br>contracting alternatively with that of opposite<br>side |

### Clinical correlation

- **Palpation of the temporomandibular joint and associated muscles:** The bilateral palpation is must to assess the entire joint and its associated muscles. First, the patient is asked to open and close the mouth several times. Then he is asked to move the opened jaw to left, and to right, and finally he is asked to move the jaw forward. For digital palpation of condyle of moving mandible place a finger into the outer portion of the external auditory meatus.
- **Dislocation of the mandible (Fig. 10.19):** The mandible is dislocated only anteriorly. When the mouth is open, the mandibular condyles move forward and lie underneath the articular eminences. This is the most unstable position of the temporomandibular joint. In this position, if there is excessive opening of mouth as during yawning, sudden violence or convulsive spasm of lateral pterygoid muscles, the head of mandible of one or both sides may slip anteriorly and get locked into the infratemporal fossa; as a result the mouth cannot be closed anymore and any passive effort to do that will invariably fracture the neck of the mandible on one or both sides.



**Fig. 10.19** Temporomandibular joint: A, normal; B, in anterior dislocation.

To reduce dislocation, the condyle must be lowered and pushed back behind the summit of articular eminence into the articular fossa. Thus the reduction is done by depressing the jaw with thumb placed on the last molar teeth, and simultaneously elevating the chin.

- **Temporomandibular joint syndrome:** This syndrome consists of group of symptoms arising from temporomandibular joints and their associated masticatory muscles. The typical presenting symptoms are:
  - *Diffuse facial pain*, due to spasm of masseter muscle.
  - *Headache*, due to spasm of temporalis muscle.
  - *Jaw pain*, due to spasm of lateral pterygoid.

These symptoms may be associated with clicking and pain in the joint. The clicking is often audible when the patient is chewing. It occurs when the posterior attachment of the disc becomes stretched or detached, allowing the disc to become temporarily or permanently trapped anteriorly. The derangement of articular disc results from an overclosure or malocclusion.

- During surgery of temporomandibular joints, the facial nerve should be preserved with utmost care.

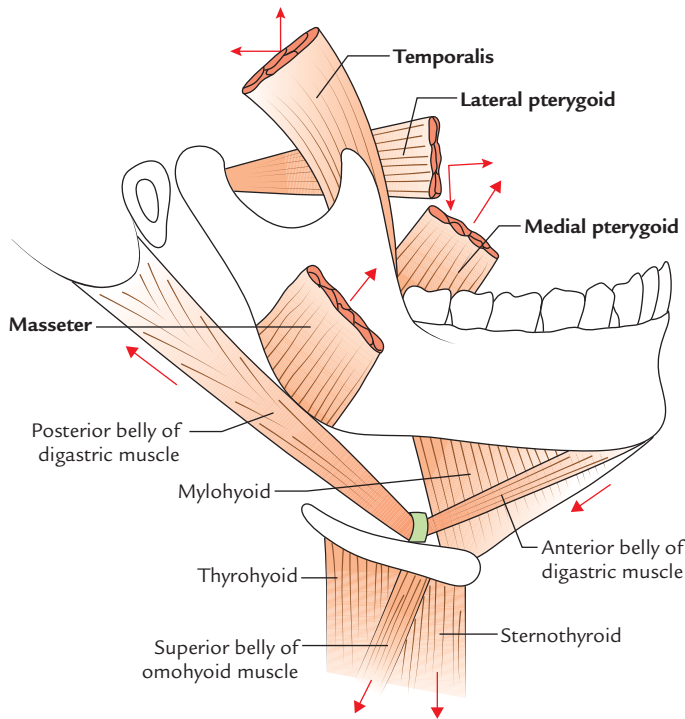
## MUSCLES OF MASTICATION

The muscles of mastication are concerned with movements of mandible at the temporomandibular joints during mastication (Fig. 10.20).

They are divided into two groups:

1. **Principal muscles**
  - (a) Temporalis
  - (b) Masseter
  - (c) Lateral pterygoid
  - (d) Medial pterygoid.
2. **Accessory muscles**
  - (a) Digastric
  - (b) Buccinator
  - (c) Mylohyoid
  - (d) Geniohyoid.

*Jayant*



**Fig. 10.20** Muscles of mastication producing movements of the temporomandibular joint. The arrows indicate the direction of their actions. Chief muscles of mastication are labelled in bold.

### Principal Muscles of Mastication

The characteristic features of the principal muscles of mastication are as follows:

1. All are located in or around the infratemporal fossa.
2. All are inserted into the ramus of the mandible.
3. All are innervated by the mandibular division of the trigeminal nerve.
4. All are concerned with movements of the mandible on the temporomandibular joints.
5. All develop from mesoderm of the first pharyngeal arch.

### Temporalis (Fig. 10.21)

It is a fan-shaped muscle located in the temporal fossa. It is covered by tough temporal fascia which is attached above to the temporal line and below to the zygomatic arch.

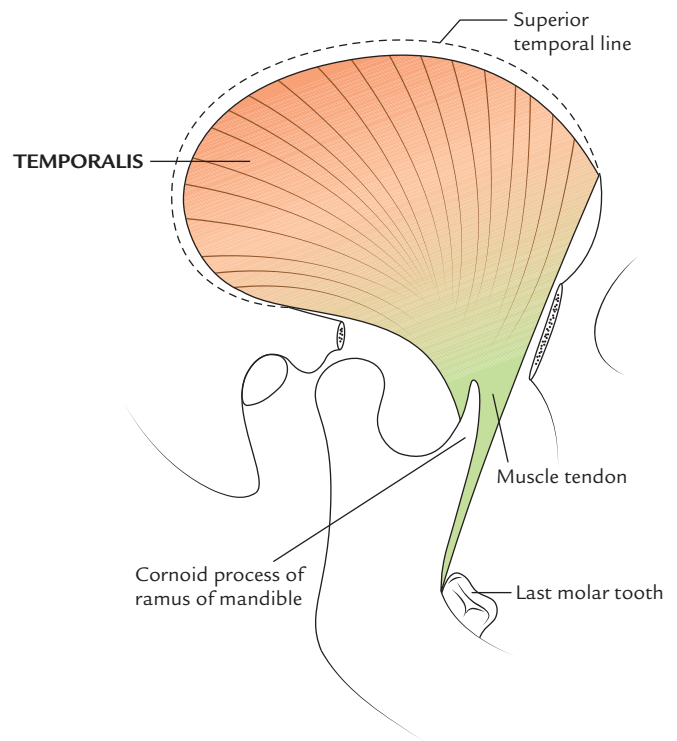
#### Origin

It arises from:

1. Whole of the floor of temporal fossa except the part formed by the zygomatic bone.
2. Deep surface of the temporal fascia.

#### Insertion

The fibres converge and descend to form a tendon, which passes through the gap between the zygomatic arch and the side of the skull. The muscle is inserted into:



**Fig. 10.21** Origin and insertion of the temporalis muscle.

1. The medial surface, apex, anterior, border of the coronoid process of ramus of mandible.
2. The anterior border of the ramus of mandible, almost up to the last molar tooth.

**N.B.** The temporalis muscle is fan shaped. The anterior fibres are oriented vertically, the posteriormost fibres are disposed almost horizontally and intervening intermediate fibres are placed obliquely.

#### Nerve supply

The temporalis is supplied by the *anterior and posterior deep temporal nerves*, the branches of the anterior division of the mandibular nerve.

#### Actions

The temporalis muscle **elevates the mandible** and so closes the mouth and approximates the teeth. This movement requires both the upward pull of the anterior fibres and backward pull of the posterior fibres.

**Posterior fibres** retract the mandible after it has been protruded.

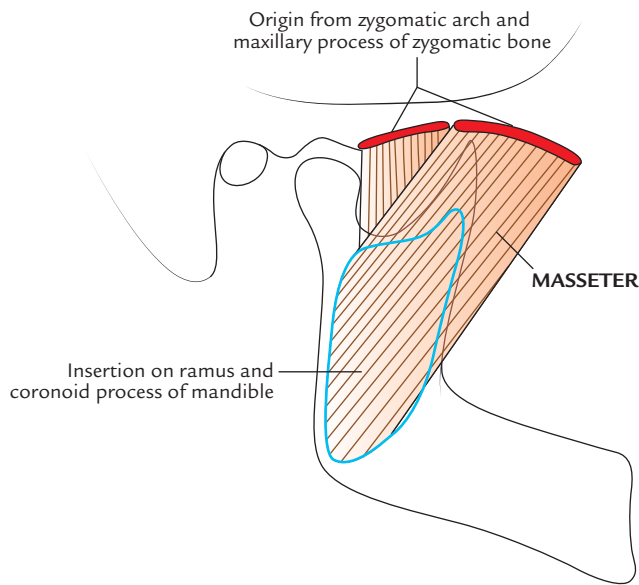
### Masseter (Fig. 10.22)

The masseter (Greek: masseter = a chewer) is a thick quadrilateral muscle covering the lateral surface of the ramus of the mandible including its coronoid process. The condylar process is left uncovered.

#### Origin

The masseter consists of the following three layers:





**Fig. 10.22** Origin and insertion of the masseter muscle.

1. Superficial layer.
2. Middle layer.
3. Deep layer.

**Superficial layer** is largest of the three layers of masseter and arises by a thick aponeurosis from: maxillary process of zygomatic bone and anterior two-third of the inferior border of the zygomatic arch.

**Middle layer** arises from lower border of the posterior one-third of the zygomatic arch.

**Deep layer** arises from deep surface of the zygomatic arch.

#### Insertion

1. **Superficial fibres** pass downwards and backwards at 45° to be inserted into the angle and lower posterior half of the lateral surface of the ramus of the mandible.
2. **Middle fibres** pass vertically downwards to insert into the central part of the ramus.
3. **Deep fibres** pass vertically downwards to insert into the upper part of the mandibular ramus and its coronoid process.

#### N.B.

- Intramuscular tendinous septa in the superficial layer are responsible for producing ridges on the ramus of the mandible.
- Middle and deep fibres together constitute the deep part of the masseter.

#### Nerve supply

The masseter is supplied by a **masseteric nerve**, a branch from anterior division of the mandibular nerve.

#### Actions

The masseter muscle **elevates the mandible** to close the mouth.

#### Lateral Pterygoid

It is described on page 134.

#### Medial Pterygoid

It is described on page 135.

#### Accessory Muscles of Mastication

They are described in detail in Chapter 9.

The origin, insertion, nerve supply, and actions of chief muscles of mastication are summarized in Table 10.4.

**N.B.** The chief muscles of mastication are often termed *muscles of mastication*.

#### Clinical correlation

The muscles of mastication and their motor innervation can be tested clinically by asking the patient to clench his teeth repeatedly and then palpating the temporalis and masseter in the temporal fossa and over the ramus of mandible, respectively.

### PTERYGOPALATINE FOSSA

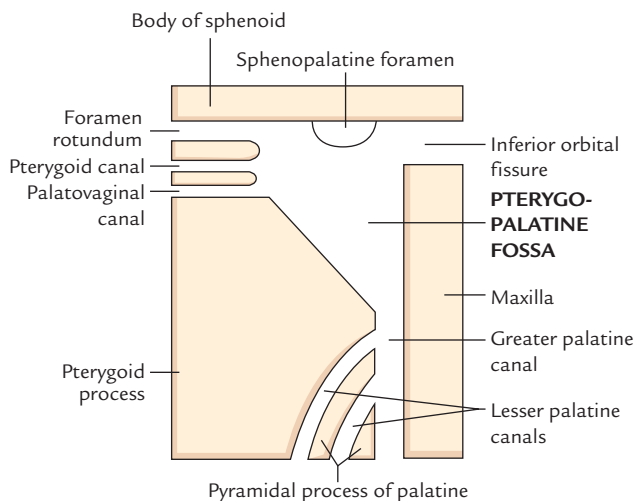
The pterygopalatine fossa is a pyramidal space situated deeply below the apex of the orbit, between the pterygoid process of sphenoid behind and the perpendicular plate of palatine in front. More laterally, the back of maxilla replaces the palatine bone as the anterior boundary of the entrance of the fossa—the pterygomaxillary fissure.

#### BOUNDARIES (Fig. 10.23)

- Anterior:** Perpendicular plate of the palatine and posterior surface of the maxilla (superomedial part).
- Posterior:** Pterygoid process and adjoining part of the anterior surface of the greater wing of the sphenoid.
- Medial:** Upper part of the perpendicular plate of palatine orbital and sphenoidal process of the palatine.
- Lateral:** The fossa opens into the infratemporal fossa through pterygomaxillary fissure.
- Superior:** Under surface of the body of the sphenoid and orbital process of the palatine. The lateral part of the roof is open and here fossa opens into the orbit through inferior orbital fissure.
- Inferior:** Pyramidal process of the palatine bone in the angle between the maxilla and the pterygoid process.

**Table 10.4** Origin, insertion, nerve supply, and actions of the muscles of the mastication

| Muscles                                  | Origin                                                                                                                                                                                            | Insertion                                                                                                                                                    | Nerve supply                            | Actions                                                                                                                                                   |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Temporalis (fan shaped)                  | <ul style="list-style-type: none"> <li>Floor of temporal fossa</li> <li>Temporal fascia</li> </ul>                                                                                                | <ul style="list-style-type: none"> <li>Tip, anterior border, and medial surface of coronoid process</li> <li>Anterior border of ramus of mandible</li> </ul> | Mandibular division of trigeminal nerve | <ul style="list-style-type: none"> <li>Elevation of mandible by anterior and middle fibres</li> <li>Retraction of mandible by posterior fibres</li> </ul> |
| Masseter (quadrilateral)                 | Zygomatic arch adjoining part of zygomatic process of maxilla                                                                                                                                     | <ul style="list-style-type: none"> <li>Lateral surface of ramus of mandible</li> <li>Coronoid process</li> </ul>                                             | Mandibular division of trigeminal nerve | Elevation of mandible to occlude the teeth for forceful bite                                                                                              |
| Lateral pterygoid (Short, thick conical) | <ul style="list-style-type: none"> <li>Upper head from infratemporal surface and crest of greater wing of sphenoid</li> <li>Lower head from lateral surface of lateral pterygoid plate</li> </ul> | <ul style="list-style-type: none"> <li>Pterygoid fovea on anterior surface of neck of mandible</li> <li>Articular disc and capsule of TMJ</li> </ul>         | Mandibular division of trigeminal nerve | <ul style="list-style-type: none"> <li>Depression of mandible by pulling the neck of mandible forward</li> <li>Protraction</li> </ul>                     |
| Medial pterygoid (quadrilateral)         | <ul style="list-style-type: none"> <li>Superficial head from tuberosity of maxilla</li> <li>Deep head from medial surface of lateral pterygoid plate</li> </ul>                                   | Medial surface of angle adjoining ramus of mandible                                                                                                          | Mandibular division of trigeminal nerve | <ul style="list-style-type: none"> <li>Elevation of mandible</li> <li>Protraction</li> </ul>                                                              |

**Fig. 10.23** Boundaries of the pterygopalatine fossa.

## COMMUNICATIONS

*Anteriorly:* With the orbit through the medial end of the inferior orbital fissure.

*Posteriorly:*

1. With the middle cranial fossa through foramen rotundum.
2. With the foramen lacerum through the pterygoid canal.

3. With the pharynx through palatovaginal canal.

*Medially:* With the nose through sphenopalatine foramen.  
*Laterally:* With the infratemporal fossa through pterygomaxillary fissure.

*Inferiorly:* With the oral cavity through greater and lesser palatine canals.

## CONTENTS

The following are the main contents of the pterygopalatine fossa:

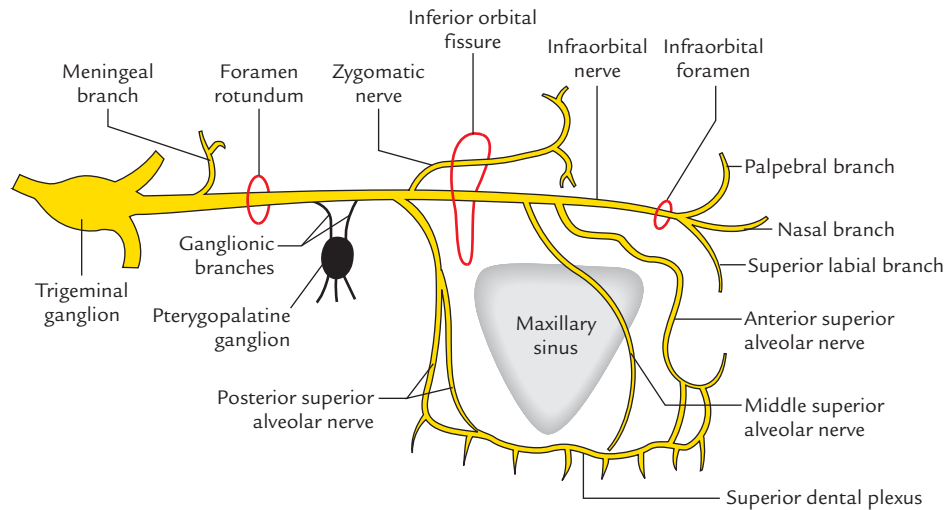
1. Maxillary nerve.
2. Pterygopalatine ganglion.
3. Third part of the maxillary artery.

## MAXILLARY NERVE (Fig. 10.24)

The maxillary nerve, the second division (V2) of the trigeminal nerve is purely sensory.

## Course and Relations

It arises from the convex anterior border of the trigeminal ganglion, pierces the trigeminal cave of dura to reach the lower part of the lateral wall of the cavernous sinus. The nerve leaves the middle cranial fossa through foramen rotundum to reach the pterygopalatine fossa. It traverses



**Fig. 10.24** Origin, course, and branches of the maxillary nerve.

straight in the upper part of the fossa and enters the orbit through inferior orbital fissure, hence it is called *infraorbital nerve*. The infraorbital nerve (in fact a continuation of maxillary nerve) runs forward along the floor of the orbit in the infraorbital groove and canal in succession and appears on the face, through infraorbital foramen. Therefore in its course the maxillary nerve traverses four regions in succession: the middle cranial fossa, the pterygopalatine fossa, the orbit and the face.

**N.B.** In the pterygopalatine fossa the pterygopalatine ganglion is suspended from the maxillary nerve by two roots.

### Branches and Distribution

The maxillary nerve gives off the following branches:

#### A. In the Middle Cranial Fossa

1. **Meningeal branch**, which supplies the dura mater of the middle cranial fossa.

#### B. In the Pterygo-palatine Fossa

1. **Ganglionic (communicating) branches**, two in number to pterygopalatine ganglion.
2. **Zygomatic nerve** enters the orbit through inferior orbital fissure and divides on the lateral wall of the orbit into:
  - (a) a zygomaticotemporal branch, which passes through a foramen in the zygomatic bone to supply the skin of the temple, and
  - (b) a zygomaticofacial branch, which passes through the foramen in the zygomatic bone to supply the skin of the face.
3. **Posterior superior alveolar nerve** enters the one or two foramina on the posterior surface of the body of maxilla

and supplies the mucus membrane of the maxillary air sinus. Then it breaks up to form superior dental plexus, which supplies the molar teeth and adjoining part of the gum.

#### C. In the Orbit (infraorbital canal)

1. **Middle superior alveolar nerve** passes downward and forward along the lateral wall of the maxillary sinus, joins superior dental plexus and supplies the premolar teeth.
2. **Anterior superior alveolar nerve** runs in the anterior wall of the maxillary sinus through a bony canal called *canalis sinuosus* and divides into dental and nasal branches:
  - (a) The dental branches join the superior dental plexus and supply the canine and incisor teeth.
  - (b) The nasal branches appear in the lateral wall of the inferior meatus and supply the mucus membrane of the lateral wall and floor of the nasal cavity.

#### D. On the Face

1. **Palpebral branches** turn upwards and supply the skin of the lower eyelid.
2. **Nasal branches** supply the skin of the side of nose and the mobile part of the nasal septum.
3. **Superior labial branches** supply the skin and mucus membrane of the upper lip.

**N.B.** The superior dental plexus is formed by *posterior, middle, and anterior superior alveolar nerves*. It is situated in the alveolar process of the maxilla above the sockets of the teeth.

The branches of maxillary nerve are summarized in the Table 10.5.

**Table 10.5** Summary of branches of the maxillary nerve

| Region                       | Branches                                                                                                                                         |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| In the middle cranial fossa  | Meningeal branch                                                                                                                                 |
| In the pterygopalatine fossa | <ul style="list-style-type: none"> <li>• Ganglionic branches</li> <li>• Posterior superior alveolar nerves</li> <li>• Zygomatic nerve</li> </ul> |
| In the infraorbital canal    | <ul style="list-style-type: none"> <li>• Middle superior alveolar nerve</li> <li>• Anterior superior alveolar nerve</li> </ul>                   |
| On the face                  | <ul style="list-style-type: none"> <li>• Palpebral branch</li> <li>• Nasal branch</li> <li>• Labial branch</li> </ul>                            |

### PTERYGOPALATINE GANGLION

The pterygopalatine ganglion (**ganglion of hay fever**) is the *largest parasympathetic peripheral ganglion*. It serves as a relay station for the secretomotor fibres to the lacrimal gland and mucus glands of the nose, palate, pharynx, and paranasal sinuses. Topographically, it is related to the maxillary nerve, but functionally it is connected to the facial nerve through greater petrosal nerve (Fig. 10.25).

#### Location

It is located deeply in the upper part of the pterygopalatine fossa, suspended from maxillary nerve by two short roots.

#### Size

Head of a small pin.

#### Relations

*Behind:* Pterygoid canal.

*Medial:* Sphenopalatine foramen.

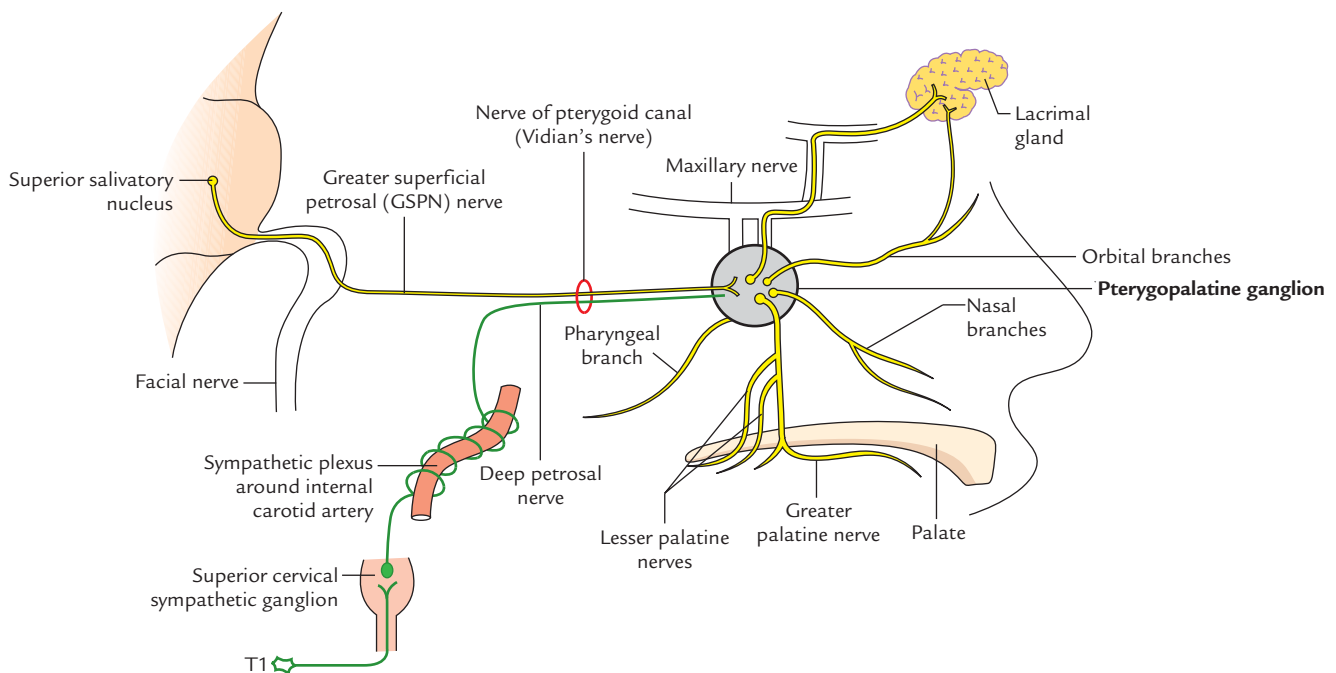
*In front:* Perpendicular plate of palatine.

*Above:* Maxillary nerve.

#### Roots

The roots of pterygopalatine ganglion are as follows:

1. **Motor or parasympathetic root:** It is derived from the nerve of pterygoid canal. It carries preganglionic parasympathetic fibres from superior salivatory nucleus (located in the lower part of the pons). These fibres relay in the ganglion. The postganglionic fibres arise from the cells in the ganglion and supply secretomotor fibres to the lacrimal gland, glands of the nose, palate, nasopharynx, and paranasal sinuses.
2. **Sympathetic root:** It is derived from sympathetic plexus around internal carotid artery via nerve of pterygoid canal. It contains postganglionic fibres from superior cervical sympathetic ganglion. These fibres pass through the ganglion without relay and provide vasomotor supply to the mucus membrane of the nose, palate, pharynx, and paranasal air sinuses.

**Fig. 10.25** Pterygopalatine ganglion, its roots and branches.

# Thyroid and Parathyroid Glands, Trachea, and Esophagus

The thyroid and parathyroid glands and cervical parts of the trachea and esophagus are closely related structures.

In the region of neck, thyroid and trachea are routinely examined by clinicians to diagnose their disorders, e.g., goitre, tracheal tug, etc. The surgical procedures are commonly performed on these structures, e.g., thyroidectomy, tracheostomy, etc. Therefore, detailed knowledge of anatomy and relations of these structures is extremely important.

## THYROID GLAND (G. THYREOS = SHIELD; EDIDOS = FORM)

The thyroid gland is the largest endocrine gland of the body. The hormones secreted by the thyroid gland include **triiodothyronine (T3)**, **tetraiodothyronine (T4; commonly called thyroxine)**, and **calcitonin**, which subserve the following functions:

1. Regulate the basal metabolic rate.
2. Stimulate the psychosomatic growth of the body.
3. Play an important role in calcium metabolism.

### Special Features

The special features of the thyroid gland are:

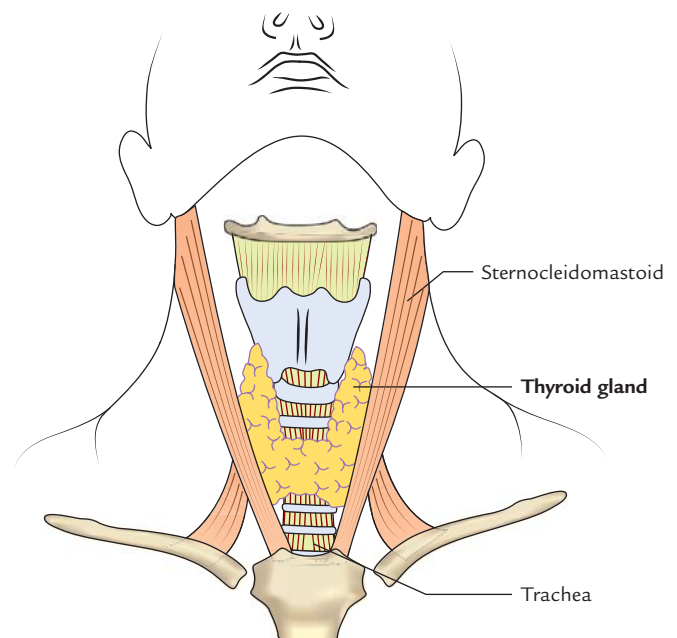
1. It is the only endocrine gland, which is located superficially in the body, hence accessible for physical examination.
2. It is the only endocrine gland that depends on external environment for raw material, iodine, to synthesize its hormones.
3. It is the only endocrine gland, which does not pour its hormones into blood immediately after formation but stores them and then releases them in blood for use as and when required.
4. It is one of the organ of the body having a very rich blood flow.

## LOCATION

The thyroid gland is located in the lower part of the front and side of the neck opposite to the C5, C6, C7, and T1 vertebrae (Figs 11.1 and 11.2), claspings the upper part of the trachea.

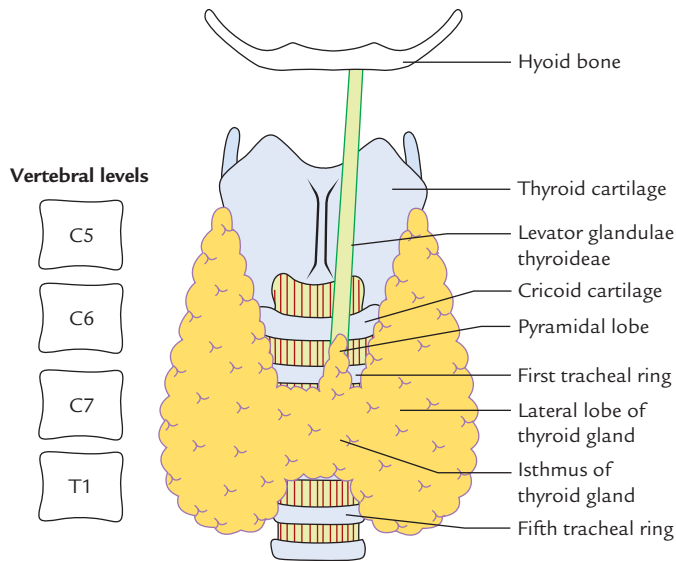
## Parts and Features

- It is H-shaped and consisting of vertical right and left lateral lobes and a horizontal isthmus connecting them across the midline. Sometimes a small **pyramidal lobe** projects upwards from the isthmus usually to the left of the midline. Not infrequently, it is connected to the body



**Fig. 11.1** Location of the thyroid gland.





**Fig. 11.2** Parts and extent of the thyroid gland.

of the hyoid bone by a fibrous or fibromuscular band called **levator glandulae thyroideae**.

- Each lateral lobe of the gland extends upwards to the oblique line of the thyroid cartilage and below up to the 5th or 6th tracheal ring.
- The isthmus extends across the midline in front of the 2nd, 3rd, and 4th tracheal rings.

### Weight and Dimensions

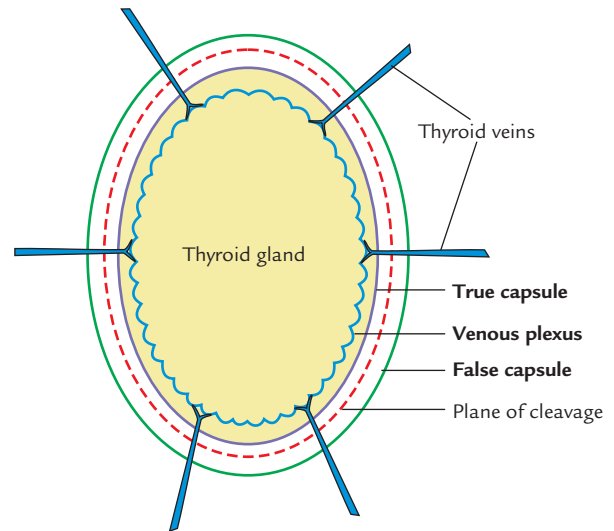
The thyroid gland weighs about 25 g. Each lobe of the thyroid gland is 5 cm long, 3 cm wide, and 2 cm thick ( $5 \times 3 \times 2$  cm). The isthmus measures about 1.25 cm in both, vertical and transverse diameters.

**N.B.** The thyroid gland is usually larger in females and further increases in size during menstruation and pregnancy.

### Capsule of the Thyroid Gland (Fig. 11.3)

The thyroid gland is invested by two capsules: an inner true capsule and an outer false capsule.

1. **True capsule:** It is formed by the peripheral condensation of the fibrous stroma of the gland.
2. **False capsule:** It is derived from the splitting of the pretracheal fascia. The important features of the false capsule are:
  - (a) It is thin along the posterior border of the lateral lobe.
  - (b) It is thick on the medial surface of the lateral lobe.
  - (c) Traced above, the pretracheal fascia after enclosing the thyroid gland is attached to the oblique lines of the thyroid cartilage, and the body of the hyoid bone in the midline. On the medial surface of thyroid lobe it thickens to form the **suspensory ligament** of



**Fig. 11.3** Capsules of the thyroid gland.

**Berry**, which connects the lobe to the cricoid cartilage.

The dense venous plexus unlike that of the prostate gland lies deep to true capsule (Fig. 11.3). Therefore, to avoid hemorrhage during thyroidectomy, the thyroid gland is removed along with the true capsule.

**N.B.** In case of prostate gland, the venous plexus lies between its true and false capsules and, therefore, during surgical removal of prostate gland (prostatectomy), prostate gland is removed; leaving behind both the capsules.

### Clinical correlation

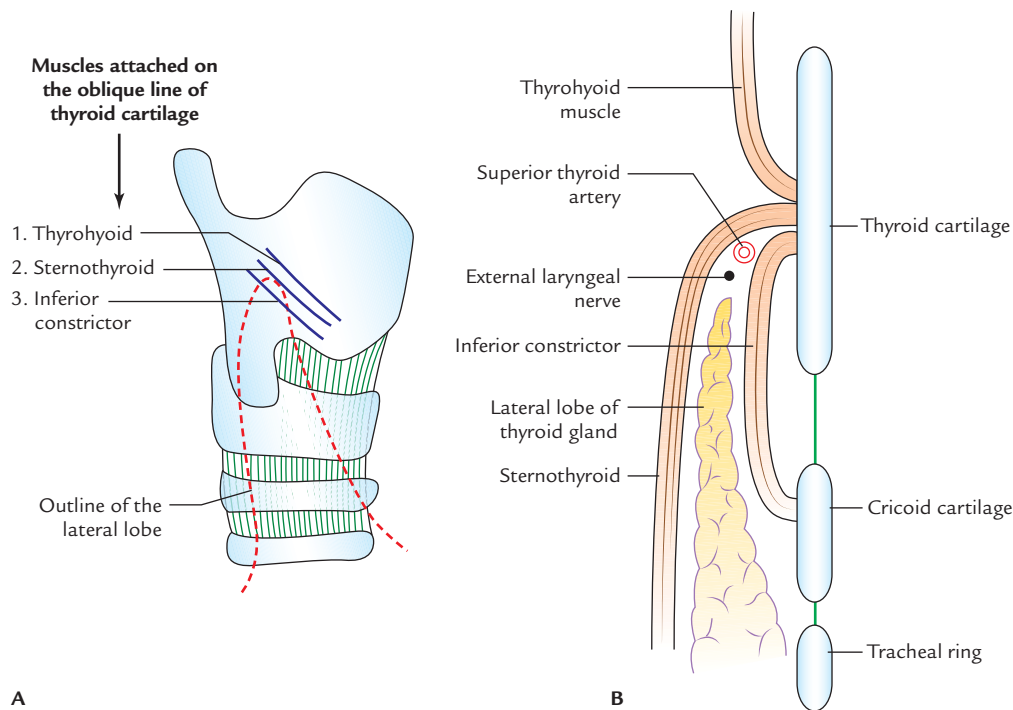
- Sometimes the thyroid tumor extends backwards, as the capsule is thin along the posterior borders of the lateral lobes, compressing the adjacent structures without forming a visible swelling on the front of the neck.
- The **thyroid gland moves up and down during swallowing** because its capsule is attached to laryngeal cartilages and hyoid bone, which moves up and down during swallowing. Thus thyroid swellings can be distinguished clinically from other swellings in the region of the neck.

### RELATIONS

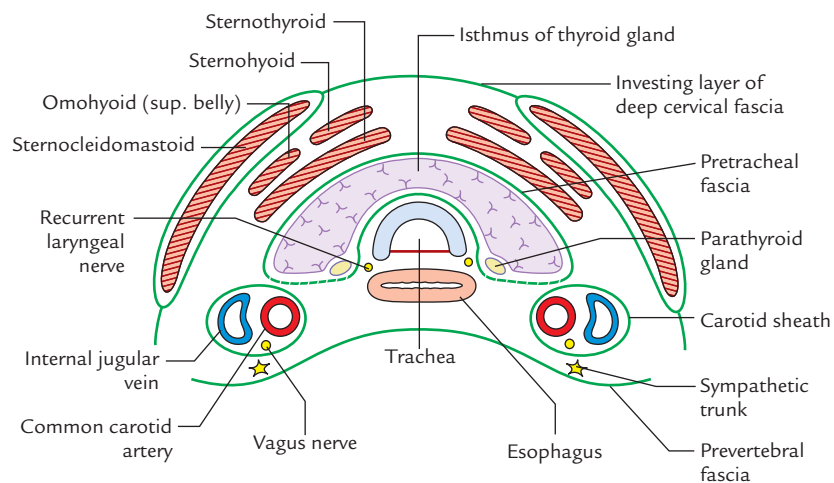
#### Relations of the Thyroid Lobe (Fig. 11.4, 11.5, and 11.6)

Each lobe of the thyroid gland is roughly pyramidal (conical) and presents **apex**, **base**, **three surfaces** (lateral, medial, and posterolateral), and **two borders** (anterior and posterior):

- **Apex:** The apex is directed upwards and slightly laterally. It extends up to the oblique line of thyroid cartilage where



**Fig. 11.4** Relations of the lateral lobe of the thyroid gland: A, upward extension of the thyroid lobe and muscles attached to the oblique line of the thyroid cartilage; B, the apex is sandwiched between the inferior constrictor and sternothyroid.



**Fig. 11.5** Transverse section of the anterior part of the neck at the level of thyroid isthmus, showing relations of the thyroid gland.

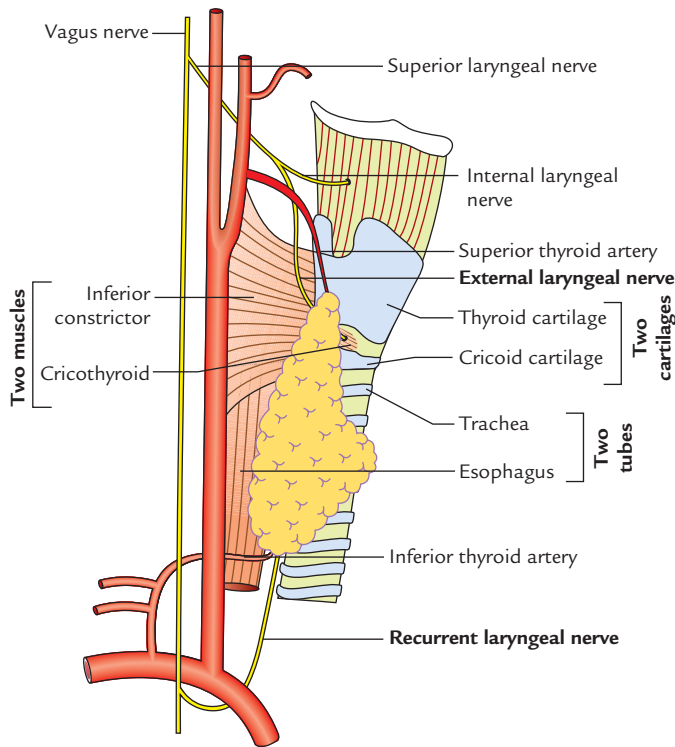
it is limited above by the attachment of sternothyroid muscle. The apex is sandwiched between the inferior constrictor medially and sternothyroid laterally (Fig. 11.4).

- **Base:** The base extends up to the 5th or 6th tracheal ring. It is related to inferior thyroid artery and recurrent laryngeal nerve.
- **Lateral (superficial) surfaces:** It is convex and is covered by (Fig. 11.5):

- three strap muscles (sternothyroid, sternohyoid, and superior belly of omohyoid), and
- anterior border of sternocleidomastoid overlapping it inferiorly.

- **Medial surface** is related to (Fig. 11.6):

- two tubes:* trachea and esophagus,
- two muscles:* inferior constrictor and cricothyroid, and
- two cartilages:* cricoid and thyroid.



**Fig. 11.6** Medial relations of the lateral lobe of the thyroid gland. The figure also shows intimate relationship of the external and recurrent laryngeal nerves with the superior and inferior thyroid arteries, respectively.

- **Posterolateral surface** is related to (Fig. 11.5) carotid sheath and its contents (common carotid artery, internal jugular vein, and vagus nerve). The ansa-cervicalis is embedded in the anterior wall of the sheath while cervical sympathetic chain lies posterior to sheath in front of prevertebral fascia.
- **Anterior border** is thin and separates superficial and medial surfaces. It is related to anterior branch of the superior thyroid artery.
- **Posterior border** is thick and rounded. It separates the medial and the posterior surfaces. It is related to (a) longitudinal arterial anastomosis between superior and inferior thyroid arteries, and (b) parathyroid glands.

### Relations of Isthmus

The isthmus is horizontal and presents **two surfaces**—anterior and posterior and **two borders**—superior and inferior.

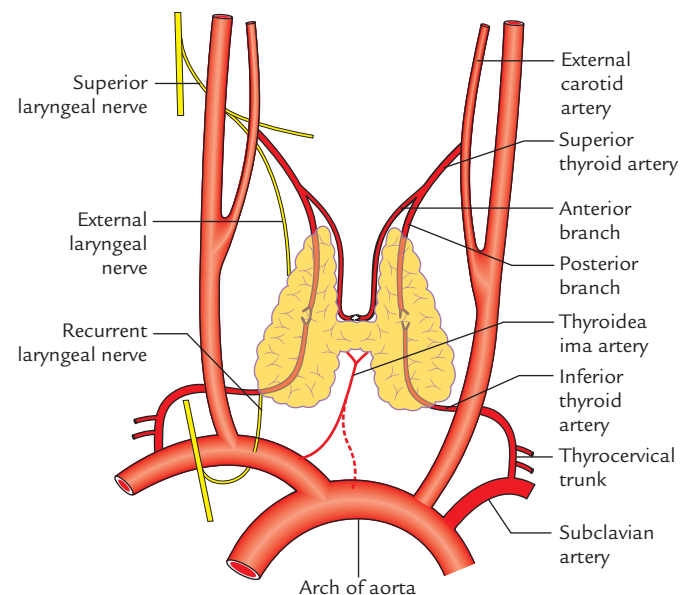
- **Anterior surface** is related to: (a) strap muscles (sternohyoid and sternothyroid) and (b) anterior jugular veins.
- **Posterior surface** is related to 2nd, 3rd, and 4th tracheal rings.

- **Superior border** is related to anastomosis between the anterior branches of two superior thyroid arteries.
- **Inferior border.** Along this border inferior thyroid vein emerge and thyroidea ima artery (when present) enters.

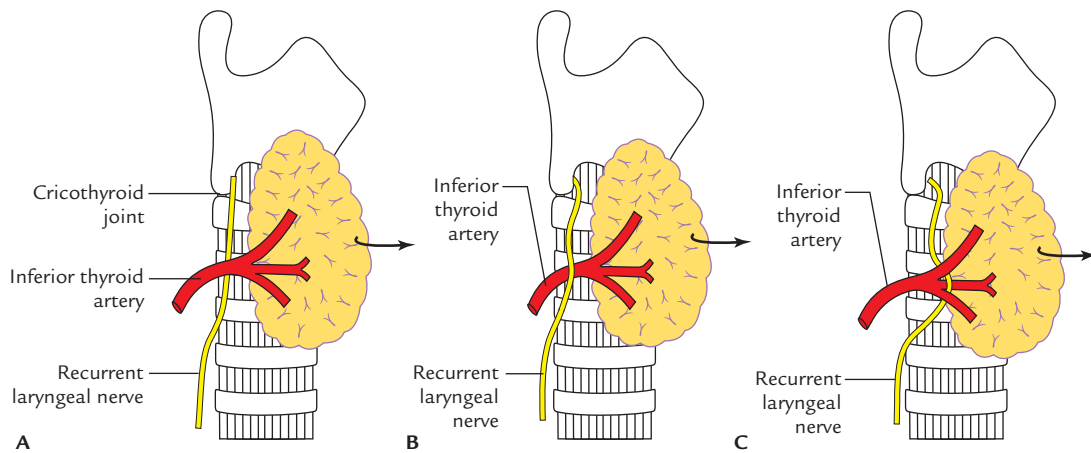
### ARTERIAL SUPPLY

The gland is highly vascular and is supplied by the following arteries (Fig. 11.7):

1. **Superior thyroid artery:** It is a branch of the external carotid artery. It runs downwards and forwards in company with the external laryngeal nerve, which it leaves near the upper pole of the thyroid lobe. At the apex of the lobe, it divides into anterior and posterior branches. The anterior branch first descends along the anterior border of the lobe and then continues along the upper border of the isthmus to anastomose with its fellows of opposite side. The posterior branch descends along the posterior border of the lobe to anastomose with the ascending branch of the inferior thyroid artery.
2. **Inferior thyroid artery:** It is a branch of thyrocervical trunk from the first part of the subclavian artery. It first runs upwards along the medial border of scalenus anterior, and then passes medially behind the carotid sheath to reach the back of the thyroid lobe, where it is intimately related to the recurrent laryngeal nerve. The *recurrent laryngeal nerve presents a variable relationship with the artery*. It may pass behind or in front of the loop of the artery or between the branches of the artery (Fig. 11.8). The artery gives 4 or 5 branches. One



**Fig. 11.7** Arterial supply of the thyroid gland.



**Fig. 11.8** Variable relationship of recurrent laryngeal nerve with the inferior thyroid artery.

ascending branch anastomoses with the posterior branch of the superior thyroid artery.

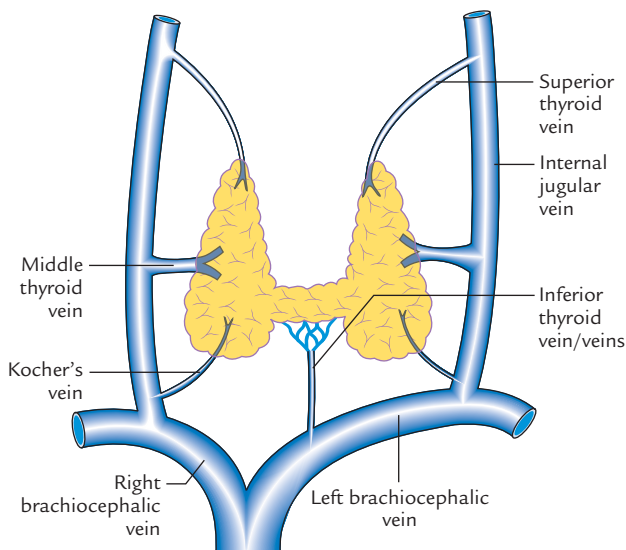
The inferior thyroid artery supplies lower two-third of the lobe and lower half of the isthmus.

3. **Thyroidea ima artery (in 30% cases):** It is a branch of the brachiocephalic trunk or may arise directly from the arch of aorta. It enters the isthmus from below.
4. **Accessory thyroid arteries:** They arise from tracheal and esophageal arteries.

## VENOUS DRAINAGE

The venous blood from the thyroid gland is drained by three set of veins (Fig. 11.9), viz.

1. **Superior thyroid vein:** It emerges at the upper pole of the thyroid lobe, runs upwards and laterally, and drains into the internal jugular vein.



**Fig. 11.9** Venous drainage of the thyroid gland.

2. **Middle thyroid vein:** This short, wide venous channel emerges at the middle of the lobe to soon enter the internal jugular vein.
3. **Inferior thyroid vein/veins:** They emerge at the lower border of the isthmus, form plexus in front of the trachea and then run downwards to drain into the left brachiocephalic vein.
4. Sometimes a fourth vein, the **thyroid vein (of Kocher)** emerges between the middle and inferior thyroid veins to drain into the internal jugular vein.

## LYMPHATIC DRAINAGE (Fig. 11.10)

The lymph vessels draining the thyroid gland are arranged into two groups, upper and lower, and they follow the arteries:

1. The **upper group** drains into the prelaryngeal (lying in front of the larynx) and upper deep cervical (jugulo-digastric) lymph nodes.
2. The **lower group** drains into pretracheal and lower deep cervical lymph nodes and group of lymph nodes along the recurrent laryngeal nerves. Those from lower part of isthmus drain into retrosternal or brachiocephalic nodes lying in the superior mediastinum.

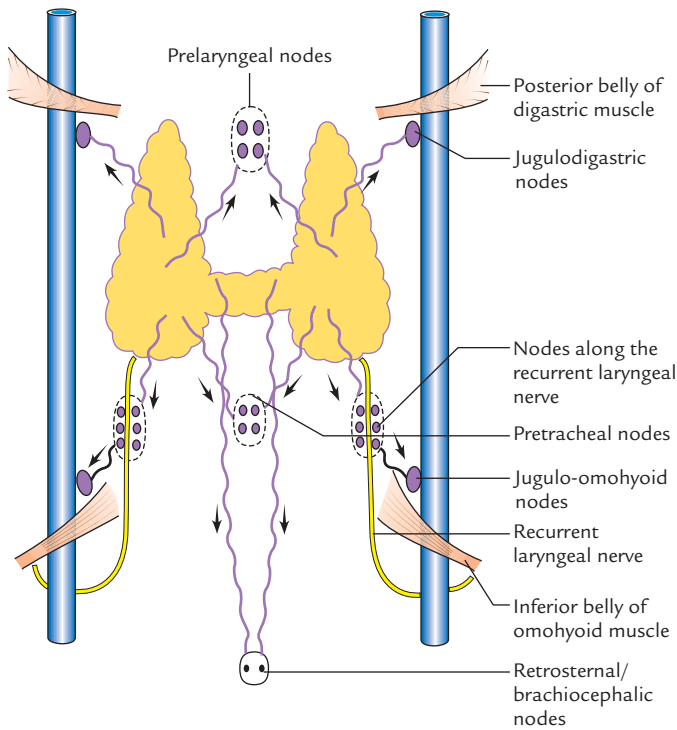
The upper lymphatics follow superior thyroid artery and lower lymphatics follow the inferior thyroid arteries.

## NERVE SUPPLY

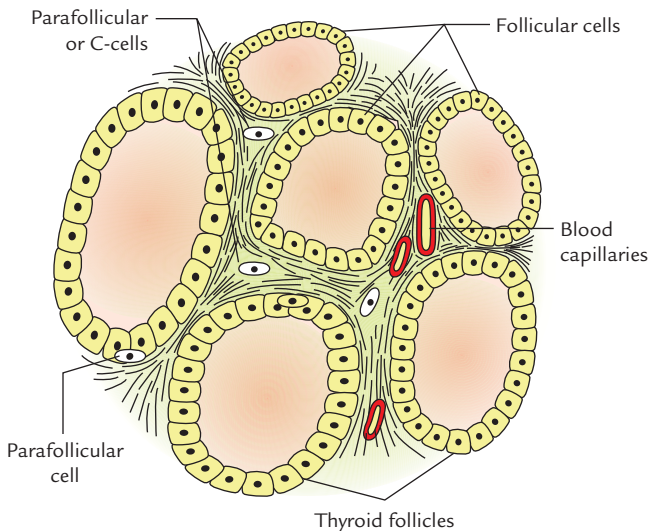
The thyroid gland is supplied by both sympathetic and parasympathetic nerve fibres:

1. The **parasympathetic supply** is derived from the vagus and recurrent laryngeal nerves.
2. The **sympathetic supply** is derived from the superior, middle, and inferior cervical sympathetic ganglia, but mainly from the middle one.





**Fig. 11.10** Lymphatic drainage of the thyroid gland.



**Fig. 11.11** Microscopic structure of the thyroid gland.

### MICROSCOPIC STRUCTURE (Fig. 11.11)

The thyroid gland consists of two types of secretory cells: follicular and parafollicular.

The **follicular cells** are cuboidal epithelial cells forming the wall of spherical thyroid follicles. They secrete two hormones: thyroxine and triiodothyronine.

These hormones are essential for normal growth and development, specially of the skeletal and nervous system.

The **parafollicular cells** or **C-cells** lie mainly between the basement membrane and the follicular cells. Some of them also lie in the spaces between the follicles. They secrete a hormone called **thyrocalcitonin** (also called **calcitonin**). It plays an important role in calcium metabolism. It reduces the blood calcium level by reducing the reabsorption of the calcium by the renal tubules. Its effects are opposite to that of parathormone.

### Clinical correlation

- **Goitre:** Any enlargement of the thyroid gland *except* that during menstruation and lactation is called *goitre*. It may be associated with hypofunction or hyperfunction of the gland.

*Hypothyroidism* occurs when there is insufficient secretion of the T3 and T4. The decreased levels of T3 and T4 cause increased secretion of TSH.

Hypothyroidism causes *cretinism* in children and myxedema in adults (for details see *General Anatomy* by Vishram Singh).

*Hyperthyroidism* (or *thyrotoxicosis*) occurs due to excess secretion of T3 and T4. The thyrotoxicosis clinically presents as: tachycardia, tremors, and systolic bruit. The main effects are due to increased BMR.

- **Simple goitre/puberty goitre:** It is an enlargement of the thyroid gland without signs of hyperthyroidism. It occurs due to deficiency of iodine in diet. Remember iodine is essential for synthesis of thyroid hormone. The decreased levels of T3 and T4 lead to increased secretion of TSH from the pituitary gland. The TSH causes hyperplasia of the thyroid gland. Simple goitre commonly occurs in females at the age of puberty, hence it is also called *puberty goitre*.

- **Effects of massive enlargement of thyroid gland (goitre; Fig. 11.12):** The thyroid gland can enlarge backwards or downwards but it cannot enlarge upwards due to attachment of its fascial sheath and sternothyroid muscle to the thyroid cartilage.

The backward enlargement is common because the thyroid capsule is relatively thin posteriorly. In backward enlargement, the gland buries itself around the sides of trachea and esophagus.

This results in three characteristic pressure symptoms, viz.

- **Dyspnea** (difficulty in breathing), due to pressure on trachea.
- **Dysphagia** (difficulty in swallowing), due to pressure on esophagus.
- **Dysphonia** (hoarseness of voice), due to pressure on the recurrent laryngeal nerve, which lies in the tracheoesophageal groove.

*Mnemonic = 3D*

The downward expansion behind the sternum is called retrosternal goitre. It can compress trachea leading to dangerous dyspnea.

It can also cause severe venous compression leading to venous congestion (Fig. 19.10).



The esophagus begins in the neck just below the cricopharyngeus at the lower border of C6 vertebra. It lies on longus colli behind the trachea. The thoracic duct lies on its left lateral border before arching over the triangle of the vertebral artery on the left side (for details see chapter 10).

## BRACHIOCEPHALIC VEINS

These veins collect blood from the head and neck, upper limbs, thoracic wall, and anterior abdominal wall. Each vein is formed by the union of corresponding internal jugular and subclavian veins behind the medial end of the clavicle. They end by joining to form superior vena cava behind the

lower border of the right first costal cartilage at the margin of the sternum.

The **right brachiocephalic vein** descends almost vertically. In the neck, it lies on the cervical pleura lateral to the brachiocephalic trunk. The phrenic nerve and internal thoracic artery lie posterior to it whereas anterior to it lie sternohyoid and sternothyroid muscles.

The **left brachiocephalic vein** runs obliquely to pass behind the upper part of the manubrium sterni to join the right brachiocephalic vein and is therefore much longer.

Both veins receive the vertebral, highest intercostal and inferior thyroid veins. The right lymphatic duct, right jugular trunk, and right subclavian trunk enter the right vein. The thoracic duct and left superior intercostal vein enter the left vein.

### Golden Facts to Remember

- |                                                                                         |                                                                                         |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| ▶ Longest prevertebral muscle                                                           | Longus cervicis                                                                         |
| ▶ Longest and largest scalene muscle                                                    | Scalenus medius                                                                         |
| ▶ Scalenus medius arises from the transverse processes of all cervical vertebrae except | Transverse process of 1st cervical vertebra                                             |
| ▶ Key muscle at the root of the neck                                                    | Scalenus anterior                                                                       |
| ▶ Largest cervical sympathetic ganglion                                                 | Superior cervical sympathetic ganglion                                                  |
| ▶ Cervical rib                                                                          | Enlarged costal element of transverse process of C7 vertebra                            |
| ▶ Root of the neck                                                                      | Area of neck immediately above the thoracic inlet                                       |
| ▶ Stellate ganglion                                                                     | Cervicothoracic ganglion (p. 174)                                                       |
| ▶ Smallest prevertebral muscle                                                          | Scalenus posterior                                                                      |
| ▶ Accessory phrenic nerve                                                               | A branch of nerve to subclavius containing C5 fibres, which join the main phrenic nerve |

### Clinical Case Study



A 19-year-old dental student complained of pain, tingling, and numbness along the medial side of her right forearm and hand. She also had difficulty in gripping the instruments firmly by the right hand. The physical examination revealed weak right radial pulse. The examining physician suspected the presence of cervical rib and performed Adson's test, which turned out to be positive. The presence of cervical rib was confirmed by plain X-ray. She was diagnosed as a case of *scalene syndrome*. She completely recovered of her symptoms after the surgical removal of the cervical rib.

#### Questions

1. What is cervical rib?
2. What are the boundaries of scalene triangle?
3. Name the structures passing through scalene triangle.
4. Mention the cause of symptoms told by the patient.
5. What is Adson's test?
6. What precautions should be taken when cervical rib is removed surgically?

#### Answers

1. It is an enlarged costal element of the transverse process of the 7th cervical vertebra.
2. It is a narrow triangular space bounded anteriorly by scalenus anterior, posteriorly by scalenus medius and inferiorly by the upper surface of the first rib.
3. Subclavian artery and lower trunk of the brachial plexus.
4. Compression of lower trunk of brachial plexus (C8 and T1) and subclavian artery.
5. This test is performed to assess the functional status of the arterial supply of the upper limb. The patient sitting on a stool is asked to take a deep breath and at the same time turn his/her face towards the affected side. Now the physician can feel the radial pulse. If the radial pulse becomes feeble or absent the test is said to be positive.
6. The periosteum of the rib must be removed to prevent the regeneration of the rib.

# Oral Cavity

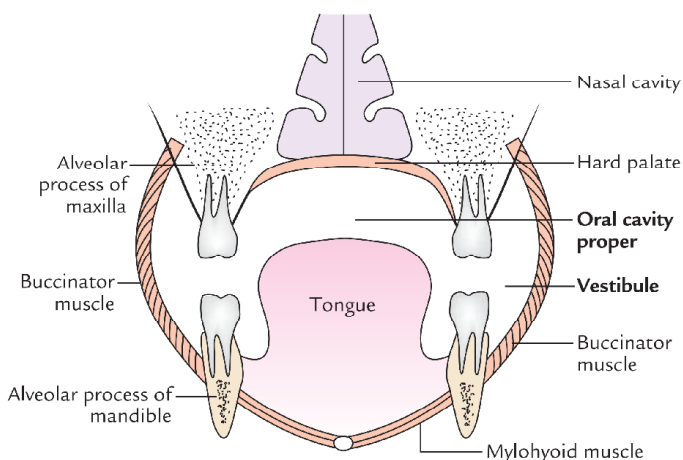
The oral cavity (mouth) is the first part of digestive tube. It extends anteroposteriorly from lips to the oropharyngeal isthmus, commonly used for ingestion of food and fluids.

It is divided into two parts (Fig. 13.1), *viz.*

1. A smaller slit-like space between lips/cheeks and teeth/gingivae (gums) called **vestibule**.
2. A larger space inside the teeth and gums called **oral cavity proper**.

## VESTIBULE OF THE MOUTH

The vestibule of the mouth is a narrow space that lies outside the teeth and gums, and inside the lips and cheeks. It is limited above and below by the reflection of mucus membrane from the lips and cheeks to the gums. When the mouth is open, it freely communicates with the oral cavity proper but when it is closed, i.e., when the teeth are occluded, it communicates on each side with the oral cavity proper through a small gap behind the third molar teeth and ramus of the mandible called **retromolar region**. Except teeth, the entire vestibule is lined by mucus membrane.



**Fig. 13.1** Subdivisions of the oral cavity as seen in its coronal section.

The lateral wall of the vestibule is formed by cheek made up of buccinator muscle. The tone of buccinator muscle and muscles of lips keep the walls of the vestibule in contact with the teeth and gums. In facial palsy, due to paralysis of these muscles, the lips and cheeks fall away from the teeth and gums, and food tends to collect in the vestibule of the mouth.

The openings in the vestibule of the mouth are as follows:

1. Opening of parotid duct; the parotid duct opens into the vestibule opposite the crown of the upper second molar tooth.
2. Openings of labial and buccal mucus glands.
3. Openings of 4 or 5 molar glands (mucus) situated on the buccopharyngeal fascia.

## LIPS

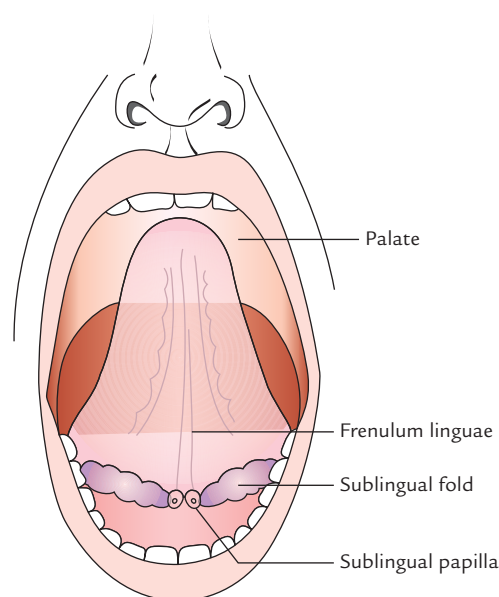
The lips are two mobile musculofibrous folds that surround the opening of the mouth. The upper and lower lips meet laterally at an angle of the mouth, usually in front of first premolar tooth.

The lips are lined externally by the skin and internally by the mucus membrane. The mucocutaneous junction lines the edge of the lip. Only a part of the mucosal surface is normally seen. The red portion of lip is called **vermillion zone**, the red hue is due to rich vascular bed visible through the thin moist epithelium. The skin and vermillion zone meet at the **vermillion border**. The internal aspect of each lip is connected to the corresponding gum by a median fold of the mucus membrane called **frenulum of the lip**.

## Structure

From superficial to deep, the lip is composed of the following structures:

1. Skin.
2. Superficial fascia.
3. Orbicularis oris muscle.
4. Submucosa containing mucus glands.
5. Mucus membrane.



**Fig. 13.10** Sublingual region seen when the tongue is turned upwards. Features on inferior surface of the tongue also seen.

- On each side of the lower end of frenulum, there is an elevation called **sublingual papilla**, on the summit of which opens the submandibular duct.
- The sublingual gland projects up into the floor of the mouth and produces an elevation in the mucus membrane on each side of the frenulum called **sublingual fold**. Most of the sublingual ducts open on this fold.

Many structures in the oral cavity are termed by their relationship to the tongue, palate, cheeks, and lips (Fig. 13.10). The structures closest to the tongue are termed **lingual**, those closest to palate **palatal**, those closest to cheeks **buccal**, and those closest to lips **labial**.

### Clinical correlation

**Ludwig's angina** is a cellulitis of the floor of the mouth, usually due to infection from a carious molar tooth, causing inflammatory edema of the floor of the mouth. It spreads to the submandibular and submental regions producing diffuse swelling in these regions also. The tongue is pushed upwards due to edema of the floor of the mouth, resulting in difficulty in swallowing.

## ROOF OF THE MOUTH

The roof of the mouth is formed by palate. The anterior two-third of palate, made up of bones is called **hard palate**, while posterior one-third made of soft issue is called **soft palate**. From the posterior-free margin of the soft palate a small conical projection called **uvula** hangs down in the median region. A poorly marked median raphe extends

from uvula to the **incisive papilla**—a slight elevation behind the incisive fossa.

The mucus membrane in the anterior part of the hard plate is thrown into 3 or 4 **transverse palatine folds** but posteriorly it is comparatively smooth. The palate is described in detail with pharynx in Chapter 14.

## TONGUE

The tongue is a mobile muscular organ in the oral cavity, which bulges upwards from the floor of the mouth and its posterior part forms the anterior wall of the oropharynx.

It is essentially a mass of skeletal muscle covered by mucus membrane. The muscle mass is separated into right and left halves by a midline fibrous septum. The tongue is separated from teeth by a deep **alveololingual sulcus**, which is filled in by palatoglossal fold/arch posterior to the last molar tooth.

### Functions

The tongue performs the following functions:

- Taste.
- Speech.
- Mastication.
- Deglutition.

### Shape

The tongue is conical in shape being elongated postero-anteriorly and flattened dorsoventrally.

### External Features

The tongue exhibits the following external features:

- A root.
- A tip.
- A body.

#### Root

The root of the tongue is attached to the mandible and hyoid bone by muscles. It is because of these attachments that the tongue is not swallowed during deglutition. The nerve and vessels of the tongue enter through its root.

#### Tip

It is the anterior free end of the tongue, which comes into contact with the central incisors.

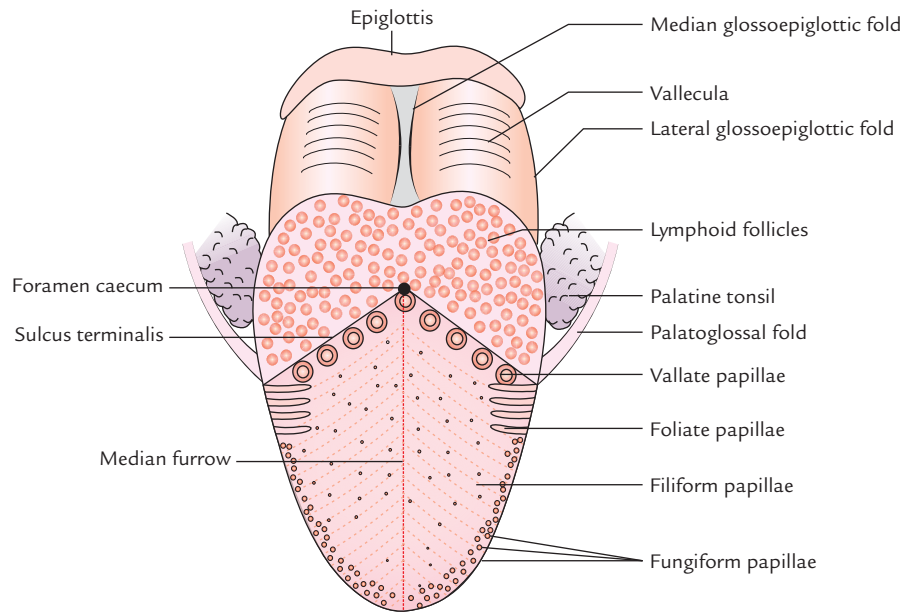
#### Body

The bulk of tongue between the root and tip is called *body*. It has dorsal and ventral surfaces and right and left lateral margins.

#### Dorsal surface (Fig. 13.11)

The dorsal surface is convex on all the sides. It is divided by a V-shaped sulcus, the **sulcus terminalis** into two parts, *viz.*

- Anterior two-third or **oral part**.
- Posterior one-third or **pharyngeal part**.



**Fig. 13.11** Features on the dorsal surface of the tongue.

The apex of the sulcus terminalis is marked by a blind foramen, the **foramen caecum**, which indicates the point of origin of the median thyroid diverticulum (thyroglossal duct) in the embryonic life.

The features differ markedly in the oral and pharyngeal parts.

The **oral part** presents the following features:

1. A median furrow, representing the bilateral origin of the tongue.
2. Large number of papillae.

The **pharyngeal part** presents the following features:

1. A large number of lymphoid follicles, which together constitute the **lingual tonsil**.
2. A large number of mucus and serous glands.

**N.B.** The oral and pharyngeal parts of the tongue are different in their embryological origin for mucosa of oral two-third develops from the 1st and 2nd pharyngeal arches while that of pharyngeal part develops from the 3rd or 4th pharyngeal arches.

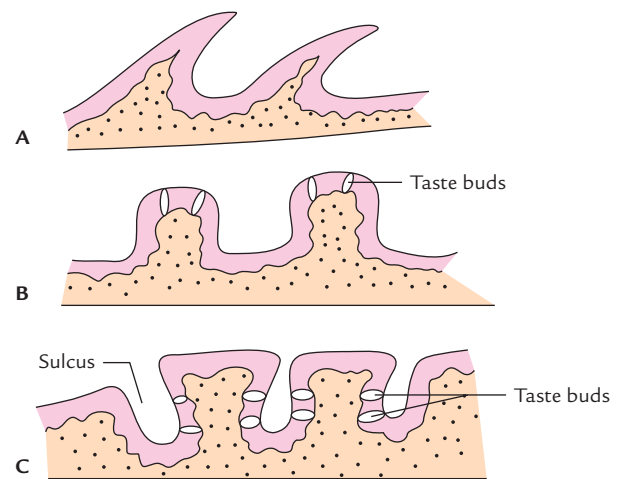
**Oral part:** The dorsum of oral part presents a shallow median furrow/groove. The mucus membrane is moist and pink and appears velvety due to the presence of numerous papillae.

### Clinical correlation

The furring or coating of tongue bears no relation to digestive disturbances as generally thought, but is usually due to smoking, respiratory tract infection, fever, or oral infection.

**Papillae of the tongue (Lingual papillae):** They are projections of lamina propria (corium) of mucus membrane covered with epithelium (Figs 13.11 and 13.12). The following four chief types of papillae are found:

1. **Vallate papillae:** The vallate papillae (known formerly as *circumvallate papillae*) are largest (1–2 mm in diameter) and vary in number from 8–12 and are arranged in a V-shaped row in front of **sulcus terminalis**. Each papilla is like a truncated cone surrounded by a circular sulcus, which is bounded on its periphery by a wall or **vallum**. The duct of serous glands open into the sulcus (moat) and taste buds are found in the papilla and its vallum.



**Fig. 13.12** Characteristic features of different types of lingual papillae: A, filiform; B, fungiform; C, vallate.



2. **Filiform papillae:** These are narrowest and most numerous. They are minute conical projections with sharply pointed tips. Filiform papillae are located abundantly on the dorsum of tongue and are largely responsible for its velvety appearance.
3. **Fungiform papillae:** They have a red rounded head (about 1 mm in diameter) and a narrower base, mostly at the apex and margins of the tongue, while some are scattered over the dorsum of the tongue. They are visible as discrete pink pinheads.
4. **Foliate papillae:** They consist of inconstant vertical grooves and ridges near the margin in front of sulcus terminalis. Foliate papillae are more prominent in the tongue of rabbits. They are rudimentary in humans.

**N.B.** Another type of papillae, called *papillae simplex*, are known. These are surface projections and can be seen only under the microscope.

**Pharyngeal part:** The dorsum of pharyngeal part faces posteriorly and forms the base of tongue.

*The base of tongue constitutes the anterior wall of the oropharynx and can be inspected only by the use of a mirror or by a downward pressure on the tongue with a tongue spatula.*

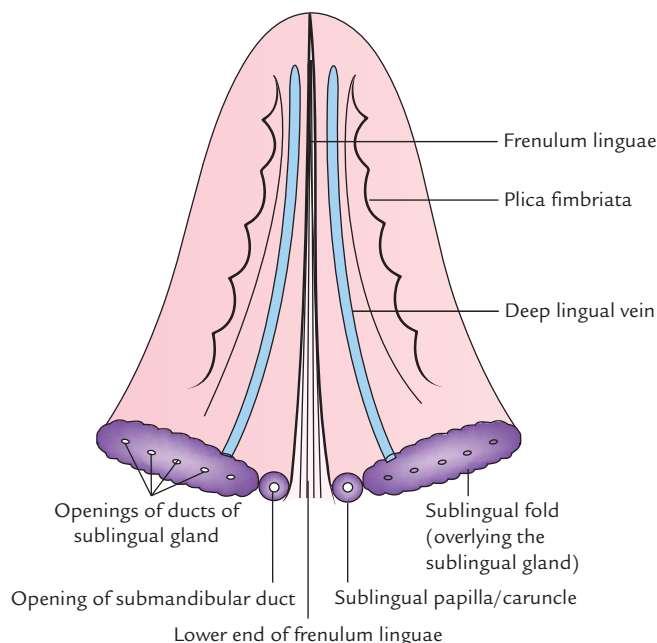
The mucus membrane over the dorsum of pharyngeal part is devoid of papillae. It, however, appears uneven due to the presence of numerous lymphatic follicles in the underlying submucosa. These follicles are collectively termed **lingual tonsil**.

The mucus membrane in this part is continuous with mucus membrane covering the palatine tonsils and the pharynx. Posteriorly, it is reflected onto the front of the epiglottis as the median **glossoepiglottic fold** and onto the lateral wall of pharynx as **lateral glossoepiglottic folds**. The space on each side of the median glossoepiglottic fold is termed **epiglottic vallecula**.

### Ventral (inferior) surface of the tongue (Fig. 13.13)

The inferior surface of tongue is situated in the oral cavity only. The mucus membrane lining this surface is thin, smooth, and purplish. It is reflected onto the floor of the mouth. The under aspect of the tongue presents the following features:

1. **Frenulum linguae**, a median-fold of mucus membrane connecting the tongue to the floor of the mouth.
2. **Deep lingual veins**, may be seen through mucous membrane on either side of frenulum linguae (the lingual nerve and lingual artery are medial to the vein but not visible).
3. **Plica fimbriata**, a fringed fimbriated fold of mucous membrane lateral to the lingual vein directed forwards and medially towards the tip of the tongue.



**Fig. 13.13** Features on the ventral (inferior) surface of the tongue and floor of the mouth.

### Clinical correlation

- **Tongue tie:** If frenulum extends too far towards the tip of the tongue, it is called *tongue tie*. It inhibits normal movements of the tongue and may interfere with normal speech. This can be corrected by cutting the frenulum surgically.
- Certain drugs, which are lipid soluble can diffuse through the thin mucous membrane lining the sublingual region of oral cavity and can be quickly absorbed into the circulation. A leading example is nitroglycerin (Sorbitrate), a vasodilator used in cases of *angina pectoris*. The drug (tablet) is placed under the tongue, where in less than 1 minute, it dissolves and passes through the thin oral mucosa into the lingual veins.
- **Carcinoma of the tongue:** The tongue is a common site of carcinoma. It mostly involves lateral margins of anterior two-thirds of the tongue. The relative frequency of various sites of involvement is as follows:
  - Anterior two-thirds: 64% (lateral margin, 47%; ventral surface and frenulum linguae, 9%; dorsum, 6.5%; tip, 11.5%)
  - Posterior one-third: 20%
  - Posterio-most (faciolingual) part: 6%

### Muscles of the Tongue

The musculature of tongue consists of extrinsic and intrinsic muscles. The intrinsic muscles are within the tongue and have no attachment outside the tongue whereas extrinsic muscles take origin from structures outside the tongue and enter the

tongue to be inserted in it. The intrinsic muscles change the shape of tongue whereas extrinsic muscles move the tongue (such as protrusion, retraction and side-to-side movements) as well as alter its shape.

The tongue is divided into symmetrical right and left halves by a medial fibrous septum, which separates the muscles of two sides. Each half of the tongue contains four intrinsic and four extrinsic muscles. These are as follows:

**Intrinsic muscles**

- 1. Superior longitudinal.
- 2. Inferior longitudinal.
- 3. Transverse.
- 4. Vertical.

**Extrinsic muscles**

- 1. Genioglossus.
- 2. Hyoglossus.
- 3. Styloglossus.
- 4. Palatoglossus.

**Intrinsic muscles (Fig. 13.14):** They are confined to the tongue and are not attached to the bone. They occupy the upper part of the tongue and alter its shape. The intrinsic muscles are arranged in several planes. They run in three directions: longitudinal, horizontal, and vertical. The complex interlacing of fibres of these muscles is responsible for the astonishing way in which the tongue can change its shape, becoming wide and flat, narrow and thick, or rolled up laterally to become gutter shaped. The latter shape cannot be achieved by a small number of people and this inability is genetically determined. Intrinsic muscles occupy the upper part of the tongue and are attached to the submucous fibrous layer and to the median fibrous

**Table 13.3** Features of the intrinsic muscles

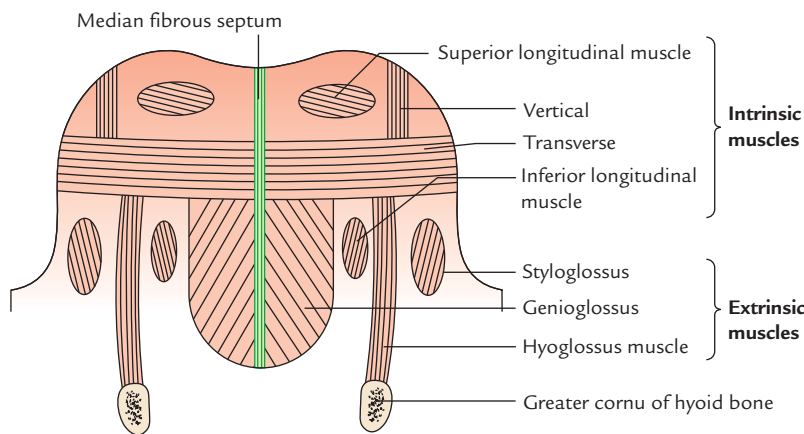
| Intrinsic muscle      | Location                                                      | Actions                                                                                                  |
|-----------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Superior longitudinal | Beneath the mucous membrane                                   | <ul style="list-style-type: none"><li>• Shortens the tongue</li><li>• Makes the dorsum concave</li></ul> |
| Inferior longitudinal | Close to inferior surface between genioglossus and hyoglossus | <ul style="list-style-type: none"><li>• Shortens the tongue</li><li>• Makes the dorsum convex</li></ul>  |
| Transverse            | Extends from median septum to the margin                      | Makes the tongue narrow and elongated                                                                    |
| Vertical              | At the border of the anterior part of the tongue              | Makes the tongue broad and flattened                                                                     |

septum. Their location and actions are enumerated in the Table 13.3.

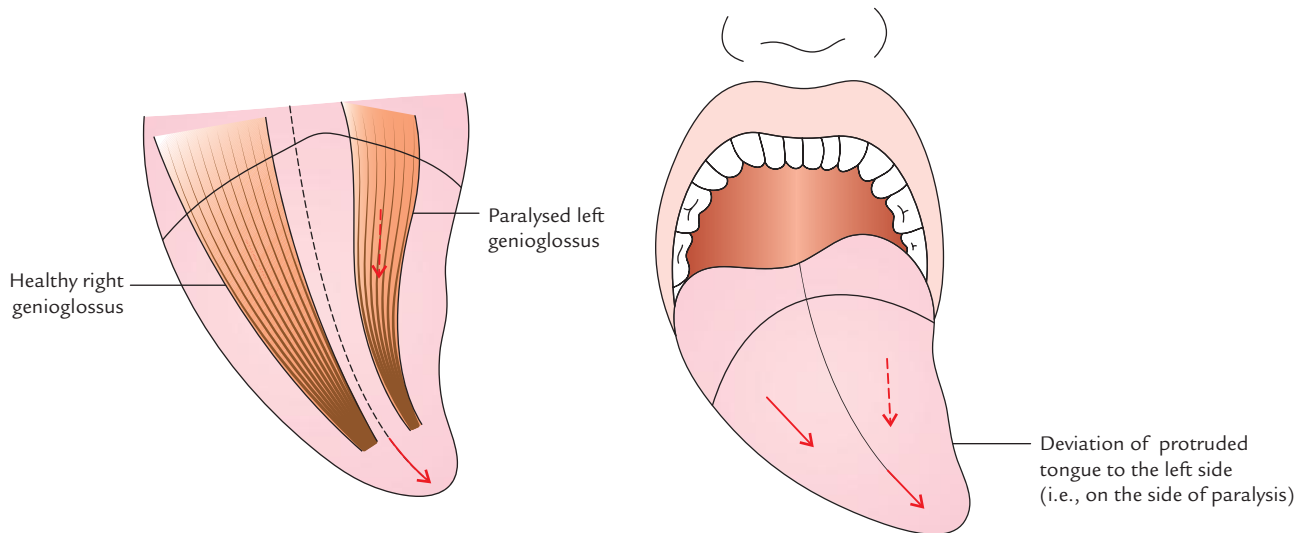
**Extrinsic muscles (Fig. 13.15):** They attach the tongue to the mandible (genioglossus), the hyoid (hyoglossus), the styloid process (styloglossus), and the palate (palatoglossus) on each side. They are described in detail in submandibular region (Chapter 9). The summary of their origin, insertion, and actions is presented in Table 13.4.

**Movements of the Tongue (Fig. 13.16)**

The movements of tongue and muscles producing them are listed in Table 13.5.



**Fig. 13.14** Coronal section of the tongue showing arrangement of intrinsic and extrinsic muscles of the tongue.

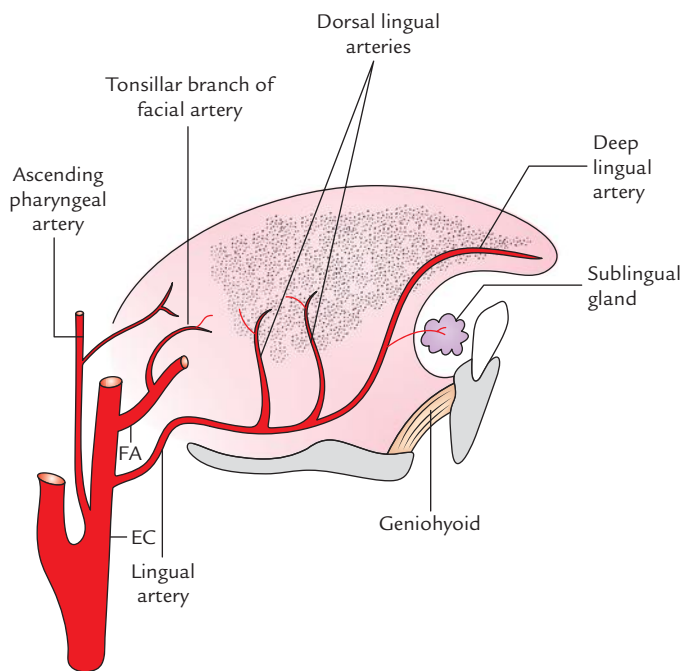


**Fig. 13.17** Effect of paralysis of the hypoglossal nerve (left). A. Paralysis of the genioglossus on the left side. B. Deviation of the tongue on the left side i.e. on the paralysed side.

### Arterial Supply (Fig. 13.18)

The tongue is supplied by the following arteries:

1. Branches of lingual artery (chief artery of tongue); the deep lingual arteries to the anterior part and dorsal lingual arteries to the posterior part.
2. Tonsillar branch of the facial artery.
3. Ascending pharyngeal artery.



**Fig. 13.18** Arterial supply of the tongue (FA = facial artery, EC = external carotid artery).

**N.B.** The deep lingual artery anastomoses with its fellow of the opposite side near the tip of the tongue. It is the only significant anastomoses across the midline of the tongue.

### Venous Drainage

It is by the following veins:

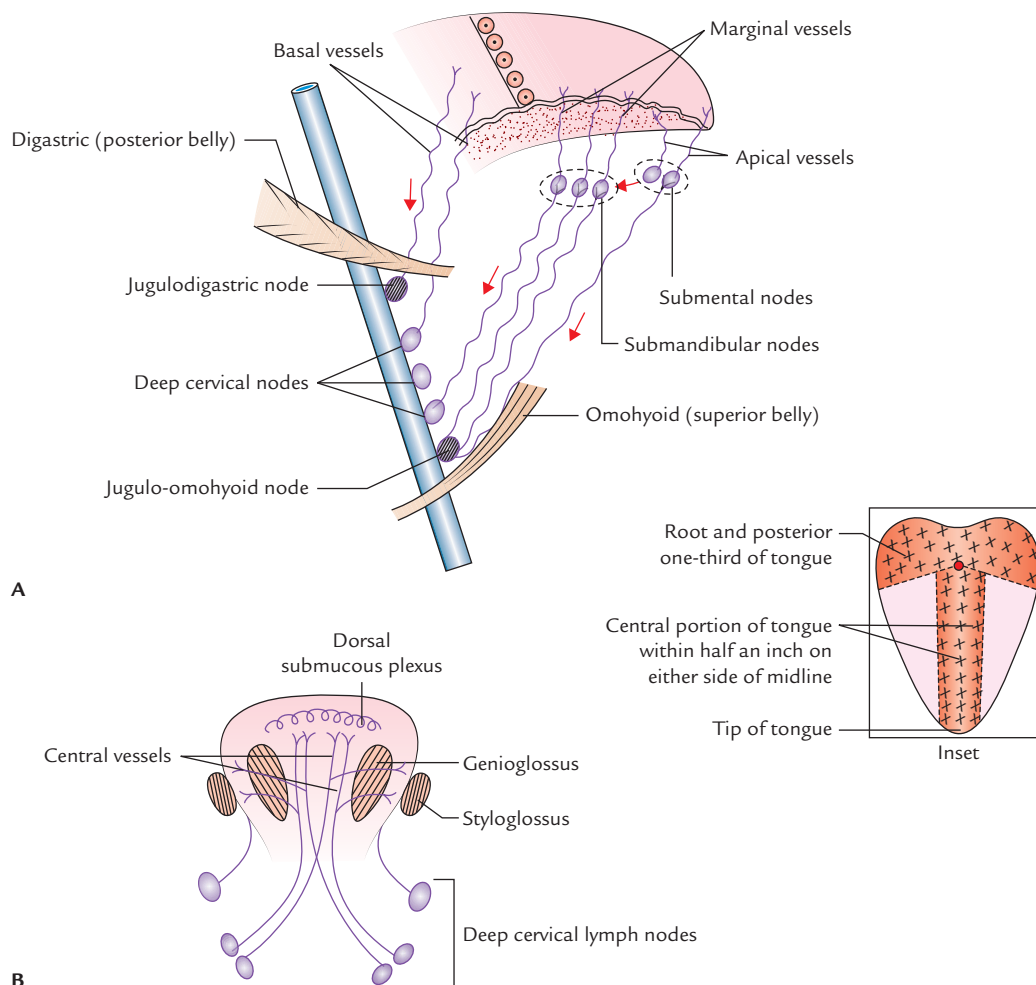
1. **Deep lingual vein** is the *principal vein of the tongue* and is visible on the inferior surface of the tongue near the median plane through thin mucous membrane in life.
2. **Venae comitantes** accompanying the lingual artery. They are joined by *dorsal lingual veins*.
3. **Venae comitantes** accompanying the hypoglossal nerve.

These veins unite at the posterior border of the hyoglossus to form the **lingual vein**, which drains into either common facial vein or internal jugular vein.

### Lymphatic Drainage

The lymphatics emerging from the tongue are grouped into the following four sets (Fig. 13.19):

1. **Apical vessels:** They drain the tip and inferior surface of the tongue into submental lymph nodes after piercing the mylohyoid muscle. Their efferents go to the submandibular nodes mainly, some cross the hyoid bone to reach the jugulo-omohyoid nodes.
2. **Marginal vessels:** They drain the marginal portions of the anterior two-third of the tongue—unilaterally into submandibular lymph nodes and then to the lower deep cervical lymph nodes, including jugulo-omohyoid.
3. **Central vessels:** They drain the central portion of the anterior two-third of the tongue (i.e., area within 0.5 inch on either side of midline). They pass vertically



**Fig. 13.19** Lymphatic drainage of the tongue: **A**, showing course and direction of apical, marginal, and basal lymph vessels; **B**, showing course and direction of central lymph vessels. Figure in the inset shows areas (in red) having bilateral lymphatic drainage.

downwards in the midline of the tongue between the genioglossus muscles and then drain bilaterally into the deep cervical lymph nodes.

4. **Basal vessels:** They drain the root of the tongue and posterior one-third of the tongue bilaterally into upper deep cervical lymph nodes, including jugulodigastric.

### Clinical correlation

**Prognosis of tongue cancer:** There is rich anastomosis across the midline between the lymphatics of the posterior one-third of the tongue; therefore, a cancer on one side readily metastasizes to ipsilateral as well as the contralateral lymph nodes. In contrast, there is little cross communication of lymphatics of the anterior two-third of the tongue where cancer, more than 0.5 inches (12 mm) away from the midline, does not metastasize to the contralateral lymph nodes till very late stage. For this reason, cancer in the posterior one-third of the tongue has poor prognosis.

### Nerve Supply (Fig. 13.20)

The nerves supplying the tongue are as follows:

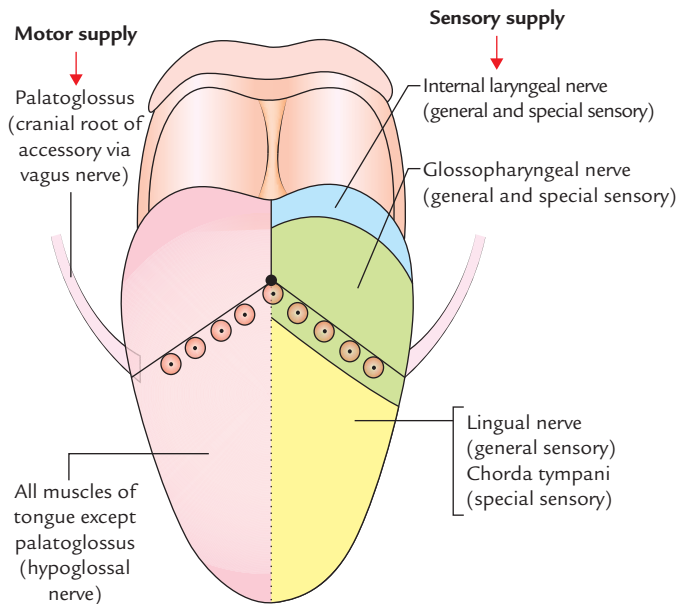
**Motor supply:** All the muscles of tongue (intrinsic and extrinsic) are supplied by the hypoglossal nerve *except* palatoglossus which is supplied by cranial root of accessory via pharyngeal plexus.

**Sensory supply:** Anterior two-third of the tongue is supplied by:

- (a) lingual nerve carrying general sensations, and
- (b) chorda tympani nerve carrying special sensations of taste.

Posterior one-third of the tongue is supplied by:

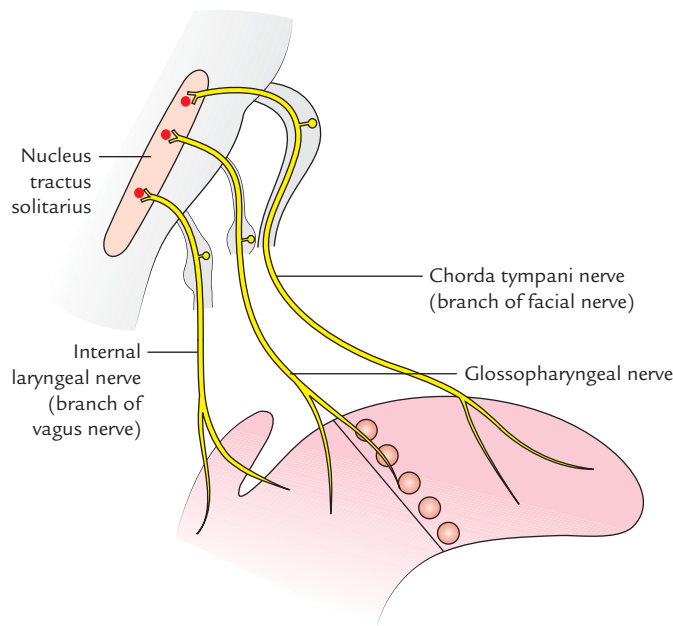
- (a) glossopharyngeal nerve, carrying both general and special sensations of taste, and
- (b) posteriormost part (base of the tongue), supplied by the internal laryngeal branch of the superior laryngeal carrying special sensations of taste.



**Fig. 13.20** Nerve supply of the tongue. Right half of the figure shows motor supply and left half shows sensory supply.

**N.B.** Nerves carrying taste sensations from the tongue are as follows (Fig. 13.21):

- *Chorda tympani* nerve (a branch of the facial nerve) from anterior two-third of the tongue.
- *Glossopharyngeal* from posterior one-third of the tongue.
- *Internal laryngeal* nerve from superior laryngeal branch of the vagus nerve, from posteriormost part of the tongue.



**Fig. 13.21** Nerves carrying taste sensations from the tongue.

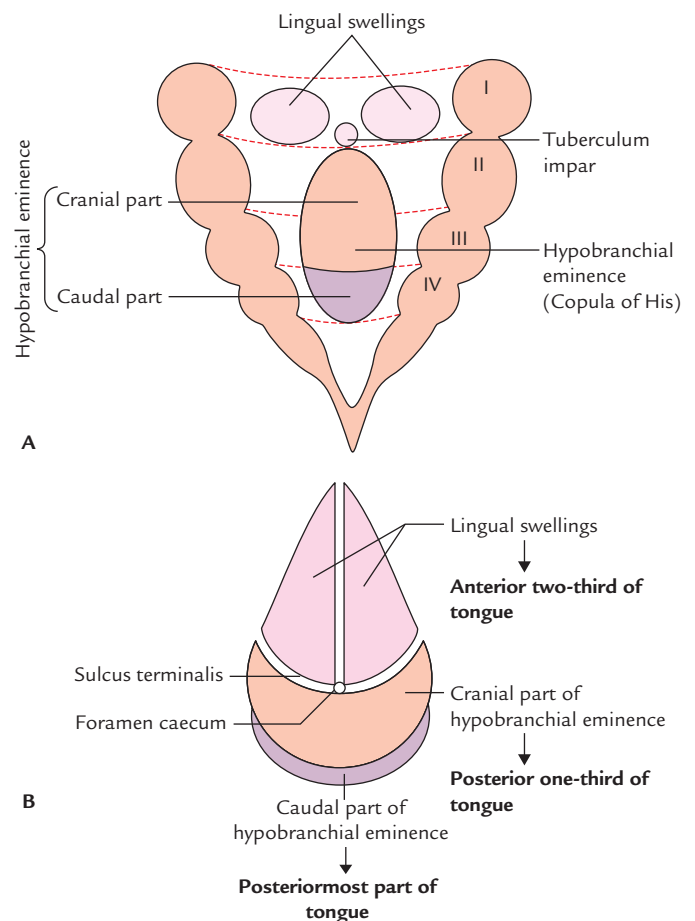
### Clinical correlation

**Referred pain of cancer tongue:** The patients with cancer tongue often complains of pain in ear, temporomandibular joint, temporal fossa, and/or lower teeth.

This is due to *referred pain*. It is important to note that pain is frequently referred from one branch of the mandibular nerve to the other. Carcinoma commonly involves anterior 2/3rd of tongue. Thus if the sensations carried from anterior 2/3rd of the tongue by the lingual nerve are referred to auriculotemporal nerve, the patient feels pain in the ear, TMJ, and temporal fossa. On the other hand, if the pain from lingual nerve is referred to the inferior alveolar nerve, the pain is felt in the lower teeth.

### Development of the Tongue (Fig. 13.22)

The tongue develops from the floor of the primitive pharynx in relation to the pharyngeal arches.



**Fig. 13.22** Development of the tongue: A, appearance of four swellings in the floor of primitive pharynx; B, showing formation of anterior two-third by the growth and fusion of two lingual swellings and posterior one-third by the growth and fusion of cranial part of the hypobranchial eminence.



**Floor:** It is formed by:

- Soft palate (sloping upper surface).
- Pharyngeal isthmus, an opening in the floor between the free edges of soft palate and posterior pharyngeal wall.

**Anterior wall:** It is formed by posterior nasal apertures separated by the posterior edge of nasal septum.

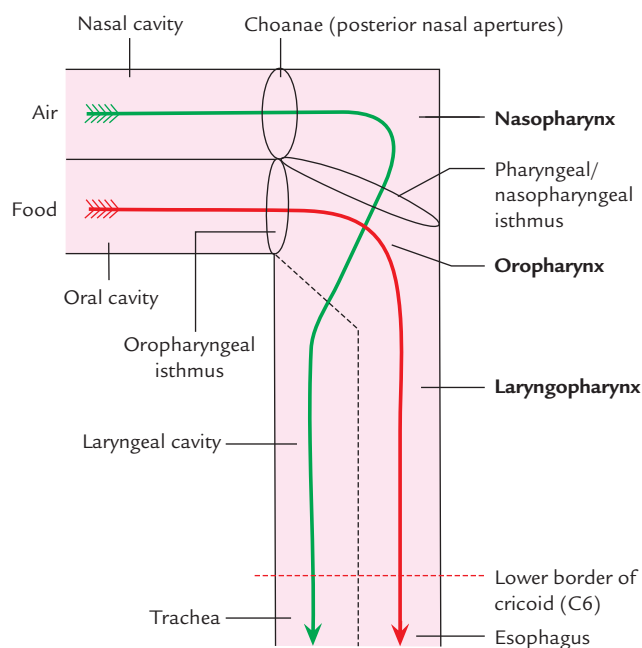
**Posterior wall:** It forms continuous sloping surface with roof. It is supported by anterior arch of C1 vertebra.

**Lateral wall:** Medial pterygoid plate of sphenoid.

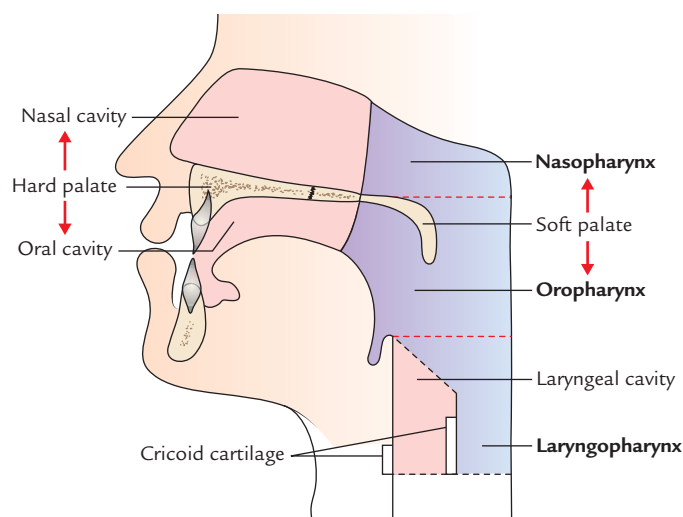
### Features

The features seen in the nasopharynx are:

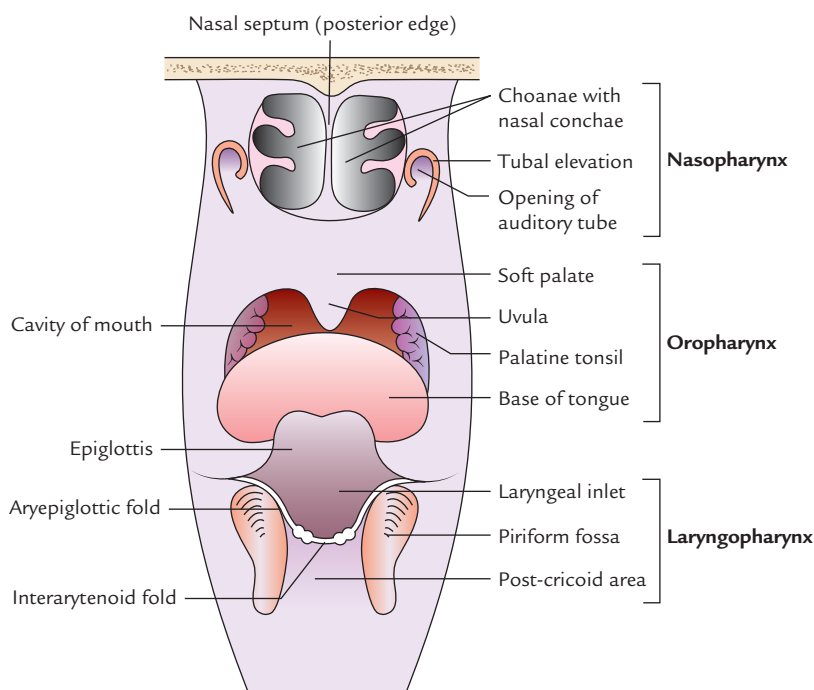
- Nasopharyngeal (pharyngeal) tonsil:** It is a collection of lymphoid tissue beneath the mucous membrane at



**Fig. 14.2** Pathways for food (red arrow) and air (green arrow) through the pharynx.



**Fig. 14.3** Subdivisions of the pharynx.



**Fig. 14.4** The pharynx opened from behind showing features in the anterior walls of the nasopharynx, oropharynx, and laryngopharynx.

(b) **Anterior wall presents:**

- (i) *Lingual tonsil*, formed by numerous nodules of lymphoid tissue underneath the mucous lining of the pharyngeal part of the dorsum of the tongue.
- (ii) *Upper free end of epiglottis*, behind the tongue.
- (iii) *Median and lateral glossoepiglottic folds*, connecting the anterior surface and edges of the epiglottis, respectively to the tongue.
- (iv) *Epiglottic valleculae* are shallow fossae between the median and lateral glossoepiglottic folds.

**Oropharyngeal Isthmus (Fig. 14.5)**

It is an arched opening between the two palatoglossal folds through which the oral cavity communicates with the oropharynx. Its boundaries are as under:

*Above:* Soft palate.

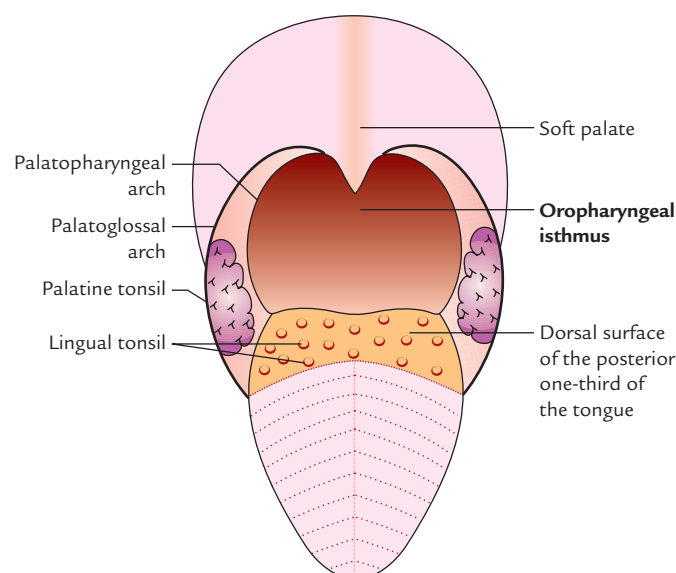
*Below:* Dorsal surface of the posterior one-third of the tongue.

*Lateral:* Palatoglossal fold/arch on either side containing palatoglossus muscle.

The oropharyngeal isthmus is closed during deglutition to prevent regurgitation of food from the pharynx to the mouth.

**Clinical correlation**

Since the pathways for food and air cross each other in the oropharynx, the food sometimes may enter into the respiratory tract and cause choking. Similarly the air often enters the digestive tract producing gas in the stomach, which results in eructation (*belching*).



**Fig. 14.5** Boundaries of the oropharyngeal isthmus.

**LARYNGOPHARYNX**

The laryngopharynx lies behind the laryngeal inlet and posterior wall of the larynx. It lies behind the larynx and extends from the upper border of the epiglottis to the lower border of cricoid cartilage anteriorly and lower border of C6 vertebra posteriorly. It communicates anteriorly with the laryngeal cavity through **laryngeal inlet** and inferiorly with the esophagus at the pharyngoesophageal junction (the narrowest part of the GIT except appendix).

**Boundaries**

*Anterior wall:* It is formed by:

- (a) Laryngeal inlet.
- (b) Posterior surface of the larynx.

*Posterior wall:* It is supported by the bodies of C3, C4, C5, and C6 vertebrae.

*Lateral wall:* It is supported by thyroid cartilage and thyrohyoid membrane.

**Features**

The features seen in the laryngopharynx are:

- (a) **Anterior wall** presents laryngeal inlet and below the inlet it is supported by cricoid and arytenoid cartilages.
- (b) **Lateral wall** presents piriform fossa one on each side of laryngeal inlet. The piriform fossa is described in detail below.

**PIRIFORM FOSSA**

It is a deep recess broad above and narrow below in the anterior part of lateral wall of the laryngopharynx, on each side of the laryngeal inlet. These recesses are produced due to bulging of larynx into laryngopharynx.

**Boundaries (Fig. 14.6)**

*Medial:* Aryepiglottic fold and quadrangular membrane of larynx.

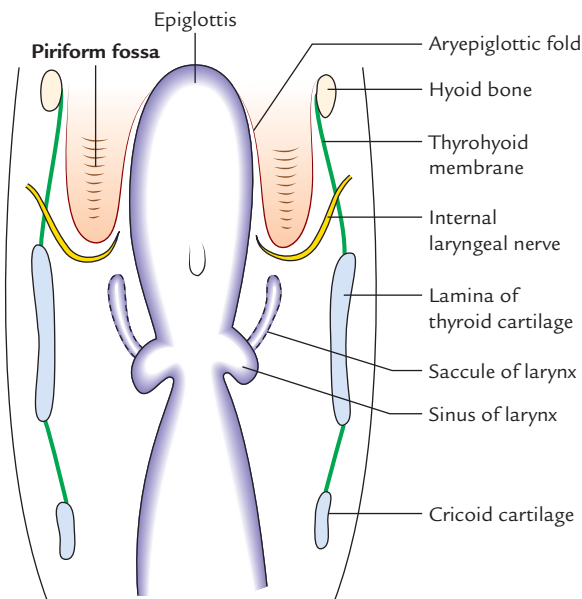
*Lateral:* Mucous membrane covering the medial surface of the lamina of thyroid cartilage and thyrohyoid membrane.

The internal laryngeal nerve and superior laryngeal vessels pierce the thyrohyoid membrane and traverse underneath the mucous membrane of the floor of the fossa to reach the medial wall.

*Above:* Piriform fossa is separated from epiglottic vallecula by lateral glossoepiglottic fold.

**N.B.**

- The piriform fossa is deep in ruminating animals in which it acts as *lateral food channel* to convey the bolus of food during deglutition by the side of closed laryngeal inlet.



**Fig. 14.6** Schematic coronal section through larynx showing the location and boundaries of the piriform fossa.

- It is sometimes, artificially deepened by smugglers using lead balls to hide precious materials such as diamonds. For this reason, the piriform fossa is also called *smuggler's fossa*.

### Clinical correlation

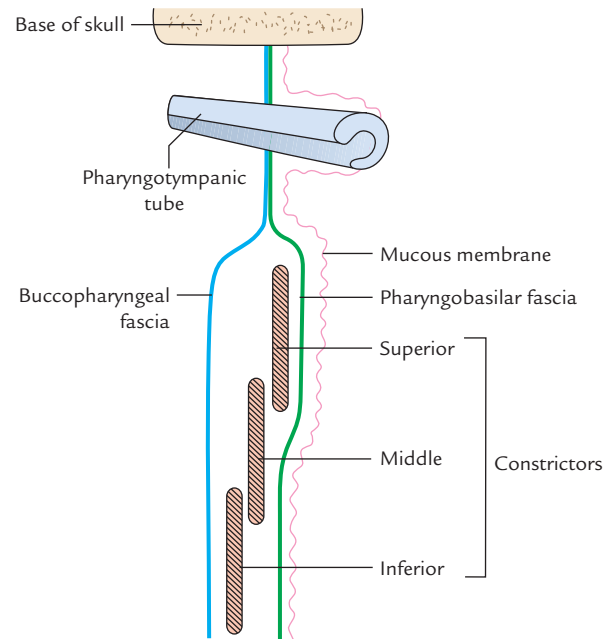
- The piriform fossae are dangerous sites for perforation by an endoscope.
- A malignant tumor of the laryngopharynx (hypopharynx) may grow in the space provided by the piriform fossa without producing symptoms until the patient presents with metastatic lymphadenopathy.
- The ingested foreign bodies (for example, fish bones, safety pins) are sometimes lodged into the piriform fossa. If care is not taken, the removal of foreign bodies may damage the internal laryngeal nerve leading to anesthesia in the supraglottic part of the larynx and subsequent loss of *protective cough reflex*.

### PHARYNGEAL WALL (Fig. 14.7)

The wall of the pharynx consists of four layers; from within outwards these are as follows:

- Mucous membrane/mucous coat.
- Pharyngobasilar fascia (pharyngeal aponeurosis).
- Muscular coat (pharyngeal muscles).
- Buccopharyngeal fascia (loose areolar sheath).

**Mucous membrane/mucous coat:** The mucous membrane lining the pharynx contains a considerable amount of elastic tissue and is continuous with the mucous lining of



**Fig. 14.7** Structure of the pharyngeal wall.

eustachian tubes, nasal cavities, mouth, larynx, and esophagus. It is lined by non-keratinized stratified squamous epithelium except in the region of the nasopharynx, where it is lined by ciliated columnar epithelium (respiratory epithelium).

**N.B.** There are many subendothelial collections of lymphoid tissue around the commencement of food and air passages, into which epithelium tends to invaginate in the form of narrow clefts (crypts).

These collections of lymphoid tissue form pharyngeal and tubal tonsils in the nasopharynx and palatine, and lingual tonsils in the oropharynx.

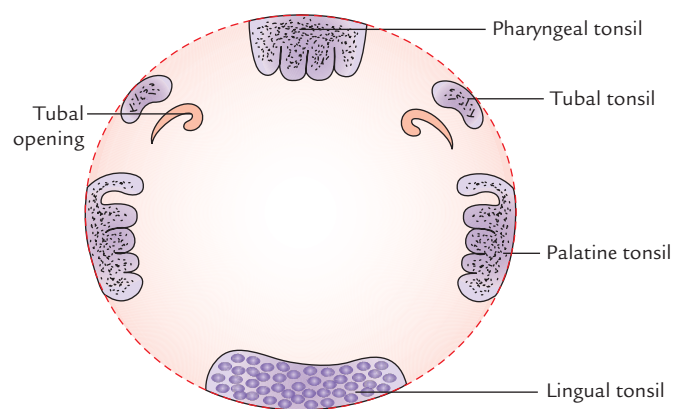
**Pharyngobasilar fascia:** It is a fibrous thickening of the submucosa. It lines the muscular coat and is thick near the base of the skull but thin and indistinct inferiorly. The pharyngobasilar fascia is thickest:

- in the upper part where it fills the gap between the upper border of superior constrictor and the base of the skull, and
- posteriorly where it forms the *pharyngeal raphe*.

**Muscular coat:** The muscular coat consists of the following two layers of striated muscles:

- The **outer layer** comprises three pairs of circular muscles called *constrictors*.
- The **inner layer** comprises three pairs of longitudinal muscles.

The pharyngeal muscles are described in detail under the heading 'muscles of the pharynx'.



**Fig. 14.8** Waldeyer's ring (an interrupted 'circle of tonsils' at the upper end of the respiratory and alimentary tracts).

**Buccopharyngeal fascia:** It is an inconspicuous fascia, which covers the outer surface of constrictor muscles. In the upper part, it is also prolonged forwards to cover the buccinator muscles, hence the name *buccopharyngeal fascia*. Above the upper border of the superior constrictor, it blends with the pharyngobasilar fascia.

### Clinical correlation

**Waldeyer's ring:** The aggregations of lymphoid tissue underneath the epithelial lining of pharyngeal wall called tonsils, surround the commencement of air and food passages. These aggregations together constitute an interrupted circle called *Waldeyer's ring*, which forms the special feature of the interior of the pharynx.

The Waldeyer's ring is formed by (Fig. 14.8):

1. Pharyngeal tonsil (nasopharyngeal tonsil), postero-superiorly.
2. Lingual tonsil, anteriorly.
3. Tubal and palatine tonsils, laterally.

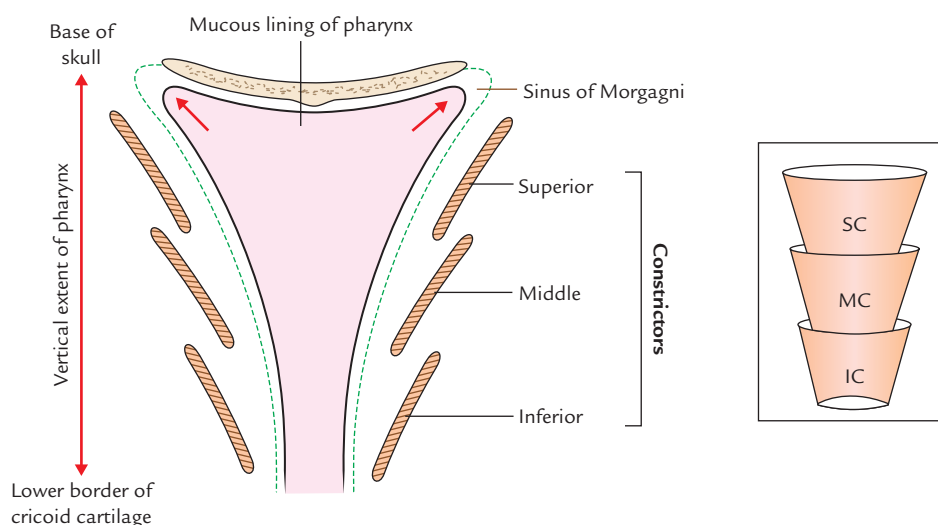
It is thought that, Waldeyer's ring prevents the invasion of microorganisms from entering the air and food passages and this helps in the defense mechanism of the respiratory and alimentary systems.

## MUSCLES OF THE PHARYNX

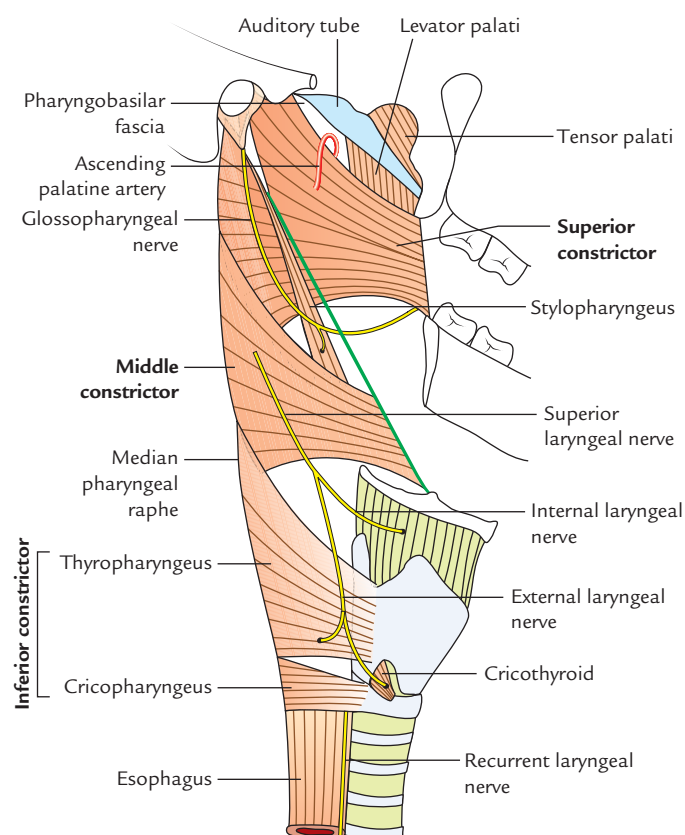
### CONSTRICTOR MUSCLES (Figs 14.9 and 14.10)

The *three constrictor muscles of the pharynx* (superior, middle, and inferior) are arranged like a flowerpot without base, placed one above the other and open in front for communication with the nasal, oral, and laryngeal cavities. Thus inferior constrictor overlaps the middle, which in turn overlaps the superior constrictor (Fig. 14.9 inset).

The **constrictor muscles** form bulk of the muscular coat of the pharyngeal wall. They arise in front from the margins of posterior openings of the nasal, oral, and laryngeal cavities. The fibres pass backwards, in a fan-shaped manner into the lateral and posterior walls of the pharynx to be inserted into the median fibrous raphe on the posterior aspect of the pharynx, extending from the base of the skull (pharyngeal tubercle of occipital bone) to the esophagus. The origin and insertion of constrictors of the pharynx is shown in Figure 14.10.



**Fig. 14.9** Overlapping arrangement of the constrictor muscles of the pharynx. The figure in the inset shows flowerpot arrangement of the constrictors (SC = superior constrictor, MC = middle constrictor, IC = inferior constrictor).



**Fig. 14.10** Origin and insertion of the constrictors of the pharynx.

The origin, insertion, nerve supply, and actions of the constrictor muscles are presented in Table 14.1.

### Clinical correlation

**Pharyngeal Pouch** (also called *Zenker's diverticulum*): Inferior constrictor muscle has two parts: *thyropharyngeus* made up of oblique fibres and *cricopharyngeus* made up of transverse fibres.

The potential gap posteriorly between the thyropharyngeus and cricopharyngeus is called *pharyngeal dimple* or *Killian's dehiscence* (Fig. 14.11). The mucosa and submucosa of the pharynx may bulge through this weak area to form a pharyngeal pouch or diverticulum (Fig. 14.12).

The formation of pharyngeal pouch in the region of Killian's dehiscence is attributed to the *neuromuscular incoordination* in this region, which may be because the two parts of the inferior constrictor have different nerve supply. The propulsive thyropharyngeus is supplied by the pharyngeal plexus and the sphincteric cricopharyngeus is supplied by the recurrent laryngeal nerve. If the cricopharyngeus fails to relax when the thyropharyngeus contracts, the bolus of food is pushed backwards and tends to produce a diverticulum.

### Gaps in the Pharyngeal Wall

The four gaps exist on either side in the pharyngeal wall in relation to constrictor muscles. The gaps and structures

**Table 14.1** Origin, insertion, nerve supply, and actions of the constrictor muscles of the pharynx

| Muscle                                           | Origin                                                                                                                                                                        | Insertion                                                                | Nerve supply                                                                                | Action               |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------|
| Superior constrictor<br>(Quadrilateral in shape) | (a) Pterygoid hamulus<br>(b) Pterygomandibular raphe<br>(c) Medial surface of the mandible at the upper end of mylohyoid line<br>(d) Side of the posterior part of the tongue | (a) Pharyngeal tubercle on the base of skull<br>(b) Median fibrous raphe | Pharyngeal branch of the vagus nerve carrying fibres of cranial root of the accessory nerve | Helps in deglutition |
| Middle constrictor<br>(Fan shaped)               | (a) Lower part of the stylohyoid ligament<br>(b) Lesser cornu of hyoid<br>(c) Upper border of greater cornu of hyoid                                                          | Median fibrous raphe                                                     | Pharyngeal branch of the vagus nerve carrying fibres of cranial root of the accessory nerve | Helps in deglutition |
| Inferior constrictor<br>(a) Thyropharyngeus      | (a) Oblique line on lamina of the thyroid cartilage<br>(b) Tendinous band between the thyroid (inferior) tubercle and cricoid cartilage                                       | Median fibrous raphe                                                     | (a) Pharyngeal plexus and<br>(b) External laryngeal nerve                                   | Helps in deglutition |
| (b) Cricopharyngeus                              | Cricoid cartilage                                                                                                                                                             | Median fibrous raphe                                                     | Recurrent laryngeal nerve                                                                   |                      |



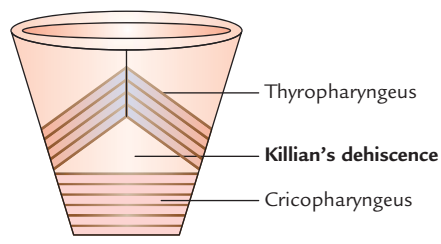


Fig. 14.11 Killian's dehiscence.

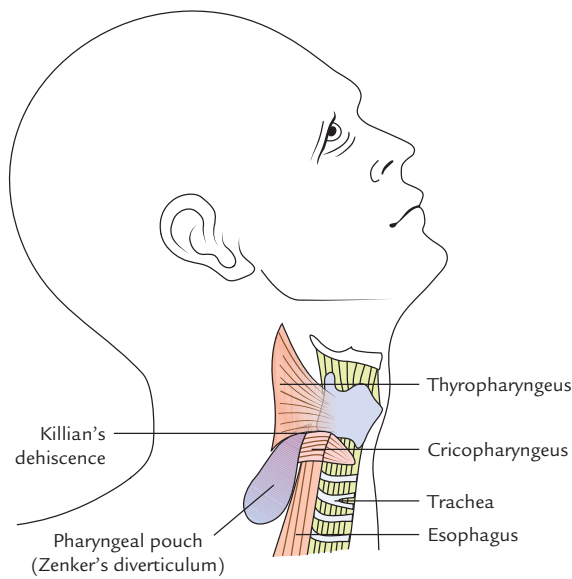


Fig. 14.12 Pharyngeal diverticulum.

**Table 14.2** The gaps in the pharyngeal wall and structures passing through them

| Gap                                                                                                   | Structures passing through them                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Between the base of skull and the upper concave border of superior constrictor (sinus of Morgagni)    | <ul style="list-style-type: none"><li>• Auditory tube</li><li>• Levator palati muscle</li><li>• Ascending palatine artery</li><li>• Palatine branch of the ascending pharyngeal artery</li></ul> |
| Between the superior and middle constrictors                                                          | <ul style="list-style-type: none"><li>• Stylopharyngeus muscle</li><li>• Glossopharyngeal nerve</li></ul>                                                                                        |
| Between the middle and inferior constrictors                                                          | <ul style="list-style-type: none"><li>• Internal laryngeal nerve</li><li>• Superior laryngeal vessels</li></ul>                                                                                  |
| Between the lower border of inferior constrictor and the esophagus (in the tracheo-esophageal groove) | <ul style="list-style-type: none"><li>• Recurrent laryngeal nerve</li><li>• Inferior laryngeal vessels</li></ul>                                                                                 |

passing through these gaps (Fig. 14.13) are presented in Table 14.2.

**LONGITUDINAL MUSCLES (Fig. 14.14)**

These muscles run longitudinally from above downwards to form the longitudinal muscle coat (Table 14.3). The origin,

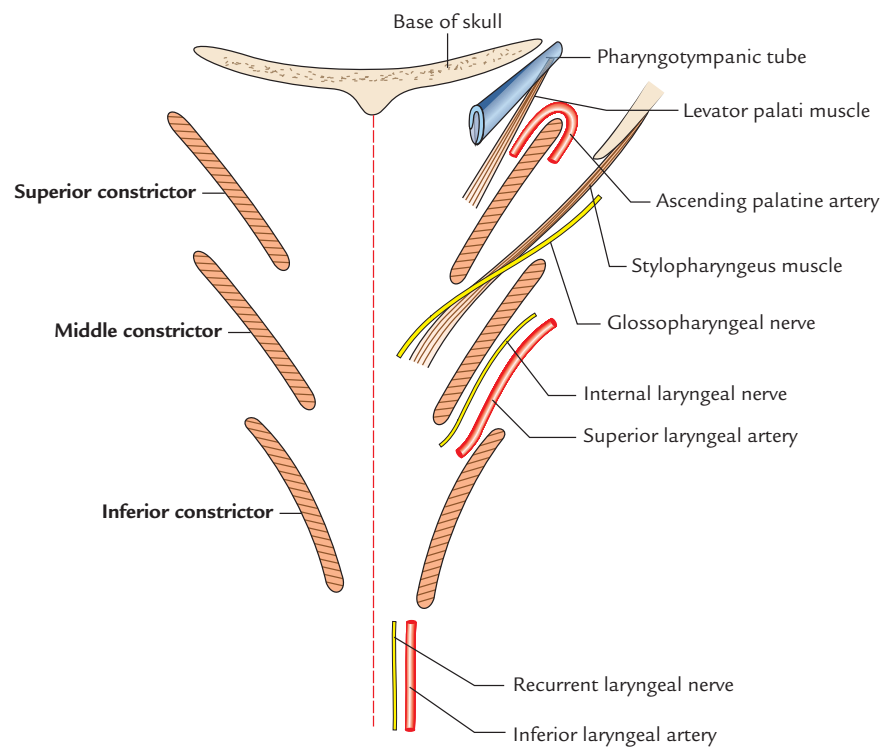


Fig. 14.13 Structures passing through the gaps in the pharyngeal wall.

**N.B.** A triangular fold of mucous membrane extends from anterior pillar to the anteroinferior part of the tonsil. It encloses a potential space called **anterior tonsillar space**.

The tonsil is separated from the tongue by a sulcus called **tonsillogingual sulcus**.

**Surface anatomy:** An oval area 1.25 cm above and in front of the angle of the mandible, marked in the parotid region of face indicates the location of tonsil on the surface.

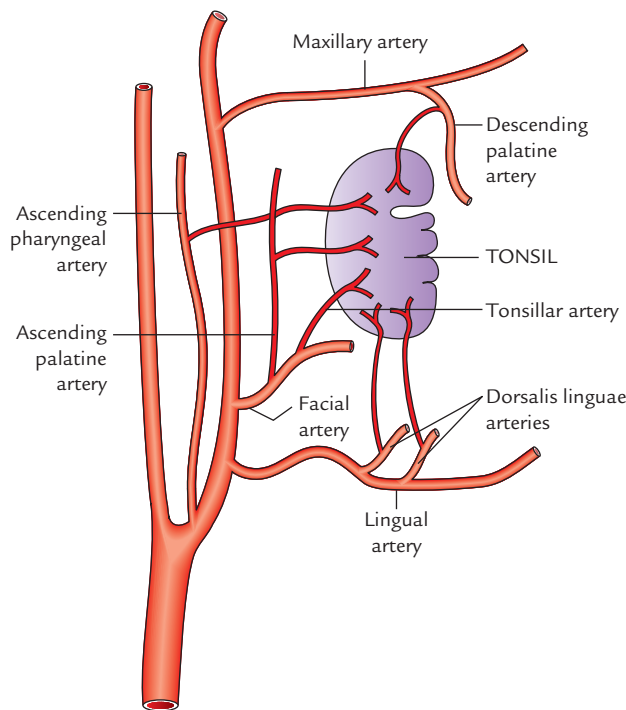
### Arterial Supply of the Tonsil (Fig. 14.17)

The following arteries supply the tonsil:

1. Tonsillar branch of facial artery (it is the principal artery and enters the lower pole of the tonsil by piercing the superior constrictor).
2. Dorsalis linguae branches of lingual artery.
3. Ascending palatine, a branch of facial artery.
4. Ascending pharyngeal, a branch of external carotid artery.
5. Greater palatine (descending palatine), a branch of maxillary artery.

### Venous Drainage of the Tonsil

The veins from tonsil drain into paratonsillar vein. The paratonsillar vein descends from the soft palate across the lateral aspect of the tonsillar capsule and pierces the superior constrictor to drain into **pharyngeal venous plexus**.



**Fig. 14.17** Arteries supplying the tonsil.

### Lymphatic Drainage of the Tonsil

The lymphatics of tonsil pierce the superior constrictor and drain into the upper deep cervical lymph nodes, particularly the jugulodigastric lymph node. It is often called **tonsillar lymph node** because it is primarily enlarged in infection of the tonsil (tonsillitis). The tonsillar lymph node is located below the angle of the mandible.

### Nerve Supply of the Tonsil

Palatine tonsil is supplied by the glossopharyngeal nerve and lesser palatine branches of the sphenopalatine ganglion.

### Clinical correlation

- **Acute tonsillitis:** The tonsils are the frequent sites of acute infection especially in school-going children. It may affect adults also. This condition is called *acute tonsillitis*. It is mostly seen in viral infection. It is rare in infants and persons above 50 years of age.
- **Acute follicular tonsillitis:** In this condition, the infection spreads into crypts, which become filled with purulent material presenting at the openings of crypts as *yellowish spots*.
- **Bleeding from tonsillar fossa after tonsillectomy:** It most commonly occurs due to damage of *paratonsillar vein*. The blood clots should be removed in order to check bleeding. If not removed, they interfere with the retraction of the vessel walls by preventing the contraction of the surrounding muscles. The postoperative edema of tonsillar bed after tonsillectomy can affect the glossopharyngeal nerve leading to loss of sensation in the posterior one-third of the tongue.

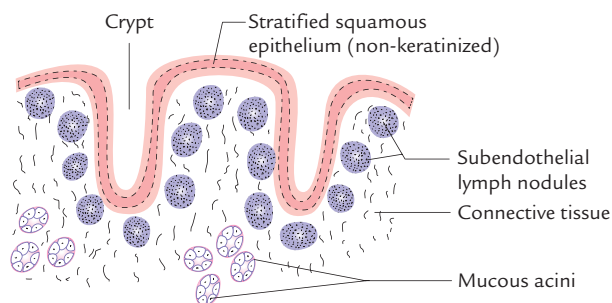
### Histological Structure (Fig. 14.18)

Histologically, the tonsil presents the following features:

1. Its oral surface is lined by stratified squamous non-keratinized epithelium, which dips into underlying tissues to form crypts.
2. Presence of lymphatic nodules on the sides of the crypts.
3. Presence of mucous glands in the deeper plane.

### Development of the Tonsil

The tonsil develops in the region of 2nd pharyngeal pouch. The cells of endodermal lining of pouch proliferate and grow out as solid columns/buds into the surrounding mesenchyme. The central portions of these cell columns are canalized and form tonsillar clefts. The lymphoid cells from the surrounding mesenchyme accumulate around the crypts and differentiate into lymphoid follicles. The remnant of 2nd pharyngeal pouch is seen as supratonsillar/intratonsillar cleft at the upper pole of the tonsil.



**Fig. 14.18** Histological structure of the palatine tonsil. (Source: Fig. 14.6, Page 147, *Selective Anatomy Prep Manual for Undergraduates*, Vol. I, Vishram Singh, Copyright Elsevier 2014, All rights reserved.)

## PHARYNGOTYMPANIC TUBE (SYN. EUSTACHIAN TUBE/AUDITORY TUBE)

It is a mucous-lined osseocartilaginous channel, which connects the nasopharynx with the tympanic cavity (Fig. 14.19). It maintains the equilibrium of air pressure on either side of the tympanic membrane for its proper vibration.

In an adult, it is about 36 mm long and runs downwards, forwards, and medially from its tympanic end.

### Parts (Fig. 14.9)

The pharyngotympanic tube is divided into two parts, *viz.*

1. **Osseous or bony part:** It is posterolateral part and forms one-third (12 mm) of the total length of the tube. It lies between the tympanic and petrous parts of the temporal bone and opens into the middle ear cavity.
2. **Cartilaginous part:** It is anteromedial part and forms two-third (24 mm) of the tube.

The two parts meet at isthmus, which is the narrowest part of the tube. The cartilaginous part is made of a single piece of elastic fibrocartilage, which is folded upon itself in such a way that it forms the whole of the medial wall, roof, and a

part of the lateral wall. The rest of the lateral wall is filled by the fibrous membrane.

The cartilaginous part lies in the groove between the petrous part of the temporal bone and the posterior border of the greater wing of the sphenoid bone.

### Ends of the Tube

The **tympanic end** of the tube is small and bony. It is situated in the anterior wall of the middle ear, a little above its floor.

The **pharyngeal end** is relatively large and slit-like (vertically). It is situated in the lateral wall of the pharynx, about 1.25 cm behind the posterior end of inferior nasal concha. The pharyngeal orifice is the widest part of the tube.

### Lining of the Tube

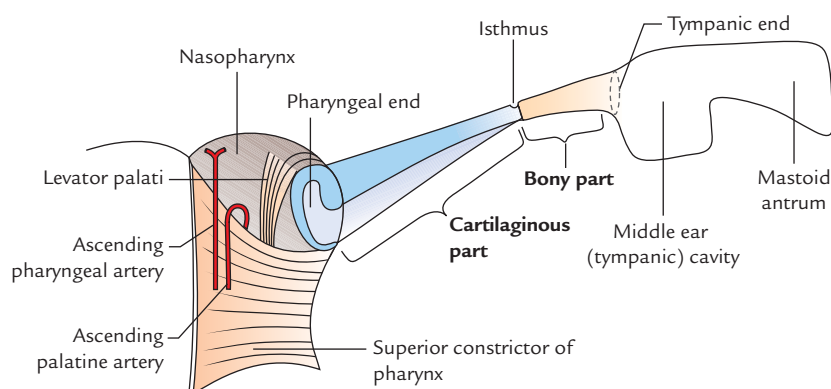
The tube is lined by pseudostratified ciliated columnar epithelium with interspersed goblet cells. The cilia beat in the direction of nasopharynx and thus help to drain the secretions and fluid from the middle ear into the nasopharynx.

### Pharyngotympanic Tubes of an Infant and an Adult

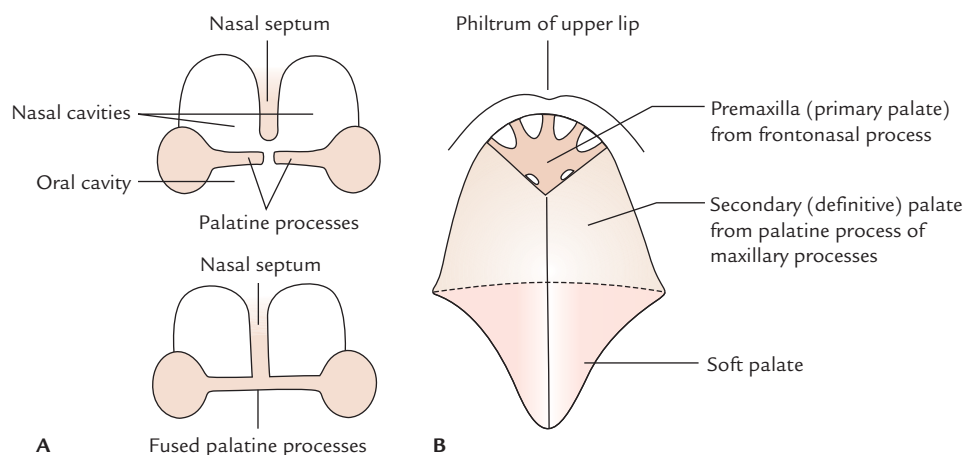
The features of the tube differ in infants and adults. These are enumerated in Table 14.4.

**Table 14.4** Differences between the eustachian tube of an infant and an adult

|                       | Infant                                                                    | Adult                                                                                                |
|-----------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Length                | 18 mm                                                                     | 36 mm                                                                                                |
| Direction             | More or less horizontal (makes an angle of 10° with the horizontal plane) | Oblique, directed downwards, forwards and medially (makes an angle of 45° with the horizontal plane) |
| Angulation of isthmus | No angulation                                                             | Angulation present                                                                                   |



**Fig. 14.19** Bony and cartilaginous parts, isthmus, tympanic, and pharyngeal ends of the pharyngotympanic tube.



**Fig. 14.24** Development of the palate: **A**, separation of the nasal cavities from each other and from oral cavity; **B**, embryological subdivisions of the palate and their source of development.

The primary palate (also called *premaxilla*) develops from the frontonasal process. The secondary palate develops from the palatine process of the maxillary processes.

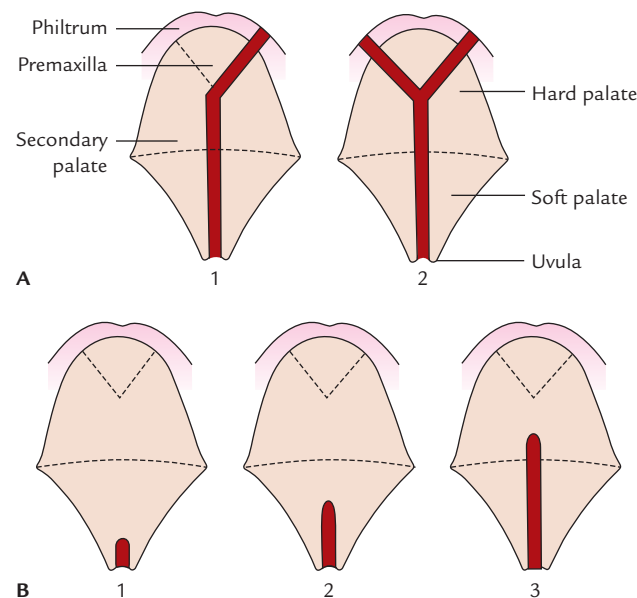
Two palatine processes (one on each side) develop from the inner aspects of the maxillary processes and fuse in the midline to form the **secondary palate**, which is soon joined by nasal septum (Fig. 14.24).

The hard palate is formed by the fusion of the secondary palate with the primary palate. The incisive foramina mark the junction of the two components of the palate. The fusion takes place from anterior to posterior parts.

The soft palate develops from two folds that grow posteriorly from the posterior edge of the palatal processes. Thus uvula is the last structure to develop. The two folds unite to form the soft palate.

(b) **Incomplete or partial cleft:** The following stages may occur

- Bifid uvula—cleft involving only uvula. It is of no clinical importance.
- Cleft of soft palate—involving uvula and soft palate.
- Cleft of soft palate—extending into the hard palate.



**Fig. 14.25** Varieties of the cleft palate: **A**, complete cleft: 1 = unilateral, 2 = bilateral; **B**, partial cleft: 1 = bifid uvula, 2 = cleft soft palate, 3 = cleft of soft palate extending into the hard palate.

### Clinical correlation

**Cleft palate** (Fig. 14.25): The defective fusion of various segments of the palate gives rise to clefts in the palate. These vary considerably in degree, leading to varieties of cleft palate, namely,

(a) **Complete cleft**

- **Unilateral complete cleft** occurs if maxillary process on one side does not fuse with the premaxilla. It is always associated with the cleft lip.
- **Bilateral complete cleft** occurs if both the maxillary processes fail to fuse with the premaxilla. In this type, secondary palate is divided into two equal halves by a median cleft with an anterior V-shaped cleft separating the premaxilla completely.

# Larynx

The larynx is the upper expanded part of the lower respiratory tract, which is modified for producing voice, hence it is also called **voice box/organ of phonation**. It acts as a sphincter at the inlet of lower respiratory tract to protect the trachea and the bronchial tree from entry of any material other than the air. If this protective role is deranged, laryngeal incompetence results, and food and fluid may be aspirated into trachea. Further, the upward and downward movements of the larynx help in swallowing.

The **functions** of larynx include the following:

1. Phonation.
2. Respiration.
3. Protection.
4. Deglutition.

**N.B.** The primary (most important) function of the larynx is protection of the lower respiratory tract. The phonation has developed later with evolution and is related to the motor speech area of the brain.

## Location and Extent

The larynx is situated in the anterior midline of the upper part of the neck in front of laryngopharynx. It extends from the root of the tongue to the trachea and lies in front of the 3rd, 4th, 5th, and 6th cervical vertebrae. However, in children and females it lies at a little higher level.

## Size

The average measurements of larynx in males and females are as follows:

|        | Vertical | Transverse | Anteroposterior |
|--------|----------|------------|-----------------|
| Male   | 44 mm    | 43 mm      | 36 mm           |
| Female | 36 mm    | 41 mm      | 26 mm           |

Till puberty the size of larynx in both males and females is more or less same but at puberty male larynx grows rapidly and becomes larger than the female larynx. The pubertal

growth of the larynx in adult female does not differ much from a child, for this reason the pitch of voice is high in both females and children. In males, the characteristic pubertal growth of angle of the thyroid cartilage (*Adam's apple*) makes the voice louder and low pitched.

## SKELETON (Fig. 15.3 )

The skeletal framework of the larynx consists of a series of cartilages, which are connected to one another by ligaments, and fibrous membranes, and moved by a number of muscles.

**N.B.** The hyoid bone is closely associated to the larynx with distinctive functional roles. However, it is usually considered as a separate structure.

## CARTILAGES

The larynx is composed of nine cartilages, of which three are unpaired and three are paired:

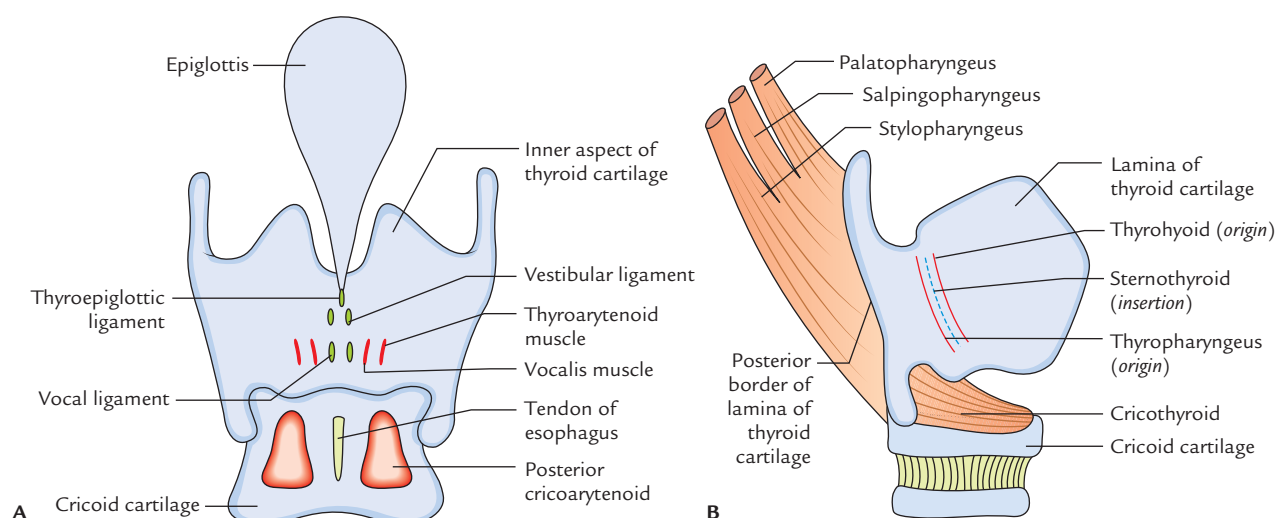
1. **Unpaired cartilages:** The unpaired cartilages are large and comprise:
  - (a) Thyroid
  - (b) Cricoid
  - (c) Epiglottis
2. **Paired cartilages:** The paired cartilages are small and comprise:
  - (a) Arytenoid
  - (b) Corniculate
  - (c) Cuneiform

The **principal cartilages of the larynx** are, cricoid, thyroid, and two arytenoids (Fig. 15.1).

### Thyroid cartilage

It is most prominent and acts as a shield to protect the larynx from the front. It consists of two quadrilateral laminae, which meet in front at an angle called **thyroid angle**, which is acute in males and obtuse in females (like subpubic angle). The angle measures 90° in males and 120° in females.





**Fig. 15.2** Attachments of the muscles and ligaments on the thyroid cartilage: **A**, on the posterior surface; **B**, on the outer surface and posterior border of the lamina.

above downwards and from medial to lateral sides, these are (Fig. 15.2B) as follows:

1. Thyrohyoid.
2. Sternothyroid.
3. Inferior constrictor (thyropharyngeus part only).

### Cricoid cartilage

This is a signet-shaped ring of cartilage with a narrow anterior arch and a broad posterior lamina. The cricoid cartilage is situated at the level of C6 vertebra and **completely encircles the lumen of the larynx**. It is considered as the **foundation stone of the larynx**. The posterior surface of lamina presents a median ridge and two depressed areas on each side of this ridge.

### Epiglottis (Fig. 15.3)

It is leaf-like and extends up behind the hyoid bone and the base of the tongue. Its broad upper end is free and forms the upper boundary of the laryngeal inlet, while the lower end (stalk) is pointed and connected to the posterior surface of the angle of the thyroid by *thyroepiglottic ligament*.

The anterior surface of epiglottis is connected with the base of the tongue by median and lateral glossoepiglottic folds. The depression on each side of the median fold is called **vallecula**.

The posterior surface of epiglottis presents a tubercle in its lower part.

**N.B.** The epiglottis is rudimentary in human beings but in macrosomatic animals it is elongated and extends beyond the soft palate in the nasopharynx.

### Arytenoid cartilages

The paired arytenoid cartilages articulate with the lateral parts of the upper border of cricoid lamina. Each arytenoid cartilage is pyramidal and presents an apex, base, three

surfaces (posterior, anterolateral, and medial), and two processes—muscular and vocal (Fig. 15.1).

The **muscular process** projects laterally and backwards whereas the **vocal process** is directed forwards.

The **base** of arytenoid cartilage is concave and articulates with the upper border of the lamina of cricoid cartilage. The base is prolonged anteriorly to form the *vocal process* and laterally to form the *muscular process*.

The **apex** is curved posteromedially and articulates with the corniculate cartilage.

### Corniculate cartilages (of Santorini)

These are two small conical nodules, which articulate with the apices of the arytenoid cartilages. They are directed posteromedially and lie in the posterior parts of the aryepiglottic folds.

### Cuneiform cartilages (of Wrisberg)

They are tiny rod-shaped cartilages lying in the posterior parts of the aryepiglottic folds just above the corniculate cartilages.

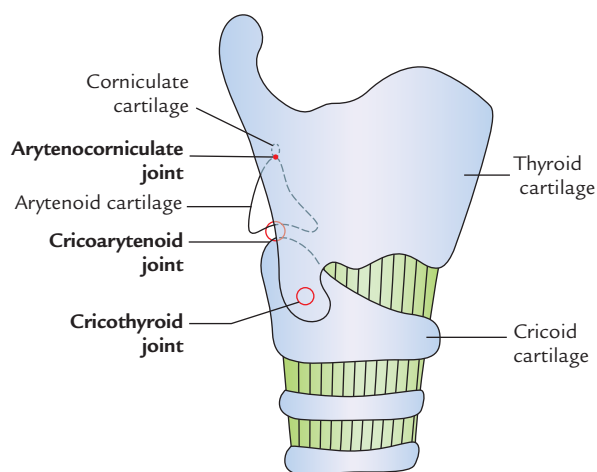
## TYPES OF LARYNGEAL CARTILAGES

The thyroid, cricoid, and basal parts of arytenoid cartilages are composed of **hyaline cartilage** and tend to ossify after 25 years of age and can be seen in radiographs.

The apices of arytenoid cartilages and other cartilages—epiglottis, corniculate, and cuneiform are made up of **elastic cartilage** and do not ossify.

## JOINTS

The laryngeal joints include paired cricothyroid, cricoarytenoid, and arytenocorniculate joints (Fig. 15.4).



**Fig. 15.4** Joints of the larynx.

### ARYTENOCORNICULATE JOINT

It is a synovial joint between the arytenoid and corniculate cartilages. It is of no functional significance.

### LIGAMENTS AND MEMBRANES

The skeletal framework of the larynx is interconnected by a number of ligaments and fibrous membranes. The most significant fibrous membranes connecting skeletal framework of the larynx are thyrohyoid, cricothyroid, quadrangular, and cricovocal membranes.

### EXTRINSIC

The extrinsic ligaments and membranes are outside the inner tube of the fibroelastic tissue of laryngeal cavity (Fig. 15.3):

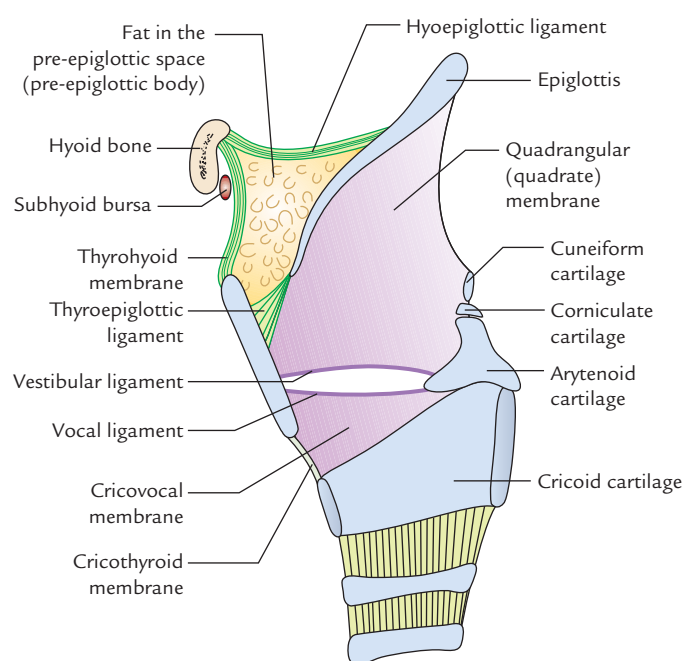
1. **Thyrohyoid membrane and ligaments:** The thyrohyoid membrane extends from the upper border of the thyroid cartilage to the upper border of the hyoid bone. It ascends behind the concave posterior surface of the hyoid bone. Between posterior aspect of hyoid and membrane lies the *subhyoid bursa*. In the median and lateral parts, the thyrohyoid membrane thickens to form *median and lateral thyrohyoid ligaments*. The lateral thyrohyoid ligament on each side contains a small nodule of elastic cartilage called **cartilago-triticea**. The thyrohyoid membrane is pierced on either side by internal laryngeal nerve and superior laryngeal vessels.
2. **Cricotracheal ligament:** It connects the cricoid cartilage with the first tracheal ring.
3. **Thyroepiglottic ligament:** It attaches the lower narrow end of epiglottis to the posterior surface of thyroid angle.
4. **Hyoepiglottic ligament:** It connects the posterior aspect of hyoid with the anterior surface of the upper end of epiglottis.

5. **Cricothyroid ligament:** It connects the lower border of the thyroid cartilage to the cricoid cartilage in the midline.

### INTRINSIC

The intrinsic ligaments and membranes are the parts of a broad sheet of fibroelastic tissue, which forms the inner tube of the laryngeal cavity outside its mucous lining. This fibroelastic tube is, however, interrupted on each side by the sinus of the larynx. The part above the sinus is called **quadrangle** or **quadrangular membrane** and part below the sinus is called **cricovocal membrane** or **conus elasticus** (Fig. 15.5):

1. **Cricovocal membrane** extends upwards and medially from the upper border of the arch of the cricoid cartilage. Its upper edge is free and attached anteriorly to the posterior surface of the thyroid cartilage and posteriorly to the vocal process of the arytenoid cartilage. It is slightly thickened to form the **vocal ligament**. The fold of mucous membrane over this ligament forms the **vocal fold**.
2. **Vocal ligament** is made up of yellow elastic tissue and extends anteroposteriorly from posterior surface of the thyroid cartilage to the vocal process of arytenoids cartilage.
3. **Quadrangular membrane** extends from sides of epiglottis to the arytenoids. Its lower edge is free and attached anteriorly to the posterior surface of the thyroid cartilage (above the cricothyroid membrane) and posteriorly to



**Fig. 15.5** Sagittal section of the larynx showing ligaments and membranes. Note the location of quadrangular and cricovocal membranes.

**Table 15.1** Extrinsic and intrinsic membranes and ligaments of the larynx

|           | Extrinsic                                                                                                                          | Intrinsic                                                                                                         |
|-----------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Membranes | <ul style="list-style-type: none"> <li>• Thyrohyoid</li> <li>• Cricotracheal</li> </ul>                                            | <ul style="list-style-type: none"> <li>• Cricovocal (conus elasticus)</li> <li>• Quadrate/Quadrangular</li> </ul> |
| Ligaments | <ul style="list-style-type: none"> <li>• Median and lateral thyrohyoid</li> <li>• Cricothyroid</li> <li>• Cricotracheal</li> </ul> | <ul style="list-style-type: none"> <li>• Vocal</li> <li>• Vestibular</li> </ul>                                   |

the lateral surface of the arytenoid cartilage (in front of muscular process). Its lower edge is thickened to form the **vestibular ligament**.

4. **Vestibular ligament** is made up of fibrous tissue and extends anteroposteriorly from posterior surface of the thyroid cartilage to the lateral surface of the arytenoid cartilage.

Extrinsic and intrinsic membranes and ligaments are summarized in Table 15.1.

## MUSCLES

They are of two types: extrinsic and intrinsic.

### EXTRINSIC

They attach the larynx to the surrounding structures and are responsible for the movement of the larynx as a whole.

All the extrinsic muscles are paired and include:

1. Palatopharyngeus.
2. Salpingopharyngeus.
3. Stylopharyngeus.
4. Thyrohyoid.
5. Sternothyroid.

All these muscles elevate the larynx except sternothyroid, which depresses the larynx.

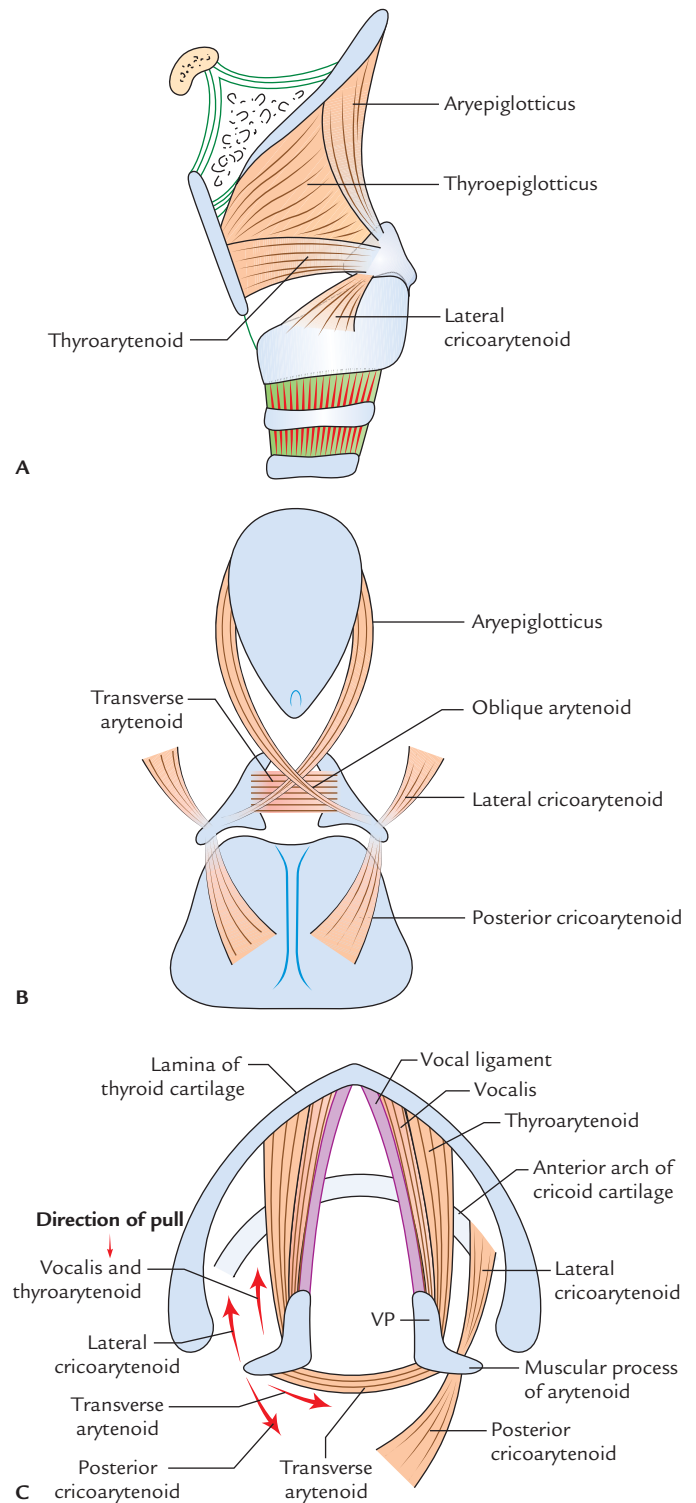
The first three muscles are discussed in detail in Chapter 14 and the last two in Chapter 6.

### INTRINSIC (Fig. 15.6)

They attach the laryngeal cartilages to each other and are responsible for their movements. Their main functions are to:

- (a) open or close the laryngeal inlet,
- (b) adduct and abduct the vocal cords, and
- (c) increase or decrease the tension of the vocal cords.

Thus according to their actions, intrinsic muscles of the larynx are arranged into the following groups:



**Fig. 15.6** Intrinsic muscles of the larynx: **A**, lateral view; **B**, posterior view; **C**, direction of pull of some intrinsic muscles (VP = vocal process of arytenoid).

### Muscles that Open or Close the Laryngeal Inlet

1. Oblique arytenoids
  2. Aryepiglotticus
  3. Thyroepiglotticus: opens the inlet of larynx.
- Closes the inlet of larynx.

## CAVITY

It extends from inlet of larynx, where it communicates with the lumen of laryngopharynx to the lower border of the cricoid cartilage, where it is continuous with the lumen of the trachea. The anterior wall of laryngeal cavity is longer than the posterior wall.

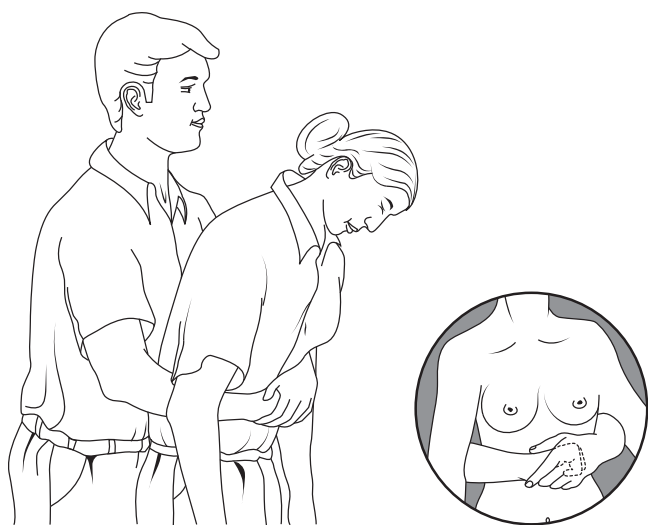
**Laryngeal inlet** is obliquely placed, sloping downwards and backwards. It opens into the laryngopharynx.

### Boundaries

|                                |                                             |
|--------------------------------|---------------------------------------------|
| <i>Anterior:</i>               | Epiglottis.                                 |
| <i>Posterior:</i>              | Interarytenoid fold of the mucous membrane. |
| <i>Lateral (on each side):</i> | Aryepiglottic fold of the mucous membrane.  |

Within the laryngeal cavity, **two pairs of folds** of the mucous membrane extend (on each side) posteroanteriorly from arytenoid cartilage to the thyroid cartilage:

1. The upper folds are produced by vestibular ligament and called **vestibular folds** or **false vocal cords**. The space between vestibular folds is called **rima vestibuli**. When the vestibular folds come together, they prevent food and liquids from entering the larynx and air from leaving the lungs, as when a person holds his breath.
2. The lower folds are produced by the vocal ligaments and vocalis muscle, and called **vocal folds** or **true vocal cords**. They extend from the middle of the thyroid angle to the vocal processes of arytenoids. The space between the right and left vocal folds is called **rima glottidis**. The rima glottidis is the narrowest part of the laryngeal cavity.



**Fig. 15.9** Heimlich maneuver. Figure in the inset shows the position of hands in the epigastric region of the victim.

## Clinical correlation

**Laryngeal obstruction:** The mucous membrane of the superior part of larynx is very sensitive. When foreign body (a piece of food or a drop of water etc.) enters into the laryngeal inlet it causes immediate explosive coughing and the foreign body is expelled out. If this reflex is slowed or absent as in neurological lesion or after consuming alcohol, a foreign body (e.g., piece of meat) may enter the laryngeal cavity and cause choking (i.e., laryngeal obstruction). Choking by food is a common cause of laryngeal obstruction and asphyxia. If foreign body is not dislodged and expelled out immediately by Heimlich maneuver, person will die within minutes, almost certainly before there is time to take him to the hospital. The **Heimlich maneuver** is performed as follows (Fig. 15.9):

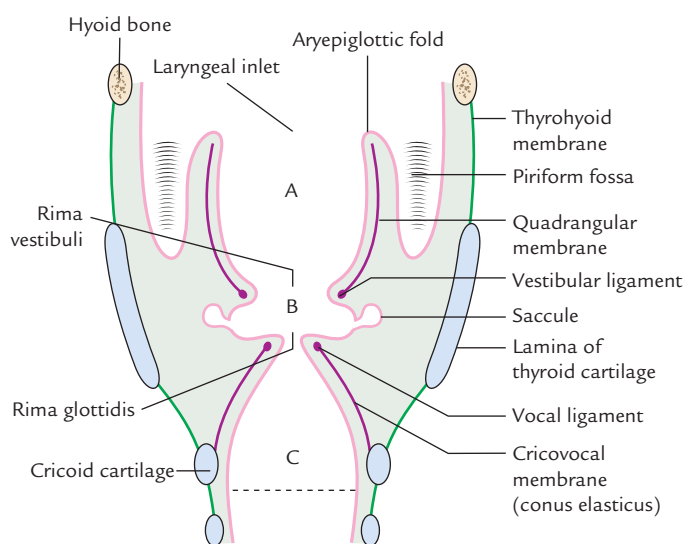
Stand behind the victim, pass your arms under his arms, place hands in front of the victim's epigastrium with one hand formed into a fist and the other hand lying over it. Now give 3 or 4 abdominal thrusts directed upwards and backwards. By doing this, the residual air in the lungs is squeezed up in trachea and larynx with force, dislodging foreign body and thus relieving laryngeal obstruction (choking). The foreign body is either expelled itself or removed.

## SUBDIVISIONS OF THE LARYNGEAL CAVITY

The laryngeal cavity is divided into three parts by two pairs of vestibular and vocal folds (Fig. 15.10), viz.

1. **Vestibule (supraglottic compartment):** It extends from laryngeal inlet to the vestibular folds.

Its **anterior wall** is formed by the mucous membrane covering the posterior surface of epiglottis, **posterior wall** by the mucous membrane covering the apices of arytenoids cartilages, and **sides** by aryepiglottic folds.



**Fig. 15.10** Coronal section of the laryngeal cavity showing its subdivisions. A=vestibule, B=ventricle of the larynx, C=infraglottic compartment.



- (b) below the vocal cords pierce the cricothyroid membrane and go to the prelaryngeal and pretracheal nodes, and then drain into lower deep cervical lymph nodes (posteroinferior group).

## RIMA GLOTTIDIS AND PHONATION

It is the narrowest anteroposterior cleft of the laryngeal cavity. The anteroposterior diameter of glottis is 24 mm in adult males and 16 mm in adult females.

### Boundaries

*In front:* Angle of thyroid cartilage.

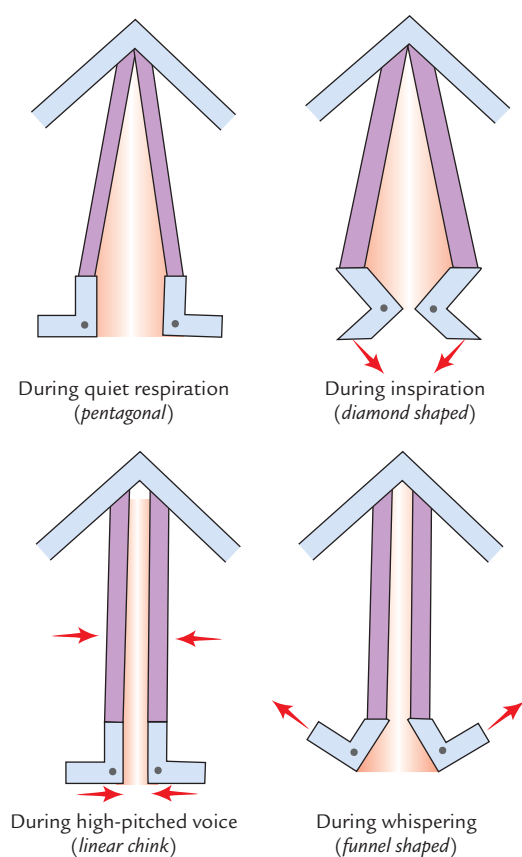
*Behind:* Interarytenoid folds of the mucous membrane.

*On each side:* Vocal fold in anterior three-fifth and vocal process of arytenoid cartilage in the posterior two-fifth.

### Subdivisions of Rima Glottidis

The rima glottidis is divided into the following two parts:

1. **Intermembranous part** in the anterior three-fifth, between the vocal cords.



**Fig. 15.12** Variations in the size and shape of rima glottidis during different movements of the vocal cords.

2. **Intercartilaginous part** in the posterior one-fifth, between the vocal processes of arytenoid cartilage.

### Shape of Rima Glottidis (Fig. 15.12)

The size and shape of glottis varies with the movements of the vocal cords:

- In quiet breathing, the intermembranous part is triangular and intercartilaginous part is rectangular. As a whole the glottis is *pentagonal*.
- In full inspiration, the glottis widens and becomes *diamond shaped* due to abduction of vocal cords.
- During high-pitched voice, the rima glottidis is reduced to a *linear chink*, due to adduction of both intermembranous and intercartilaginous parts.
- During whispering, the intermembranous part is highly adducted and intercartilaginous part is separated by triangular gap, thus rendering an *inverted funnel* shape to the rima glottidis.

### Clinical correlation

**Laryngoscopy:** The interior of the larynx can be inspected directly by laryngoscope (*direct laryngoscopy*), or indirectly through a laryngeal mirror (*indirect laryngoscopy*). The following structures are viewed (Fig. 15.13):

- Base of tongue.
- Valleculae.
- Epiglottis.
- Aryepiglottic folds.
- Piriform fossae.
- False vocal cords (red and widely apart).
- True vocal cords (pearly white), medial to false vocal cords.
- Sinus of the larynx between false and true vocal cords.

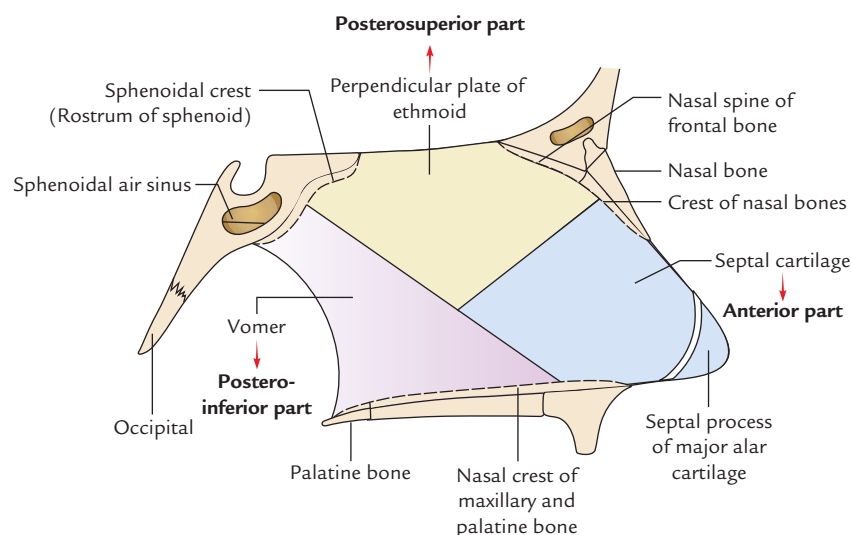
### MECHANISM OF PHONATION

The phonation (speech) is produced by the vibrations of the vocal cords. The greater the amplitude of vibration, the louder is the sound.

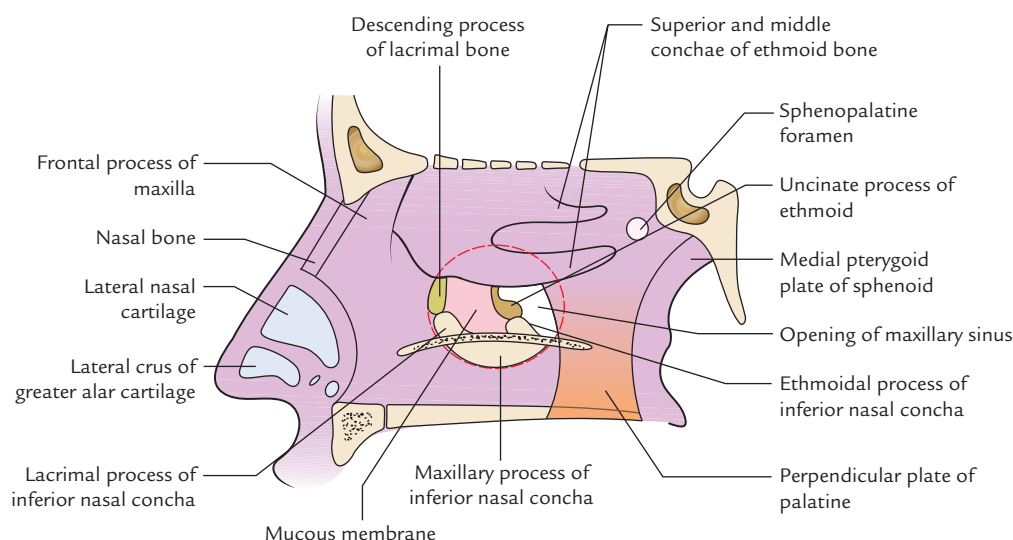
The larynx is like a wind instrument. The voice is produced in following manner:

1. Vocal cords are kept adducted.
2. Infraglottic air pressure is generated by the exhaled air from lungs by the contraction of abdominal, intercostal, and other expiratory muscles.
3. Force of air opens the cords and is released as small puffs.
4. As the moving air passes through the vocal cords it makes them to vibrate producing sound.
5. Sound is amplified by mouth, pharynx, esophagus, and nose.
6. Sound is converted into speech by the modulatory actions of lips, tongue, palate, pharynx, and teeth.





**Fig. 17.4** Formation of the nasal septum.



**Fig. 17.5** Formation of the lateral wall of the nasal cavity. Red circle indicates the position of maxillary air sinus.

### Clinical correlation

- **Deviated nasal septum (DNS):** The deviated nasal septum is not uncommon and is an important cause of nasal obstruction. The males are affected more than females. It occurs due to numerous factors such as trauma, developmental error, etc. If DNS is severe and causing mechanical obstruction, leading to difficulty in breathing, sinusitis, headache, excessive snoring, etc., it is corrected by submucous resection (SMR) or septoplasty. (For details consult ENT textbooks.)
- The septal cartilage not only forms partition between the right and left nasal cavities but also provides support to

the dorsum of the anterior two-third of the nose. Therefore, its destruction due to disease or excessive removal in submucous resection leads to supratip depression of the external nose.

**Lateral wall:** The lateral wall of the nose (Fig. 17.5) is complicated. It is formed by a number of bones and cartilages (Fig. 17.5).

The **bones** forming the lateral wall are:

- nasal,
- frontal process of maxilla,
- lacrimal,

- (d) conchae and labyrinth of ethmoid,
- (e) inferior nasal concha,
- (f) perpendicular plate of palatine, and
- (g) medial pterygoid plate of sphenoid.

The **cartilages** forming the lateral wall are:

- (a) lateral nasal cartilage (upper nasal cartilage),
- (b) major alar cartilage (lower nasal cartilage), and
- (c) three to four tiny cartilages of the alae (minor alar cartilages).

**Features (Fig. 17.6):** The lateral wall is divided into the following three areas:

1. **Anterior part** presents a small depressed area, the **vestibule**. It is lined by the skin containing **vibrissae** (short, stiff curved hair).
2. **Middle part** is known as **atrium of the middle meatus**. It is limited above by a faint ridge of mucous membrane, the **agger nasi**. The curved mucocutaneous junction between the atrium and vestibule is known as **limen nasi**.
3. **Posterior part** presents three scroll-like projections, the **conchae or turbinates**. The spaces separating the conchae are called **meatuses**.

The **conchae and meatuses** form the main features of the lateral wall:

**Conchae** (also called **turbinates**) are the curved bony projections directed downwards and medially. Below and lateral to each concha is a corresponding **meatus**. From above downwards the conchae are superior, middle, and inferior nasal conchae. Sometimes a 4th concha, the concha suprema is also present.

**Conchae:**

1. Superior and middle nasal conchae are the projections from the medial surface of the ethmoidal labyrinth.

2. Inferior concha is an independent bone.
3. The superior concha is smallest and inferior concha is largest in size.

**Meatuses:**

Meatuses are the passages (recesses) beneath the overhanging conchae. They are visualized once conchae are removed.

1. **Inferior meatus** is the largest and lies underneath the inferior nasal concha.
2. **Middle meatus** lies underneath the middle concha. It presents following features:
  - (a) *Ethmoidal bulla* (bullae ethmoidalis), a round elevation produced by the underlying middle ethmoidal sinuses.
  - (b) *Hiatus semilunaris*, a deep semicircular sulcus below the bulla ethmoidalis.
  - (c) *Infundibulum*, a short passage at the anterior end of middle meatus.
3. **Superior meatus** is the smallest and lies below the superior concha.

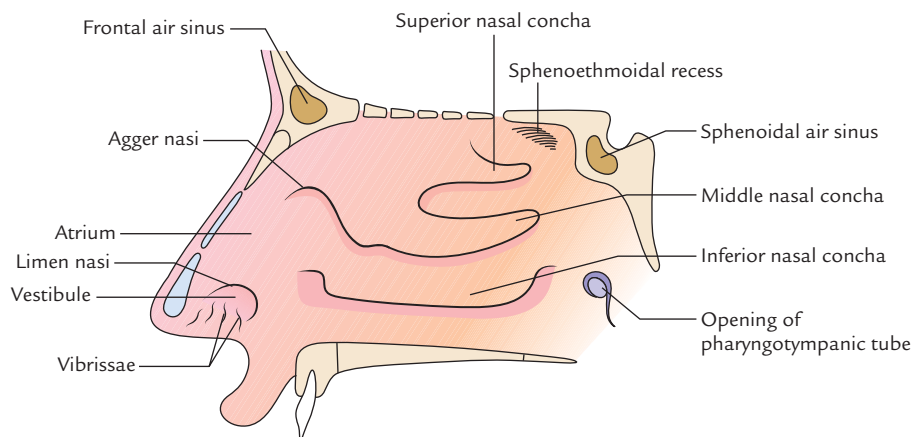
A triangular depression, above and behind the superior concha is known as the **sphenoethmoidal recess**.

**Openings (Fig. 17.7):** The lateral wall of the nose has number of openings. These are listed in Table 17.1.

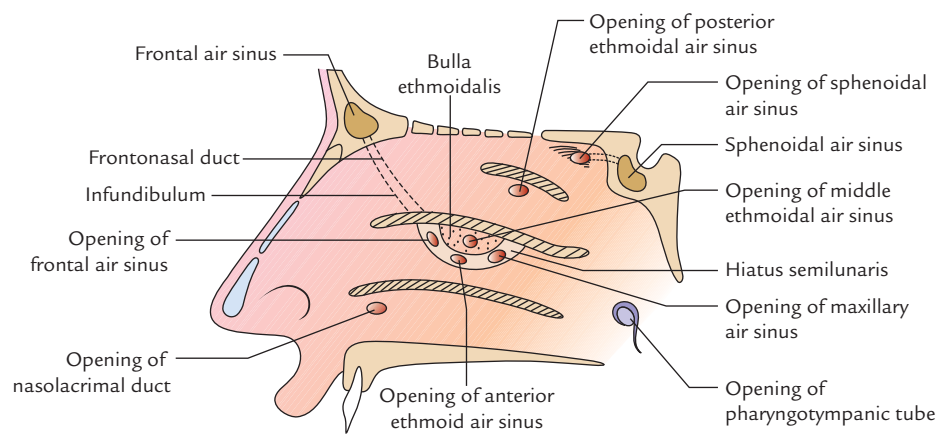
### Lining of Nasal Cavity

The lining of the various regions of nasal cavity (Fig. 17.8) are as follows:

1. **Vestibule:** It is lined by the skin containing a large number of sebaceous glands and interlacing coarse hair—the **vibrissae**. As the air passes through the nostrils the large particles of dust in the air are trapped by the vibrissae.



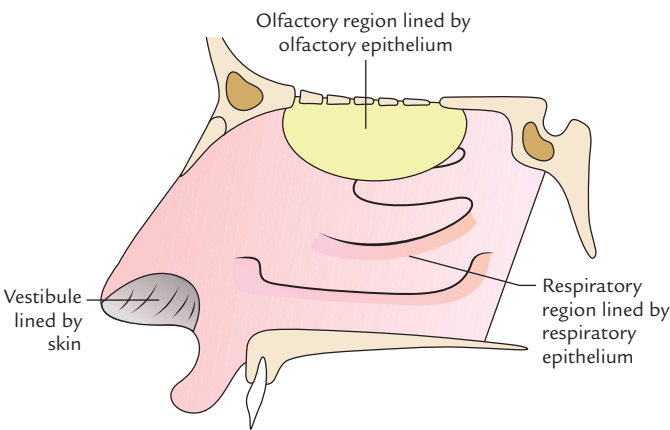
**Fig. 17.6** Features of the lateral wall of the nasal cavity.



**Fig. 17.7** Lateral wall of the nose with conchae removed showing openings of various sinuses and nasolacrimal duct.

**Table 17.1** The openings in the lateral wall of the nose

| Sites                   | Openings                                                          |
|-------------------------|-------------------------------------------------------------------|
| Sphenoethmoidal recess  | Opening of the sphenoidal air sinus                               |
| Superior meatus         | Opening of the posterior ethmoidal air sinuses                    |
| Middle meatus           | Opening of the middle ethmoidal air sinuses                       |
| • On bulla              |                                                                   |
| • In hiatus semilunaris |                                                                   |
| – Anterior part         | Opening of the frontal air sinus                                  |
| – Middle part           | Opening of the anterior ethmoidal air sinuses                     |
| – Posterior part        | Opening of the maxillary air sinus                                |
| Inferior meatus         | Opening of the nasolacrimal duct (in the anterior part of meatus) |



**Fig. 17.8** Lining of the nasal cavity.

2. **Olfactory region:** The upper third of the nasal cavity bounded above by cribriform plate of ethmoid, laterally by superior nasal concha and medially by upper one-third of the nasal septum forms the olfactory region. It is lined by the olfactory epithelium, which contains receptor cells for smell. Here mucous membrane is paler in color.

3. **Respiratory region:** The rest of the nasal cavity (lower two-third) is lined by the respiratory epithelium, i.e., pseudostratified ciliated columnar epithelium with goblet cells. The respiratory mucosa is highly vascular and contains a large number of cavernous spaces and sinusoids to warm the air. In addition, it contains a large number of serous and mucous glands. The secretion of the serous glands makes the air moist while the secretion of the mucous glands traps the dust and other particles. Further, the cilia on the surface of the mucous membrane sweep the mucus posteriorly into the pharynx where it is swallowed and eliminated by the GIT.

**N.B.** Due to the presence of cilia in the epithelial lining and secretions of the serous and mucous glands in the nasal cavity, the air while passing through the nasal cavity is not only filtered but also moistened and humidified, a mechanism that protects the lower respiratory tract.

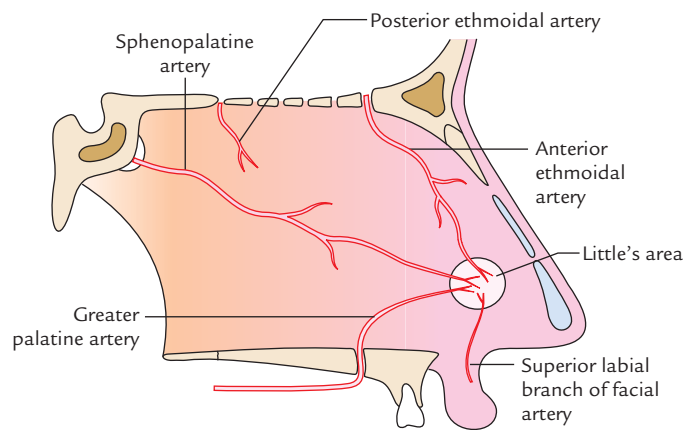
### Arterial Supply of Nasal Cavity

The nasal cavity has rich arterial supply.

### Arterial Supply of Nasal Septum

The nasal septum is supplied by the following arteries (Fig. 17.9):

1. Septal branch of the anterior ethmoidal artery (a branch of ophthalmic artery).
2. Septal branch of the posterior ethmoidal artery (a branch of ophthalmic artery).
3. Septal branch of the sphenopalatine artery (a branch of maxillary artery).
4. Septal branch of the greater palatine artery (a branch of maxillary artery).
5. Septal branch of the superior labial artery (a branch of facial artery).



**Fig. 17.9** Arterial supply of the nasal septum.

### Clinical correlation

**Little's area:** It is an area in the anteroinferior part of the nasal septum just above the vestibule. It is highly vascular. Here the septal branches of the anterior ethmoidal, sphenopalatine, greater palatine, and superior labial arteries anastomose to form a vascular plexus called *Kiesselbach's plexus*. This area of nasal septum is the commonest site of **epistaxis (nose bleeding)** in children and young adults usually due to finger nail trauma following picking of the nose.

3. **Posterosuperior quadrant**, by sphenopalatine artery, a branch of maxillary artery.
4. **Posteroinferior quadrant**, by branches of greater palatine artery, which pierces the perpendicular plate of palatine.

### Venous Drainage of Nasal Cavity

The veins draining the nasal cavity form plexus beneath the mucosa and in general accompany the arteries. The veins of nasal cavity drain into facial vein, pterygoid venous plexus, and pharyngeal venous plexus.

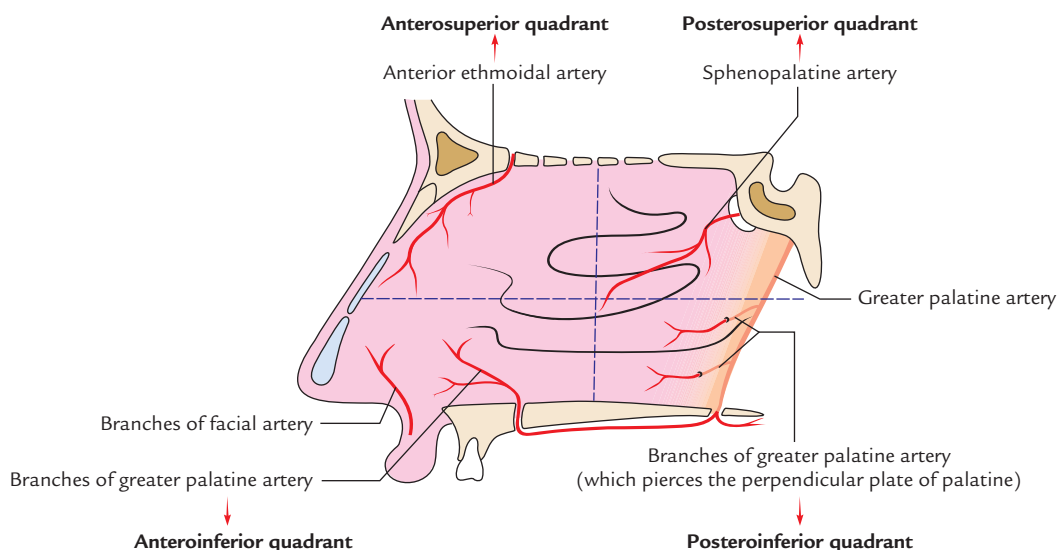
#### N.B.

- The submucous venous plexus is more marked in the region of little's area.
- The *retrocolumellar* vein runs vertically downwards, crosses the floor of nasal cavity to join the venous plexus on the lateral wall. This is the common site of venous bleeding in young individuals.

### Arterial Supply of Lateral Wall

The arterial supply (Fig. 17.10) of the various parts of the lateral wall is as follows:

1. **Anterosuperior quadrant**, by the anterior ethmoidal artery, a branch of ophthalmic artery.
2. **Anteroinferior quadrant**, by branches of facial and greater palatine arteries.



**Fig. 17.10** Arterial supply of the lateral wall.

## Lymphatic Drainage of Nasal Cavity

Lymph from anterior half of nasal cavity (both medial and lateral walls) is drained into *submandibular lymph nodes* and from posterior half into *retropharyngeal lymph nodes*.

## Nerve Supply of Nasal Cavity

The nasal cavity is supplied by the following nerves:

1. **Olfactory nerves:** They carry sense of smell from olfactory region of nasal cavity (for details see Chapter 22).
2. **Nerves of general sensation:** These are:
  - (a) anterior ethmoidal nerve,
  - (b) branches of sphenopalatine ganglion, and
  - (c) branches of infraorbital nerve.

They carry sensations of pain, touch, and temperature from respiratory region and vestibule of nasal cavity.

3. **Autonomic nerves:** Both parasympathetic and sympathetic fibres supplying nasal cavity are derived from Vidian's nerve (nerve of pterygoid canal) through pterygopalatine ganglion.

The parasympathetic fibres supply nasal glands and control nasal secretion.

The sympathetic fibres, on stimulation, cause vasoconstriction.

### Clinical correlation

**Rhinitis:** It is the inflammation of mucus membrane lining the nasal cavity. The hypertrophy of mucosa over inferior concha is a common feature of *allergic rhinitis*. Clinically it presents as: nasal blockage, sneezing, and water discharge from nose (*rhinorrhea*).

The *excessive rhinorrhea* due to *vasomotor and allergic rhinitis* can be controlled by sectioning the Vidian nerve.

## Nerve Supply of Nasal Septum (Fig. 17.11)

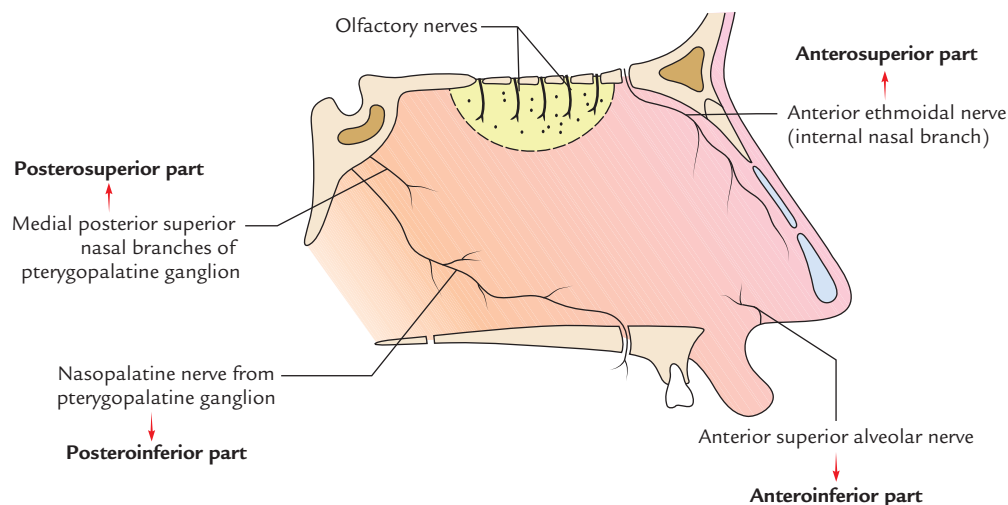
The nasal septum receives supply from the following nerves:

1. **Olfactory nerves**—supply the upper part (one-third) just below the cribriform plate.
2. **Internal nasal branch** of the anterior ethmoidal nerve, a branch from nasociliary—supplies the anterosuperior part.
3. **Nasopalatine nerve**, a branch of pterygopalatine ganglion—supplies the posteroinferior part.
4. **Medial posterior superior nasal branches** of pterygopalatine ganglion—supply the posterosuperior part.
5. **Nasal branch of greater palatine nerve**—supplies the posterior part.
6. **Anterior superior alveolar nerve**, a branch of maxillary nerve—supplies the anteroinferior part.

## Nerve Supply of Lateral Wall

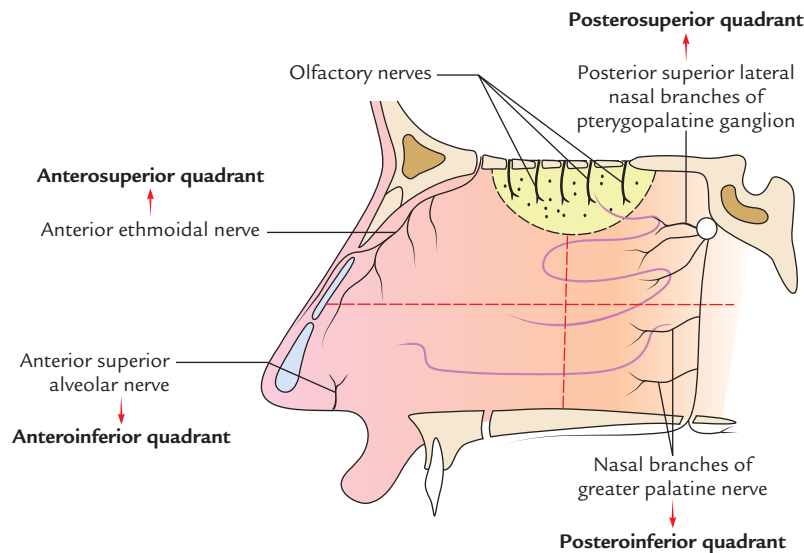
The following nerves supply (Fig. 17.12) the different parts of the lateral wall:

1. **Olfactory nerves**—supply the upper part (one-third) just below the cribriform plate of ethmoid up to the superior concha.
2. **Anterior ethmoidal nerve** (from ophthalmic)—supplies the anterosuperior quadrant.
3. **Anterior superior alveolar nerve**, a branch of infraorbital nerve (from maxillary)—supplies the anteroinferior quadrant.
4. **Posterior superior lateral branches**, of pterygopalatine ganglion—supply posterosuperior quadrant.
5. **Nasal branches of greater palatine nerve**, (from pterygopalatine ganglion)—supply posteroinferior quadrant.

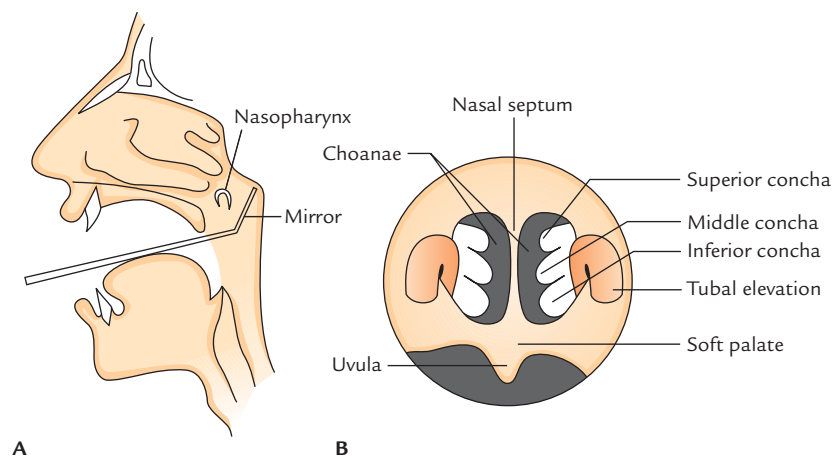


**Fig. 17.11** Nerve supply of the nasal septum.





**Fig. 17.12** Nerve supply of the lateral wall of the nose.



**Fig. 17.13** Posterior rhinoscopy: **A**, showing placement of mirror in the nasopharynx; **B**, structures seen in the mirror.

### Clinical correlation

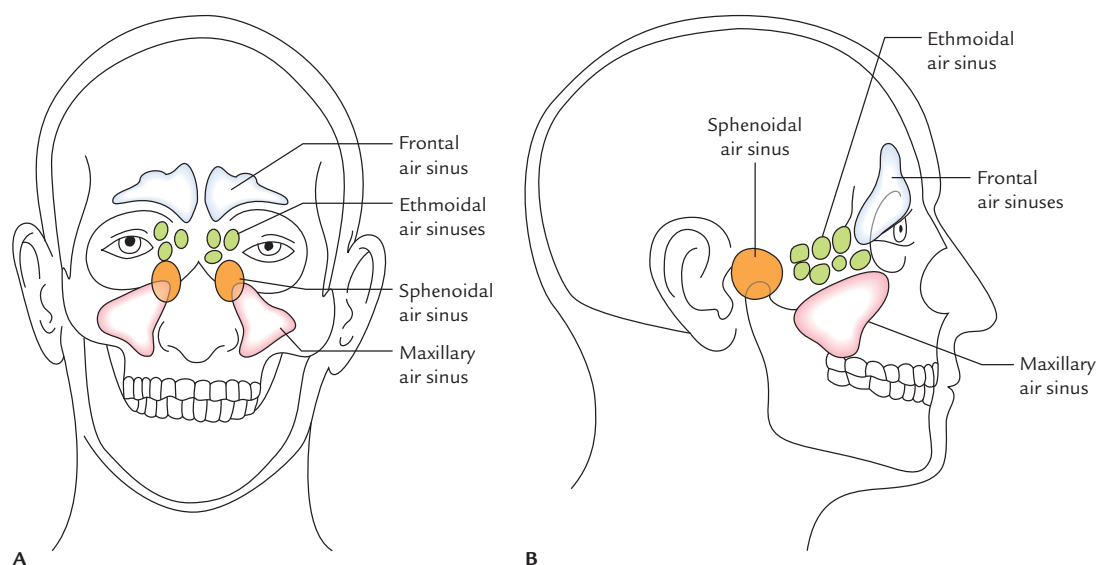
**Examination of the nasal cavity (Rhinoscopy):** The nasal cavity can be examined in the living individual either through the nostril (**anterior rhinoscopy**) or through the pharynx (**posterior rhinoscopy**).

- Anterior rhinoscopy:** It is carried out by inserting a nasal speculum through a nostril. The following features are visualized by this method:
  - Middle and inferior conchae.
  - Superior middle and inferior meatuses.
  - Nasal septum.
  - Floor of the nasal cavity.
- Posterior rhinoscopy:** It is carried out by inserting a mirror into the pharynx (Fig. 17.13A). The following features can be visualized by this method (Fig. 17.13B):

- Choanae.
- Conchae.
- Posterior border of the nasal septum.

## PARANASAL AIR SINUSES

The paranasal air sinuses are air-containing cavities in the bones around the nasal cavity. The paranasal air sinuses develop as mucosal diverticula of the main nasal cavity invading the adjacent bones. They are lined by a **pseudostratified ciliated columnar epithelium** as in the nasal cavity (Fig. 17.14).



**Fig. 17.14** Positions of paranasal air sinuses on the surface: A, front view; B, side view.

There are four paranasal air sinuses on each side and are named after the bones containing them, *viz.*

1. **Frontal air sinuses** present in the frontal bone.
2. **Ethmoidal air sinuses** present in the ethmoid bone.
3. **Maxillary air sinuses** present in the maxilla.
4. **Sphenoidal air sinuses** present in the sphenoid bone.

The paranasal air sinuses are arranged in pairs *except* the ethmoidal air sinuses, which are arranged in three groups, *viz.* anterior, middle, and posterior on each side.

The orbit of eye serves as a landmark to appreciate the location of various paranasal air sinuses.

The relationship of the paranasal sinuses to the orbit (Fig. 17.15) is as follows:

- the frontal air sinus *above*,
- the maxillary air sinus *below*,
- the ethmoidal air sinus *medial* to, and

- the sphenoidal air sinus *behind the orbit*.

The sinuses exhibit two spurts of growth, *viz.*

1. First around 7–8 years (during the eruption of the teeth).
2. Second at puberty.

**N.B.** All the paranasal air sinuses are present in rudimentary form at birth except the frontal air sinuses, which start developing 2 or 3 years after birth.

## FUNCTIONS

Paranasal air sinuses perform the following functions:

1. Make the skull lighter.
2. Add resonance to the voice.
3. Act as air conditioning chambers by adding humidity and temperature to the inspired air.
4. Aid in the growth of facial skeleton after birth.

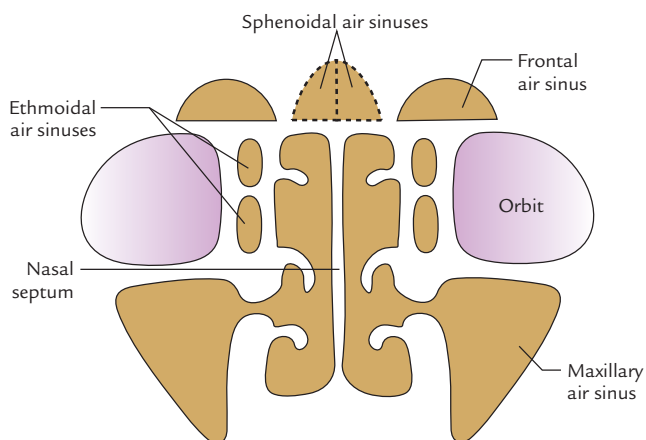
## Clinical correlation

**Sinusitis:** The infection of a paranasal air sinus is called sinusitis. Clinically it presents as: headache and persistent thick purulent discharge from the nose.

In standard radiological images, the normal paranasal air sinuses are radiolucent, whereas diseased paranasal sinuses show varying degree of opacity. The newer imaging techniques, *viz.* CT scan provide very clear images of the sinuses, which significantly aid in diagnosis (Fig. 17.16).

## CLASSIFICATION

Clinically the sinuses are divided into the following two main groups:



**Fig. 17.15** Schematic diagram to show the location of various paranasal sinuses in relation to the orbit.

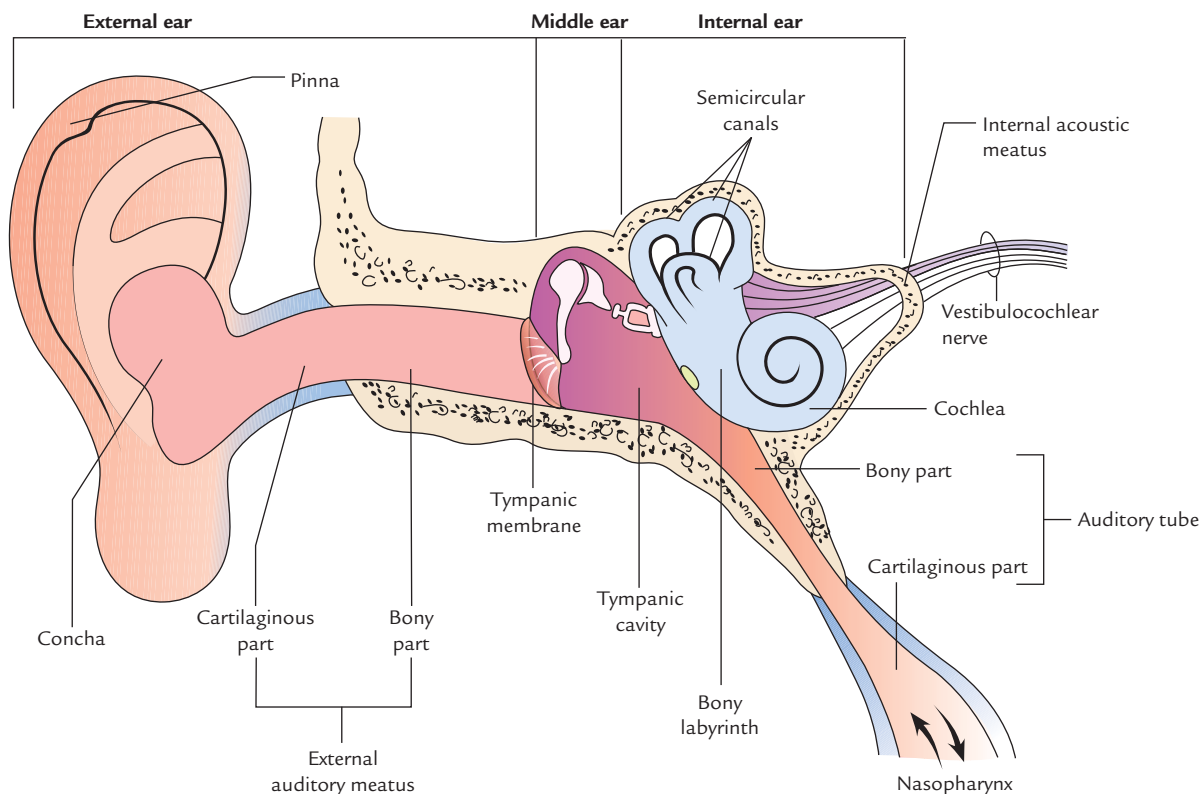
The ear is the organ of hearing and plays an important role in maintaining the balance (equilibrium) of the body. The ear is divided into three parts (Fig. 18.1), *viz.*

1. External ear.
2. Middle ear.
3. Internal ear.

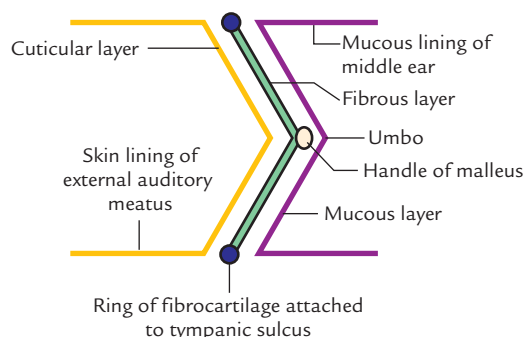
The external ear consists of the auricle or pinna and external auditory meatus at the medial end of which lies the tympanic membrane or ear drum, separating the external ear from the middle ear. The middle ear or tympanic cavity is a small

air-filled space within the petrous part of the temporal bone containing the auditory ossicles. It communicates with the nasopharynx through auditory tube. By its medial wall, the middle ear adjoins the internal ear. The internal ear consists of a bony labyrinth, a complicated space, also in the petrous part of the temporal bone containing fluid-filled membranous labyrinth. The membranous labyrinth contains sensory receptors for hearing and balancing.

The sensations of hearing and balancing from these receptors are carried by vestibulocochlear nerve to the cerebral cortex for interpretation.



**Fig. 18.1** The ear and its subdivisions (external, middle, and internal ear).



**Fig. 18.4** Layers of the tympanic membrane.

### Structure

The tympanic membrane is made of three layers (Fig. 18.4); from lateral to medial these are as follows:

1. **Outer cuticular layer** of stratified squamous epithelium, which is continuous with the skin lining the external auditory meatus.
2. **Middle fibrous layer**, which encloses the handle of the malleus. It contains outer radiating and inner circular fibres.
3. **Inner mucosal layer** is lined by low columnar epithelium, which is continuous with the mucous lining of the middle ear.

### Parts

The tympanic membrane is divided into two parts: pars tensa and pars flaccida.

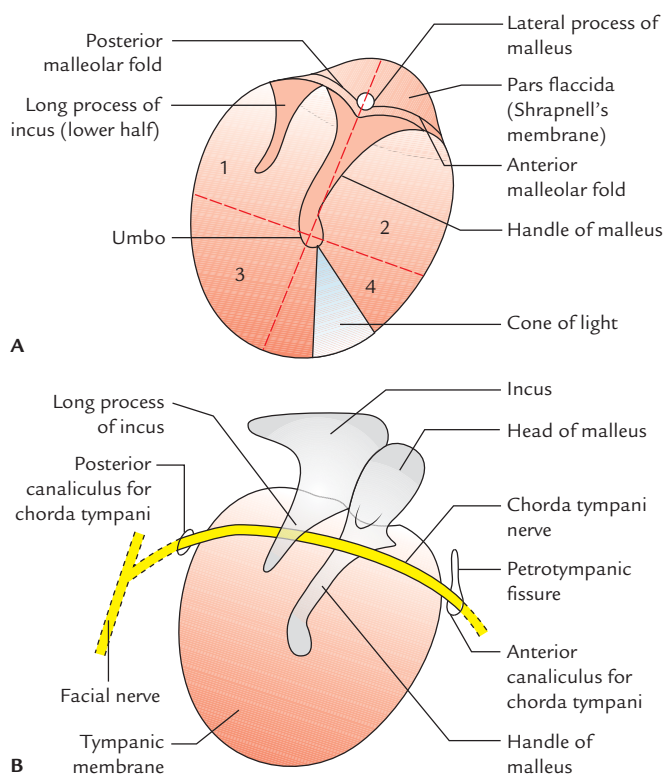
**Pars tensa** forms most of the tympanic membrane. Its periphery is thickened to form a fibrocartilaginous rim called **annulus tympanicus**, which fits into the tympanic sulcus. The fibrocartilaginous rim presents a notch above. From the margins of the notch the anterior and posterior malleolar folds in mucous membrane of tympanic cavity pass to gain attachment to the lateral process of the malleus. The handle of the malleus is firmly attached to the inner surface of the pars tensa. This part is rendered tough by the inward pull of the tensor tympani muscle, attached to the root of handle of the malleus and radial fibres.

**Pars flaccida (Shrapnell's membrane)** is a small triangular area above the lateral process of malleus between *anterior and posterior malleolar folds* (now called *malleal folds*). This part is thin and lax because intermediate fibrous layer here is replaced by loose areolar tissue. It appears slightly pinkish.

### Surfaces

**Lateral surface** of the tympanic membrane is concave towards the meatus and directed downwards, forwards, and laterally.

**Medial surface** is convex and bulges towards the middle ear. The point of maximum convexity is called **umbo** (Fig. 18.5). When the tympanic membrane is illuminated for



**Fig. 18.5** Tympanic membrane (schematic diagram): **A**, external surface of tympanic membrane as seen through otoscope: 1 = posterosuperior quadrant, 2 = anterosuperior quadrant, 3 = posteroinferior quadrant, 4 = anteroinferior quadrant; **B**, medial surface of tympanic membrane showing the course of the chorda tympani nerve.

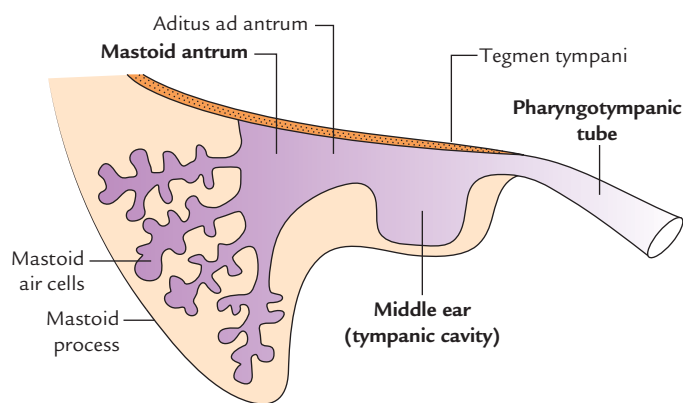
inspection, the concavity of the membrane produces a '*cone of light*' radiating from the umbo over the anteroinferior quadrant. This surface receives the attachment of malleus up to the center of the membrane. Here the handle of the malleus is crossed medially by chorda tympani nerve, which runs forwards between the fibrous and mucosal layer at the junction of pars flaccida and pars tensa.

### Arterial Supply

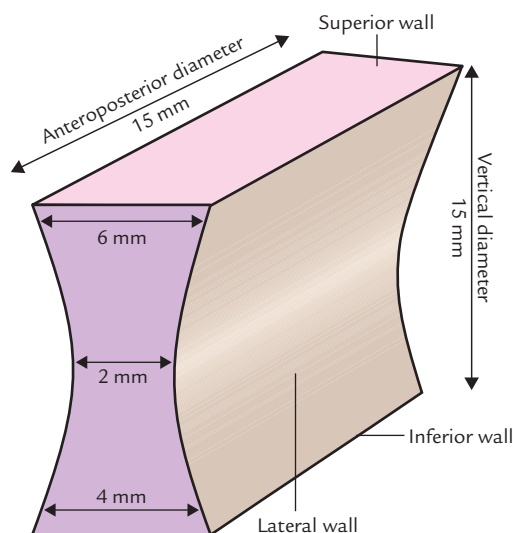
1. The outer surface is supplied by deep auricular artery, a branch from first part of maxillary artery.
2. The inner surface is supplied by (a) anterior tympanic artery, a branch from first part of maxillary artery, and (b) posterior tympanic artery, a branch from stylomastoid artery arising from posterior auricular artery.

### Venous Drainage

1. Veins from outer surface drain into external jugular vein.
2. Veins from inner surface drain into transverse sinus and pterygoid venous plexus.



**Fig. 18.7** Section through long axis of the petromastoid bone showing pharyngotympanic tube, middle ear (tympanic cavity), and mastoid antrum.



**Fig. 18.8** The shape and dimensions of the middle ear cavity.

- *Transverse diameters*
  - (a) At roof: 6 mm.
  - (b) In the center: 2 mm.
  - (c) At the floor: 4 mm.

### Communication

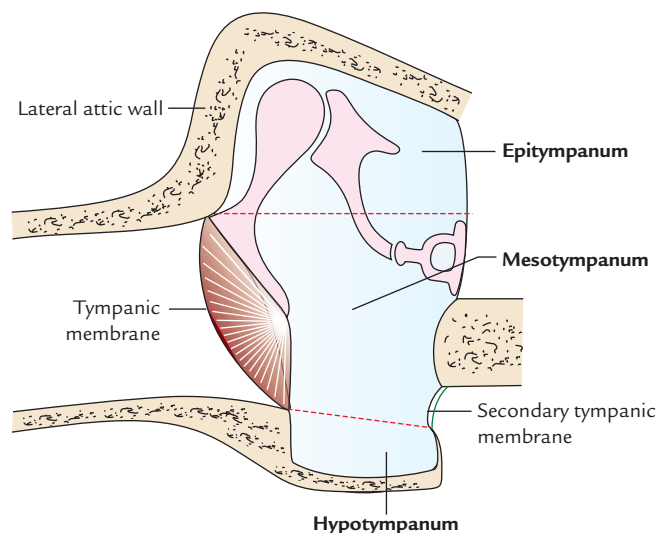
The middle ear communicates:

- Anteriorly* with nasopharynx through pharyngotympanic tube.
- Posteriorly* with mastoid antrum and mastoid air cells through aditus to antrum called **aditus ad antrum**.

### Contents of the Middle Ear

*Inside the mucous lining:*

1. Air.



**Fig. 18.9** Subdivisions of the middle ear.

*Outside the mucous lining:*

1. Three small bones called ear ossicles: malleus, incus, and stapes.
2. Two muscles: tensor tympani and stapedius.
3. Two nerves: chorda tympani and tympanic plexus.
4. Vessels supplying and draining the middle ear.
5. Ligaments of the ear ossicles.

**N.B.** The mucous membrane lining of the middle ear invests all the structures within it and forms several folds, which project into the cavity giving it a honey-comb appearance.

Thus strictly speaking, the middle ear contains only air.

### Subdivisions of the Middle Ear (Fig. 18.9)

The tympanic cavity extends much beyond the limits of tympanic membrane, which forms its lateral boundary. It is divided into three parts, viz.

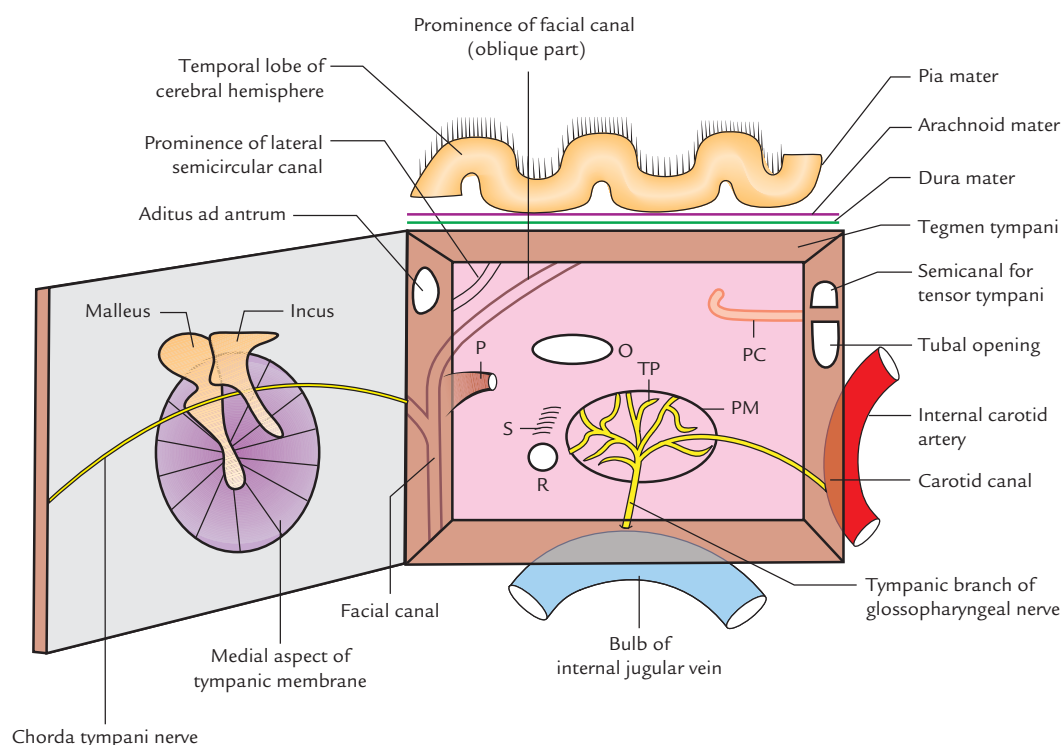
1. **Epitympanum** (attic), a part above the tympanic membrane containing head of malleus, body, and short process of incus.
2. **Mesotympanum**, a part opposite to tympanic membrane containing handle of malleus, long process of incus, and stapes. It is the narrowest part of the middle ear.
3. **Hypotympanum**, a part below the tympanic membrane.

### Boundaries (Fig. 18.10)

The middle ear is likened to a six-sided box and thus presents six walls, namely,

1. Roof.
2. Floor.
3. Anterior wall.
4. Posterior wall.
5. Medial wall.
6. Lateral wall.





**Fig. 18.10** Schematic diagram to show the boundaries (and their relations) of the middle ear. The middle ear is likened to a six-sided box and its lateral side is opened out (O = oval window, P = pyramid, PC = processus cochleariformis, PM = promontory, R = round window, S = sinus tympani, TP = tympanic plexus).

**Roof:** It is formed by a thin plate of bone called **tegmen tympani**. It separates the tympanic cavity from the middle cranial fossa. The tegmen tympani also extends posteriorly to form the roof of **aditus ad antrum**.

**Floor:** The floor is also formed by a thin plate of bone, which separates the tympanic cavity from the jugular bulb. Sometimes it is congenitally deficient and the jugular bulb then projects into the middle ear, being separated from cavity only by mucosa.

The **tympanic branch** of glossopharyngeal nerve pierces the floor between the jugular fossa and lower opening of the carotid canal and enters the tympanic cavity to take part in the formation of **tympanic plexus**.

**Anterior wall:** It is formed by a thin plate of bone. In the lower part it separates the cavity from internal carotid artery. The upper part of anterior wall presents two openings or canals, the upper one for the tensor tympani muscle and the lower one for the auditory tube.

The bony partition between the two canals extends backwards along the medial wall in the tympanic cavity as a curved lamina called **processus cochleariformis**.

**Posterior wall:** The posterior wall separates the tympanic cavity from mastoid antrum and mastoid air cells, and presents the following features:

1. **Aditus ad antrum**, an opening in the upper part through which tympanic cavity communicates with the mastoid antrum.
2. **Fossa incudis**, a small depression close to the aditus, lodging the short process of the incus.
3. **Pyramid**, a hollow conical bony projection below the aditus containing stapedius muscle whose tendon appears through its summit, passes forwards to be attached to the neck of the stapes.
4. **Vertical part of facial canal** runs in the posterior wall just behind the pyramid and descends up to the stylomastoid foramen.
5. **Posterior canaliculus for chorda tympani**, a small aperture for emergence of this nerve.

**Medial wall:** It separates the tympanic cavity from the internal ear; thus it is actually formed by the bony lateral wall of the internal ear. The medial wall presents the following features:

1. **Promontory**, a rounded prominence in the centre produced by first (basal) turn of the cochlea. The tympanic branch of the glossopharyngeal nerve ramifies on it to form **tympanic plexus**.
2. **Oval window (fenestra vestibuli)**, a reniform aperture located above and behind the promontory. It is closed by the base of stapes and annular ligament.

# Orbit and Eyeball

## ORBIT

The orbits are a pair of pyramidal-shaped bony cavities, located one on either side of the root of the nose and provides sockets for rotatory movements of the eyeballs. Each orbit is a four-sided pyramid with apex directed behind at the optic canal and base forward, represented by the orbital margin. The medial walls of two orbits are parallel to each other but the lateral walls are set at right angle to each other (Fig. 19.1).

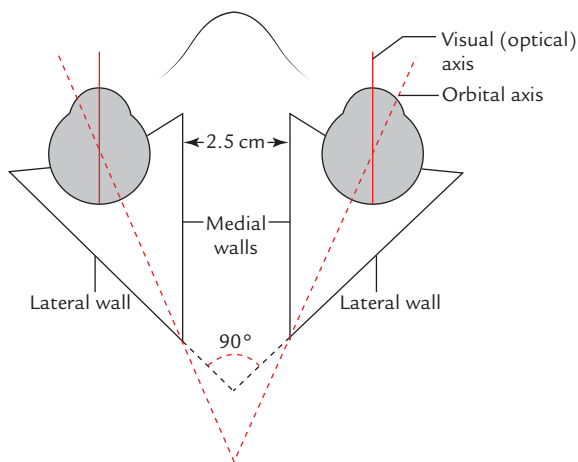
The long axis of each orbit (**orbital axis**) passes backwards and medially.

## BOUNDARIES

The boundaries of the orbit (Fig. 19.2) are as follows:

**Medial wall (thinnest):** Formed by four bones; from before backwards these are as follows:

1. Frontal process of maxilla.
2. Lacrimal process of maxilla.



**Fig. 19.1** Diagrammatic horizontal section through orbits showing: angle formed between lateral walls, distance between medial walls and their direction (parallel to each other), visual and orbital axes.

3. Orbital plate of ethmoid.
4. Body of sphenoid.

**Lateral wall (strongest):** Formed by two bones, viz.

1. Orbital surface of the zygomatic bone in front.
2. Orbital surface of greater wing of sphenoid behind.

**Floor:** Formed by three bones:

1. Orbital surface of the body of maxilla.
2. Orbital surface of the zygomatic bone, anterolaterally.
3. Orbital process of the palatine bone, posteromedially.

**Roof:** Formed by two bones, viz.

1. Orbital plate of the frontal bone in front.
2. Lesser wing of the sphenoid behind.

## Presenting Features in Four Boundary Walls of Orbit (Fig. 19.3)

**Medial wall:** Presents two features, viz.

1. Lacrimal fossa is bounded in front by the anterior lacrimal crest of maxilla and behind by the posterior lacrimal crest of the lacrimal bone. The lacrimal fossa communicates with the nasal cavity through nasolacrimal canal. The lacrimal fossa and nasolacrimal canal lodge lacrimal sac and nasolacrimal duct, respectively.
2. Anterior and posterior ethmoidal foramina, lie at the junction between medial wall and the roof of the orbit.

**Lateral wall:** Presents two features, viz.

1. Zygomatic foramen. Sometimes there are two small foramina, for zygomaticofacial and zygomaticotemporal nerves.
2. Whitnall's tubercle, a small bony tubercle just behind the lateral orbital margin and slightly below the frontozygomatic suture.

**Floor:** Presents two features, viz.

1. Infraorbital groove and canal.
2. Small rough impression in anteromedial angle for origin of inferior oblique muscle.

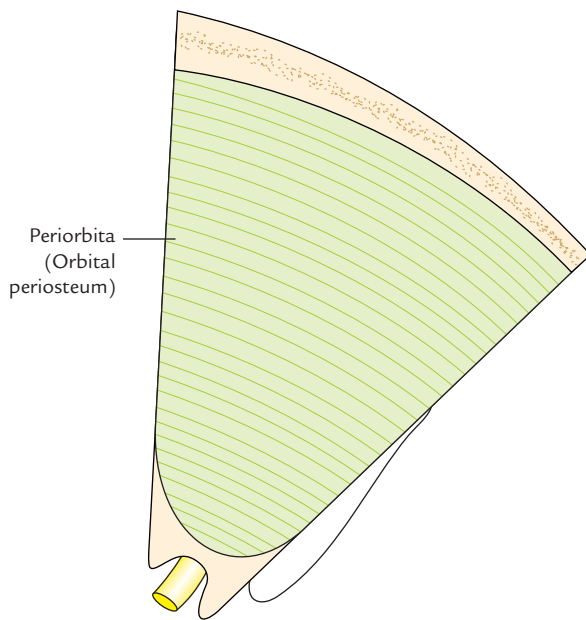


Fig. 19.5 Periorbital.

## ORBITAL FASCIA OR PERIORBITA

It is the periosteum of the bony orbit, which lines the bony boundaries of the orbit and forms a funnel-shaped fascial sheath that encloses the orbital contents (Fig. 19.5). It is loosely attached to the bones, hence can be easily stripped off especially from roof and medial wall of the orbit.

At the optic canal and superior orbital fissure, it becomes continuous with the periosteum lining the interior of the skull (*endocranium*). At the infraorbital fissure and orbital margins, it becomes continuous with the periosteum covering the external surface of the skull (*periosteum*).

## CONTENTS

Contents of the orbit includes the following structures (Fig. 19.4):

1. Eyeball (most important content).
2. Muscles.
3. Fascia bulbi.
4. Nerves
  - (a) Optic
  - (b) Oculomotor
  - (c) Trochlear
  - (d) Abducent
  - (e) Ophthalmic
  - (f) Ciliary ganglion.
5. Ophthalmic artery.
6. Ophthalmic veins.
7. Lacrimal gland.
8. Orbital fat.

All these structures lie within the orbital periosteum that lines the bony walls of the orbit.

**N.B.** Infraorbital nerve and vessels and zygomatic nerve lie outside the orbital periosteum, hence not included in the contents of the orbit.

## EXTRAOCULAR MUSCLES

The extraocular muscles are classified into two groups: voluntary and involuntary.

### VOLUNTARY MUSCLES

There are seven voluntary muscles in the orbit. Of these, six muscles move the eyeball and one muscle moves the upper eyelid.

The muscles moving the eyeball are four recti and two oblique muscles. The one which moves the upper eyelid is called *levator palpebrae superioris* (Fig. 19.6):

1. **Four recti muscles**
  - (a) Superior rectus,
  - (b) Inferior rectus,
  - (c) Medial rectus, and
  - (d) Lateral rectus.
2. **Two oblique muscles**
  - (a) Superior oblique,
  - (b) Inferior oblique.
3. **One levator palpebrae superioris.**

### INVOLUNTARY MUSCLES

There are three involuntary/smooth muscles, viz. **superior tarsal or Muller's muscle, inferior tarsal and orbitalis.**

The voluntary muscles are described in detail because of their functional significance in the following text:

#### *Recti muscles*

**Origin:** All the recti arise from the corresponding margins of the common tendinous ring. The lateral rectus arises by two heads.

The common tendinous ring encloses the optic canal and middle part of the superior orbital fissure. It is attached medially to apex of the orbit and laterally to a small tubercle (tubercle of Zinn) on the lower border of superior orbital fissure.

**Insertion:** All the recti are inserted into sclera little posterior to the limbus (corneoscleral junction) in front of the equator of the eyeball. Average distance from limbus is:

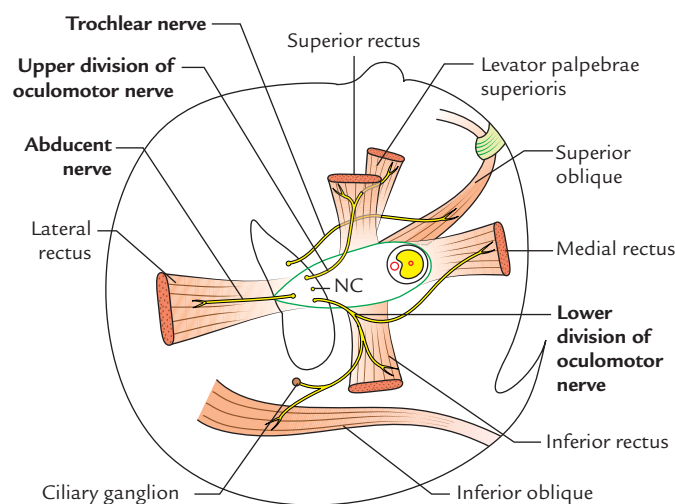
Medial rectus, 5 mm

Inferior rectus, 6 mm

Lateral rectus, 7 mm

Superior rectus, 8 mm

The origin and insertion of recti muscles are shown in Figure 19.6.



**Fig. 19.7** Nerve supply of the extraocular muscles (NC = nasociliary nerve).

## ACTIONS OF MUSCLES ON THE EYEBALL

The two eyes face forwards and their long axes (visual optic axes) lie in sagittal plane, parallel with each other and with medial walls of the orbits (Fig. 19.1). The lateral walls of orbits slope backwards and medially making a right (90°) angle with each other. The ocular muscles and optic nerve come from the apex of the orbit near the back of the medial wall and pass forwards and laterally to be attached to the eyeball. The actions of superior and inferior recti are, therefore, not straight, despite their name. The superior and inferior oblique muscles, therefore, have to act in concert with two recti in order to produce direct upward and downward movements of the eyeball.

### Axis of Movements of the Eyeball

- **Elevation and depression:** around the transverse axis passing through the equator.
- **Adduction and abduction:** around the vertical axis passing through the equator.
- **Rotation (torsion):** around the anteroposterior axis extending from anterior pole to posterior pole of the eyeball.

When 12 o'clock position of the cornea rotates medially, it is called *intorsion* and when it rotates laterally, it called *extorsion*.

### Actions of Individual Muscles

The actions of individual muscles are easily deduced from Starling's diagram (Fig. 19.8).

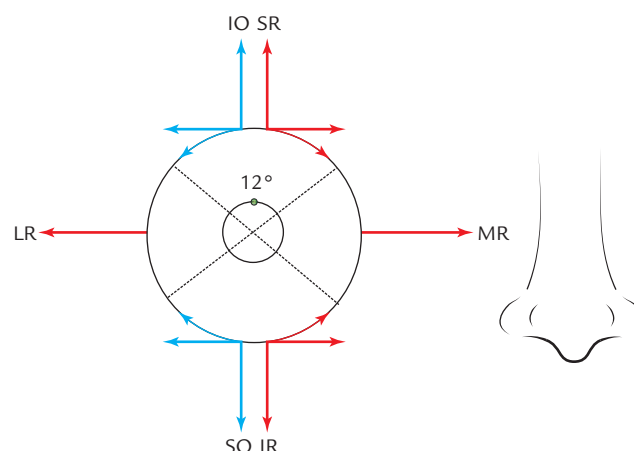
They are as follows:

*Superior rectus (SR):* elevation, adduction, and intorsion.

*Medial rectus (MR):* adduction.

*Inferior rectus (IR):* depression, adduction, and extorsion.

*Lateral rectus (LR):* abduction.



**Fig. 19.8** Schematic diagram (modified after Starling) showing actions of various extraocular muscles (IO = inferior oblique, IR = inferior rectus, LR = lateral rectus, MR = medial rectus, SR = superior rectus, SO = superior oblique).

*Superior oblique (SO):* depression, abduction, and intorsion.

*Inferior oblique (IO):* elevation, abduction, and extorsion.

### Associated Movements of the Eyeball

1. **Conjugate movements:** when both the eyes move in the same direction with visual axes being parallel.
2. **Disconjugate movements:** when the axes of both eyes converge or diverge.

### Clinical correlation

**Strabismus/squint:** Unilateral paralysis of an individual muscle due to involvement of the nerves produces strabismus or squint (deviation of eye to the opposite side) and may result in diplopia (double vision). In diplopia, the light from an object is not focused on the identical areas of both retinæ. The real image falls on the macula of the unaffected eye and false image falls on the peripheral part of retina of the paralyzed eye. The examples are as under:

| Muscle paralyzed | Nerve involved | Effects        |
|------------------|----------------|----------------|
| Lateral rectus   | Abducent       | Medial squint  |
| Medial rectus    | Oculomotor     | Lateral squint |

### Levator palpebrae superioris (LPS) muscle

**Origin:** from undersurface of the lesser wing of sphenoid at the apex of the orbit, above the common tendinous ring by a narrow tendon.

**Insertion (Fig. 3.22):** the muscle broadens as it passes forwards from its origin and divides into three lamellæ:

1. Upper lamella consisting of skeletal muscle penetrates the orbital septum passes through the fibres of orbicularis oculi to be inserted into the skin of upper eyelid.

the optic nerve and pass forwards in the space between the sclera and choroid to reach the target organs.

### Clinical correlation

The ciliary ganglion is blocked to produce dilatation of pupil before cataract extraction.

### Infraorbital and Zygomatic Nerves (Fig. 19.16)

They lie outside the periosteum of the orbit. They are described in detail in Chapter 10.

## OPHTHALMIC ARTERY

It is branch of the internal carotid artery. It arises from internal carotid artery as it emerges from the roof of the cavernous sinus medial to anterior clinoid process close to the optic canal. The artery enters the orbit through optic canal inferolateral to the optic nerve, both lying in a common dural sheath (Fig. 19.17).

The artery pierces the dura mater, ascends over the lateral side of the optic nerve to cross it superiorly from lateral to medial side along with the nasociliary nerve. It then runs forwards along the medial wall of the orbit tortuously to allow the movements of the eyeball. Near the medial angle of the eye it terminates by dividing into two branches: **supratrochlear** and **dorsal nasal**.

### Branches

The branches of the ophthalmic artery are as follows:

1. **Central artery of the retina** (first and most important) arises from ophthalmic artery (still in dural sheath) below the optic nerve, runs forwards in the dural sheath

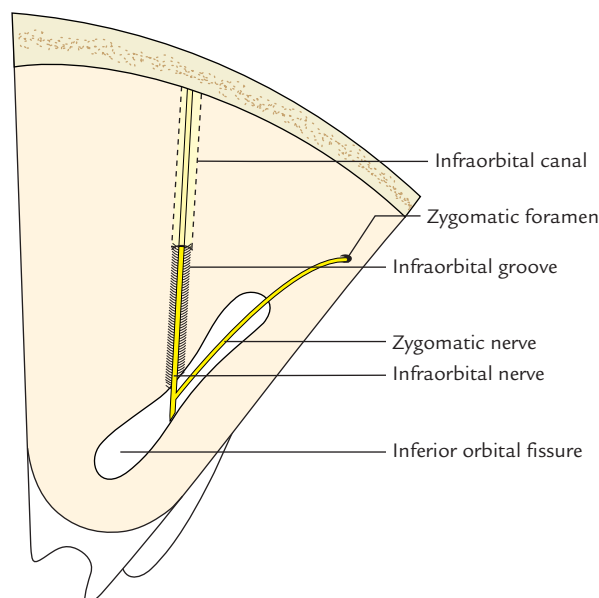


Fig. 19.16 Infraorbital and zygomatic nerves.

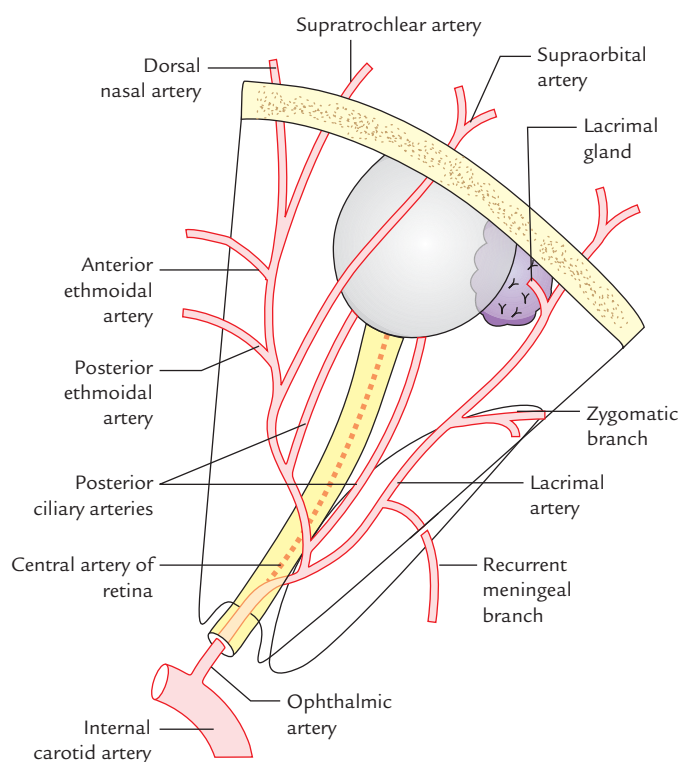


Fig. 19.17 Ophthalmic artery.

and pierces the optic nerve inferomedially about 1.25 cm behind the eyeball. The central artery reaches the optic disc through the central part of the nerve. It supplies the optic nerve and inner 6/7 layers of the retina.

### Clinical correlation

The central artery of retina is an example of a typical end artery. Its damage produces sudden total blindness on the side of the lesion.

2. **Lacrimal artery** arises from ophthalmic artery just before it crosses the optic nerve. It passes forwards along the upper border of lateral rectus and supplies the lacrimal gland, eyelids, and conjunctiva. The lacrimal artery gives off the following branches:
  - (a) Glandular branches to lacrimal gland.
  - (b) Two lateral palpebral arteries—one to each eyelid.
  - (c) Two zygomatic branches: zygomaticofacial and zygomaticotemporal.
  - (d) Recurrent meningeal branch runs backwards to enter the middle cranial fossa through the superior orbital fissure.
  - (e) Muscular branches.
3. **Posterior ciliary arteries** consist of two sets: long and short, both of which pierce the sclera of the eyeball around the optic nerve and chiefly supply the choroid and sclera. Long ciliary arteries are usually two and short